

# Physics Performance & Assessment Tasks

The complete book, translated and solved — step by step

كتاب الأداءات والتقييمات — ترجمة كاملة مع الحل  
بالخطوات

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Every lesson of the ministry assessments book, rewritten in scientific English, with every figure redrawn from scratch and every question worked through in full. The complete answer key is gathered at the back of the book.

19 handouts

483 questions

167 original figures

290 pages

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**Mr. Gemy**

PHYSICS · ENGLISH SECTION

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QUESTIONS

# Unit One

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Mechanics

الوحدة الأولى · الميكانيكا

# Velocity Vectors and Relative Velocity

Classwork, home assignments and the weekly assessment — fully worked, step by step.

Grade 11 · Egyptian Baccalaureate

35 questions

Step-by-step solutions

 $g = 10 \text{ m/s}^2$ 

**Mr. Gemy**  
PHYSICS

## The toolkit for this lesson

$$v_x = v \cos \theta$$

horizontal component

$$v_y = v \sin \theta$$

vertical component

$$v = \sqrt{v_x^2 + v_y^2}$$

magnitude from components

$$\tan \theta = v_y / v_x$$

direction of the vector

$$R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

resultant of two vectors

$$v_{AB} = v_A - v_B$$

velocity of A relative to B

**Before you start.** Relative velocity is a **vector subtraction**, never a plain arithmetic one. Fix a positive direction first, give every velocity its sign, and only then subtract. For perpendicular velocities use Pythagoras.

## Resolving vectors and finding the resultant

- 1 The figure shows eight velocity vectors drawn from the origin. **Resolve each vector** into its horizontal (x) component and its vertical (y) component.

RESOLUTION

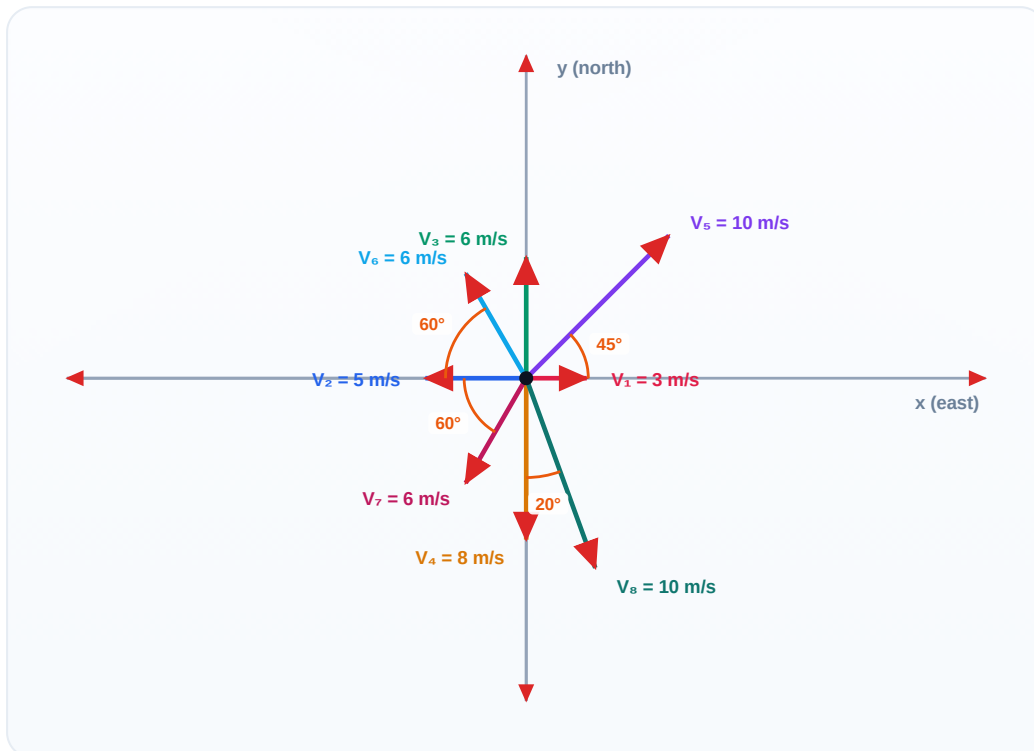


FIGURE 1 · EIGHT VELOCITY VECTORS

- 2 Find the **resultant** of each pair of displacement vectors below, giving its magnitude and its direction, and show it on a sketch.

RESULTANT

(A)

(B)

(C)

## Components, resultants and the angle of projection

COMPONENTS

- 3 An aircraft takes off with a **constant** velocity of 100 m/s directed at  $60^\circ$  above the horizontal.

- Calculate the horizontal component of its velocity.
- Calculate the vertical component of its velocity.
- Calculate the height it reaches 10 s after take-off.

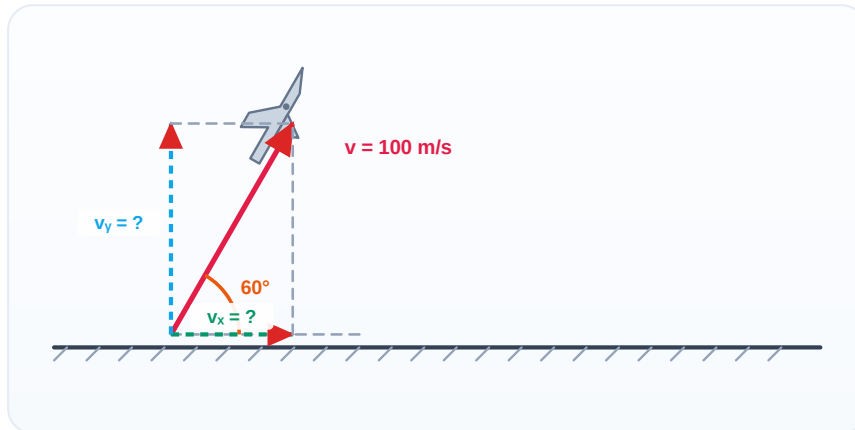


FIGURE 2

COMPONENTS

- 4 A body moves with a velocity  $v$  that makes an angle  $\theta$  with the horizontal. The horizontal component of the velocity is 3 m/s and the vertical component is 4 m/s. Find, with a sketch:

- the magnitude of the velocity  $v$ .
- the angle  $\theta$ .

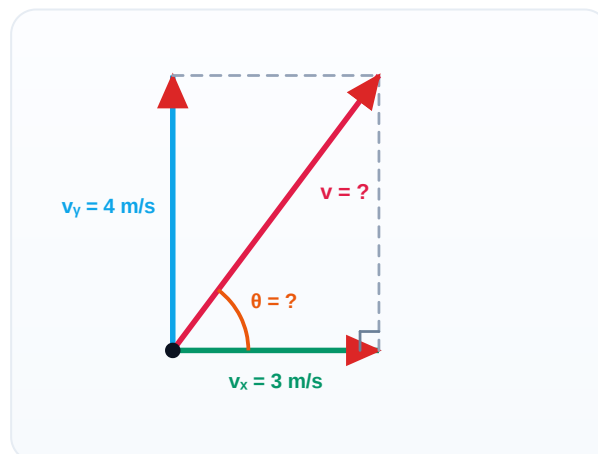


FIGURE 3

- 5 An aircraft climbs with a constant velocity of 100 m/s at an angle  $\theta$  to the horizontal, and reaches a height of 500 m after 10 s from take-off. **What is the value of  $\theta$ ?**

ANGLE

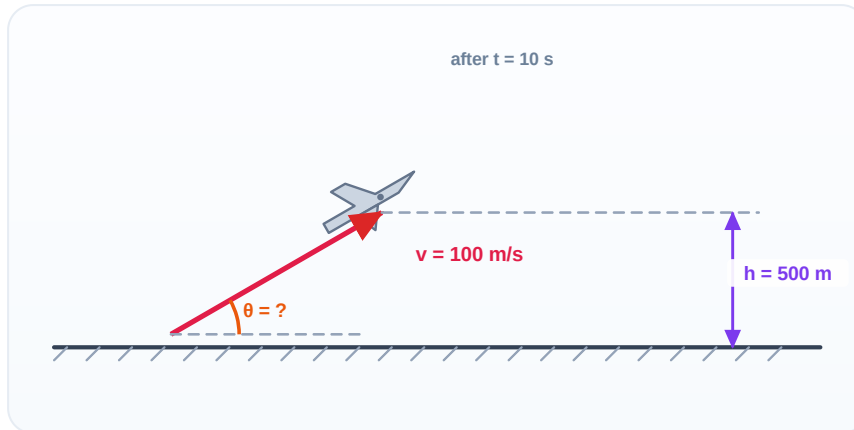
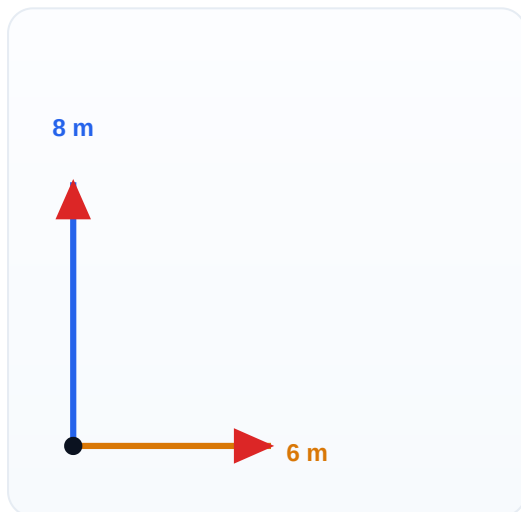


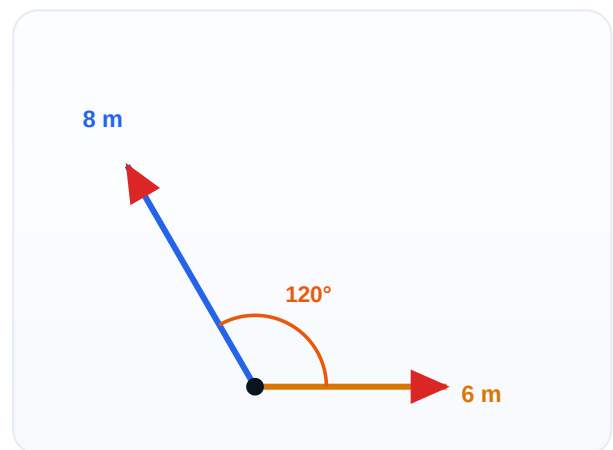
FIGURE 4

- 6 Find the resultant of each of the following pairs of displacement vectors, with a sketch.

RESULTANT



(A)



(B)

- 7 A body is projected at  $30^\circ$  above the horizontal and reaches a maximum height of 45 m. Its initial speed equals ..... ( $g = 10 \text{ m/s}^2$ )

MCQ

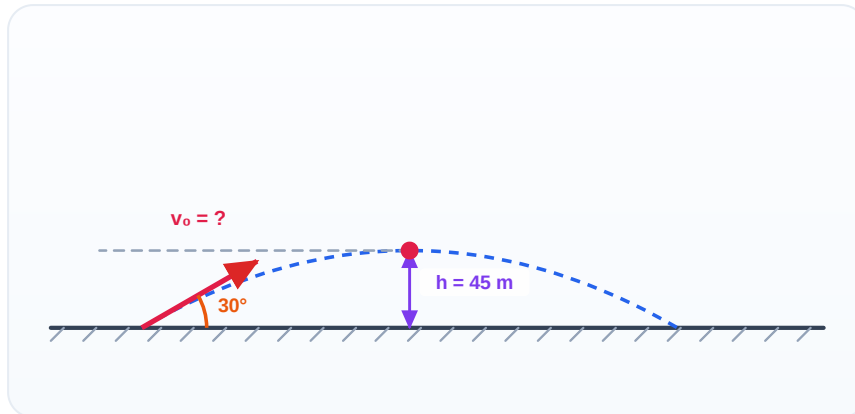


FIGURE 5

A 30 m/s

B 45 m/s

C 60 m/s

D 90 m/s

## CLASSWORK 2

### C Relative velocity

- 8 Two cars A and B move with speeds  $v_A$  and  $v_B$  respectively. **Write the expression for the magnitude of the velocity of B relative to A** in each of the following cases:

CONCEPT

**First:** the two cars move in the same direction.

**Second:** the two cars move in opposite directions.

**Third:** the velocity of A is perpendicular to the velocity of B.

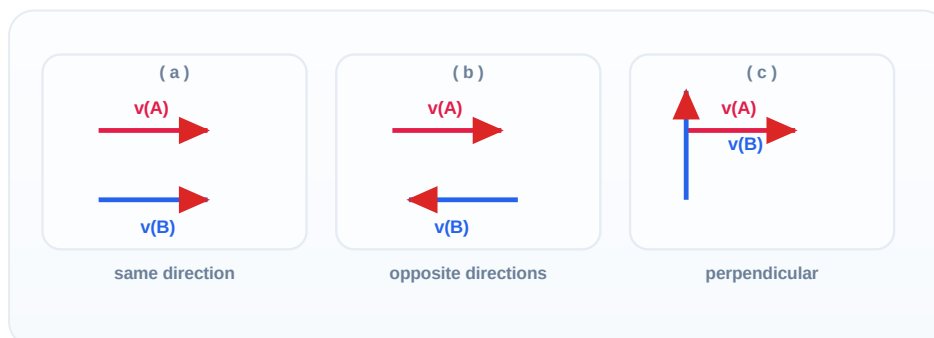


FIGURE 6 · THE THREE CASES

- 9 If the speed of body A ( $v_A$ ) is greater than the speed of body B ( $v_B$ ) and both move in the same direction, **how does the motion of A appear to an observer on B?** Explain your answer.

EXPLAIN

- 10 Car A moves at 60 m/s towards the **west**, while car B moves at 80 m/s towards the **north**. Find the magnitude and the direction of the velocity of car A relative to car B, and show your answer on a sketch.

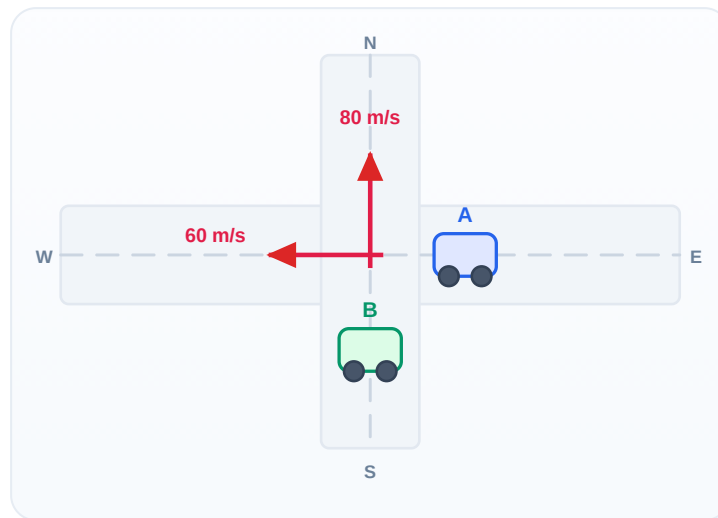


FIGURE 7

## Relative velocity — problems and multiple choice

- 11 If the speed of body A ( $v_A$ ) is **twice** the speed of body B ( $v_B$ ) and the two bodies move in **opposite** directions, what is the speed of B relative to A? Explain your answer.

EXPLAIN

- 12 A passenger sits beside the window of a bus moving along a straight horizontal road at 20 m/s. The rain, which is falling vertically, appears to this passenger to fall at an angle of  $40^\circ$  to the vertical.

PROBLEM

- a) Find the velocity of the raindrops relative to the ground.  
b) Explain why the rain does not appear to fall vertically to this passenger.

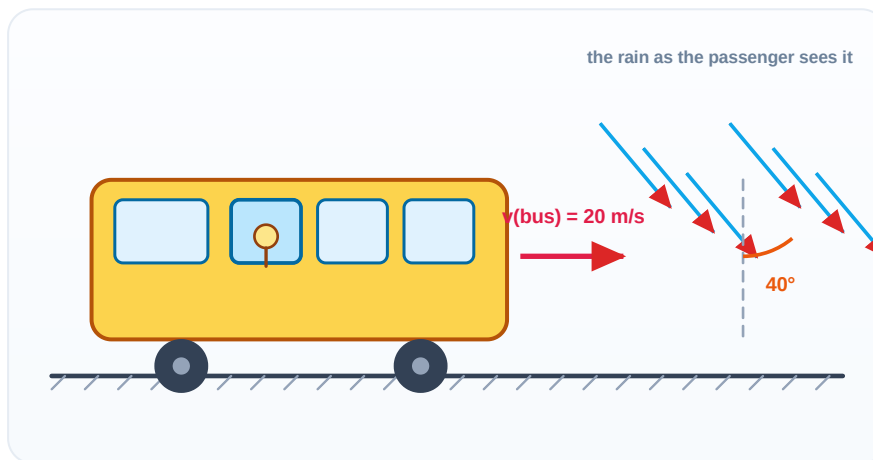


FIGURE 8 · THE BUS AND THE RAIN

- 13 Two bodies A and B move in the same direction with speeds  $v_A$  and  $v_B$  respectively. Which of the following relations gives the magnitude of the velocity of A relative to B ( $v_{AB}$ )?

MCQ

A  $v_{AB} = v_A - v_B$

B  $v_{AB} = \sqrt{(v_A^2 - v_B^2)}$

C  $v_{AB} = v_A + v_B$

D  $v_{AB} = \sqrt{(v_A^2 + v_B^2)}$

- 14 If the velocity of a body relative to another body equals zero, this means that the two bodies:

MCQ

A move with different speeds in the same direction

B move with equal speeds in the same direction

C move with equal speeds in opposite directions

D move with different speeds in opposite directions

15 Two cars move in opposite directions; the speed of the first is 30 m/s and that of the second is 20 m/s. The velocity of the first car relative to the second equals .....

MCQ

A 10 m/s

B 20 m/s

C 50 m/s

D 600 m/s

16 Car A moves at 40 m/s while car B moves at 25 m/s, both in the same direction. To the driver of car B, car A appears to:

MCQ

A move in the same direction at 15 m/s

B move in the opposite direction at 15 m/s

C move in the same direction at 65 m/s

D move in the opposite direction at 65 m/s

17 Body A moves at 30 m/s to the right while body B moves at 20 m/s to the left. Relative to body A, how does body B appear to move?

MCQ

A to the right at 10 m/s

B to the left at 10 m/s

C to the right at 50 m/s

D to the left at 50 m/s

18 Body A moves at 30 m/s towards the east while body B moves at 40 m/s towards the north. What is the **direction** of the velocity of B relative to A?

MCQ

A 35° north of west

B 53.1° north of west

C 35° north of east

D 53.1° north of east

19 If body A is faster than body B and both move in the same direction, then relative to B, body A appears to:

MCQ

A move in the opposite direction at a speed less than its actual speed

B move in the same direction at a speed greater than its actual speed

C move in the opposite direction at a speed greater than its actual speed

D move in the same direction at a speed less than its actual speed

20 Body A moves at 30 m/s to the right while body B moves at 40 m/s vertically upwards. What is the magnitude of the velocity of B relative to A?

MCQ

A 10 m/s

B 50 m/s

C 70 m/s

D 120 m/s

## Groups A, B and C

## Group A

- 21 Two bodies A and B move in the same direction. The speed of A is 20 m/s and the speed of B is 21 m/s. What is the magnitude of the velocity of A relative to B?

MCQ

A 8 m/s

B 32 m/s

C 10 m/s

D 1 m/s

- 22 The figure shows a resultant vector **R** produced by two vectors A and B. Which diagram correctly represents the pair of vectors A and B that form this resultant?

MCQ

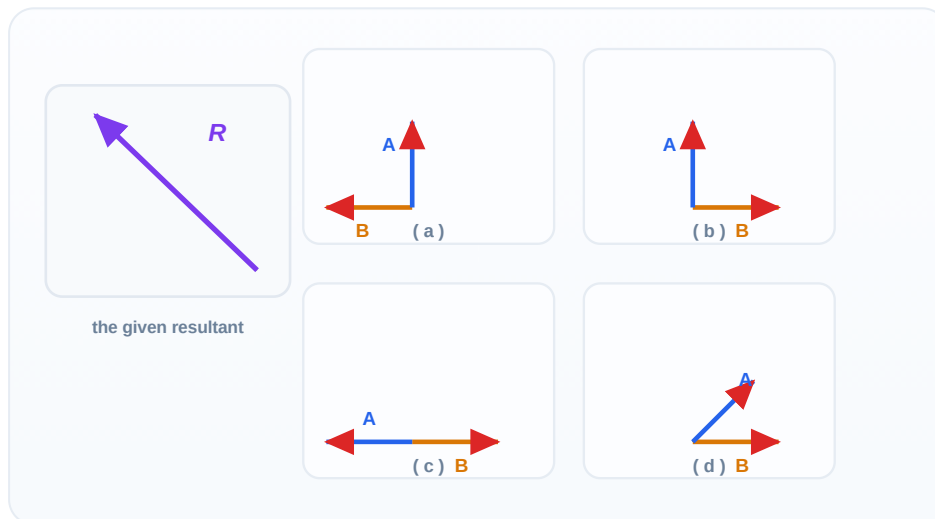


FIGURE 9 · THE RESULTANT AND THE FOUR OPTIONS

- 23 Two projectiles are launched with the **same initial speed** but at different angles to the horizontal. Projectile A is launched at a larger angle than projectile B. **Which one reaches the greater maximum height?** Explain your answer.

EXPLAIN

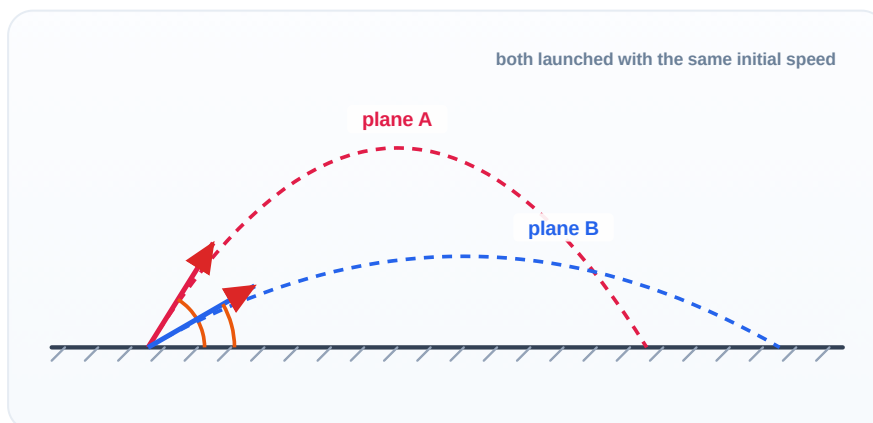


FIGURE 10

- 24 A ferry captain wants his ferry to land at the point exactly opposite its starting point on the other bank of a flowing river. **How must he direct the ferry?** Explain.

EXPLAIN

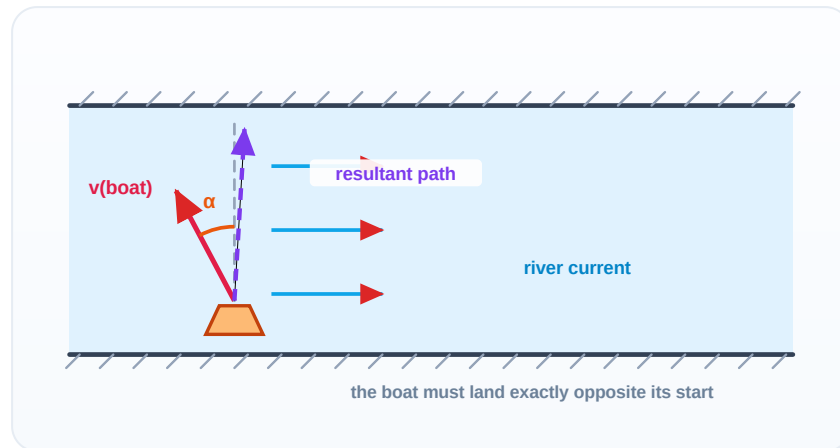


FIGURE 11

- 25 Car A moves at 30 m/s towards the **east** while car B moves at 40 m/s towards the **south**. Find the magnitude and the direction of the velocity of car A relative to car B.

PROBLEM

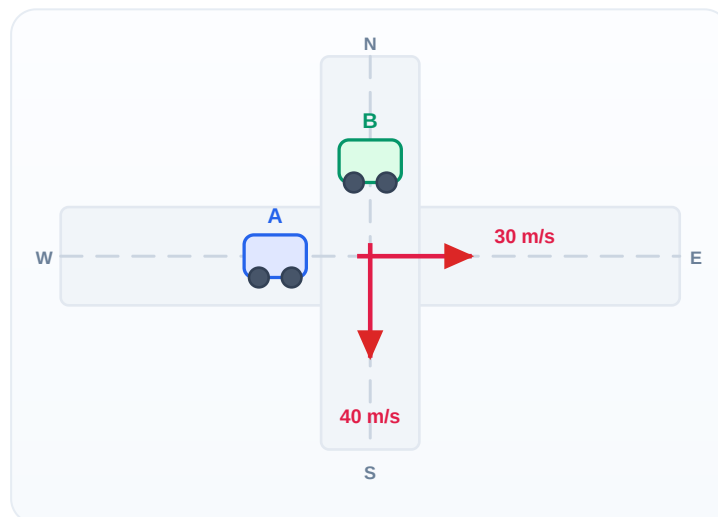


FIGURE 12

## Group B

- 26 A body is projected with an initial speed of 40 m/s and reaches a maximum height of 20 m. Its angle of projection above the horizontal is ..... ( $g = 10 \text{ m/s}^2$ )

MCQ

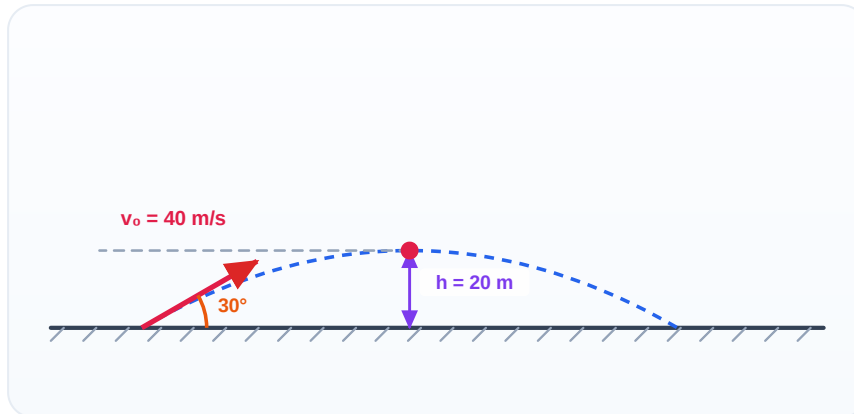


FIGURE 13

- A  $15^\circ$       B  $30^\circ$   
C  $45^\circ$       D  $60^\circ$

- 27 A body moves with a velocity  $v$  making an angle  $\theta$  with the horizontal. The horizontal component of the velocity is 5 m/s and the vertical component is 5 m/s. Find, with a sketch:

COMPONENTS

- 1) the magnitude of the velocity  $v$ .
- 2) the angle  $\theta$ .

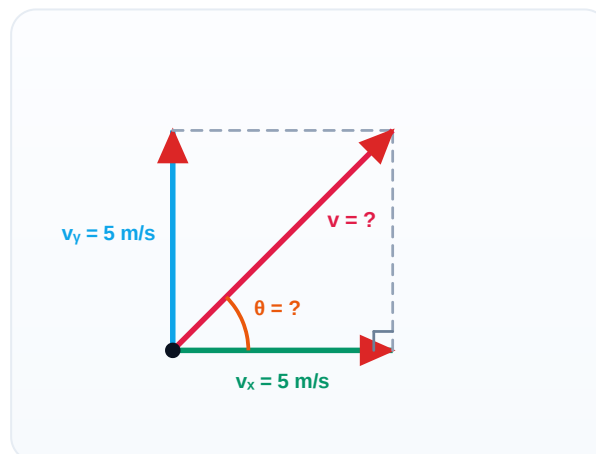


FIGURE 14

- 28 If the speed of body A ( $v_A$ ) is **half** the speed of body B ( $v_B$ ) and both move in the same direction, how does the motion of A appear relative to B? Explain.

EXPLAIN

- 29 What happens to the relative velocity between two bodies moving with **equal speeds in the same direction**? Explain your answer.

EXPLAIN

- 30 When does the resultant of two vectors that are equal in magnitude and opposite in direction vanish? Explain your answer.

EXPLAIN

### Group C

- 31 If two bodies move with equal speeds in opposite directions, the magnitude of the relative velocity between them is:

MCQ

- A equal to the speed of one of them  
B equal to half the speed of one of them  
C equal to twice the speed of one of them  
D equal to four times the speed of one of them

- 32 Body A moves at 30 m/s and body B moves at 20 m/s in the same direction. Relative to body A, how does body B appear to move?

MCQ

- A in the same direction at 10 m/s  
B in the opposite direction at 10 m/s  
C in the same direction at 50 m/s  
D in the opposite direction at 50 m/s

- 33 An aircraft climbs with a constant speed of 80 m/s at an angle of  $30^\circ$  above the horizontal.

COMPONENTS

- a) Calculate the horizontal component of its velocity.  
b) Calculate the vertical component of its velocity.

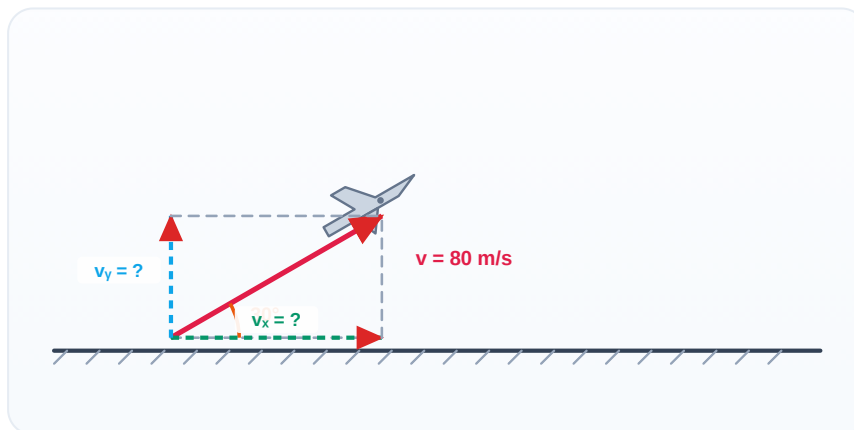


FIGURE 15

- 34 The current of a river stays constant, but the ferry increases its speed relative to the water while its bow is kept pointing straight across the river. **What happens to the drift of the ferry along the direction of the current?** Explain.

EXPLAIN

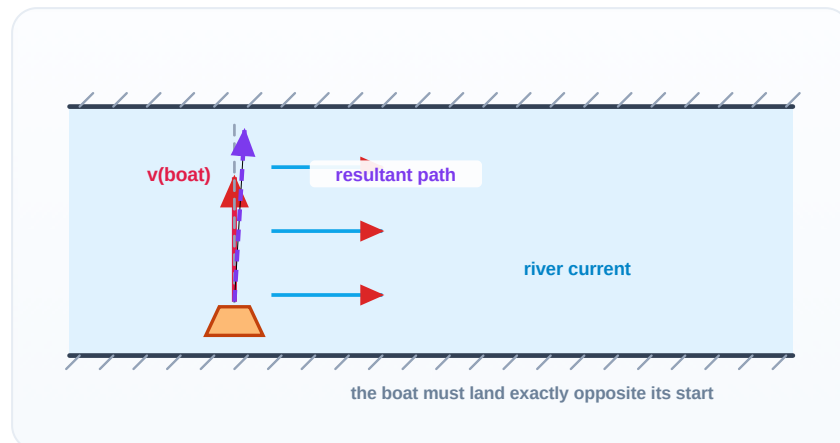


FIGURE 16

- 35 A body is projected with a speed  $v$  at an angle  $\theta$  above the horizontal, and reaches a height  $h$  after a time  $t$  from the instant of projection. **Explain, in two different ways,** how the height reached in the same time interval can be increased.

EXPLAIN

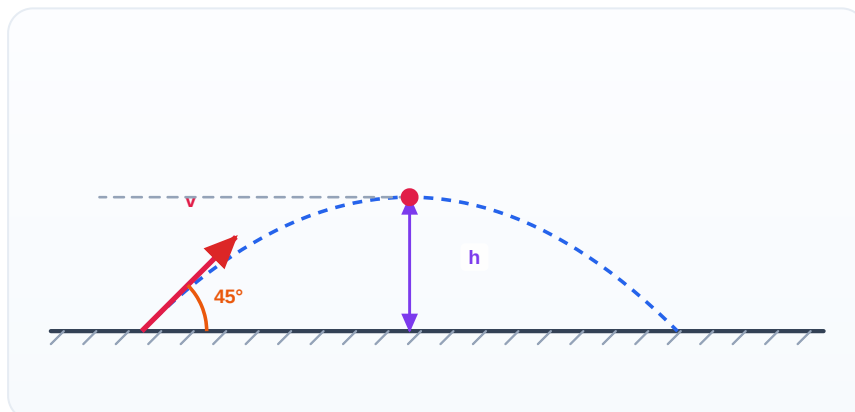


FIGURE 17

# Horizontal Projectile Motion

Classwork, home assignments and the weekly assessment — fully worked, step by step.

Grade 11 · Egyptian Baccalaureate

28 questions

Step-by-step solutions

 $g = 10 \text{ m/s}^2$  unless stated
**Mr. Gemy**  
PHYSICS

## The toolkit for this lesson

$$a_x = 0, a_y = g$$

the two motions are independent

$$h = \frac{1}{2} g t^2$$

vertical distance fallen

$$t = \sqrt{2h / g}$$

time of flight — height only

$$x = v t$$

horizontal range

$$v_y = g t$$

vertical component at any instant

$$v = \sqrt{v^2 + v_y^2}$$

speed at any instant

**The one idea behind the whole lesson.** A horizontal projectile falls exactly like a body dropped from rest, while at the same time it travels horizontally with a **constant** velocity. So the **time of flight depends on the height only**, never on the launch speed.

## Horizontal projectile motion

CONCEPT

- 1 The figure shows the path of a projectile fired horizontally from a cannon with an initial velocity  $v$  until it reaches the point B.
- At which of the points 1, 2 and 3 is the **horizontal** component of the velocity the greatest? Explain.
  - At which of the points 1, 2 and 3 is the **vertical** component of the velocity the greatest? Explain.
  - Write the relation used to calculate the horizontal distance AB.
  - Write the relation used to calculate the vertical distance CB.
  - If the projectile is fired from the same height with a doubled speed ( $2v$ ), what happens to: (1) the time it takes to reach the ground? (2) its horizontal range?

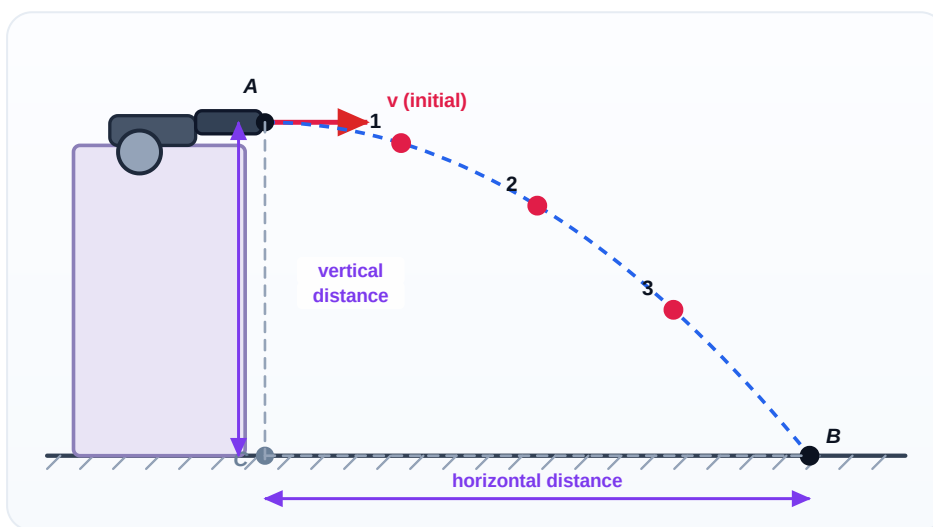


FIGURE 1 · THE PATH OF A HORIZONTAL PROJECTILE

PROBLEM

- 2 A body is projected horizontally with a velocity of  $14.7 \text{ m/s}$  from a height of  $19.6 \text{ m}$  above the ground. Taking  $g = 9.8 \text{ m/s}^2$ , find:
- the time the body takes to reach the ground, in seconds.
  - the horizontal range of the body, in metres.
  - the speed of the body just before it strikes the ground, in  $\text{m/s}$ .

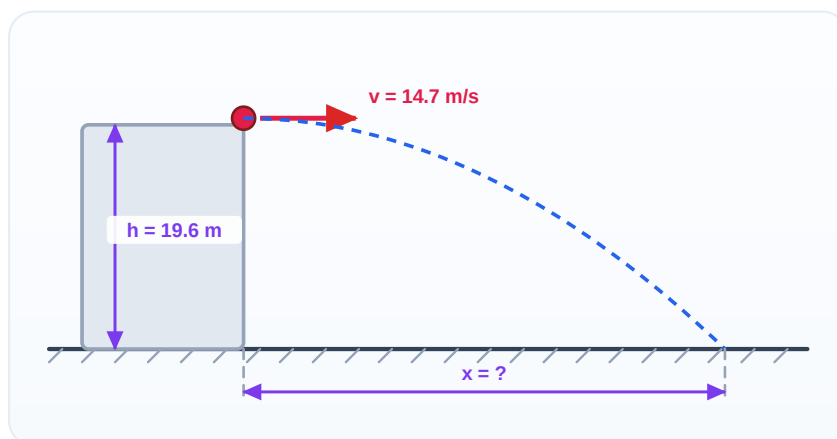


FIGURE 2

## B The two independent motions

- 3 The **horizontal** motion of a projectile launched horizontally from a certain height is characterised by:

MCQ

- A a constant acceleration
- B a constant velocity
- C an increasing acceleration
- D an increasing velocity

- 4 Doubling the initial horizontal velocity of a horizontal projectile, while keeping the height constant, leads to:

MCQ

- A doubling the time of flight
- B halving the time of flight
- C doubling the horizontal range
- D increasing the horizontal range four times

- 5 A body is projected horizontally with a velocity of  $4.0 \text{ m/s}$  from a certain height and reaches the ground after  $2.0 \text{ s}$ . Calculate the height from which it was projected and its horizontal range.

PROBLEM

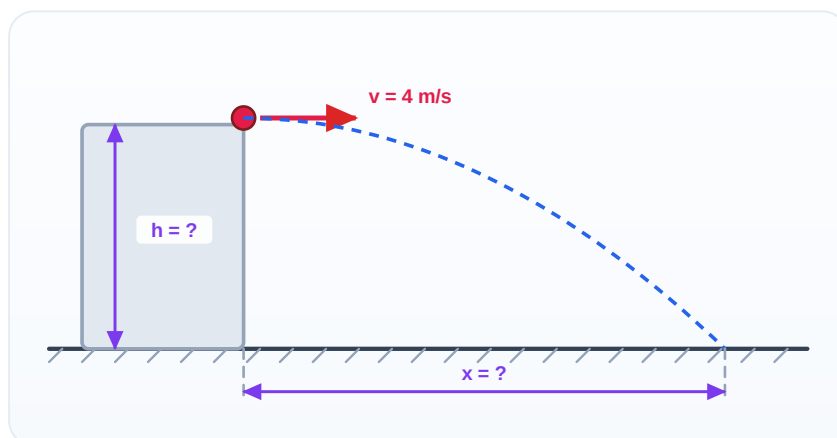


FIGURE 3

- 6 What happens to the horizontal range of a horizontally projected body if the **height** of the point of projection increases while the initial horizontal velocity stays constant? Explain.

EXPLAIN

- 7 A body is projected horizontally from the top of a building of height 45 m with an initial horizontal velocity of 20 m/s, neglecting air resistance and taking  $g = 10 \text{ m/s}^2$ . Calculate:

- 1) the time the body takes to reach the ground.
- 2) the horizontal distance covered before it reaches the ground.

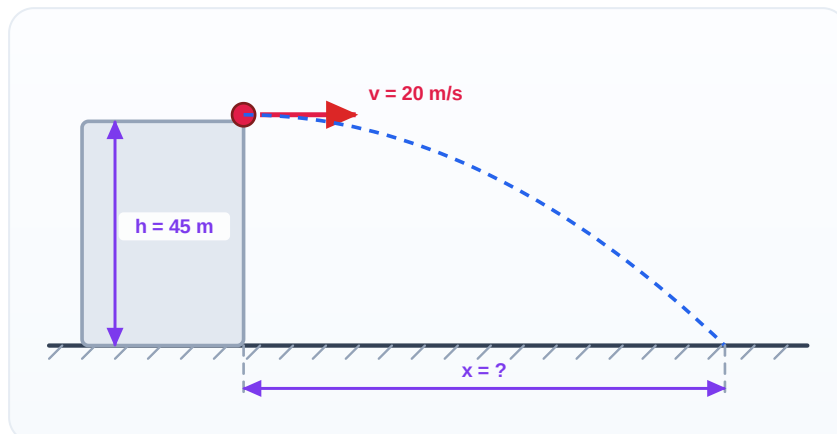


FIGURE 4

## CLASSWORK 2

C

## The laws of horizontal projectiles

- 8 Two projectiles are fired horizontally from two different heights as follows:

RATIOS

- the first from a height of 20 m with a horizontal velocity of 10 m/s.
- the second from a height of 80 m with a horizontal velocity of 20 m/s.

Neglecting air resistance and taking  $g = 10 \text{ m/s}^2$ , **find the ratio between:**

- 1) the time of flight of the first projectile and that of the second.
- 2) the horizontal range of the first projectile and that of the second.

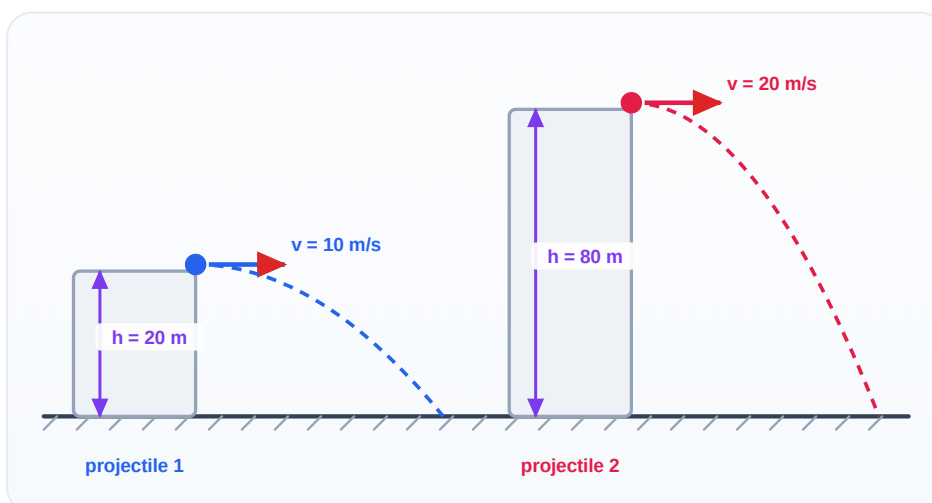


FIGURE 5

- 9 What happens to the time of flight and to the horizontal range of a body if it is projected horizontally from the **same height** but with a **greater** horizontal velocity? Explain.

EXPLAIN

## More on the laws of horizontal projectiles

10 The motion of a body projected horizontally from a certain height is characterised by:

MCQ

- ☐ A zero acceleration during the horizontal motion
 ☐ B zero acceleration during the vertical motion
 ☐ C an acceleration equal to  $g$  during the horizontal motion
 ☐ D an acceleration equal to  $\frac{1}{2}g$  during the vertical motion

11 What happens when the initial horizontal velocity of a horizontal projectile increases while the height is kept constant?

MCQ

- ☐ A the time of flight increases
 ☐ B the time of flight stays constant
 ☐ C the horizontal range stays constant
 ☐ D the horizontal range decreases

12 Two projectiles are fired horizontally from two different heights: the first from a height  $h$  with an initial horizontal velocity  $v$ , and the second from a height  $4h$  with an initial horizontal velocity  $2v$ . Taking  $g$  as constant, find the ratio between:

RATIOS

- 1) the time of flight of the first projectile and that of the second.
- 2) the horizontal range of the first projectile and that of the second.

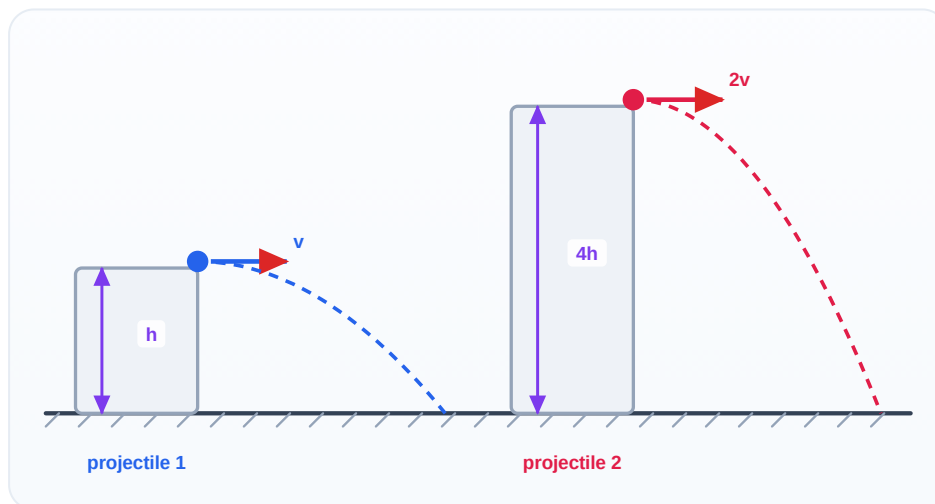


FIGURE 6

- 13 A billiard ball rolls off a table of height 0.60 m with an initial horizontal velocity of 2.4 m/s. Find the time the ball takes to hit the floor, and the horizontal distance between the edge of the table and the point where the ball lands.

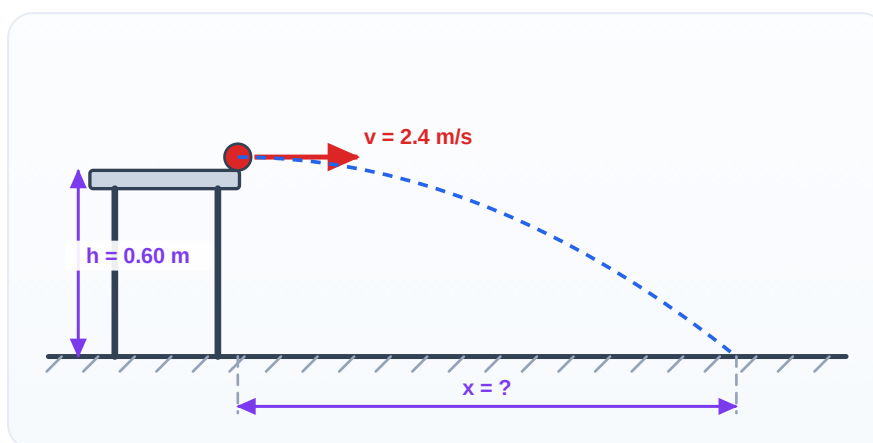


FIGURE 7

## WEEKLY ASSESSMENT 2

E

## Groups A, B and C

## Group A

- 14 A body is projected horizontally from the **fourth** floor of a building of ten floors. If the same body is projected horizontally from the **tenth** floor with the same initial horizontal velocity, then the time to reach the ground and the horizontal range respectively:

MCQ

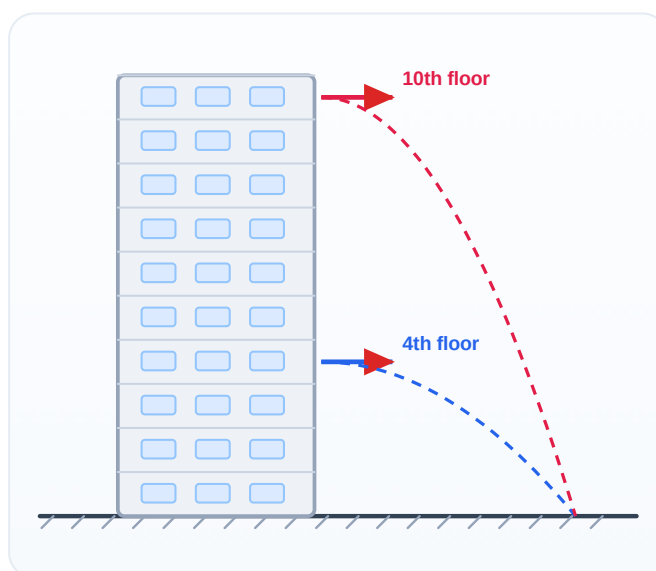


FIGURE 8

A stay constant · stay constant

B stay constant · increase

C increase · increase

D increase · stay constant

- 15 A helicopter flies horizontally with a velocity  $v$  at a height  $h$  and drops a food package so that it lands on a man standing on the ground. What is the distance between the helicopter and the man at the instant the package is released?

MCQ

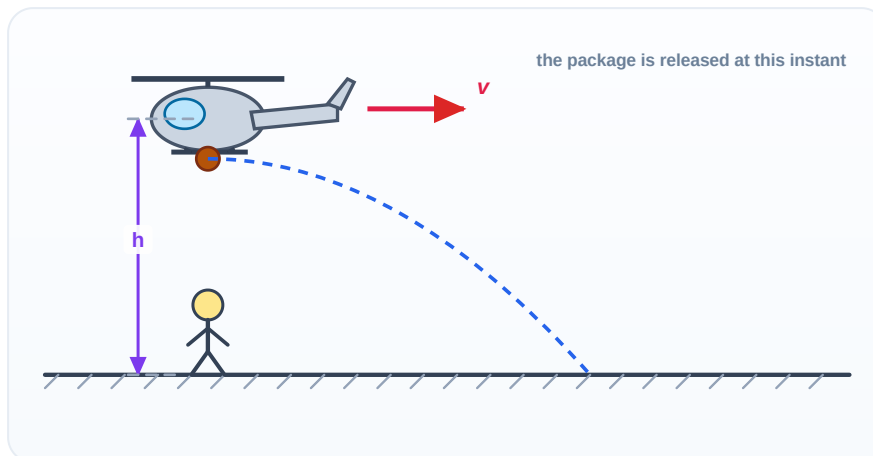


FIGURE 9

- A  $\sqrt{(2gh / v^2 + h^2)}$
- B  $\sqrt{(2v^2h / g + h^2)}$
- C  $\sqrt{(2ghv^2 + h^2)}$
- D  $\sqrt{(2ghv^2 + 1) / h^2)}$

- 16 A body is projected horizontally with a velocity of 6.0 m/s from a certain height and reaches the ground after 3.0 s. Calculate the height from which it was projected and its horizontal range.

PROBLEM

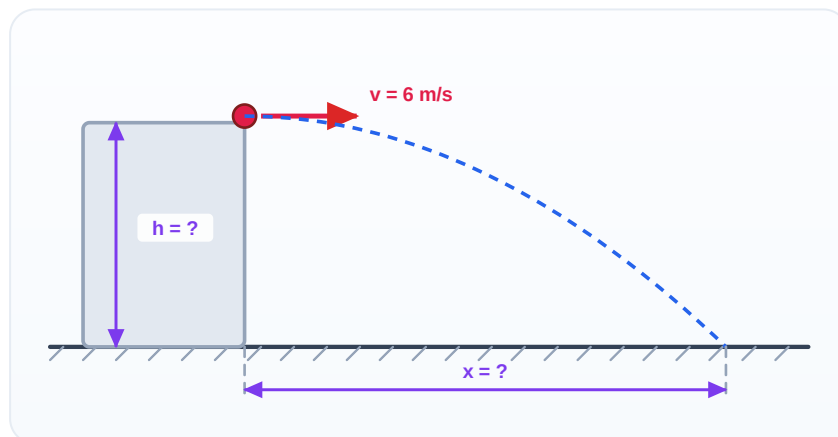


FIGURE 10

- 17 What happens to the time of flight and to the horizontal range of a body if it is projected horizontally from the same height but with a **smaller** horizontal velocity? Explain.

EXPLAIN

- 18** A billiard ball rolls off a table of height 0.75 m with an initial horizontal velocity of 1.5 m/s. What is the horizontal distance between the edge of the table and the point where the ball lands?

PROBLEM

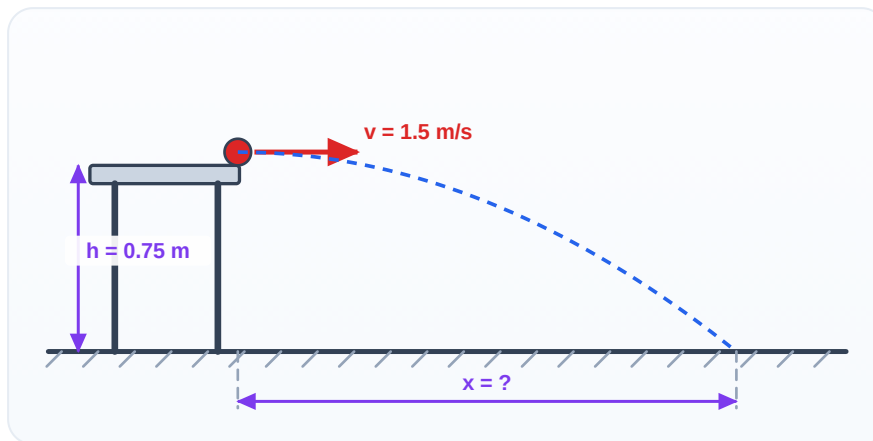


FIGURE 11

### Group B

- 19** A body is projected horizontally from a certain height with an initial velocity  $v$ . If the initial horizontal velocity is doubled while the height is kept constant, the time the body takes to reach the ground:

MCQ

- A** is doubled
- B** is halved
- C** does not change
- D** becomes four times as large

- 20** The motion of a body projected horizontally from a certain height is characterised by:

MCQ

- A** a positive acceleration during the horizontal motion
- B** a negative acceleration during the horizontal motion
- C** an acceleration equal to  $g$  during the vertical motion
- D** an acceleration equal to  $\frac{1}{4}g$  during the vertical motion

- 21** A body is projected horizontally from a height  $h$  with an initial horizontal velocity  $v$ , neglecting air resistance. What happens to the time it takes to reach the ground and to its horizontal range if the height is increased to  $4h$  while the initial horizontal velocity stays constant? Explain.

EXPLAIN

- 22 The figure shows two coins A and B. Coin A is projected horizontally with an initial velocity  $v_0$ , while coin B is released to fall freely from the same height at the same instant. **Which coin reaches the ground first?** Explain.

EXPLAIN

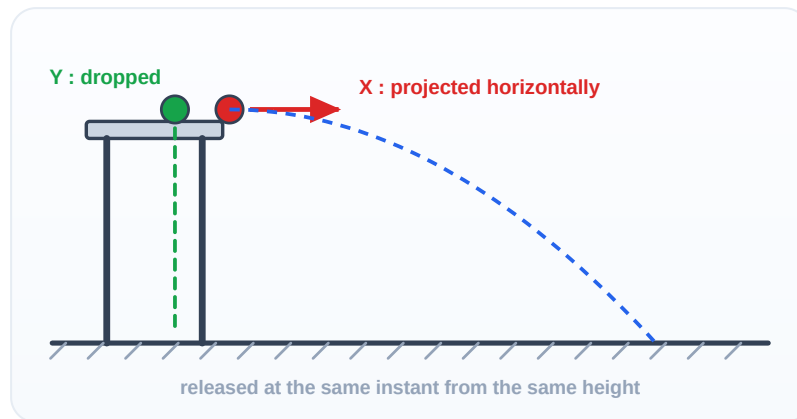


FIGURE 12

- 23 A body is projected horizontally with a velocity of 8 m/s from a height of 30 m above the ground. Taking  $g = 10 \text{ m/s}^2$ , find:

PROBLEM

- a) the time the body takes to reach the ground, in seconds.  
b) the horizontal range of the body, in metres.

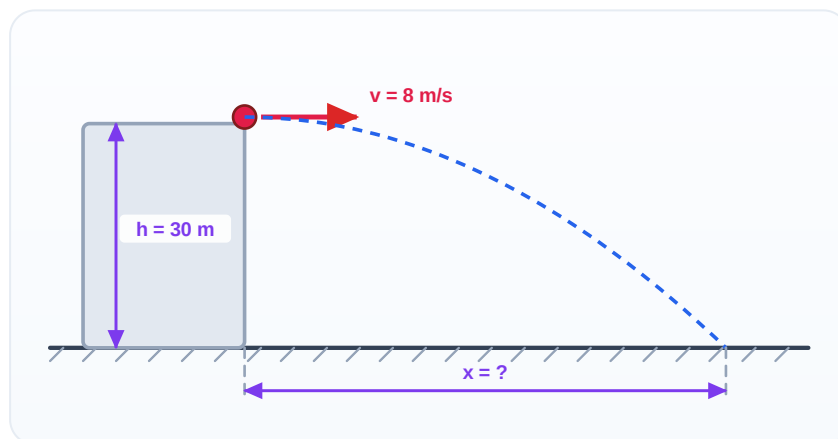


FIGURE 13

### Group C

- 24 Halving the initial horizontal velocity of a horizontal projectile, while keeping the height constant, leads to:

MCQ

- A halving the time of flight
- B doubling the time of flight
- C halving the horizontal range
- D doubling the horizontal range

25 The motion of a body projected horizontally from a certain height, while it is falling towards the ground, is characterised by:

MCQ

- A its horizontal velocity increases with time
- B its horizontal velocity decreases with time
- C its vertical velocity decreases with time
- D its vertical velocity increases with time

26 A ball is kicked horizontally from the top of a hill of height 22.0 m and lands at a horizontal distance of 35.0 m from the edge of the hill. **What is its initial horizontal velocity?**

PROBLEM

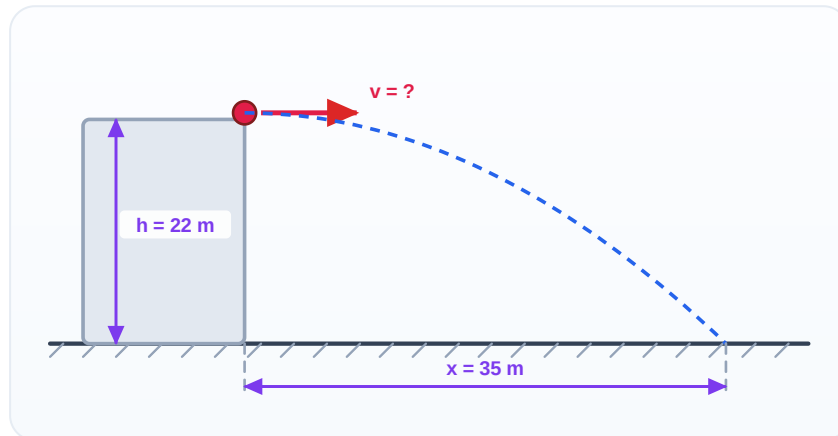


FIGURE 14

27 What happens to the time the body takes to reach the ground and to its horizontal range if it is projected horizontally from a **greater height** while its initial horizontal velocity stays constant? Explain.

EXPLAIN

28 A body is projected horizontally with a velocity of 3.0 m/s from a certain height and reaches the ground after 3.0 s. Find its horizontal range ( $x$ ) and the height ( $h$ ) from which it was projected.

PROBLEM

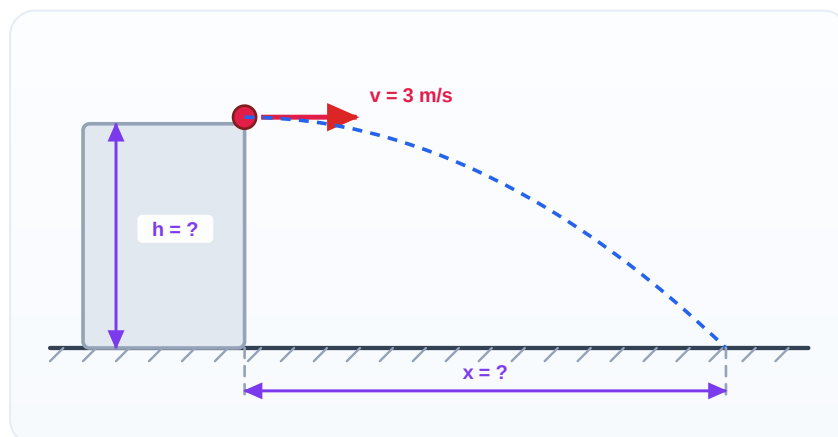


FIGURE 15

# Projectile Motion at an Angle

Two-dimensional motion — analysis, applications and the weekly assessment, fully worked.

Grade 11 · Egyptian Baccalaureate

27 questions

Step-by-step solutions

g = 10 m/s<sup>2</sup> unless stated
**Mr. Gemy**  
PHYSICS

## The toolkit for this lesson

$$v_{0x} = v_0 \cos \theta$$

constant all the way

$$v_{0y} = v_0 \sin \theta$$

changes by g each second

$$t_{\text{up}} = v_0 \sin \theta / g$$

time to the highest point

$$T = 2 v_0 \sin \theta / g$$

total time of flight

$$H = v_0^2 \sin^2 \theta / 2g$$

maximum height

$$R = v_0^2 \sin 2\theta / g$$

horizontal range

**The rule of the lesson.** Split the initial velocity into a horizontal part that never changes and a vertical part that behaves exactly like a body thrown straight up. At the highest point  $v_y = 0$ , but the body is still moving horizontally with  $v_0 \cos \theta$ .

## Analysing the motion of a projectile fired at an angle

DERIVATIONS

- 1 The figure shows a body projected from the ground with an initial velocity  $v_0$  at an angle  $\theta$  to the horizontal. Write the relation used to calculate each of the following:

a) the horizontal component of the initial velocity ( $v_{0x}$ ).

b) the vertical component of the initial velocity ( $v_{0y}$ ).

At position (1): c) the horizontal component  $v_{x1}$  · d) the vertical component  $v_{y1}$  · e) the magnitude of the resultant velocity  $v_1$ .

f) the horizontal distance  $X$  covered from the start up to position (1).

g) the vertical height  $Y$  risen from the start up to position (1).

h) the maximum height  $H$  reached by the body.

i) the horizontal range  $R$  of the body.

j) at position (2): the two components  $v_{x2}$  and  $v_{y2}$ .

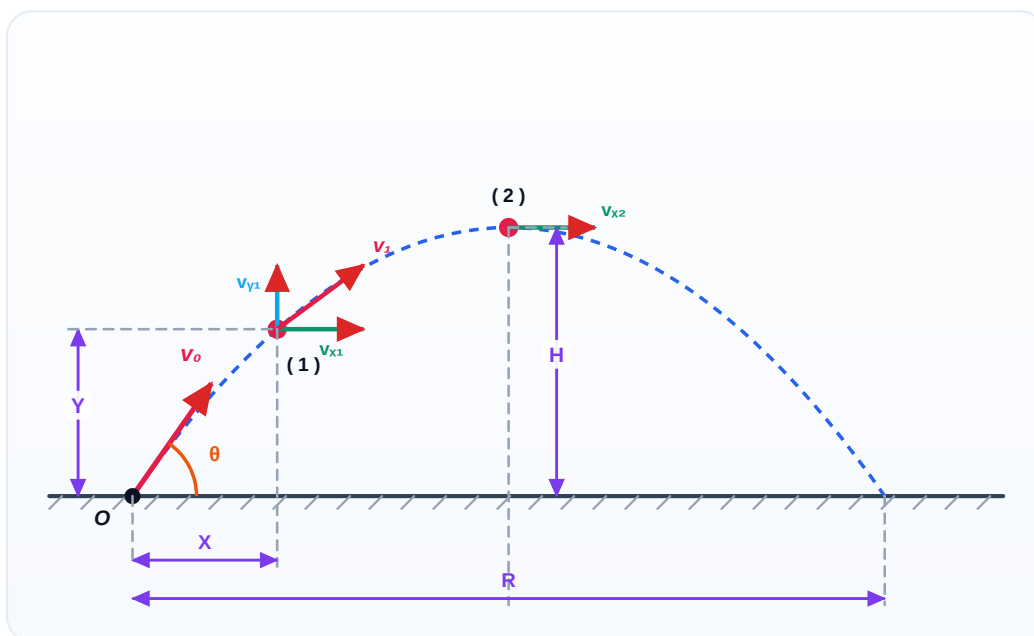


FIGURE 1 · THE COMPLETE PATH OF A PROJECTILE FIRED AT AN ANGLE

## HOME ASSIGNMENT 1

## The two motions of an angled projectile

MCQ

- 2 The **horizontal** motion of a projectile fired at an angle is characterised by:

A a constant acceleration

B a constant velocity

C an increasing acceleration

D an increasing velocity

3 The horizontal acceleration of a projectile fired at an angle equals:

MCQ

- A zero
- B half the acceleration due to gravity
- C the acceleration due to gravity
- D twice the acceleration due to gravity

4 A small ball is projected from the ground with an initial velocity of 15 m/s at  $60^\circ$  to the horizontal. Neglecting air resistance and taking  $g = 10 \text{ m/s}^2$ , and knowing that the ball returns to the same level of projection, calculate:

PROBLEM

- a) the time needed to reach the maximum height.
- b) the total time of flight.
- c) the horizontal range of the ball.
- d) the maximum height reached by the ball.

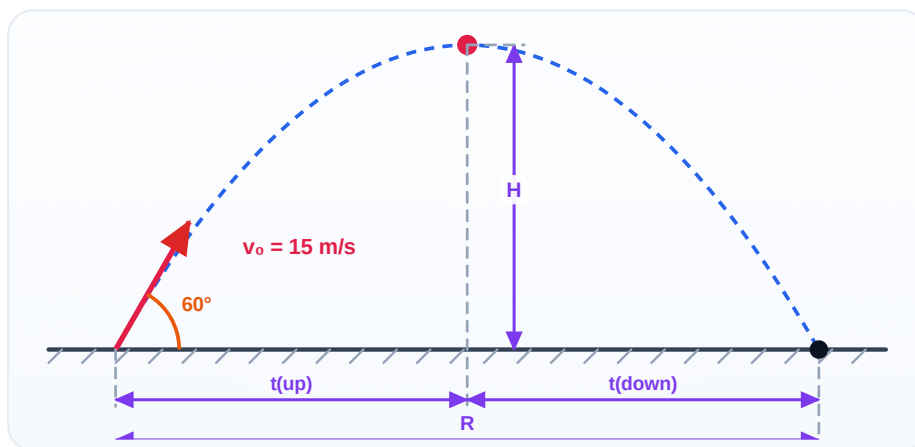


FIGURE 2

## CLASSWORK 2

### C Applications on projectiles fired at an angle

5 The figure shows a body of mass  $m$  projected from the ground with an initial velocity  $v_0$  at  $60^\circ$  to the horizontal, neglecting air resistance. **Write two different ways to increase the total time of flight** of the body.

EXPLAIN

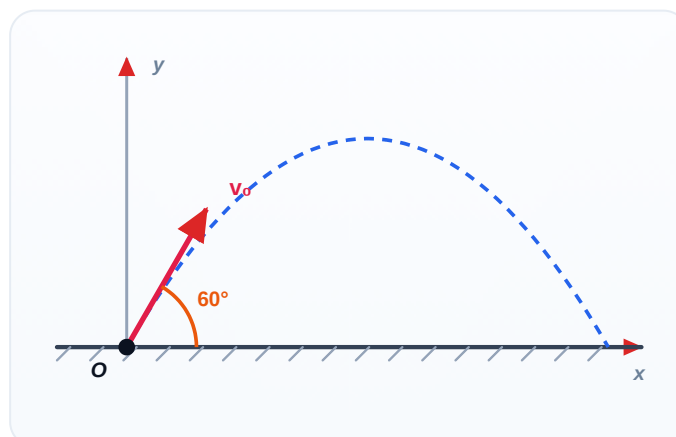


FIGURE 3

6 For the same body of the previous question, write **two different ways** to increase the maximum height reached by the body.

EXPLAIN

7 For the same body, write **two different ways** to increase the horizontal range of the body.

EXPLAIN

8 What is the angle of projection needed for the body to reach its **greatest possible horizontal range**, keeping the initial speed constant? Explain.

EXPLAIN

9 What other angle of projection, used instead of  $60^\circ$ , gives the body the **same horizontal range** shown in the figure, keeping the initial speed constant? Explain.

EXPLAIN

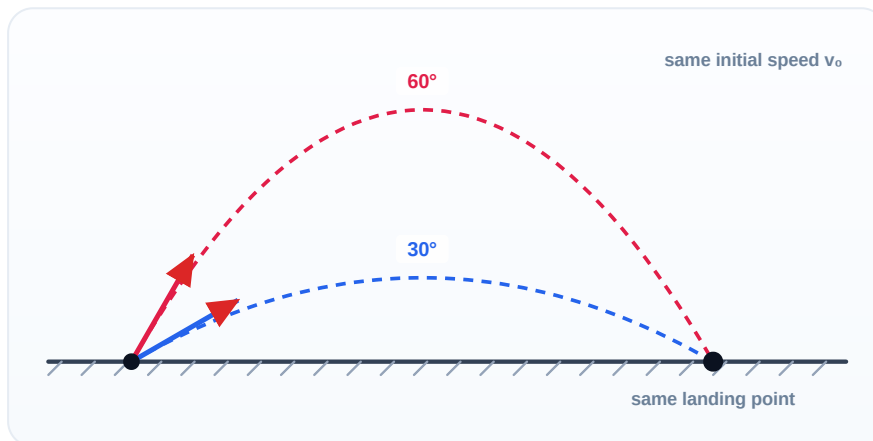


FIGURE 4

D

## The vertical motion and a full calculation

10 The **upward vertical** motion of a projectile fired at an angle is characterised by:

MCQ

A a decreasing acceleration

B a decreasing velocity

C an increasing acceleration

D an increasing velocity

11 At the maximum height, the vertical component of the velocity of a projectile fired at an angle equals:

MCQ

A zero

B a maximum value

C half the initial vertical component

D the same as the initial vertical component

12 A small ball is projected from the ground with an initial velocity of 20 m/s at  $60^\circ$  to the horizontal. Neglecting air resistance and taking  $g = 9.8 \text{ m/s}^2$ , and knowing that the ball returns to the same level of projection, calculate:

PROBLEM

- the time needed to reach the maximum height.
- the total time of flight.
- the horizontal range of the ball.
- the maximum height reached by the ball.

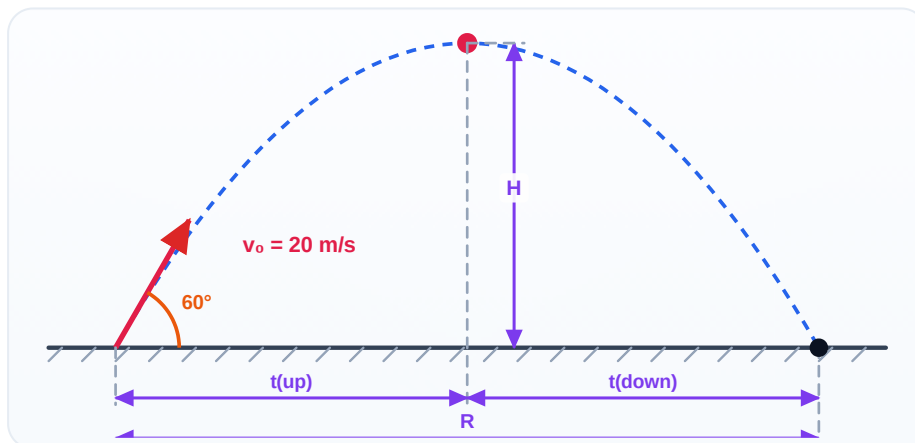


FIGURE 5

E

## Groups A, B and C

## Group A

13 At the maximum height, the magnitude of the velocity of a projectile fired at an angle equals:

MCQ

- A zero
- B a maximum value
- C the initial vertical component of the velocity
- D the horizontal component of the initial velocity

14 A body is projected from the ground with a constant initial velocity  $v_0$  at  $30^\circ$  to the horizontal. At which other angle, with the same initial velocity, does it reach the same horizontal range?

MCQ

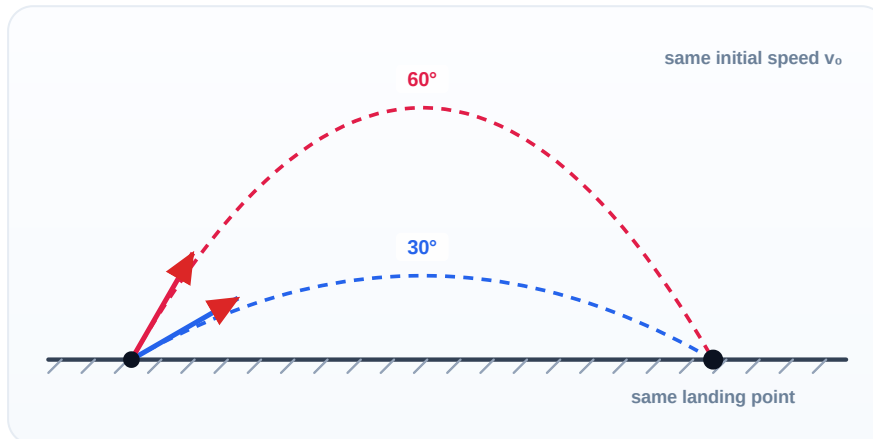


FIGURE 6

- A  $30^\circ$
- B  $45^\circ$
- C  $60^\circ$
- D  $75^\circ$

15 The figure shows a body of mass  $m$  projected from the ground with an initial velocity  $v_0$  at  $60^\circ$  to the horizontal. **Write a way to double the total time of flight** of the body, and explain it.

EXPLAIN

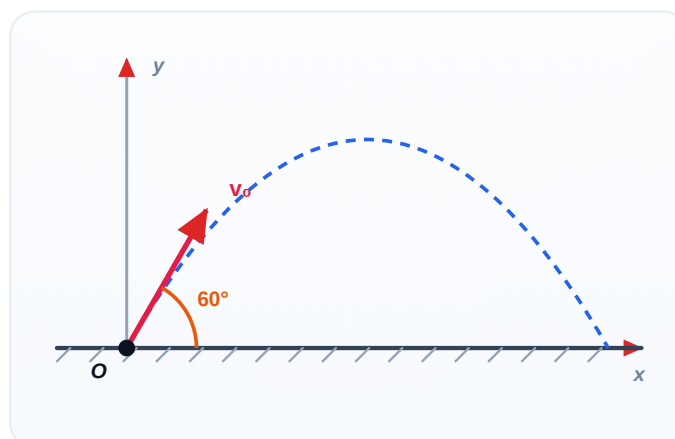


FIGURE 7

- 16 A football is kicked with an initial velocity of 30 m/s at  $60^\circ$  to the **vertical**. Find the horizontal range of the ball.

PROBLEM

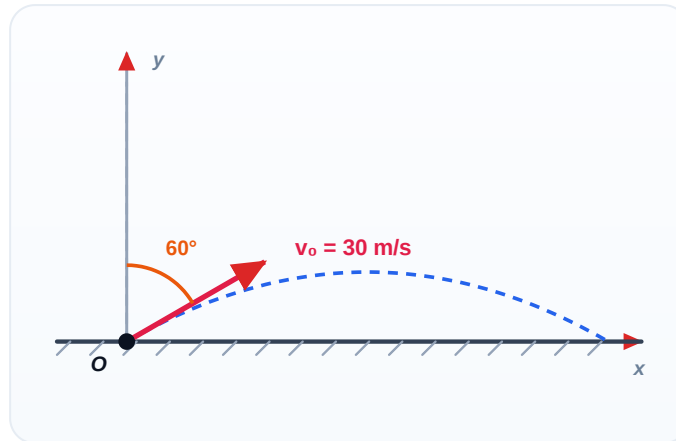


FIGURE 8

- 17 The figure shows a body projected from the top of a building of height 50 m with a velocity of 15 m/s at  $30^\circ$  to the horizontal. **What is the total time of flight** until the body reaches the ground?

PROBLEM

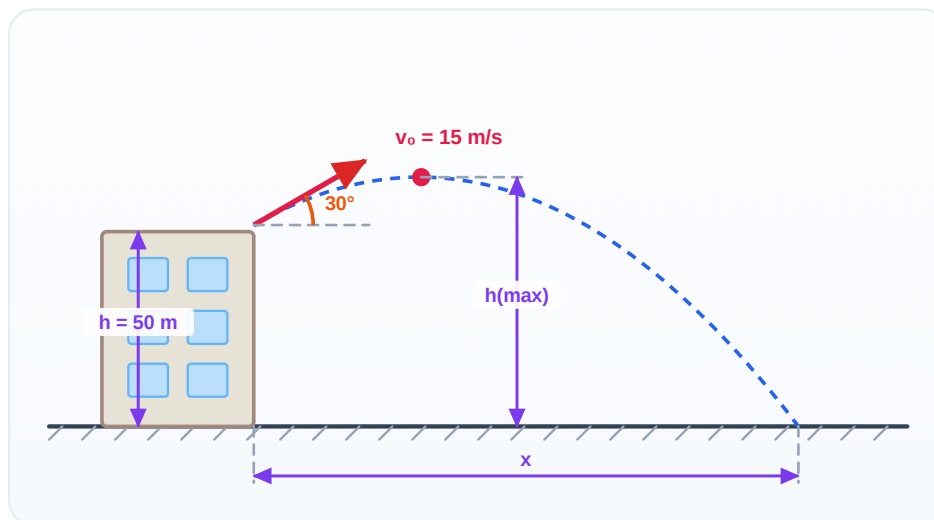


FIGURE 9

## Group B

- 18 When a body is projected at an angle to the horizontal, and air resistance is neglected, its **vertical** motion is characterised by:

MCQ

- A a constant velocity and zero acceleration
- B a varying velocity and a constant acceleration
- C an increasing velocity and an increasing acceleration
- D a constant velocity and a constant acceleration

- 19 A body is projected from the ground with an initial velocity  $v_0$  at an angle  $\theta$  to the horizontal. At which angle  $\theta$  is the **maximum height equal to the horizontal range**?

MCQ

- A  $24^\circ$       B  $45^\circ$   
C  $76^\circ$       D  $90^\circ$

- 20 For the body shown in Figure 7, write a way to **double the maximum height** reached by the body, and explain it.

EXPLAIN

- 21 A football is kicked with an initial velocity of 25 m/s at  $60^\circ$  to the horizontal. Find its total time of flight.

PROBLEM

- 22 A body is projected from the top of a building of height 50 m with a velocity of 15 m/s at  $30^\circ$  to the horizontal. **What is the maximum height reached above the ground?**

PROBLEM

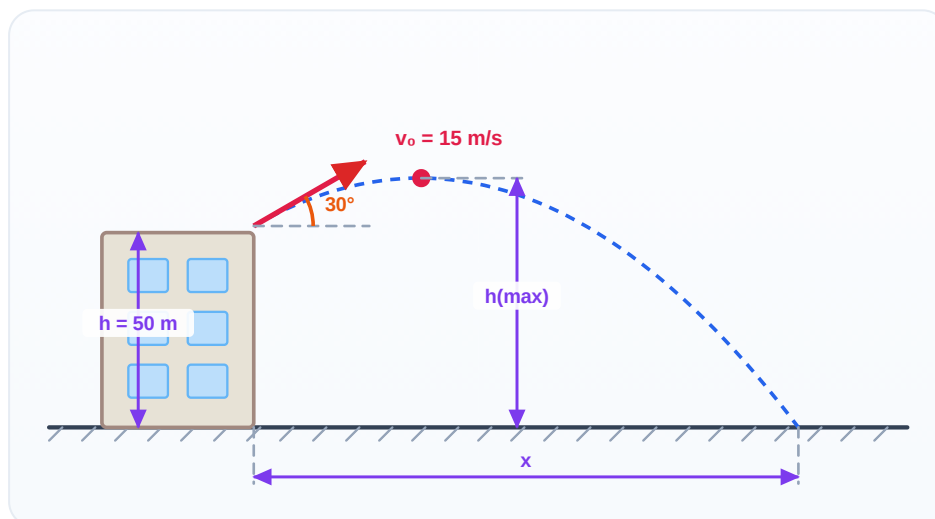


FIGURE 10

### Group C

- 23 A body is projected with an initial velocity  $v_0$  at an angle  $\theta$  to the horizontal, neglecting air resistance. **What happens to the vertical velocity of the projectile while it rises?**

MCQ

- A it increases continuously      B it decreases gradually until it becomes zero at the highest point  
C it stays constant      D it becomes equal to the horizontal velocity

- 24 A body is projected from the ground with an initial velocity  $v_0$  at an angle  $\theta$  to the horizontal. Which angle of projection gives the projectile the **greatest horizontal range** without changing the speed of projection?

MCQ

A  $30^\circ$

B  $45^\circ$

C  $60^\circ$

D  $90^\circ$

- 25 For the body shown in Figure 7, write a way to **double the horizontal range** reached by the body, and explain it.

EXPLAIN

- 26 A football is kicked with an initial velocity of 20 m/s at  $30^\circ$  to the **vertical**. Find the maximum height reached by the ball.

PROBLEM

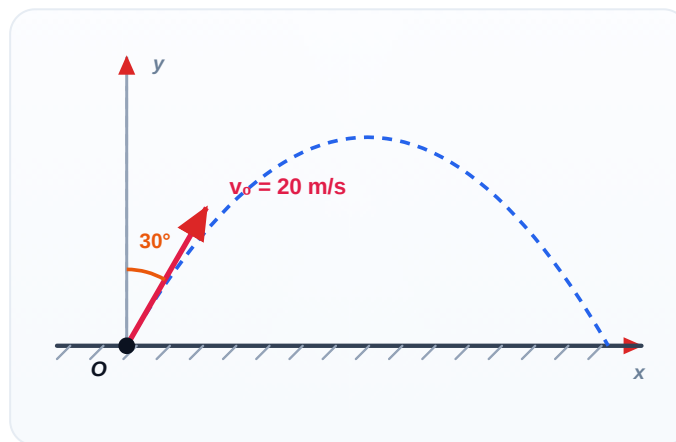


FIGURE 11

- 27 A body is projected from the top of a building of height 50 m with a velocity of 15 m/s at  $30^\circ$  to the horizontal. **What is its horizontal range** measured from the base of the building?

PROBLEM

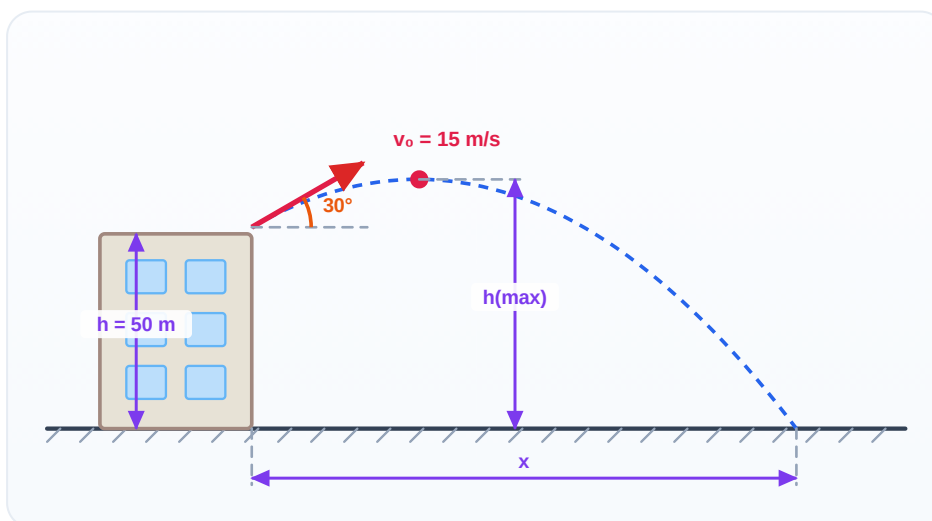


FIGURE 12

# Moment of a Force and Equilibrium

Two lessons in one sheet: the turning effect of a force, and the equilibrium of a body under several forces.

Mr. Gemy

PHYSICS

Grade 11 · Egyptian Baccalaureate

30 questions

Step-by-step solutions

 $g = 9.8 \text{ m/s}^2$  where stated

## The toolkit for these two lessons

$$M = F \times d$$

force  $\perp$  to the arm

$$M = F L \sin \theta$$

force inclined at  $\theta$  to the arm

$$\text{unit : N}\cdot\text{m}$$

moment is a vector quantity

$$\Sigma F = 0$$

condition of translational equilibrium

$$\Sigma F_x = 0, \Sigma F_y = 0$$

applied on both axes separately

$$T \sin \theta = W$$

the vertical component carries the weight

**Two ideas, one sheet.** A moment measures how strongly a force **turns** a body about a point, and it needs the **perpendicular** distance. Equilibrium means the forces cancel on **each axis separately**, which is why an inclined force must always be resolved first.

## LESSON 1-4 · CLASSWORK 1

### A The moment of a force

1 The expression for the moment  $M$  of a force when the force is **perpendicular** to the arm is:

MCQ

A  $M = F / d$

B  $M = F d$

C  $M = F + d$

D  $M = F - d$

2 The moment of a force is a measure of:

MCQ

A the magnitude of the force only

B the turning effect of the force about a point

C the speed of the body

D the energy stored in the body

3 The SI unit of the moment of a force is:

MCQ

A newton (N)

B joule (J)

C newton · metre (N·m)

D metre / second (m/s)

4 The law of the lever was formulated in antiquity by:

MCQ

A Newton

B Archimedes

C Galileo

D Kepler

5 A force of 5.0 N acts **perpendicular** to an arm of length 2.0 m measured from the point of support. Calculate the moment of the force.

PROBLEM

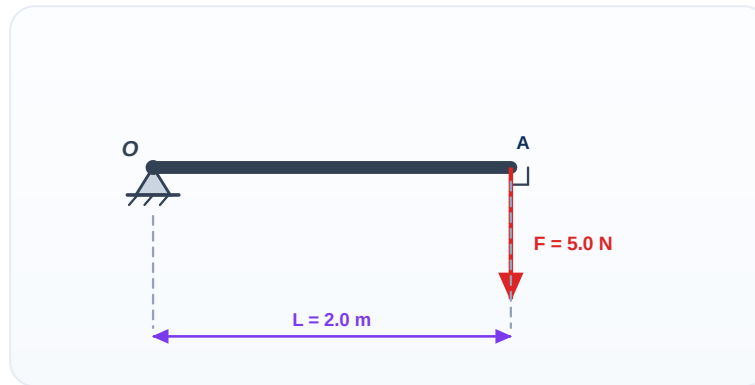


FIGURE 1

LESSON 1-4 · HOME ASSIGNMENT 1

B Using the moment relation

6 If the perpendicular distance is doubled while the force stays constant, then the moment of the force:

MCQ

A is halved

B is doubled

C stays constant

D is reduced to a quarter

7 A door opens most easily when it is pushed at:

MCQ



FIGURE 2

A the hinge itself

B the handle, far from the hinge

C the middle of the door only

D any point — it makes no difference

- 8 A moment of  $12 \text{ N}\cdot\text{m}$  is produced by a force of  $4.0 \text{ N}$  acting perpendicular to its arm. The length of the perpendicular arm equals:

MCQ

- A  $3.0 \text{ m}$       B  $8.0 \text{ m}$   
C  $48 \text{ m}$       D  $33 \text{ m}$

- 9 A force of  $3.0 \text{ N}$  acts perpendicular at the end of a horizontal rod, at a distance of  $20 \text{ m}$  from the point O. Calculate the moment of the force about O.

PROBLEM

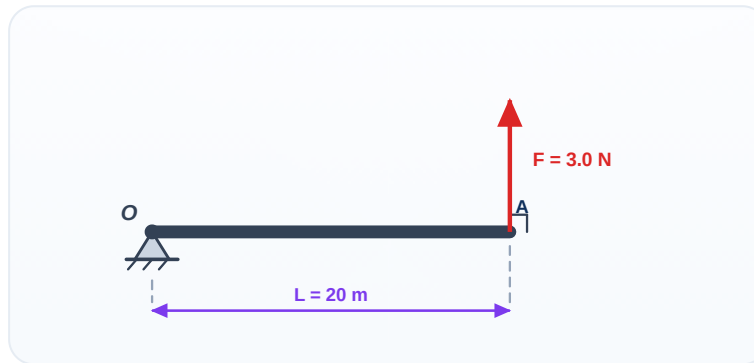


FIGURE 3

#### LESSON 1-5 · CLASSWORK 2

### C The equilibrium of a body

- 10 A body is in a state of equilibrium when:

MCQ

- A its velocity is always zero      B it stays at rest or moves with a constant velocity  
C its velocity changes continuously      D its mass vanishes

- 11 The condition of **translational** equilibrium of a body is written mathematically as:

MCQ

- A  $\Sigma F = mg$       B  $\Sigma F = 0$   
C  $\Sigma F = ma$ , where  $a \neq 0$       D  $\Sigma F = F_1 - F_2$  only

- 12 To deal with a force acting at an angle, it is resolved into two perpendicular components, and the two equilibrium equations are then applied to:

MCQ

- A one axis only      B both axes  $x$  and  $y$  :  $\Sigma F_x = 0$  and  $\Sigma F_y = 0$   
C the angle  $\theta$  only      D the mass only

- 13 The figure shows a wire fixed between two poles, carrying a weight at its middle. **Explain how the angle  $\theta$  at the two fixing points changes** if the wire is allowed to sag more at its middle, and what is the expected effect on the tension  $T$ .

EXPLAIN

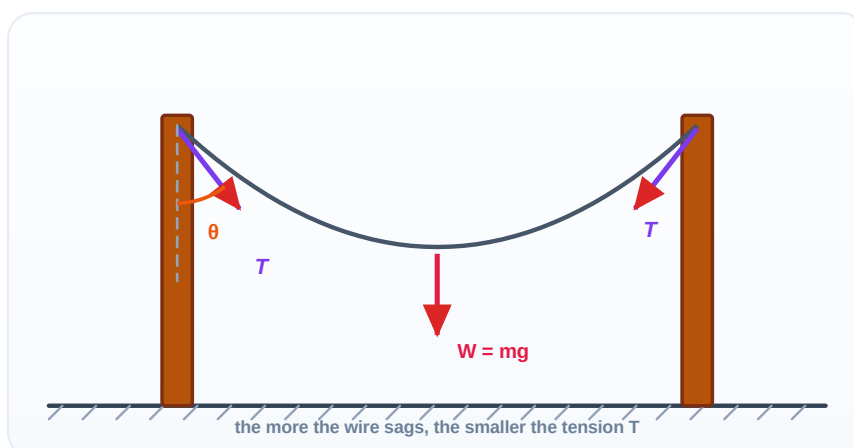


FIGURE 4

D

## Hanging bodies and the tension in cables

- 14 A lamp hangs from a cable fixed to the ceiling, together with a light horizontal rope fixed to a wall. The cable makes an angle of  $25^\circ$  with the vertical. The **vertical** component of the tension in the cable must be equal to:

MCQ

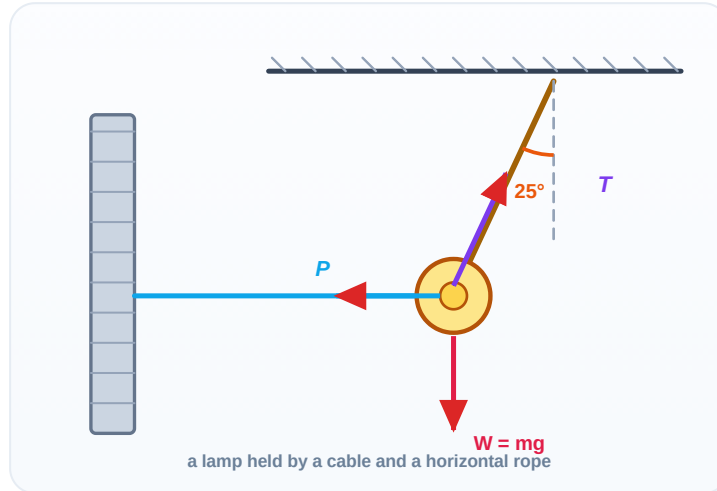


FIGURE 5

- A zero
- B the weight of the lamp
- C the tension in the horizontal rope
- D half the weight of the lamp

- 15 A lamp of mass  $0.6 \text{ kg}$  hangs from two identical cables, each making an angle of  $30^\circ$  with the horizontal. If  $g = 9.8 \text{ m/s}^2$ , then the weight of the lamp is approximately:

MCQ

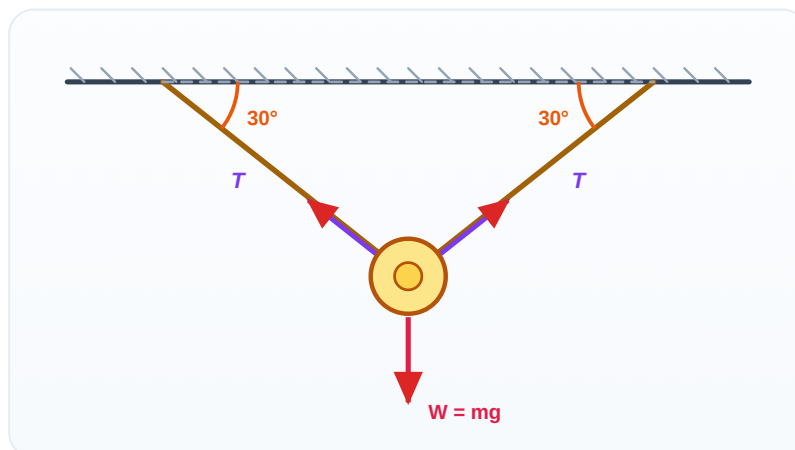


FIGURE 6

- A  $0.6 \text{ N}$
- B  $58.8 \text{ N}$
- C  $4.29 \text{ N}$
- D  $5.88 \text{ N}$

- 16 If the angle that each cable makes with the horizontal is increased (so that the two cables become closer to the vertical) while the load stays constant, then the tension in each cable:

MCQ

- ☐ A increases
- ☐ B decreases
- ☐ C stays constant
- ☐ D becomes zero

- 17 The figure shows the tension in a cable resolved into two components. **Explain which of the two components** (the horizontal one or the vertical one) is responsible for balancing the weight of the hanging body, and **why the other component is left out** of the vertical equilibrium equation.

EXPLAIN

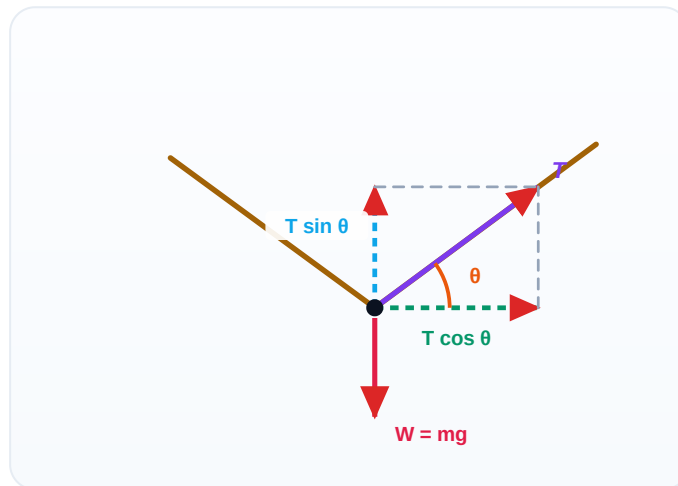


FIGURE 7

### WEEKLY ASSESSMENT 3

E

## Groups A, B and C

### Group A

- 18 If the force is doubled while the perpendicular distance stays constant, then the moment of the force:

MCQ

- ☐ A is doubled
- ☐ B is halved
- ☐ C stays constant
- ☐ D cannot be determined

19 A wheel-nut spanner loosens the nut more easily when it is:

MCQ

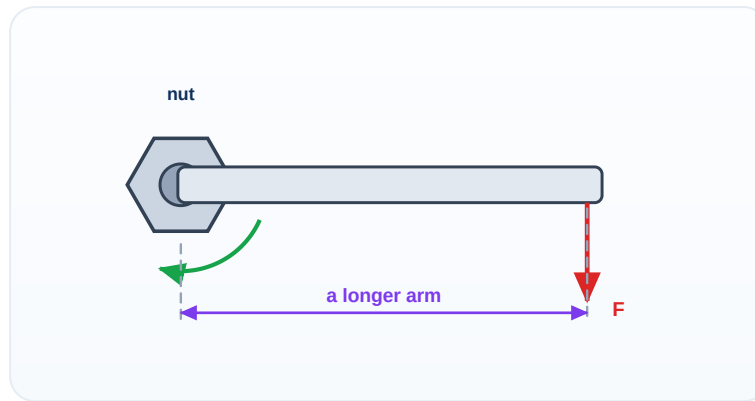


FIGURE 8

A shorter

B longer

C without an arm

D its length makes no difference

20 **Compare** the moment of a force acting perpendicular to its arm ( $\theta = 90^\circ$ ) with the moment of the **same** force acting at an angle smaller than  $90^\circ$  on the **same** arm, in terms of the magnitude of the moment and the relation used in each case.

COMPARE

21 The figure shows a force  $F$  acting at the end of a lever. **How can the moment of this force about the other end of the lever be increased in three different ways**, without changing the magnitude of the force itself?

EXPLAIN



FIGURE 9

## Group B

- 22 A force of 4.0 N acts at the end of an arm of length 0.30 m, making an angle of  $45^\circ$  with the arm. The moment of the force about the point of support is approximately: ( $\sin 45^\circ \approx 0.71$ )

MCQ

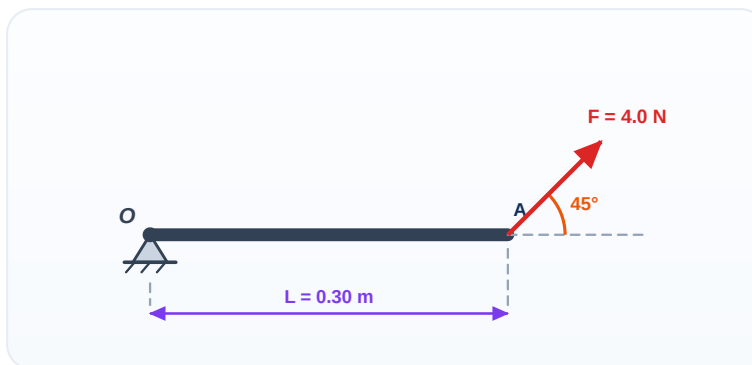


FIGURE 10

- A 0.85 N·m  
B 1.2 N·m  
C 1.7 N·m  
D 6.0 N·m

- 23 A ruler is balanced on a point of support. One coin placed at a distance of 15 cm on the right-hand side balances  $n$  identical coins placed at a distance of 5 cm on the left-hand side. The number of coins  $n$  equals:

MCQ

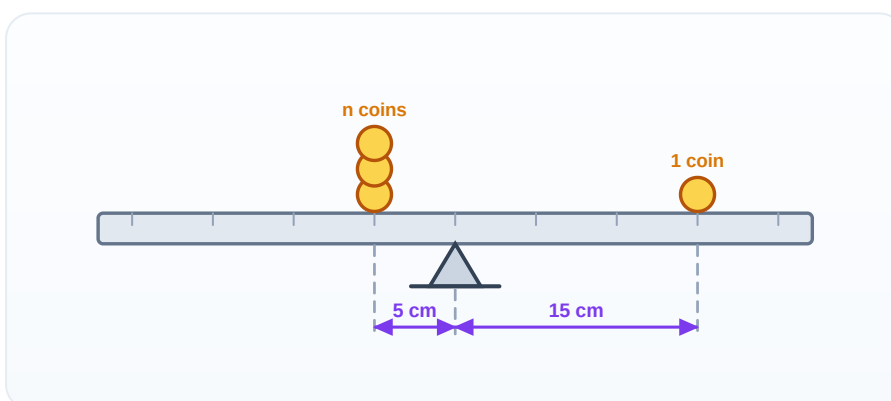


FIGURE 11

- A 1  
B 2  
C 3  
D 5

- 24 **Compare** the effect of pushing a door at the handle far from the hinge with the effect of pushing it close to the hinge, in terms of the length of the moment arm and the magnitude of the resulting moment, with the same applied force.

COMPARE

- 25 Using the figure, explain — **without any numerical calculation** — why the magnitude of the moment of the same force differs according to the point (A or B) about which the moment is calculated, although the force has the same magnitude and the same direction in both cases.

EXPLAIN

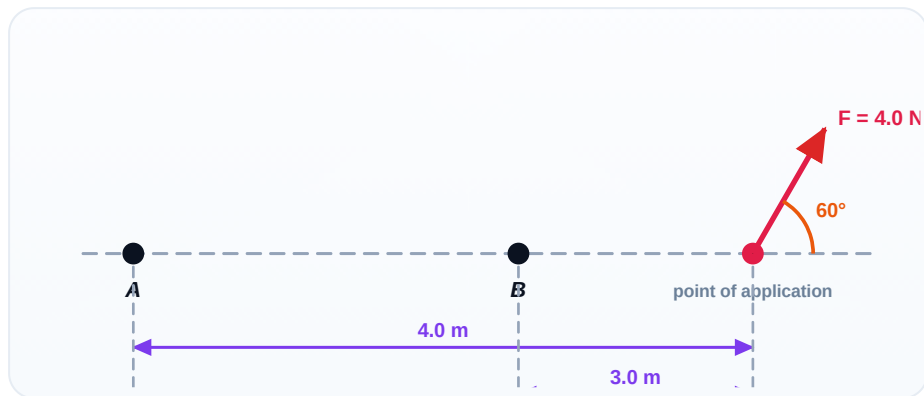


FIGURE 12

### Group C

- 26 When a spanner with a **longer** arm is used to produce the same required moment, the force needed is:

MCQ

- ☐ A larger
- ☐ B smaller
- ☐ C exactly the same
- ☐ D zero

- 27 A force of  $6.0 \text{ N}$  acts at the end of a lever of length  $0.50 \text{ m}$ , at an angle of  $30^\circ$  to the lever. The moment of the force about the other end equals approximately: ( $\sin 30^\circ = 0.5$ )

MCQ

- ☐ A  $1.5 \text{ N}\cdot\text{m}$
- ☐ B  $0.3 \text{ N}\cdot\text{m}$
- ☐ C  $0.5 \text{ N}\cdot\text{m}$
- ☐ D  $0.9 \text{ N}\cdot\text{m}$

- 28 A force  $F = 4.0 \text{ N}$  acts at an angle of  $60^\circ$  at a point which is  $4.0 \text{ m}$  from A and  $3.0 \text{ m}$  from B, where A and B lie on the same straight line. ( $\sin 60^\circ \approx 0.87$ )

MCQ

- 1) The moment of the force about A equals approximately: a)  $13.9 \text{ N}\cdot\text{m}$  anticlockwise b)  $10.4 \text{ N}\cdot\text{m}$  clockwise  
c)  $13.9 \text{ N}\cdot\text{m}$  clockwise d)  $4.0 \text{ N}\cdot\text{m}$  anticlockwise
- 2) The moment of the same force about B equals approximately: a)  $13.9 \text{ N}\cdot\text{m}$  anticlockwise b)  $10.4 \text{ N}\cdot\text{m}$  clockwise  
c)  $10.4 \text{ N}\cdot\text{m}$  anticlockwise d) zero

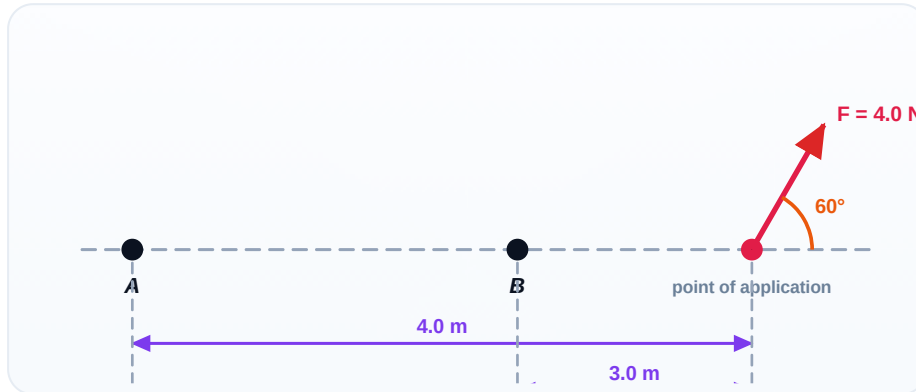


FIGURE 13

- 29 **Compare** a positive moment with a negative moment, in terms of the sense of rotation that each one represents.

COMPARE

- 30 Using the figure, explain — **without any numerical calculation** — how the moment of the force about O changes if the point of application of  $F$  slides along the rod towards O, while the direction and the magnitude of the force stay the same.

EXPLAIN

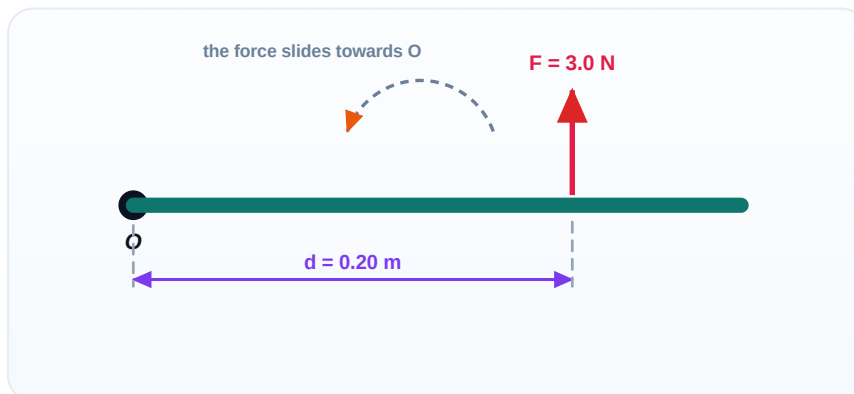


FIGURE 14

# Power and Efficiency & Equilibrium Review

The rate of doing work, the efficiency of a machine, and a full review of a body held in equilibrium by cables and springs.

Grade 11 · Egyptian Baccalaureate

22 questions

Step-by-step solutions

 $g = 9.8 \text{ m/s}^2$ 

**Mr. Gemy**  
PHYSICS

## The toolkit for this sheet

$$P = \Delta W / \Delta t$$

average power — unit : watt

$$P = F v \cos \theta$$

power of a force on a moving body

$$\eta = \text{useful output} / \text{total input}$$

efficiency, always &lt; 100 %

$$W = m g h$$

work done in raising a load

$$\Sigma F_x = 0, \Sigma F_y = 0$$

equilibrium on each axis

$$T = W / (2 \sin \alpha)$$

two symmetric cables carrying a weight

**Watch the difference.** Work tells you *how much* energy was transferred; power tells you *how fast* it was transferred. Two machines can do exactly the same work and still have very different powers.

## LESSON 1-6 · CLASSWORK 1

A

### Power and efficiency

1 Power is a measure of:

MCQ

A the amount of work done only

B the rate of doing work or of transferring energy

C the magnitude of the force only

D the total displacement of the body

2 The mathematical expression used to calculate the average power from the work and the time is:

MCQ

A  $P = W \times t$

B  $P = \Delta W / \Delta t$

C  $P = W + t$

D  $P = W / t^2$

3 The SI unit of power is:

MCQ

A the joule (J)

B the watt (W)

C the newton (N)

D the kilogram (kg)

- 4 The figure shows two lifts A and B raising the **same** load through the **same** height, but in different times. **Explain which lift has the greater power, and which one does the greater useful work.** ( $g = 9.8 \text{ m/s}^2$ )



FIGURE 1

## LESSON 1-6 · HOME ASSIGNMENT 1

B

## Using power and efficiency

- 5 When two lifts raise the same load through the same vertical height, but one of them finishes the lift in a shorter time, then this lift has:

MCQ

- A a greater useful work
- B a greater power
- C necessarily a greater efficiency
- D the same work and the same power

- 6 A machine has an efficiency of 40 %. If the total input energy supplied to it is 250 J, then the useful output energy equals:

MCQ

- A 100 J
- B 250 J
- C 62.5 J
- D 400 J

- 7 An electric water heater receives an electric power of 2400 W and produces a useful thermal power of 2040 W. Its efficiency equals:

MCQ

- A 85 %
- B 15 %
- C 117 %
- D 60 %

8

The figure shows a force  $F$  acting at an angle  $\theta$  to the direction of motion of a body moving with a velocity  $v$ . **Explain what happens to the power produced by the force as  $\theta$  gets closer to  $90^\circ$** , compared with the case in which the force acts exactly along the direction of motion ( $\theta = 0^\circ$ ).

EXPLAIN

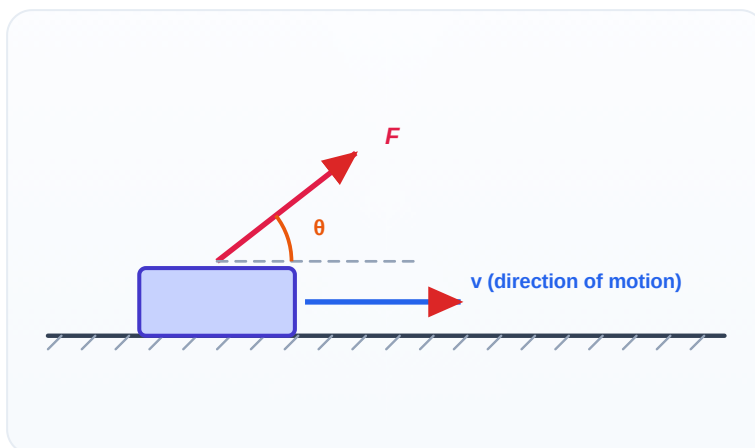


FIGURE 2

## A body in equilibrium under several forces

- 9 A rigid body is in equilibrium under the action of three forces only:  $F_1 = 10\text{ N}$  upwards and  $F_2 = 6\text{ N}$  downwards. The third force  $F_3$  needed to keep the body in equilibrium (on the same vertical line) must be:

MCQ

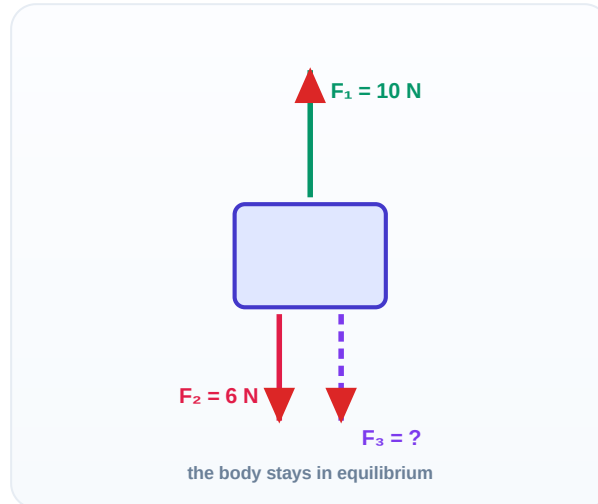


FIGURE 3

- A 4 N upwards
- B 4 N downwards
- C 16 N upwards
- D zero

- 10 A lamp of mass  $3.0\text{ kg}$  hangs from a cable fixed to the ceiling together with a light horizontal rope fixed to a wall, and the cable makes an angle of  $25^\circ$  with the vertical. The weight of the lamp is  $29.4\text{ N}$ , so the tension  $T$  in the cable is approximately:

MCQ

- A 26.8 N
- B 32.3 N
- C 12.4 N
- D 29.4 N

- 11 For a lamp hanging in equilibrium from two identical cables that make the **same** angle with the horizontal, the two horizontal components of the tensions:

MCQ

- A are equal in magnitude and opposite in direction, so they cancel each other
- B add together
- C have nothing to do with the equilibrium
- D each one equals the weight

- 12 If the angle that each cable makes with the horizontal is **decreased** (so that the two cables become closer to the horizontal) while the weight of the load stays constant, then the tension in each cable:

MCQ

- A increases
- B decreases
- C stays constant
- D becomes zero

- 13 If the angle that each cable makes with the horizontal is **increased** (so that the two cables become closer to the vertical) while the weight of the load stays constant, then the tension in each cable:

MCQ

- A increases
- B decreases
- C stays constant
- D becomes zero

- 14 A metal piece of mass 1.5 kg hangs at rest from two identical light springs, each making an angle of  $40^\circ$  with the vertical, the spring constant of each being 60 N/m. The weight of the piece is approximately: ( $g = 9.8 \text{ m/s}^2$ )

MCQ

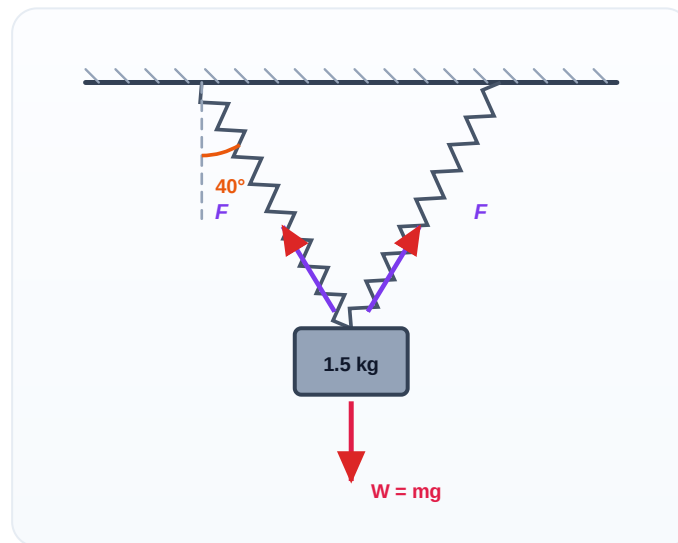


FIGURE 4

- A 1.5 N
- B 14.7 N
- C 6.0 N
- D 60 N

- 15 For the same metal piece of the previous question, calculate the force  $F$  in **each** spring. ( $\cos 40^\circ \approx 0.77$ )

PROBLEM

- 16 The figure shows a lamp of mass 4.0 kg hanging from a cable fixed to the ceiling and a light horizontal rope fixed to a wall. The tension measured in the horizontal rope is  $P = 15 \text{ N}$ . Calculate the tension  $T$  in the cable.

PROBLEM

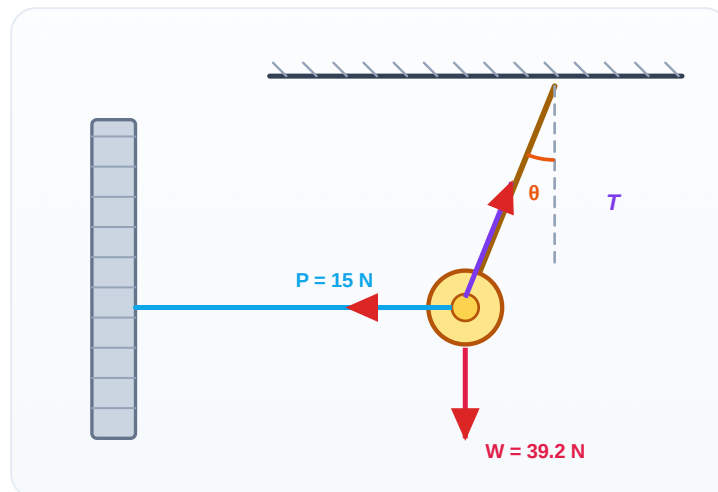


FIGURE 5

- 17 For the same lamp of the previous question, find the angle  $\theta$  that the cable makes with the vertical.

PROBLEM

- 18 Using the figure, explain how the diagram shows that the sum of the two vertical components of the tensions equals the weight of the lamp, while the two horizontal components cancel each other.

EXPLAIN

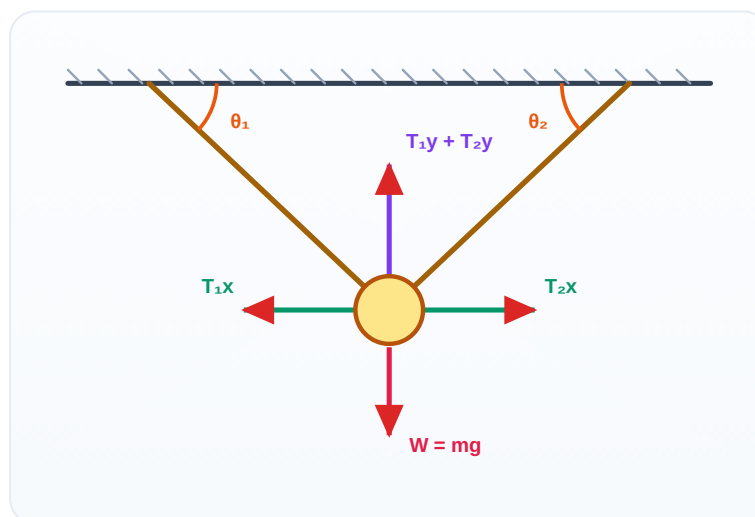


FIGURE 6

- 19 Compare the case of a lamp hanging from two identical cables making the same angle with the case of another lamp of the same weight hanging from two cables making different angles, in terms of whether the tensions in the two cables are equal or different.

COMPARE

- 20 The figure shows a shop sign of mass 10 kg hanging from two light cables attached to a horizontal beam. The first cable makes an angle of  $35^\circ$  with the horizontal and the second makes  $65^\circ$  with the horizontal. **Find the tension in each cable.** ( $g = 9.8 \text{ m/s}^2$ )

PROBLEM

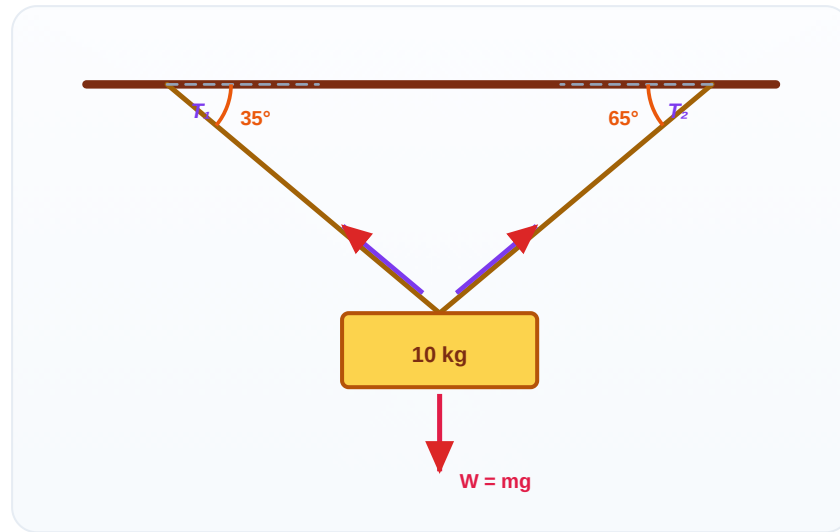


FIGURE 7

- 21 For a lamp of weight  $W$  hanging in equilibrium from two identical cables making the same angle with the horizontal, calculate the **sum of the two vertical components** of the tensions.

PROBLEM

- 22 The figure shows a book resting on a table. **What changes in this diagram** if another book of the same weight is placed on top of the first one? State clearly which of the two forces changes in value and which one stays the same, and identify the **source** of each force.

EXPLAIN

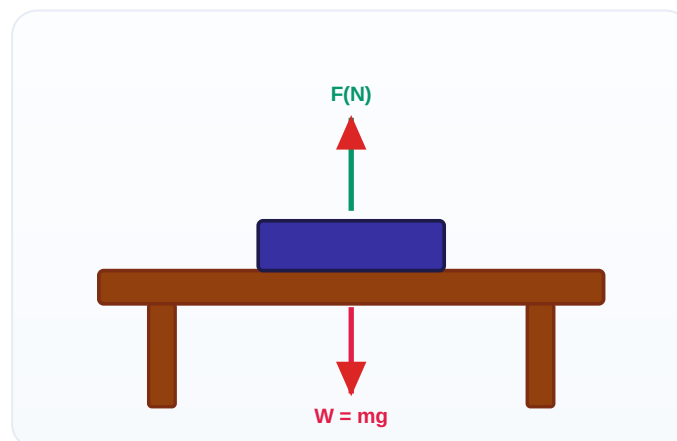


FIGURE 8

# Momentum, Impulse and Its Conservation

Two lessons in one sheet: the impulse of a force on a body, and the conservation of momentum in collisions and explosions.

Mr. Gemy

PHYSICS

Grade 11 · Egyptian Baccalaureate

22 questions

Step-by-step solutions

smooth horizontal surfaces

## The toolkit for these two lessons

$$p = m v$$

momentum — unit : kg·m/s

$$J = F \times \Delta t$$

impulse of a force — unit : N·s

$$J = \Delta p = m v_f - m v_i$$

impulse–momentum theorem

$$F = \Delta p / \Delta t$$

a longer contact means a smaller force

$$J = \text{area under the } F\text{--}t \text{ curve}$$

whatever the shape of the pulse

$$\Sigma p \text{ (before)} = \Sigma p \text{ (after)}$$

conservation of momentum

**Momentum is a vector.** Before you add or subtract any two momenta, fix a positive direction and give every velocity its sign; for motion in two dimensions, apply the conservation law to the x and y directions **separately**.

## LESSON 1–7 · CLASSWORK 1

A

### Momentum and impulse

- 1 A person A of mass 40 kg moves at 0.3 m/s towards the east, while a person B of mass 60 kg moves at 0.2 m/s towards the west along the same straight line. **Which of them has the greater momentum in magnitude?**

MCQ

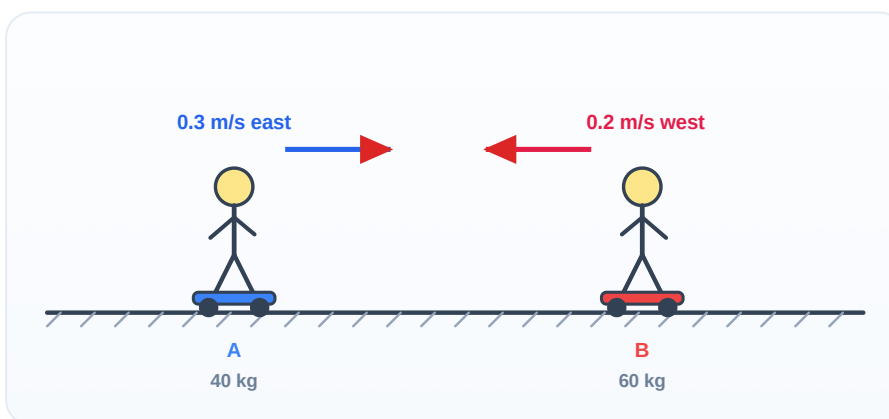


FIGURE 1

A person A

B person B

C they are equal

D it cannot be decided without choosing a positive direction

- 2 A phone falls from the same height once onto a hard floor and once onto a cushion, and it comes to rest in both cases.

EXPLAIN

- a) Explain why the average collision force on the cushion is smaller, although the change in momentum is the same in the two cases.
- b) What happens to the average force if the stopping time of the phone is doubled?

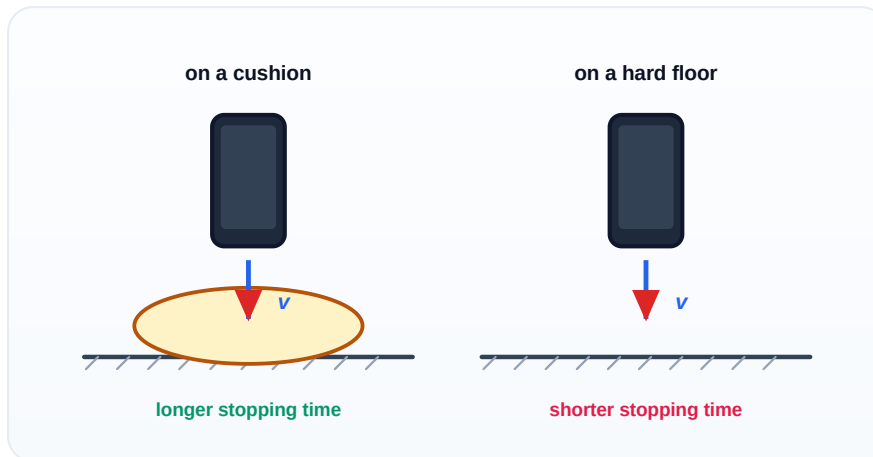


FIGURE 2

- 3 A ball of mass 0.10 kg moves eastwards at 6 m/s, then it is struck so that it moves northwards with the same speed. **What is the magnitude of the impulse given to the ball, and its direction?**

MCQ



FIGURE 3

A 0.6 N·s northwards

B  $0.6\sqrt{2}$  N·s north-west

C 1.2 N·s north-east

D  $0.6\sqrt{2}$  N·s south-west

## B

## The force–time relation

- 4 If an airbag increases the stopping time of a passenger to **three times** its value, while the change in his momentum stays the same, then the average force becomes:

MCQ

- A three times as large  
B half as large  
C one third as large  
D unchanged

- 5 A dynamics trolley collides with a spring bumper connected to a force sensor, and then rebounds in the opposite direction. The sensor records the force–time curve shown.

EXPLAIN

- a) Explain how the force–time curve is used to determine the impulse given to the trolley.  
b) How does the shape of the force pulse change when the bumper is replaced by a softer one, while the change in momentum stays the same?

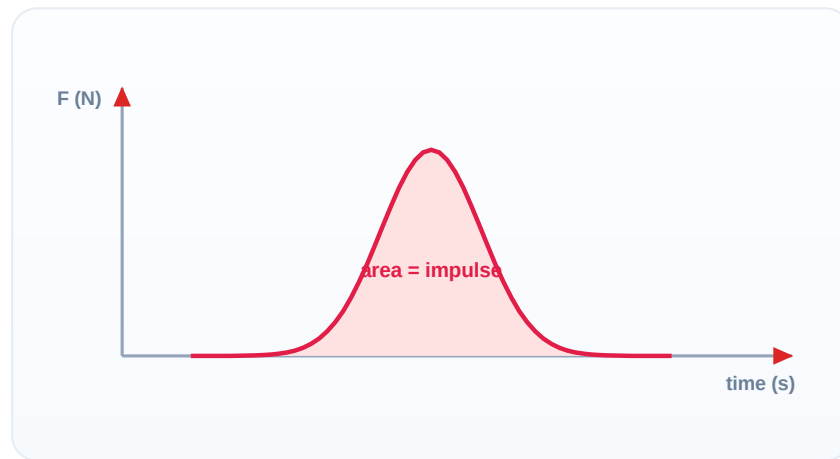


FIGURE 4

## Conservation of momentum

- 6 A passenger stands at the middle of a boat that is at rest on a calm lake, then he walks slowly towards the bow of the boat.

EXPLAIN

- a) Describe the total momentum of the system before the passenger starts to walk.  
b) Explain why the boat moves in the direction opposite to the motion of the passenger, stating the reason.

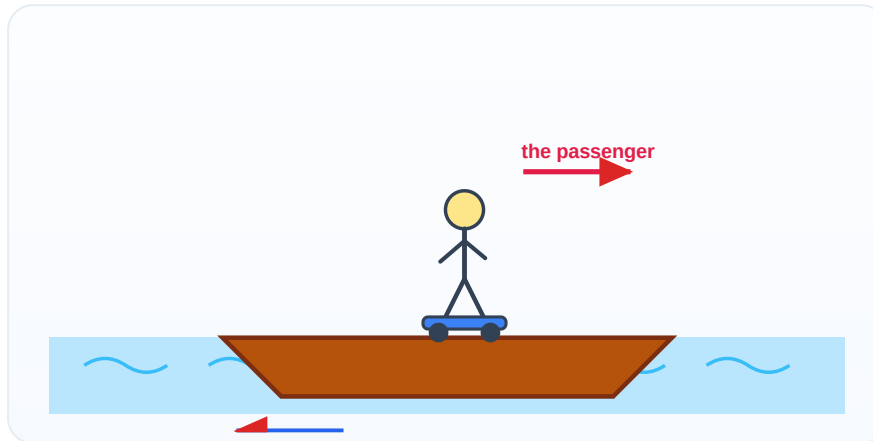


FIGURE 5

- 7 A body of mass  $m_1 = 0.50$  kg moving at  $6.0$  m/s towards the east collides with a body of mass  $m_2 = 1.00$  kg at rest, and the two bodies stick together and move as one body after the collision.

PROBLEM

- a) Calculate the common velocity of the two bodies after the collision.  
b) Compare the total momentum of the system before the collision with that after it, and state what you conclude.



FIGURE 6

D

## Collisions in one and two dimensions

- 8 A body A of mass 2.0 kg moving at + 6.0 m/s collides with a body B of mass 4.0 kg moving in the opposite direction at 2.0 m/s. If the two bodies stick together after the collision, what is their common speed and its direction?

MCQ

- A 0.67 m/s in the direction of A  
 B 0.67 m/s in the direction of B  
 C 2.0 m/s in the direction of A  
 D 2.0 m/s in the direction of B

- 9 On a smooth horizontal surface, a body A of mass 1.5 kg moves at 6.0 m/s along the positive direction of the horizontal axis and collides with a body B of mass 3.0 kg at rest. After the collision, body A is deflected through  $30^\circ$  above the original line of motion, while body B is deflected through  $60^\circ$  below it. Find the final speed of each of the two bodies.

PROBLEM

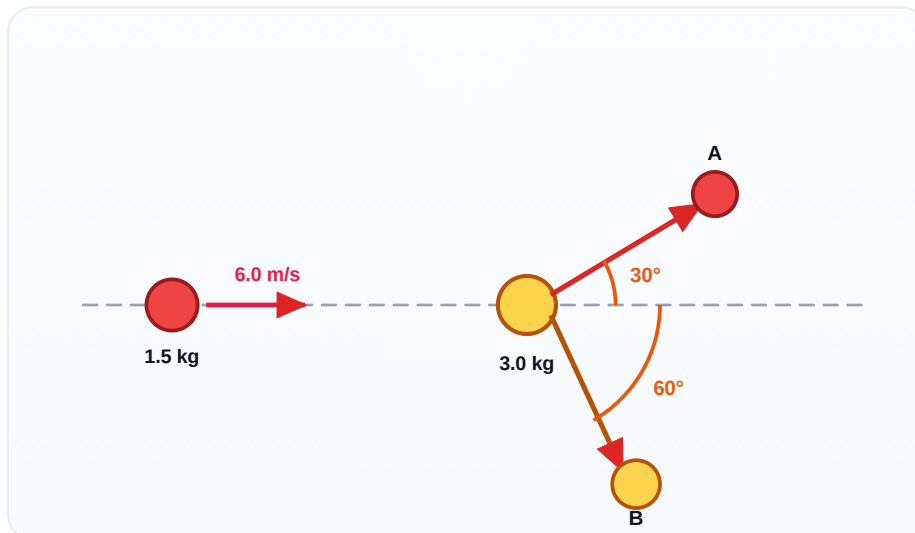


FIGURE 7

## WEEKLY ASSESSMENT

E

## Groups A, B and C

## Group A

- 10 A tennis player strikes a ball with a racket, so that the average force acts on the ball for  $4 \times 10^{-3}$  s, which is just enough to give it a speed of 50 m/s from rest in the direction of the stroke. If the mass of the ball is 0.060 kg, what is the magnitude of the impulse it receives?

MCQ

- A 3.0 N·s  
 B 5.0 N·s  
 C 7.5 N·s  
 D 15 N·s

- 11 For a constant change in momentum, if the time of the collision is doubled, then the average force:

MCQ

- A is doubled
- B is halved
- C does not change
- D becomes four times as large

- 12 A body of mass 9.0 kg at rest explodes into a body A of mass 5.0 kg and a body B of mass 4.0 kg. If body A moves towards the left at 8.0 m/s after the explosion, **find the velocity of body B** and its direction.

PROBLEM



FIGURE 8

- 13 On a smooth horizontal surface, a body A of mass 0.30 kg moving at 4.0 m/s collides with a body B of mass 0.60 kg at rest. Immediately after the collision, body A moves **perpendicular** to its original line of motion, while body B is deflected through  $60^\circ$  below that line. **Find the final speed of each body.**

PROBLEM

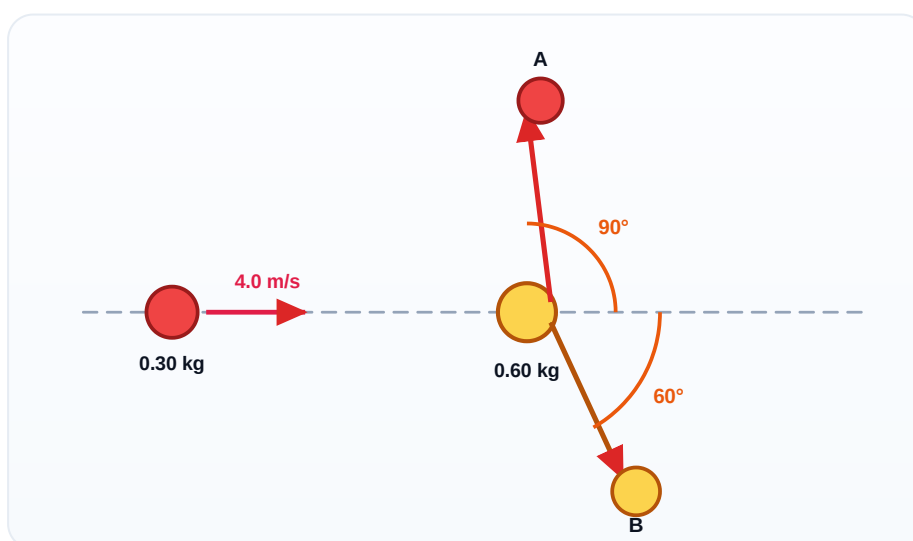


FIGURE 9

## Group B

- 14 A ball of mass 0.20 kg moves eastwards at 5 m/s, then it is struck so that it moves westwards at 3 m/s. What is the magnitude of the impulse acting on the ball and its direction?

MCQ

- A 0.4 N·s westwards  
B 1.0 N·s eastwards  
C 1.6 N·s westwards  
D 1.6 N·s eastwards

- 15 For a constant change in momentum, if the time of the collision is reduced to half, then the average force:

MCQ

- A is doubled  
B is halved  
C does not change  
D becomes four times as large

- 16 On a smooth horizontal surface, a body A of mass 0.40 kg moving at 6.0 m/s collides with a body B of mass 0.80 kg moving at 1.5 m/s in the **same** direction. After the collision, body A is deflected through  $53^\circ$  above the original line of motion and body B through  $30^\circ$  below it. **Find the final speed of each body.**

PROBLEM

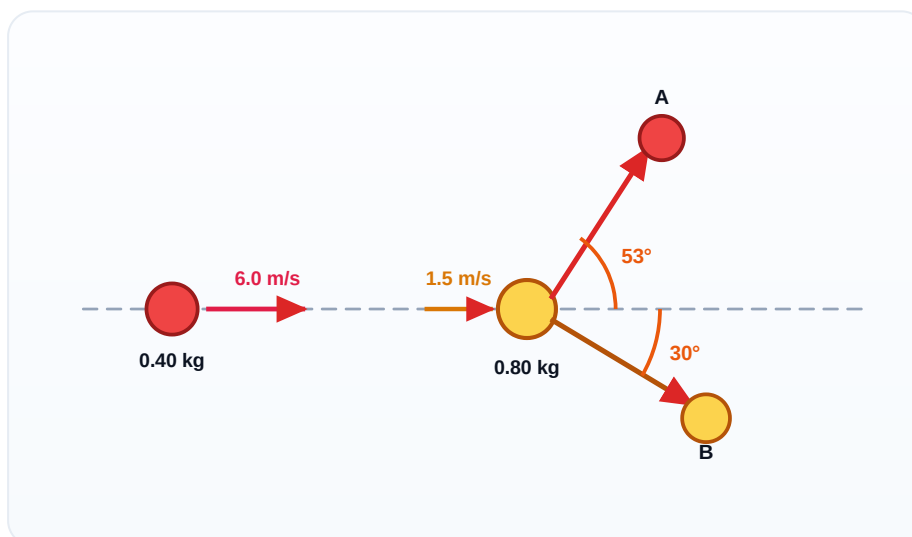


FIGURE 10

- 17 A large artillery shell is fired horizontally towards the east from a cannon standing on horizontal ground.

EXPLAIN

- a) Describe the total momentum of the system before the shell is fired.
- b) Explain the backward recoil of the cannon after firing, and state the relation between the recoil speed of the cannon and its mass.

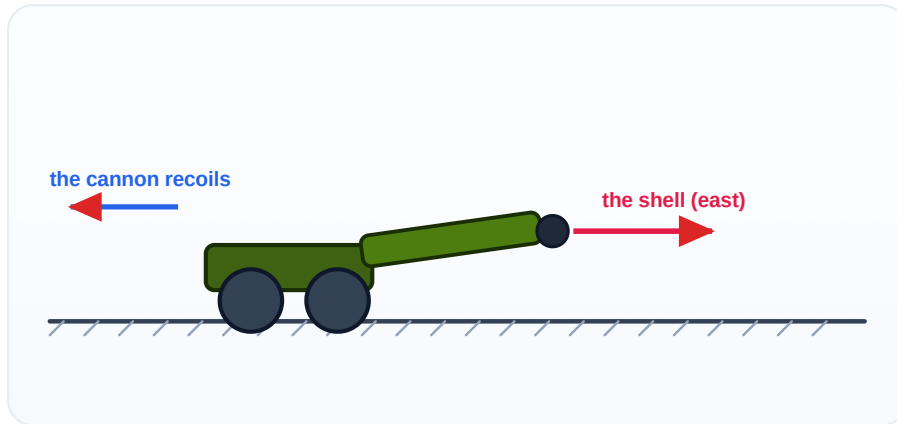


FIGURE 11

### Group C

- 18 A ball of mass  $0.10\text{ kg}$  moves at  $8\text{ m/s}$ , then it is struck so that it moves at  $6\text{ m/s}$  in a direction **perpendicular** to its original direction of motion. What is the magnitude of the impulse acting on the ball?

MCQ

A  $20\text{ N}\cdot\text{s}$

B  $0.60\text{ N}\cdot\text{s}$

C  $1.00\text{ N}\cdot\text{s}$

D  $1.40\text{ N}\cdot\text{s}$

- 19 For a constant change in momentum, if the time of the collision becomes **four times** its value, then the average force:

MCQ

A is doubled

B is halved

C does not change

D is reduced to one quarter

- 20 On a smooth horizontal surface, a body A of mass 0.30 kg moving at 5 m/s collides with a body B of mass 0.60 kg at rest. If body A moves at 3 m/s perpendicular to its original direction immediately after the collision, then the speed of body B after the collision and the angle  $\theta$  that its direction makes below the original line of motion are:

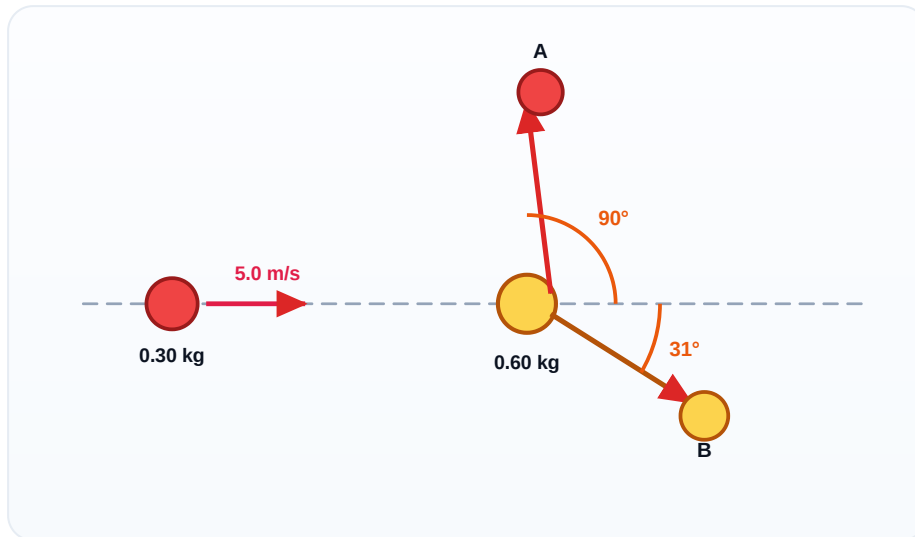


FIGURE 12

- A 3.25 m/s ,  $\theta \approx 26.6^\circ$       B 2.92 m/s ,  $\theta \approx 31.0^\circ$   
 C 3.25 m/s ,  $\theta \approx 31.0^\circ$       D 2.92 m/s ,  $\theta \approx 26.6^\circ$

- 21 The two graphs show the force–time relation for two bodies 1 and 2. Find the ratio between the impulse in the two cases ( $J_1 : J_2$ ).

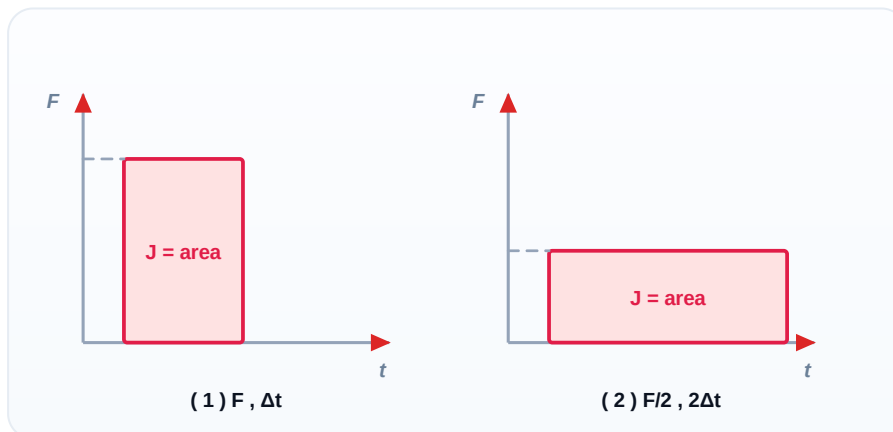


FIGURE 13

- 22 The graph shows the force acting on a body of mass 1.50 kg during the time from 0 to 6 s. If the initial velocity of the body was zero, what is its speed at  $t = 6$  s?

MCQ

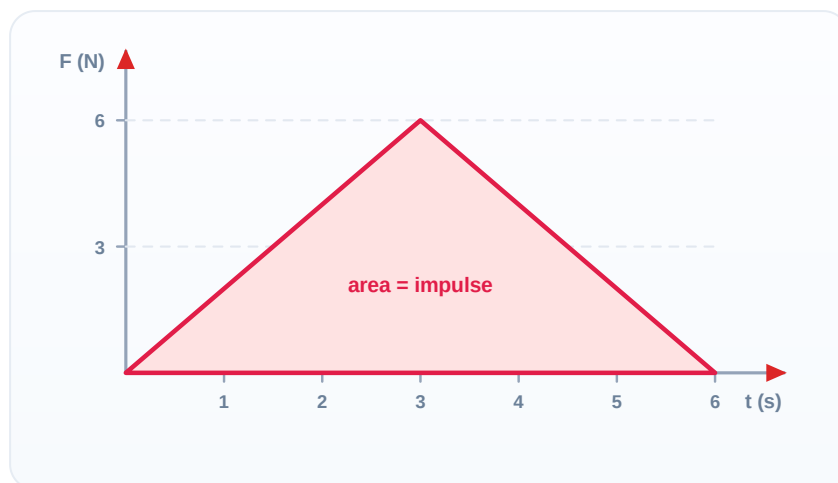


FIGURE 14

A 6.0 m/s

B 9.0 m/s

C 12 m/s

D 18 m/s

# Momentum and Mechanical Energy

Elastic and inelastic collisions, the coefficient of restitution, and the kinetic energy lost in a collision.

Mr. Gemy

PHYSICS

Grade 11 · Egyptian Baccalaureate

13 questions

Step-by-step solutions

smooth horizontal surfaces

## The toolkit for this lesson

 $\Sigma p \text{ before} = \Sigma p \text{ after}$ 

always true for an isolated system

 $KE = \frac{1}{2} m v^2$ 

kinetic energy of each body

 $e = \text{separation speed} / \text{approach speed}$ 

coefficient of restitution

 $e = 1$ 

perfectly elastic — KE conserved

 $0 < e < 1$ 

inelastic — part of the KE is lost

 $e = 0$ 

perfectly inelastic — the bodies move together

**The key distinction of this lesson.** Momentum is conserved in **every** collision of an isolated system, elastic or not. Kinetic energy is conserved only in a perfectly elastic collision; in any other collision part of it turns into internal energy, sound and permanent deformation.

## CLASSWORK 1

A

### Collisions and the coefficient of restitution

1 In an **inelastic** collision of an isolated system, which of the following statements is correct?

MCQ

A both the momentum and the kinetic energy are conserved

B the momentum is not conserved while the kinetic energy is conserved

C the momentum is conserved while the kinetic energy decreases

D both the momentum and the total energy decrease

- 2 A car A of mass 1000 kg moving at 20 m/s towards the east collides with a stationary truck B of mass 3000 kg at a traffic signal. The car and the truck lock together immediately after the collision and move as one body.
- Calculate the speed of the combined body (car + truck) just after the collision.
  - Calculate the coefficient of restitution ( $e$ ) for this collision, and deduce the type of the collision from its value.
  - Calculate the amount of kinetic energy lost as a result of this collision.

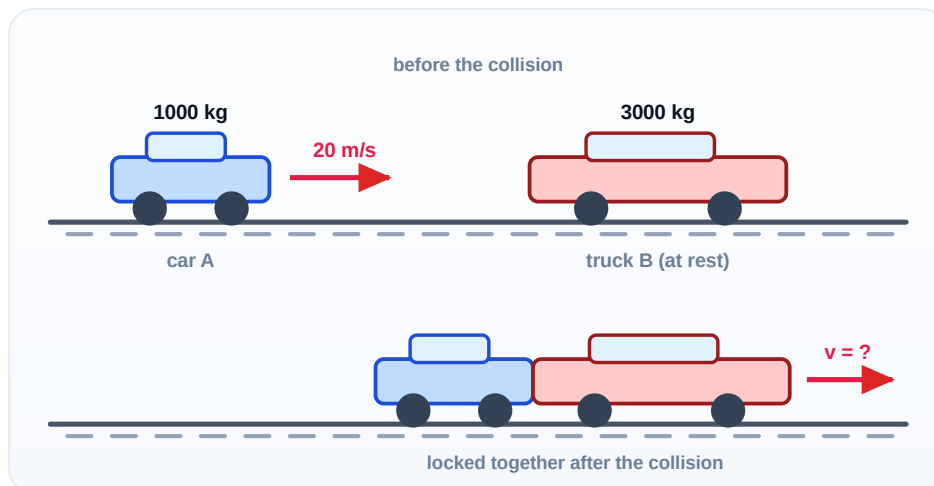


FIGURE 1

## HOME ASSIGNMENT 1

**B** Energy lost and the restitution coefficient

- 3 Two balls collide and then separate after the collision, and the sum of their kinetic energies decreases. We conclude that the collision is:

MCQ

- A** perfectly elastic, because the two balls separated
- B** inelastic, and it is not necessary for the two bodies to stick together
- C** perfectly inelastic
- D** not obeying the conservation of momentum

- 4 A ball A of mass 2.0 kg moving at 6.0 m/s towards the east collides with a ball B of mass 3.0 kg at rest. The two balls stick together after the collision and move together in the same direction.

- a) Calculate their common speed after the collision.  
b) Calculate the decrease in the total kinetic energy as a result of the collision.



FIGURE 2

- 5 A rubber ball collides perpendicularly with a smooth wall. Its speed before the collision was 10 m/s towards the wall, and it rebounded after the collision at 6 m/s in the opposite direction. **What is the value of the coefficient of restitution  $e$ ?**

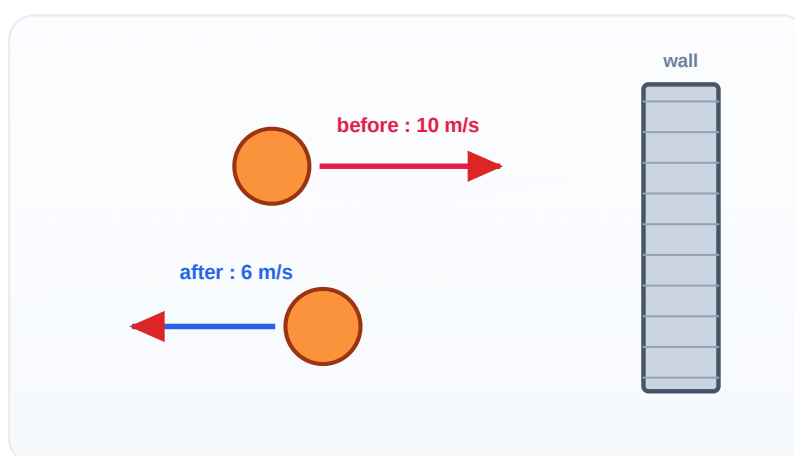


FIGURE 3

## Group A

- 6 A lump of clay of mass 0.50 kg moves at 8.0 m/s towards a stationary trolley of mass 1.50 kg on a smooth horizontal surface. After the collision the clay sticks to the trolley and the two move together. **Calculate their common speed after the collision.**

PROBLEM

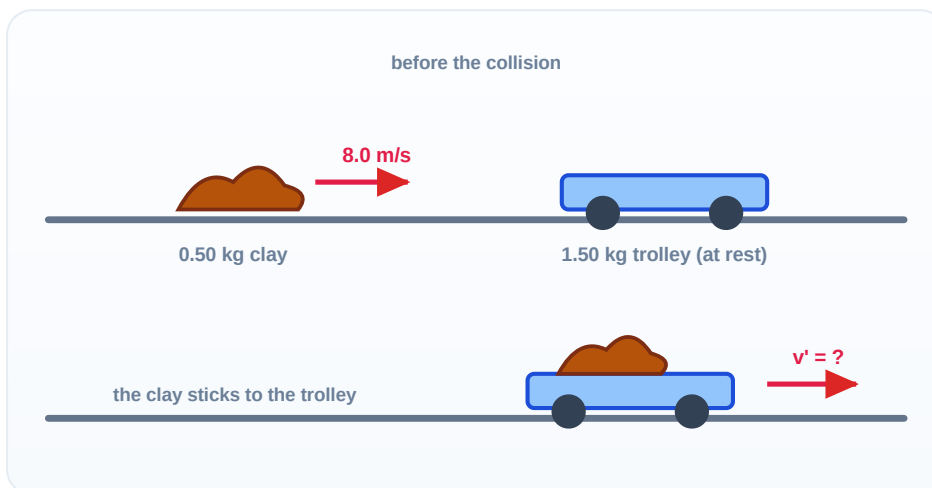


FIGURE 4

- 7 For the collision of the previous question, which statement describes it correctly?

MCQ

- A both the momentum and the kinetic energy are conserved
- B the momentum is not conserved, because the two bodies stuck together
- C the two masses move with one common speed after the collision
- D the trolley rebounds in the opposite direction and the clay stays separate

- 8 A car collides with a barrier, so that its front is deformed and the temperature of some of its parts rises. The kinetic energy that was lost was mainly converted into:

MCQ

- A potential energy only
- B internal energy, sound and deformation
- C extra momentum
- D extra mass for the car

- 9 A ball A moves at 4.0 m/s along a straight line towards a stationary ball B. Immediately after the collision the speed of ball B is 3.0 m/s in the original direction of motion, and the coefficient of restitution is 0.50. **What is the speed of ball A after the collision?**

MCQ

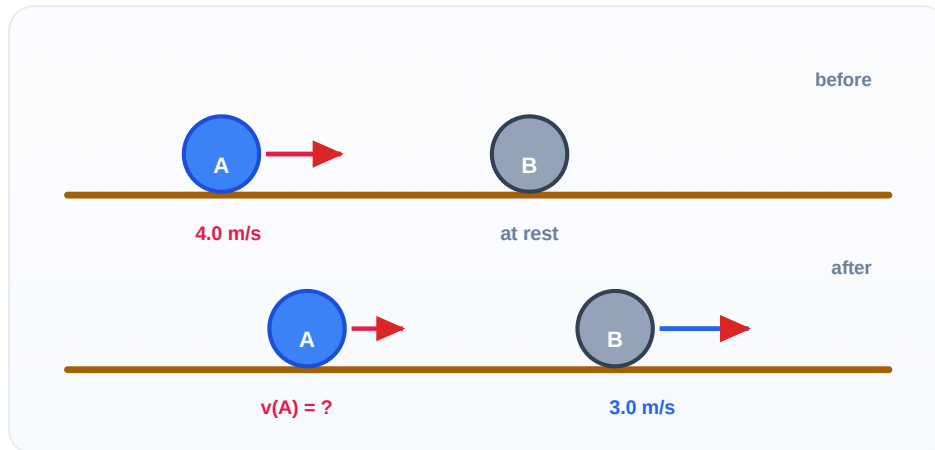


FIGURE 5

A zero

B 0.5 m/s

C 1.0 m/s

D 2.0 m/s

### Group B

- 10 For the two balls of question 4 (2.0 kg at 6.0 m/s striking 3.0 kg at rest and sticking to it), **what percentage of the original kinetic energy is lost** in the collision?

PROBLEM

- 11 Hearing a loud sound during an inelastic collision is an evidence of:

MCQ

A the destruction of part of the energy

B the conversion of part of the kinetic energy into sound energy

C the non-conservation of momentum

D the increase of the total kinetic energy

### Group C

- 12 A trolley A of mass 2.0 kg moves at 4.0 m/s towards the east and collides with a stationary trolley B of mass 3.0 kg. After the collision the two trolleys lock together and move in the same direction with a common speed  $v'$ . Find this common speed.



FIGURE 6

- 13 If the total kinetic energy of two bodies decreases by 30 J during a collision in an isolated system, then the most accurate explanation is that:

MCQ

- A 30 J of energy has disappeared
- B 30 J has been converted into other forms of energy inside the system and its near surroundings
- C the momentum has decreased by 30 kg·m/s
- D the mass of the system has decreased

# Uniform Circular Motion and Centripetal Force

Periodic time, frequency, angular velocity, centripetal acceleration and the force that keeps a body on its circular path.

Grade 11 · Egyptian Baccalaureate

21 questions

Step-by-step solutions

 $\pi = 3.14$ 

**Mr. Gemy**  
PHYSICS

## The toolkit for this lesson

**$T$  = time / number of revolutions**  
periodic time (s)

**$f = 1 / T$**   
frequency (Hz)

**$\omega = 2\pi / T = 2\pi f$**   
angular velocity (rad/s)

**$v = \omega r = 2\pi r / T$**   
linear (tangential) speed

**$a_c = v^2 / r = \omega^2 r$**   
centripetal acceleration

**$F_c = m v^2 / r = m \omega^2 r$**   
centripetal force, towards the centre

**The idea that the whole lesson turns on.** In uniform circular motion the **speed** never changes, but the **velocity** does, because its direction turns at every instant. A changing velocity means an acceleration, and that acceleration always points towards the centre.

## Describing uniform circular motion

- 1 Which of the following accurately describes uniform circular motion and the reason for the acceleration of the body that moves in this way?

MCQ

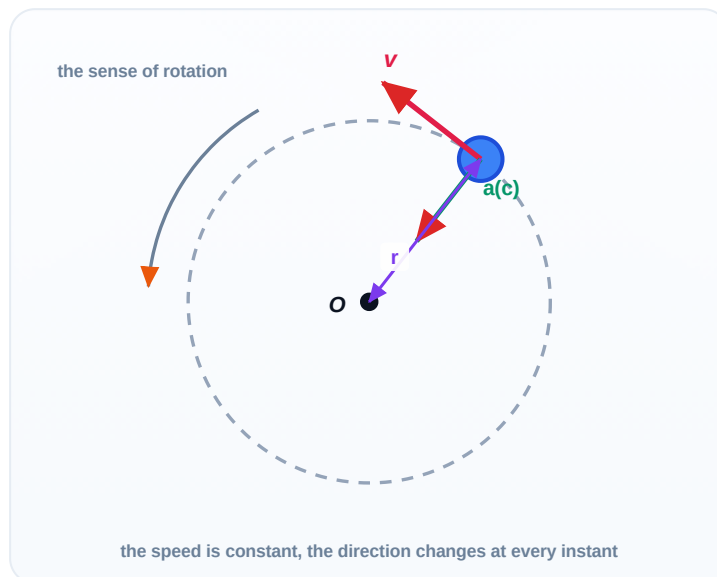


FIGURE 1

- A the velocity vector is constant, so the acceleration equals zero
- B the magnitude of the velocity changes while its direction stays constant
- C the speed is constant, but the direction of the velocity changes continuously towards the centre
- D the body is acted on by an outward centrifugal force that produces a tangential acceleration

- 2 A body moves in uniform circular motion, completing 10 revolutions in 2.0 s on a circular path of radius 0.40 m.

PROBLEM

- a) Determine the periodic time of the body.
- b) Determine the angular velocity of the body.

- 3 A small ball of mass 0.50 kg is fixed to a string of length 2.0 m and moves in uniform circular motion with an angular velocity of 3.0 rad/s. **Calculate its centripetal acceleration.**

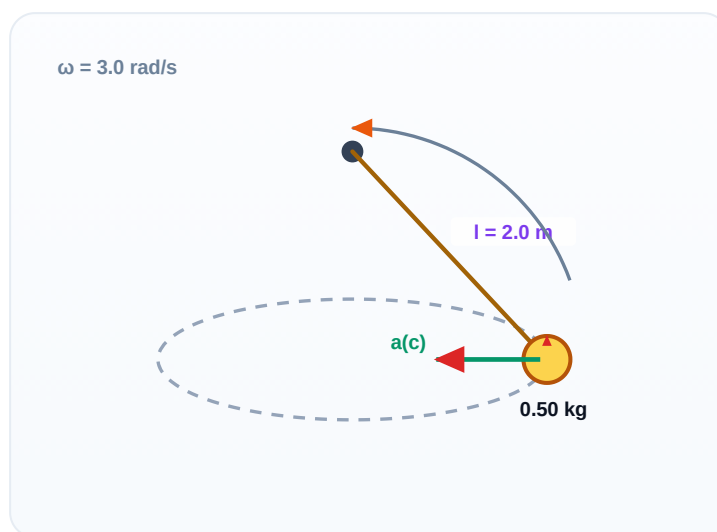


FIGURE 2

## HOME ASSIGNMENT 2

## B Speed, force and a rotating spring

- 4 A body moves in uniform circular motion, completing 20 revolutions in 5.0 s on a circular path of radius 0.20 m.
- Calculate the periodic time.
  - Calculate the linear speed.

- 5 A small ball of mass 0.50 kg is fixed to a string of length 2.0 m and moves with an angular velocity of 3.0 rad/s.
- Determine its linear speed.
  - Determine the magnitude of the force needed to maintain the circular motion.

- 6 A small ball of mass  $0.50\text{ kg}$  is fixed to a spring of natural length  $0.10\text{ m}$  and spring constant  $30\text{ N/m}$ , and moves in a circular path with an angular velocity of  $6.0\text{ rad/s}$ . Determine the extension of the spring.

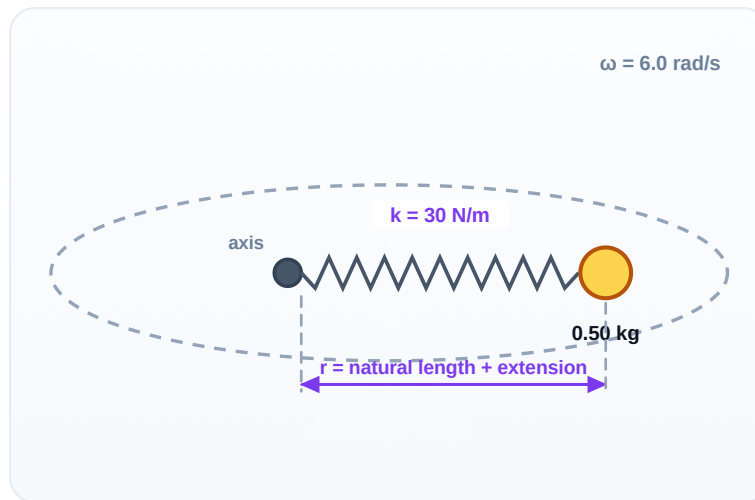


FIGURE 3

## Group A

7 The angular velocity ( $\omega$ ) of a body moving in a circular path is defined as:

MCQ

- ☐ A the linear displacement per unit time
- ☐ B the angular displacement per unit time
- ☐ C the total number of revolutions in one second
- ☐ D the time taken to complete one full revolution

8 The centripetal force acting on a body of mass  $m$  moving in a circular path of radius  $r$  with a speed  $v$  is given by the relation:

MCQ

- ☐ A  $F = m v r$
- ☐ B  $F = m v^2 / r$
- ☐ C  $F = m v / r^2$
- ☐ D  $F = m^2 v / r$

9 A horizontal disc rotates with a constant angular velocity of  $4\pi$  rad/s. The point A lies on the rim of the disc, at a distance of 0.30 m from the centre. Calculate:

PROBLEM

- a) the periodic time  $T$ .
- b) the linear speed of the point.

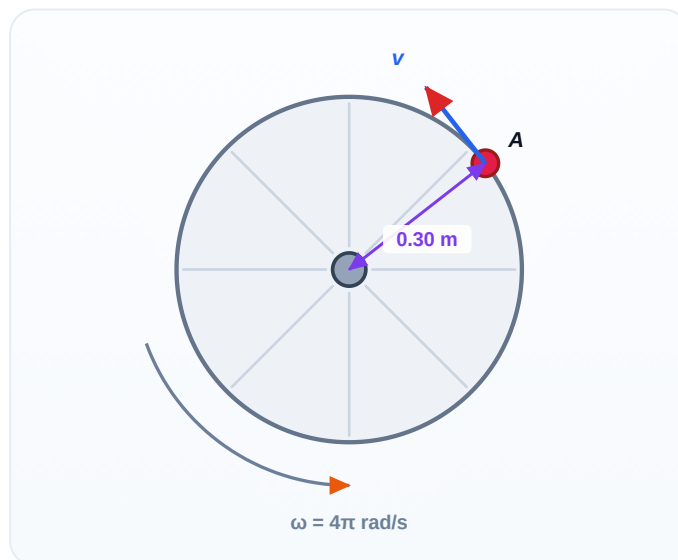


FIGURE 4

- 10** Explain the physical mechanism that makes the water leave the drum of a washing machine through the holes in its wall during the drying (fast spin) cycle.

EXPLAIN

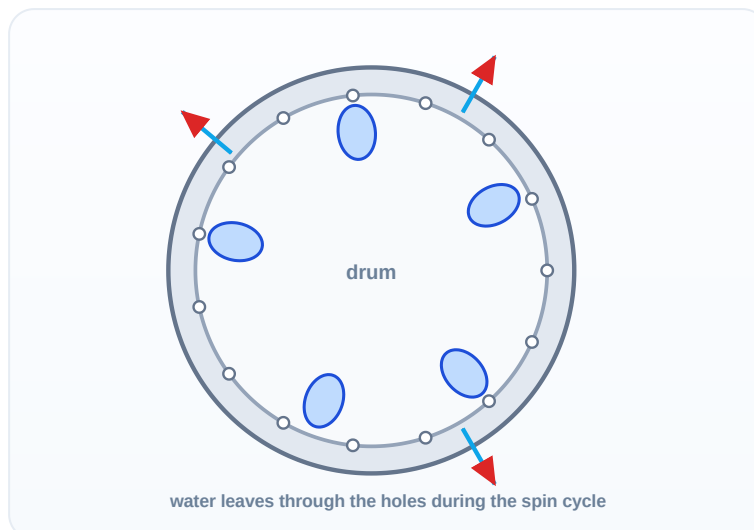


FIGURE 5

- 11** A body of mass 3.0 kg moves in a circular path of radius 2.0 m, completing one revolution in 4.0 s. ( $\pi = 3.14$ )

PROBLEM

- a) Determine the magnitude of its acceleration.  
b) Determine the magnitude of the centripetal force required.

## Group B

- 12** The periodic time (T) of a body moving in a circular path is defined as the time taken to complete:

MCQ

- A one revolution  
B  $2\pi$  revolutions  
C 10 revolutions  
D a fraction of a revolution in one second

- 13** Which of the following statements correctly describes the resultant of the forces acting on a body moving in uniform circular motion?

MCQ

- A an outward centrifugal force balances the inward force  
B there is no outward force; the resultant of the forces is always directed towards the centre  
C the resultant of the forces acts tangentially, along the direction of the velocity  
D the resultant of the forces equals zero because the speed is constant

- 14** Describe the path of a ball being whirled in a circle at the instant the string holding it breaks, and explain why it does not fly off directly outwards, away from the centre.

EXPLAIN

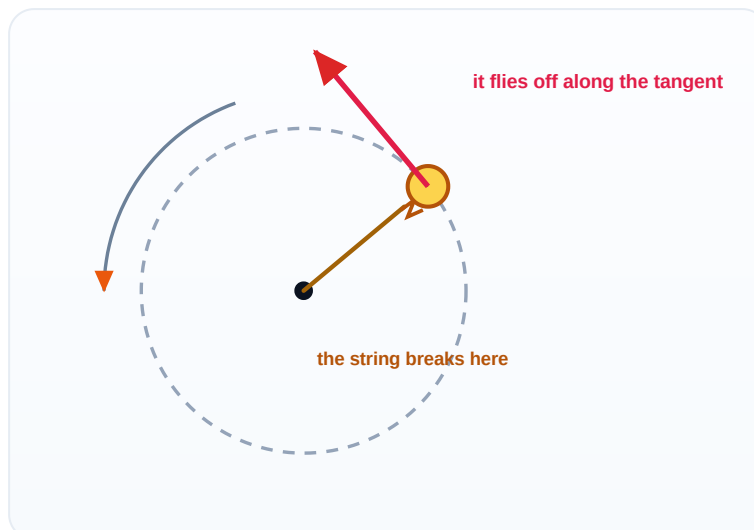


FIGURE 6

- 15** What happens in each of the following cases?

EXPLAIN

- a) a sudden break of the string that holds a ball moving in a circular path.
- b) increasing the radius of the circular path while the angular velocity of the body stays constant.

- 16** A body moves in uniform circular motion, completing 15 revolutions in 3.0 s on a circular path of radius 0.30 m.

PROBLEM

- a) Calculate the periodic time and the frequency.
- b) Calculate the linear speed.

### Group C

- 17** The centripetal acceleration ( $a_c$ ) of a body moving in a circular path can be calculated using:

MCQ

**A**  $a_c = \omega^2 r$  or  $a_c = v^2 / r$

**B**  $a_c = \omega / r$  or  $a_c = v r$

**C**  $a_c = \omega r^2$  or  $a_c = v^2 r$

**D**  $a_c = \omega^2 / r$  or  $a_c = v / r^2$

- 18 A ball moves along the inner surface of a horizontal semicircular ring as shown in the top view. Which arrow represents the direction of the average force acting on the ball?

MCQ

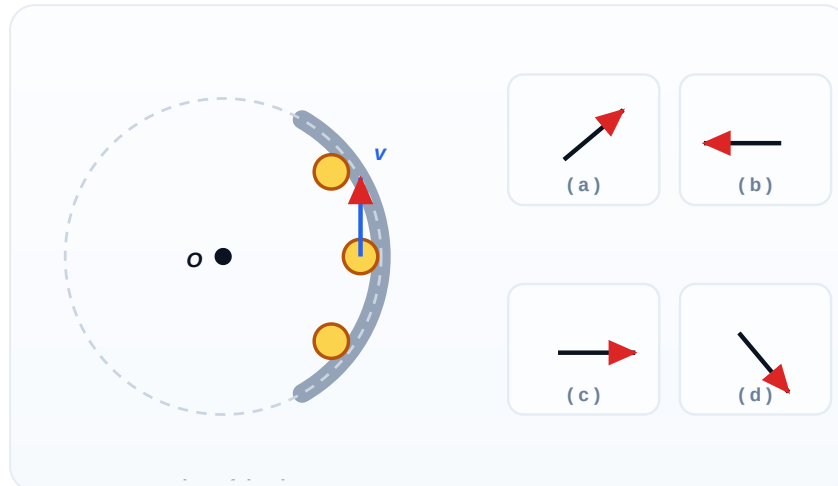


FIGURE 7

- ☒ A arrow (a)
- ☐ B arrow (b)
- ☐ C arrow (c)
- ☐ D arrow (d)

- 19 Explain why a body that moves in a circular path with a constant speed possesses an acceleration that is not equal to zero.

EXPLAIN

- 20 State the reason for each of the following:

EXPLAIN

- a) a body moving in a circular path with a constant speed is said to be in a state of acceleration.
- b) there is no "centrifugal force" directed outwards acting on a body in circular motion.

- 21 A ball of mass 0.40 kg moves in a horizontal circular path of radius 0.50 m with a speed of 2.0 m/s.

PROBLEM

- a) Determine the value of the centripetal acceleration.
- b) Determine the value of the centripetal force required.

# Horizontal and Vertical Circular Motion

The conical pendulum, motion inside a conical vessel, and a body whirled in a vertical circle.

Grade 11 · Egyptian Baccalaureate

27 questions

Step-by-step solutions

 $g = 9.8 \text{ m/s}^2$ 

**Mr. Gemy**  
PHYSICS

## The toolkit for this lesson

$$T \cos \theta = mg$$

conical pendulum — vertical

$$T \sin \theta = m v^2 / r$$

conical pendulum — horizontal

$$r = l \sin \theta$$

radius of the horizontal circle

$$\text{top : } T + mg = m v^2 / r$$

both forces point to the centre

$$\text{bottom : } T - mg = m v^2 / r$$

the tension is largest here

$$v(\text{min}) = \sqrt{g r}$$

at the top, when  $T = 0$ 

**One habit solves this whole lesson.** Draw the body at the position asked about, mark every force, then ask: *which way is the centre?* The resultant along that direction is always  $m v^2 / r$  — nothing else.

## The vertical circular path

- 1 The minimum value of the speed needed at the highest point of a vertical circular path of radius  $r$ , so that the body keeps moving on its circular path (that is, the tension in the string becomes  $T = 0$ ), is given by the relation:

MCQ

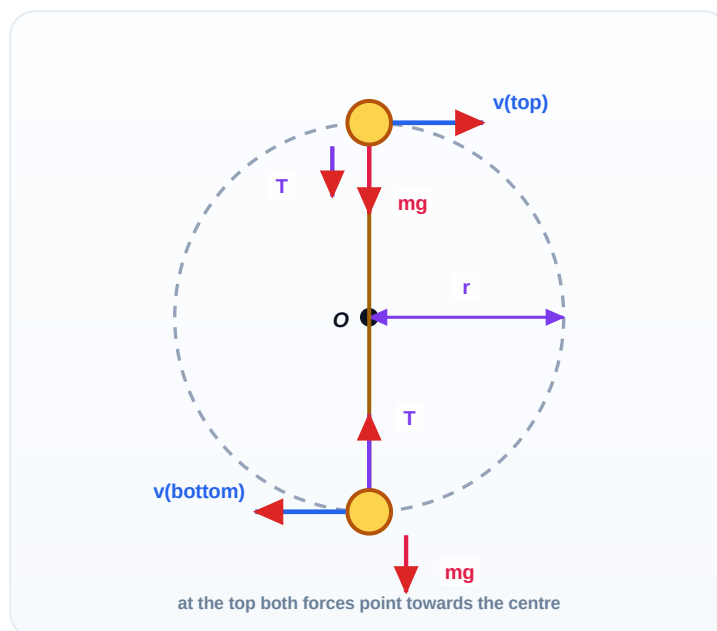


FIGURE 1

A  $v(\text{min}) = \sqrt{2gr}$

B  $v(\text{min}) = \sqrt{gr}$

C  $v(\text{min}) = gr$

D  $v(\text{min}) = \sqrt{g/r}$

- 2 Determine the reason why the water stays inside a bucket that is whirled in a vertical circular path, even at the instant when the bucket is upside down at the highest point of the path.

EXPLAIN

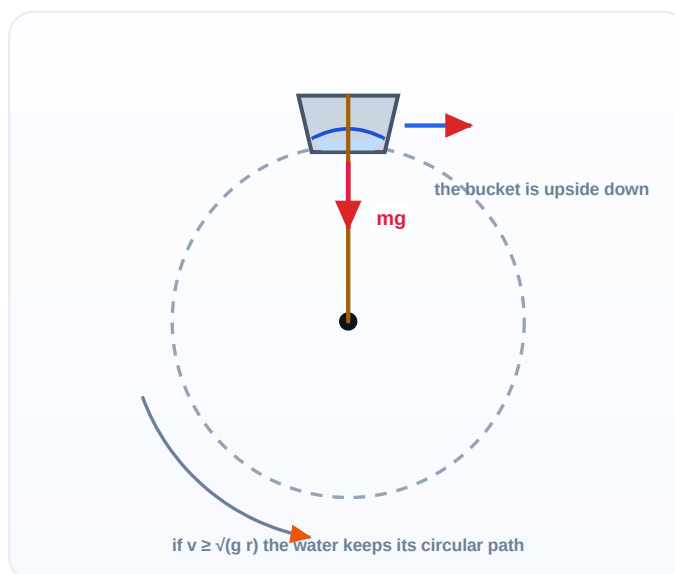


FIGURE 2

- 3 A small ball of mass 0.50 kg is tied to a string and moves in a vertical circular path of radius 0.80 m. At the lowest point of the path its speed is 4.0 m/s. **Calculate the tension in the string at this point.**

PROBLEM

#### HOME ASSIGNMENT 1

### B The conical vessel and the minimum speed

- 4 A small ball moves in uniform circular motion at a height  $h$  inside a vessel with a smooth conical inner surface, where the axis of the cone makes an angle  $\theta$  with the slant (generating) line of the cone.

PROBLEM

- a) Determine the normal force  $N$  acting on the ball, in terms of  $m$ ,  $\theta$  and  $g$ .  
b) Determine the speed of the ball, in terms of  $h$  and  $g$ .

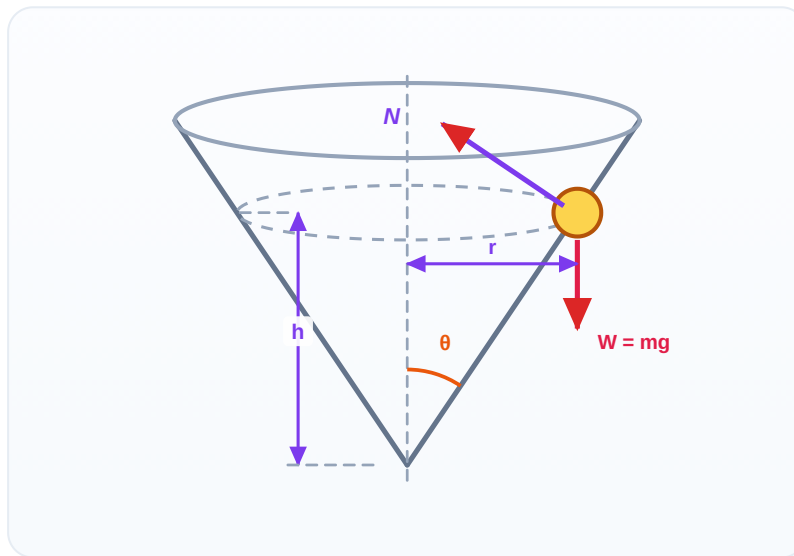


FIGURE 3

- 5 A small ball of mass 0.50 kg is tied to a string and whirled in a vertical circular path of radius 0.80 m. **Determine the minimum speed needed at the highest point of the circular path so that the string remains taut.**

PROBLEM

- 6 **Explain why** the water in the whirling-bucket experiment begins to spill first at the **upper** part of the circular path rather than at the lower part.

EXPLAIN

## The conical pendulum

- 7 A small ball of mass  $0.40 \text{ kg}$  is tied to a string and moves in a vertical circle of radius  $0.60 \text{ m}$ . At the lowest point of the circle its speed is  $4.0 \text{ m/s}$ . The tension in the string at the lowest point equals:

MCQ

A  $12.6 \text{ N}$ B  $13.4 \text{ N}$ C  $14.6 \text{ N}$ D  $15.4 \text{ N}$ 

- 8 Describe the forces acting on a conical pendulum, and determine which component of the tension force provides the centripetal force.

EXPLAIN

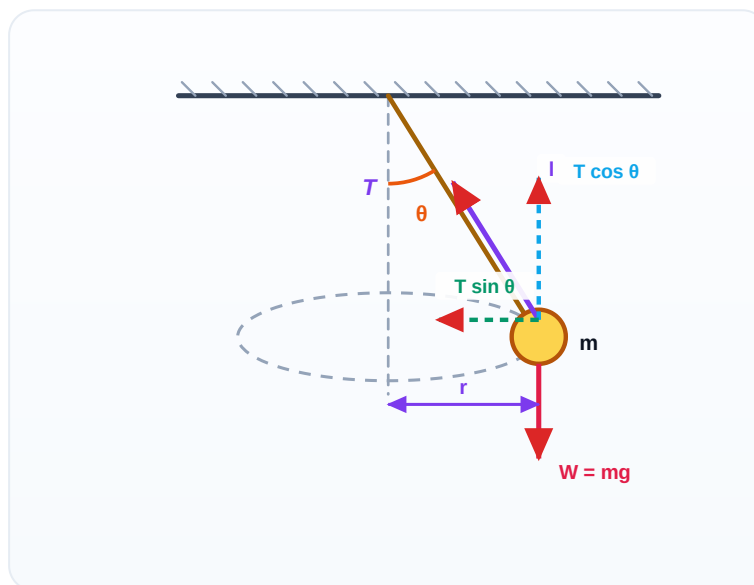


FIGURE 4

- 9 A small ball moves in uniform circular motion on the smooth inner surface of a conical vessel, along a horizontal circular path of radius  $r$ , where the slant side of the cone makes an angle  $\theta$  with the horizontal plane.

PROBLEM

- Write the equation of vertical equilibrium for the ball.
- Write the equation of the centripetal force acting on the ball.

D

## Tension, normal force and the loop

- 10 One end of a string of length 1 m is fixed to the ceiling, and a small ball of mass 0.5 kg tied to its other end moves in uniform circular motion in a smooth horizontal plane. If the angle between the string and the vertical is  $30^\circ$  and  $g = 9.8 \text{ m/s}^2$ , then the tension in the string is approximately:

MCQ

A 4.90 N

B 5.66 N

C 2.54 N

D 9.80 N

- 11 One end of a string of length  $l$  is fixed to the ceiling, and a ball of mass  $m$  tied to its other end moves in uniform circular motion in a horizontal plane, so that the string makes an angle  $\theta$  with the vertical and the angular velocity is  $\omega$ .

PROBLEM

- a) Determine the tension in the string.  
b) Determine the normal force acting on the ball if it also rests on a horizontal surface.

- 12 **Explain why** a car that almost loses contact with the track at the top of a circular loop is subjected to a very large normal force at the bottom of the loop.

EXPLAIN

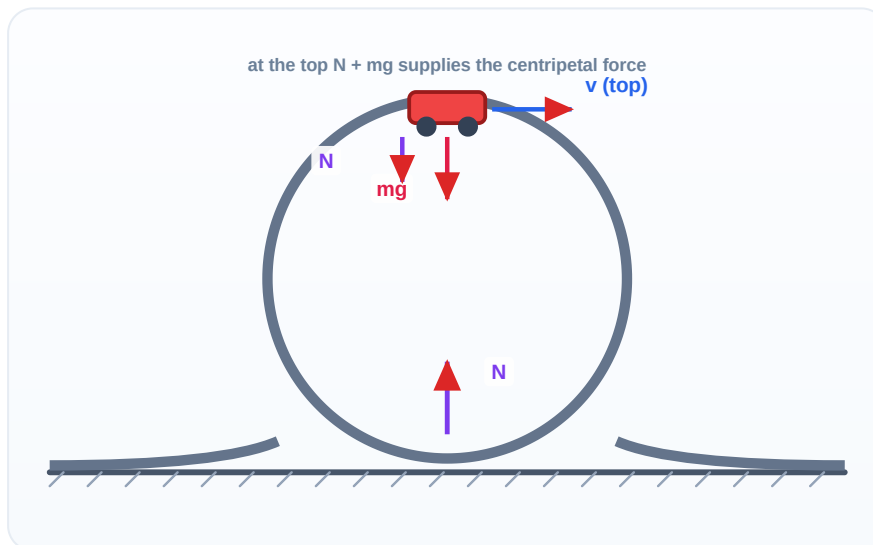


FIGURE 5

E

## Groups A, B and C

## Group A

13 At the highest point of a vertical circular path, the resultant of the tension force ( $T$ ) and the weight ( $mg$ ) must be directed:

MCQ

- A tangentially, in the direction of motion
- B directly away from the centre of the circle
- C directly towards the centre of the circle
- D vertically upwards, balancing gravity

14 The tension in a string whirling a body in a vertical circular path reaches its **maximum** value at:

MCQ

- A the highest point of the path
- B the lowest point of the path
- C the horizontal position level with the centre
- D the point at which the speed is zero

15 **Evaluate and discuss** the correctness of the following statement: "The minimum value of the speed at the top of a vertical circular path occurs when the normal force (or the tension) becomes infinite."

EXPLAIN

16 A stone tied to a string moves in a horizontal circular path with a constant speed  $v$ . The figure shows the stone at two different positions X and Y. **Show, by drawing, the direction of the change in the velocity** of the stone as it moves from position X to position Y.

EXPLAIN

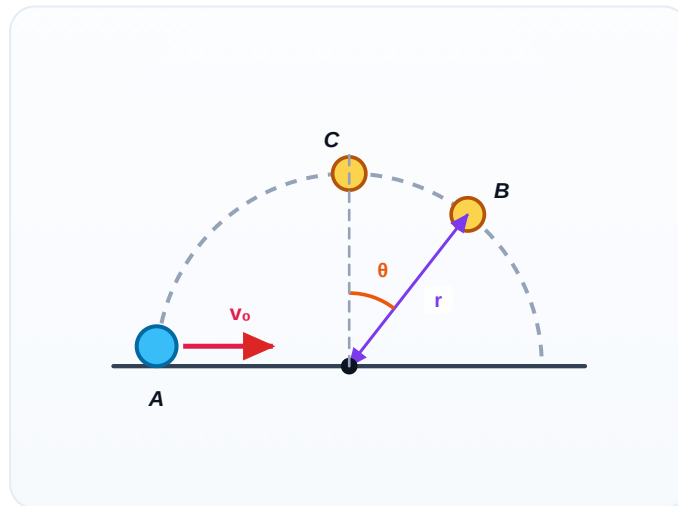


FIGURE 6

17 A body moves with an initial speed  $v_0$  along a semicircular path from A, passing through B and C as shown. If the radius of the path is reduced to half while the initial speed  $v_0$  stays constant, **what happens to the speed of the body at the point B?**

EXPLAIN

## Group B

**18** In vertical circular motion, the minimum speed at the highest point of the loop is derived from the physical condition that the tension in the string becomes:

MCQ

**A** equal to twice the weight

**B** zero ( $T = 0$ )

**C** equal to the centripetal force only

**D** infinite

**19** A real roller-coaster loop is designed as a teardrop-shaped curve rather than a complete circle, in order to:

MCQ

**A** increase the maximum speed at the top

**B** change the radius of curvature gradually and control the centripetal acceleration on the riders

**C** cancel the gravitational force completely

**D** keep the force on the track constant all along the path

**20** **Analyse the forces** acting on a conical pendulum, and explain why the **horizontal** component of the tension — and not the vertical component — is the one that provides the centripetal force.

EXPLAIN

**21** What happens in each of the following cases?

EXPLAIN

**a)** whirling a bucket of water in a vertical circular path too slowly to maintain the speed required at the highest point.

**b)** increasing the angle that the string of a conical pendulum makes with the vertical.

**22** A mass of 0.60 kg is tied to a string of length 1.0 m and rotates in a vertical circular path. **Determine the minimum speed at the highest point** of the path needed to keep the string taut.

PROBLEM

## Group C

**23** In a conical pendulum, the component of the tension force that provides the required centripetal force is:

MCQ

**A**  $T \cos \theta$  (the vertical component)

**B**  $T \sin \theta$  (the horizontal component)

**C**  $T \tan \theta$

**D** the whole tension  $T$  acting vertically

**24** **Explain the effect** of the tension force and the gravitational force acting in the **same** direction at the highest point of a vertical circular path.

EXPLAIN

**25** Discuss the wrong idea which says that “the water spills out of a bucket whirled too slowly in a vertical circle because of an outward centrifugal force”. Give the correct physical explanation, based on the concept of inertia.

EXPLAIN

**26** State the reason for each of the following:

EXPLAIN

- a) roller-coaster loops are designed as teardrop-shaped curves instead of perfect circles.
- b) the tension in the string whirling a ball in a vertical circular path has its maximum value at the lowest point of the path.

**27** A conical pendulum consists of a bob of mass 0.30 kg hanging from a string of length 0.80 m, making an angle of  $30^\circ$  with the vertical.

PROBLEM

- a) Calculate the tension in the string.
- b) Calculate the speed of the bob.

# Kepler's Laws and Universal Gravitation

Newton's law of gravitation, the gravitational field, the three laws of Kepler and the geostationary satellite.

Grade 11 · Egyptian Baccalaureate

27 questions

Step-by-step solutions

 $G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$ Mr. Gemy  
PHYSICS

## The toolkit for this lesson

$$F = G m_1 m_2 / r^2$$

Newton's law of gravitation

$$g = F / m$$

field strength = force per unit mass

$$g = G M / (R + r)^2$$

at a height  $r$  above the surface**1st law : an ellipse**

the Sun at one focus

**2nd law : equal areas** $v r = \text{constant} \rightarrow$  fastest when nearest**3rd law :  $T^2 \propto r^3$**  $T^2 / r^3$  is the same for one central body

**Two shortcuts worth memorising.** For Kepler's third law with the Earth's year and the astronomical unit,  $T \text{ (years)} = [a \text{ (AU)}]^{3/2}$ . For the second law, a comet's speeds obey  $v_1 r_1 = v_2 r_2$  at the nearest and farthest points.

## CLASSWORK 1

A

### Newton's law of gravitation

- 1** Newton's law of universal gravitation states that the magnitude of the gravitational force between two **identical** masses  $m$  separated by a distance of 1 metre is given by the relation:

MCQ

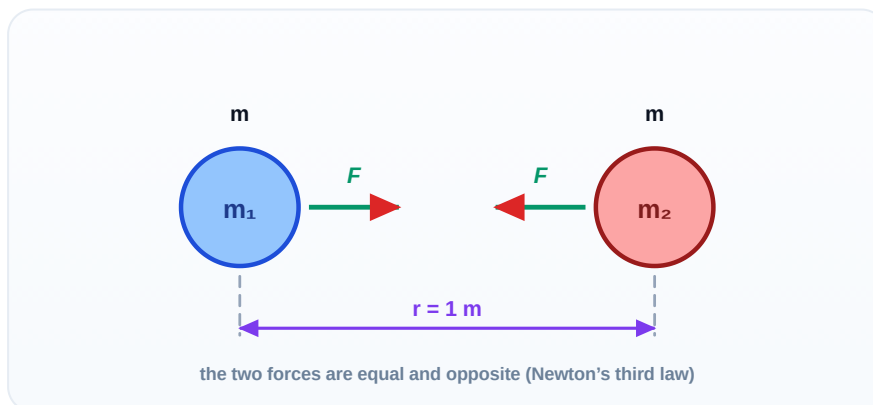


FIGURE 1

A  $F = G m^2$

B  $F = G m$

C  $F = m / G$

D  $F = G / m$

- 2 Two bodies, each of mass 60 kg, are placed so that the distance between them is 3.0 m. Using  $G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$ , **determine the magnitude of the gravitational force between them.**

PROBLEM

- 3 The length of the semi-major axis of the orbit of a planet is 9.0 astronomical units. Using the values for the Earth (a semi-major axis of 1.0 AU and a periodic time of 1.0 year), **determine the orbital periodic time of the planet.**

PROBLEM

#### HOME ASSIGNMENT 1

### B The gravitational field and the second law

- 4 **Define** the concept of the gravitational field of the Earth, and **explain** how it can be represented using field lines.

EXPLAIN

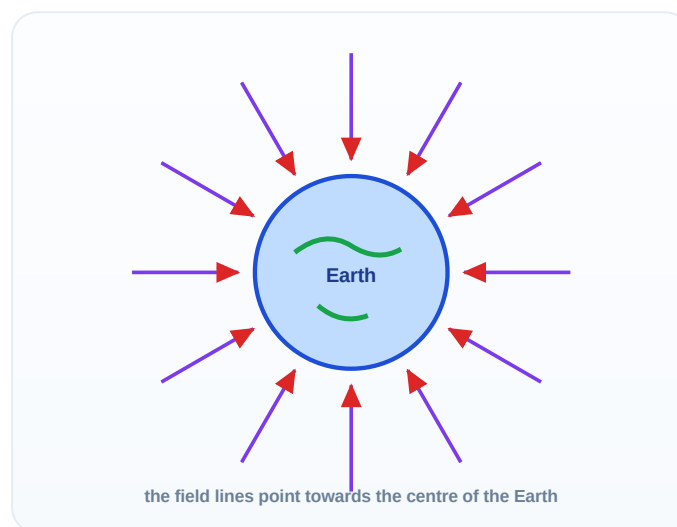


FIGURE 2

- 5 Two bodies, each of mass 40 kg, are separated by a distance of 3.0 m. Using  $G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$ , **calculate the gravitational force between them.**

PROBLEM

- 6 The distance between a comet and the Sun at its farthest point is 2.5 astronomical units, and at its nearest point it is 1.5 astronomical units. **Determine the ratio between its speed at the nearest point and its speed at the farthest point.**

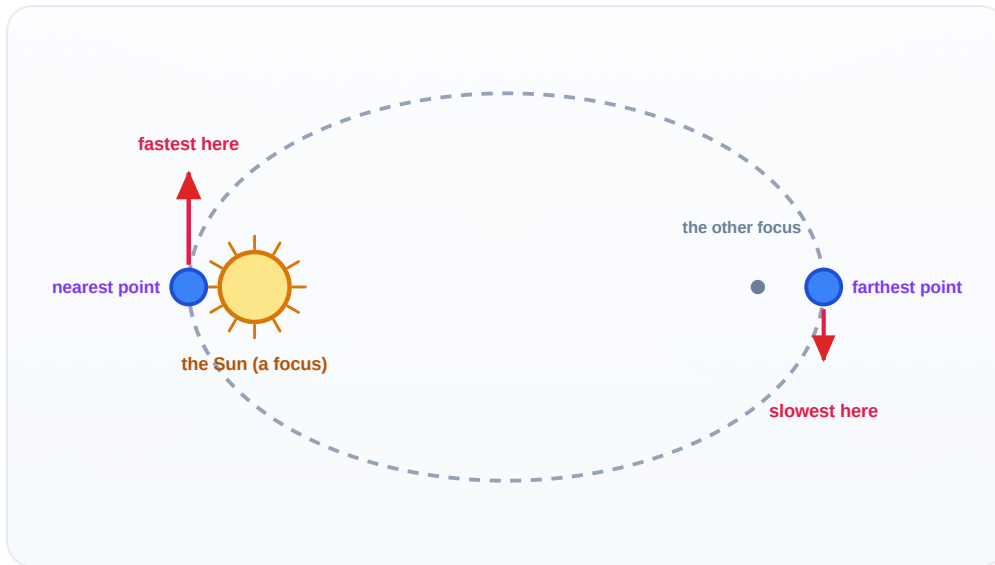


FIGURE 3

## CLASSWORK 2

C

## Kepler's third law and the geostationary satellite

- 7 Kepler's third law states that, for any satellite or planet revolving around a central body, the square of the orbital periodic time ( $T^2$ ) is directly proportional to:

MCQ

- A the semi-major axis ( $r$ )
- B the square of the semi-major axis ( $r^2$ )
- C the cube of the semi-major axis ( $r^3$ )
- D the square root of the semi-major axis ( $\sqrt{r}$ )

- 8 The length of the semi-major axis of the orbit of a planet is 7.0 astronomical units. Using the values for the Earth, **determine the orbital periodic time of the planet.**

PROBLEM

- 9 Explain why the orbital periodic time of a geostationary satellite must be equal to the periodic time of the rotation of the Earth about its own axis.

EXPLAIN

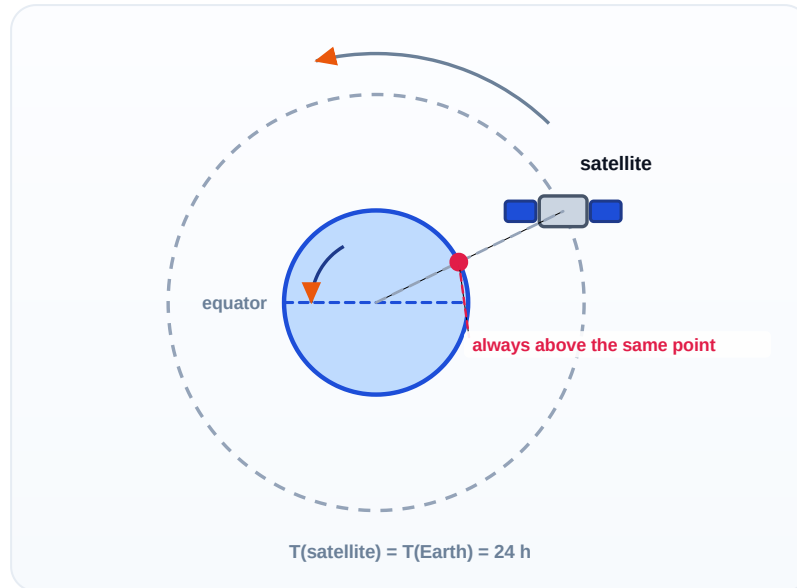


FIGURE 4

## HOME ASSIGNMENT 2

D

### Applying the laws

- 10 According to Newton's law of universal gravitation, if the distance between two bodies is increased to **three times** the original distance while their two masses stay constant, which of the following statements correctly describes the relation between the gravitational force and the distance separating the two bodies?

MCQ

- A it is directly proportional to the distance
- B it is directly proportional to the square of the distance
- C it is inversely proportional to the distance
- D it is inversely proportional to the square of the distance

- 11 Explain the reason for the constancy of the ratio  $T^2/r^3$  for the four largest moons of Jupiter, even though they revolve in orbits of different radii.

EXPLAIN

- 12** Explain how Kepler's third law helps in explaining the difference between the orbital periodic times of the planets and the asteroids that revolve around the Sun, showing why the distant bodies take a longer time to complete one revolution.

EXPLAIN

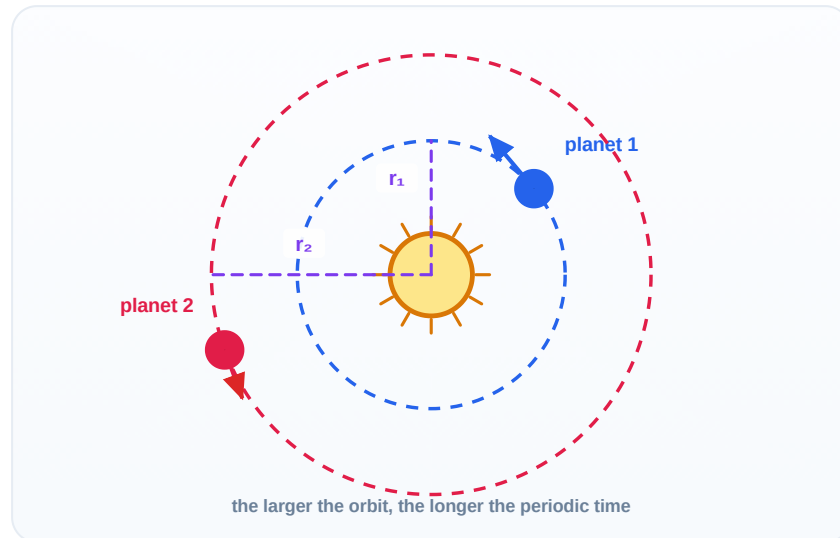


FIGURE 5

#### WEEKLY ASSESSMENT

E

### Groups A, B and C

#### Group A

- 13** According to Newton's law of universal gravitation, the force of attraction between two point masses:

MCQ

**A** is inversely proportional to the separating distance  $r$

**B** is directly proportional to the product of their masses and inversely proportional to  $r^2$

**C** is directly proportional to the sum of their masses

**D** is independent of the distance between them

- 14 The gravitational field strength  $g$  at a distance  $r$  **above the surface** of a spherical body of mass  $M$  and radius  $R$  is given by the relation:

MCQ

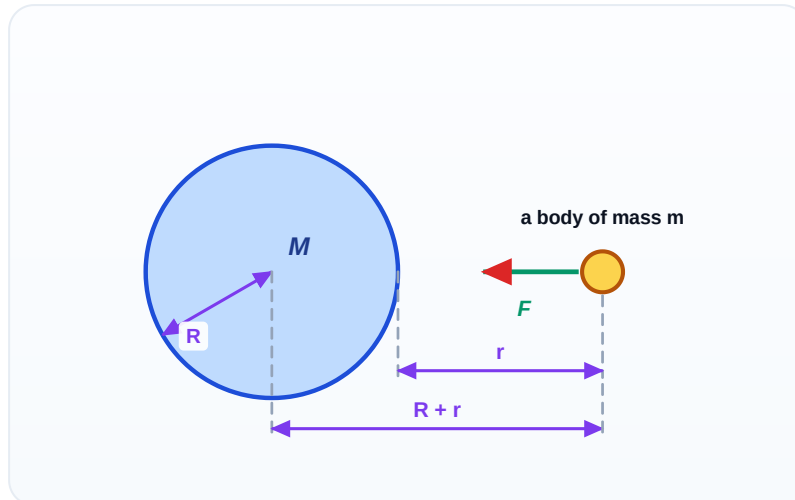


FIGURE 6

- A  $g = G M / r^2$
- B  $g = G M / (R + r)^2$
- C  $g = G M^2 / r$
- D  $g = G / (M r^2)$

- 15 **State** Kepler's second law of planetary motion, and **explain its physical origin** in terms of the principle of conservation of angular momentum.

EXPLAIN

- 16 **Explain why** the planets that have larger orbital radii have longer orbital periodic times according to Kepler's third law, comparing this with simple uniform linear motion.

EXPLAIN

- 17 A comet revolves around the Sun in an elliptical orbit. If the distance between the comet and the Sun at the farthest point is **five times** the distance at the nearest point:

PROBLEM

- a) Calculate the ratio between the speed of the comet at the farthest point and its speed at the nearest point.
- b) If the speed of the comet at the nearest point is 30 km/s, what is its speed at the farthest point?

## Group B

- 18 Kepler's third law states that the square of the periodic time ( $T^2$ ) of a planet is directly proportional to:

MCQ

- A the orbital speed ( $v$ )
- B the cube of the length of the semi-major axis ( $r^3$ )
- C the mass of the planet ( $m$ )
- D the reciprocal of the square of the orbital radius ( $1/r^2$ )

**19** A satellite that is launched around the Earth in a circular equatorial orbit with a periodic time of 24 hours, so that it appears fixed relative to an observer on the surface of the Earth, is called:

MCQ

**A** a polar satellite

**B** a geostationary satellite

**C** a low Earth orbit (LEO) satellite

**D** a satellite in an elliptical orbit

**20** **Explain why** the unit newton / kilogram (N/kg) is equivalent to the unit metre per second squared (m/s<sup>2</sup>).

EXPLAIN

**21** What happens in each of the following cases?

EXPLAIN

- a) doubling the distance between two masses, according to Newton's law of universal gravitation.
- b) moving a satellite into a larger circular orbit around the Earth.

**22** Two bodies, each of mass 80 kg, are separated by a distance of 2.0 m. **Calculate the gravitational force between them**, given that  $G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$ .

PROBLEM

## Group C

**23** The gravitational field strength (g) at a point in space is defined as:

MCQ

**A** the gravitational potential energy per unit volume

**B** the gravitational force acting per unit mass placed at that point

**C** the total gravitational force between two planets

**D** the gravitational acceleration multiplied by the mass

**24** According to Kepler's first law, where does the Sun lie in the orbit of a planet?

MCQ

**A** at the centre of the circular orbit

**B** at one of the two foci of the elliptical orbit

**C** completely outside the elliptical orbit

**D** at the midpoint of the minor axis of the orbit

**25** Two bodies, each of mass 100 kg, are separated by a distance of 5.0 m. Using  $G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$ , **calculate the gravitational force between them**.

PROBLEM

**26** Give the reason for each of the following:

EXPLAIN

- a) a geostationary satellite appears fixed above one single point on the surface of the Earth.
- b) a comet moves with its maximum speed when it is as close as possible to the Sun.

27

The length of the semi-major axis of the orbit of a planet around a star is **4.0 times** the length of the semi-major axis of the orbit of a second planet around the same star. Given that the periodic time of the second planet is 2.0 years, **calculate the periodic time of the first planet.**

PROBLEM

Mr. Gemy

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Mr. Gemy

Mr. Gemy

Mr. Gemy

Mr. Gemy

Mr. Gemy

QUESTIONS

# Unit Two

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Waves, Sound and Light

الوحدة الثانية · الموجات والصوت والضوء

# Simple Harmonic Motion

The restoring force  $F = -kx$ , the angular frequency, the periodic time of a mass–spring system and the displacement equation  $x = A \sin(\omega t)$ .

Grade 11 · Egyptian Baccalaureate

28 questions

Step-by-step solutions

 $\pi = 3.14$ 

**Mr. Gemy**  
PHYSICS

## The toolkit for this lesson

$$F = -kx$$

the restoring force, always towards  $x = 0$

$$\omega = \sqrt{k/m}$$

the angular frequency, in rad/s

$$T = 2\pi \sqrt{m/k}$$

the periodic time of a mass–spring system

$$f = 1/T$$

the frequency, in hertz

$$x = A \sin(\omega t)$$

displacement from the equilibrium position

$$v(\max) = \omega A$$

the speed at  $x = 0$ , where  $KE = E$

**Read the equation, do not memorise the numbers.** Whenever a question gives you  $x = A \sin(\omega t)$ , the number in front of the sine **is** the amplitude  $A$  and the number multiplying  $t$  **is**  $\omega$ . Everything else follows in one line:  $T = 2\pi/\omega$ ,  $f = 1/T$  and  $v(\max) = \omega A$ .

## Investigating a vibrating ruler

EXPLAIN

- 1 A student carried out the exploration shown in the figure: a steel ruler was clamped under a stack of books at the edge of a table so that most of its length projected beyond the edge. The free end was pushed downwards and released, and it vibrated. The student noticed that the **pitch** of the note given out by the ruler stayed practically constant throughout the vibration, even though the amplitude (the distance travelled by the free end) went on decreasing because of air resistance, until the ruler came to rest completely.

- a) Based on this observation, what is the scientific explanation of the constancy of the pitch of the note in spite of the continuous decrease of the amplitude?
- b) The student then moved the ruler so that the projecting part beyond the edge became **shorter** than before, and repeated the experiment. The pitch of the note given out rose (it became sharper). What is the explanation of this?

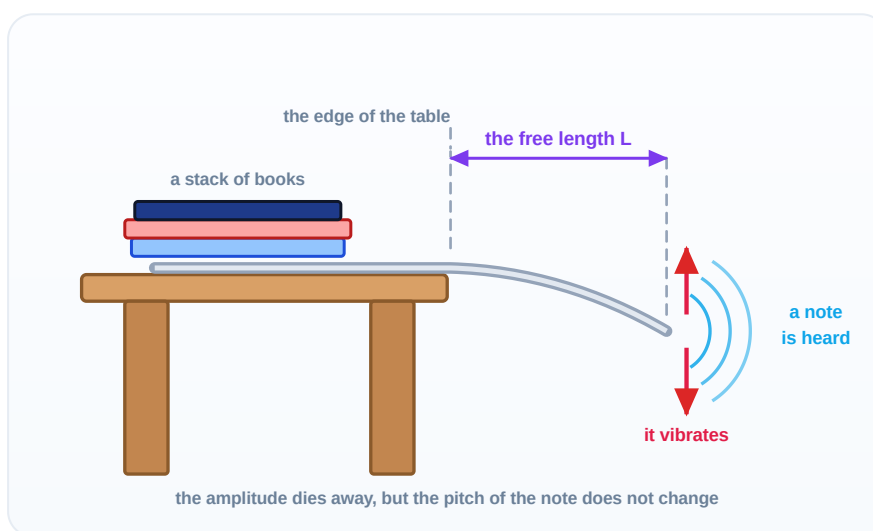


FIGURE 1

- 2 When the end of the vibrating ruler reaches the point of maximum displacement, it stops moving for one instant ( $v = 0$ ). At that very instant the restoring force and the acceleration have their maximum values. What is the direct scientific reason for this?

MCQ

A because a speed equal to zero forces the acceleration and the force to reach their greatest values

B because the restoring force depends on the magnitude of the displacement from the equilibrium position, and the displacement is greatest at this point

C because the energy of the motion is greatest at the point of instantaneous rest

D because the mass of the ruler increases momentarily on stopping, which increases the force and the acceleration

## The periodic time and the mass

- 3 A mass–spring system has a periodic time  $T_1 = 1.2$  s.

PROBLEM

- a) The student replaced the original mass by another one whose magnitude is **four times** the first mass ( $4m$ ), keeping the same spring. Predict first: will the new periodic time increase or decrease? Then find the ratio between the new periodic time  $T_2$  and the original periodic time  $T_1$  **without** needing the numerical values of  $m$  and  $k$ .
- b) Based on the ratio you reached in part (a), calculate the numerical value of the new periodic time  $T_2$ .

- 4 Can it be said that multiplying the mass by a certain number of times leads to multiplying the periodic time by the same number of times? **Explain your answer** by going back to the mathematical form of the relation between the periodic time and the mass.

EXPLAIN

- 5 A body attached to a horizontal spring vibrates with simple harmonic motion. It starts from the equilibrium position ( $x = 0$ ) at the instant  $t = 0$  moving towards the maximum positive displacement ( $+A$ ), which it reaches at the instant  $t = T/4$ ; it then returns to  $x = 0$  at the instant  $t = T/2$ , then reaches the maximum negative displacement ( $-A$ ) at the instant  $t = 3T/4$ , and finally returns to  $x = 0$  on the completion of the full periodic time at the instant  $t = T$ . **State the displacement, the speed and the acceleration of the body at each of these five instants**, and give the reason in each case.

EXPLAIN

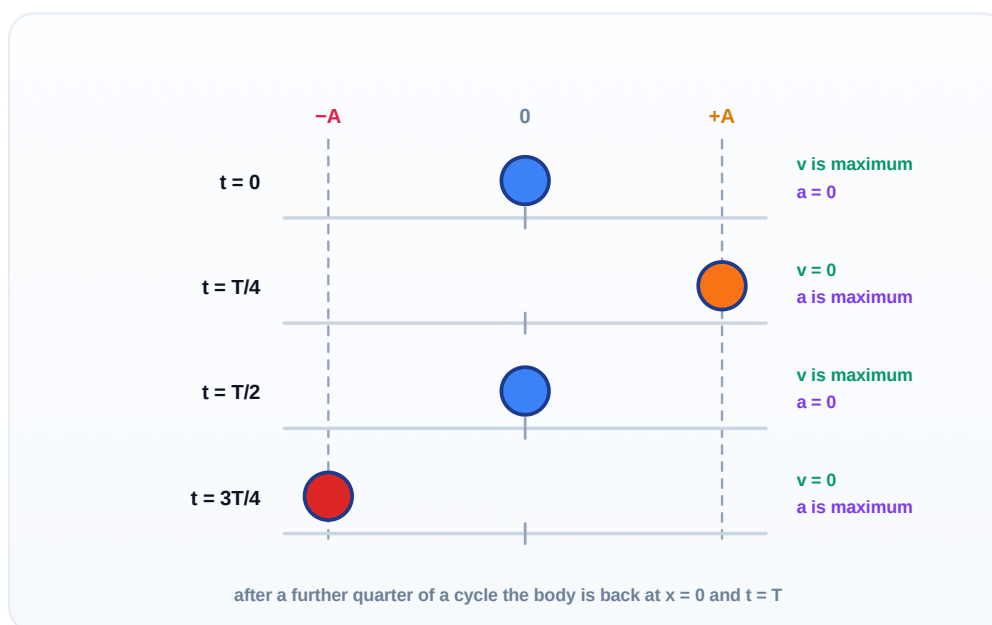


FIGURE 2

- 6 A mass attached to a horizontal spring was displaced towards the left until the spring was compressed to the greatest possible displacement, and was then released from rest at that instant. At this instant exactly (the instant of release), the restoring force acting on the mass is:



FIGURE 3

- A directed towards the left, because the mass is about to move in that direction
- B directed towards the right (towards the equilibrium position), and it has the greatest possible value at this instant exactly
- C completely zero, because the mass is still in a state of complete rest at the instant of release
- D directed towards the right, but it has the smallest possible value at this instant exactly

- 7 A student hung a mass of magnitude 0.9 kg from a vertical spring of force constant  $k = 40 \text{ N/m}$ , and left it to vibrate with simple harmonic motion about its equilibrium position. The student then **increased the amplitude** of the vibration without changing either the mass or the spring. Which of the following statements is correct?

- A the periodic time increases with the increase of the amplitude, and its value becomes about 0.94 s
- B the periodic time decreases with the increase of the amplitude, and its value becomes about 0.94 s
- C the periodic time does not change with the increase of the amplitude, and its value is about 0.94 s
- D the periodic time does not change with the increase of the amplitude, and its value is about 1.41 s

## The angular frequency and the periodic time

- 8 A small block of mass  $m = 0.50 \text{ kg}$  is attached to a spring of force constant  $k = 20 \text{ N}\cdot\text{m}^{-1}$  on a smooth horizontal path. When the block is displaced from the equilibrium position, a restoring force acts on it according to the relation  $F = -kx$ . **Calculate the angular frequency  $\omega$  of the vibration.**

PROBLEM

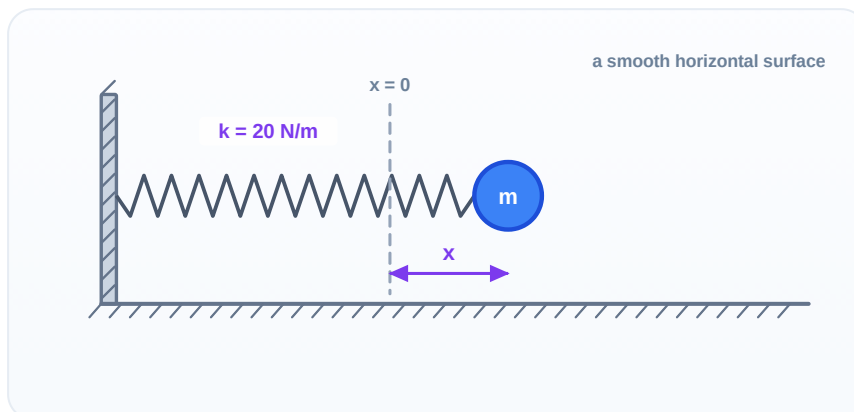


FIGURE 4

- 9 A small block of mass  $m = 0.20 \text{ kg}$  slides on a smooth horizontal path and is attached to a spring of force constant  $k = 80 \text{ N}\cdot\text{m}^{-1}$ . A restoring force of magnitude  $F = -kx$  acts on the block when its displacement from the equilibrium position is  $x$ . Given that  $\pi = 3.14$ , **find the periodic time of the vibration** using the relation  $T = 2\pi\sqrt{(m/k)}$ .

PROBLEM

## HOME ASSIGNMENT 2

## Applying the relations

- 10 A mass-spring system has a mass  $m = 0.30 \text{ kg}$  and a spring constant  $k = 30 \text{ N/m}$ . Taking  $\pi = 3.14$ , **calculate the periodic time  $T$**  using the relation  $T = 2\pi\sqrt{(m/k)}$ .

PROBLEM

- 11 A body moves with simple harmonic motion according to the relation  $x = A \sin(\omega t)$ . If the displacement is  $x = 0.15 \text{ m}$  at the time  $t = \pi/4 \text{ s}$ , and the angular frequency is  $\omega = 2 \text{ rad/s}$ , **find the amplitude  $A$ .**

PROBLEM

- 12 In a mass–spring system moving with simple harmonic motion, what is the kinetic energy KE at the equilibrium position  $x = 0$ ? MCQ

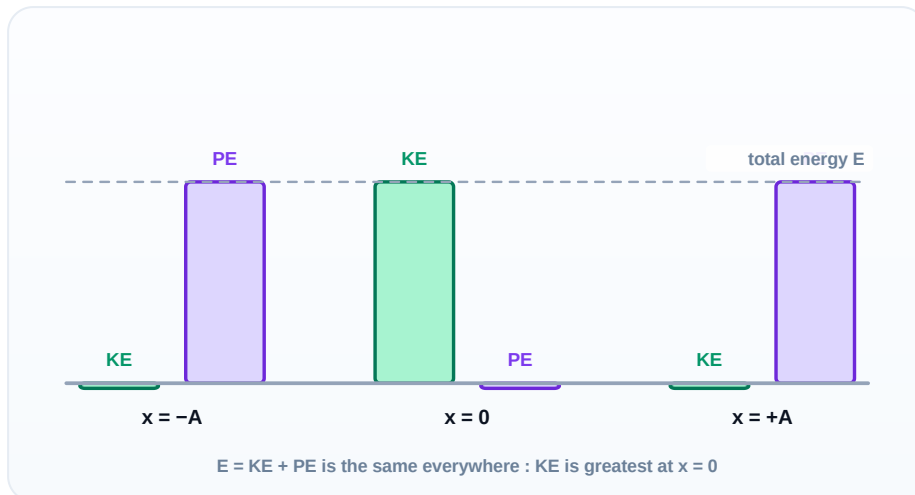


FIGURE 5

- A KE = 0
- B KE =  $E / 4$
- C KE =  $E / 2$
- D KE = E

- 13 In an experiment on a small block attached to a spring on a smooth horizontal path, the spring constant was  $k = 20 \text{ N/m}$  and the periodic time of the vibration was  $T = 1.256 \text{ s}$ . If the relation is  $T = 2\pi\sqrt{(m/k)}$  and  $\pi = 3.14$ , what is the value of the mass  $m$  of the block? MCQ

- A 0.8 kg
- B 1.0 kg
- C 1.2 kg
- D 1.6 kg

#### WEEKLY ASSESSMENT

### E Groups A, B and C

#### Group A

- 14 When a body moving with simple harmonic motion reaches the maximum displacement from the equilibrium position, its speed is: MCQ

- A the greatest value it has
- B zero
- C constant and not equal to zero
- D equal to the acceleration

- 15 A body moves with simple harmonic motion, and the periodic time of its motion is  $T = 0.25$  s. What is its frequency  $f$ ?

MCQ

- A 0.25 Hz      B 2 Hz  
C 4 Hz      D 8 Hz

- 16 If the displacement of a body moving with simple harmonic motion is given by the relation  $x(t) = 0.06 \sin(8\pi t)$  m, **find the amplitude of the motion  $A$ , the periodic time  $T$ , the frequency  $f$  and the maximum speed  $v_{\max}$ .**

PROBLEM

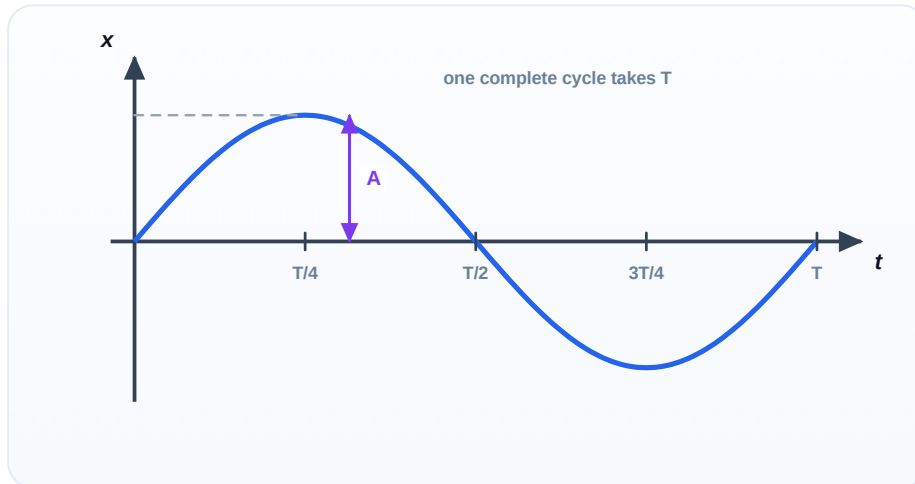


FIGURE 6

- 17 A small block of mass  $m = 0.20$  kg attached to a smooth horizontal spring oscillates, and the spring constant is  $k = 80 \text{ N}\cdot\text{m}^{-1}$ . **Find the periodic time  $T$  and then the frequency  $f$  of the motion, given that  $\pi = 3.14$ .**

PROBLEM

- 18 A body of mass  $m = 0.50$  kg moves with simple harmonic motion under the action of a restoring force that obeys the relation  $F = -kx$ , where  $k = 40 \text{ N}\cdot\text{m}^{-1}$ . If its displacement is  $x = +0.10$  m, **find the restoring force  $F$  and the acceleration  $a$** , and determine the direction of each of them with respect to the equilibrium position.

PROBLEM

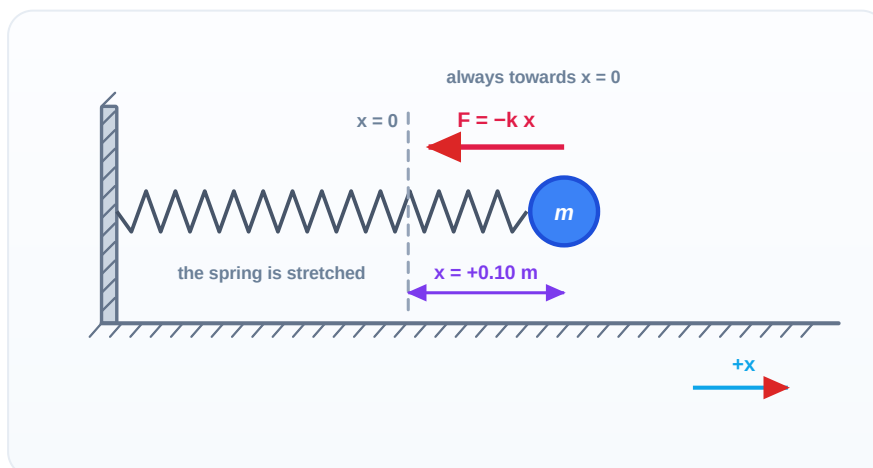


FIGURE 7

## Group B

- 19 When a body moving with simple harmonic motion passes through the equilibrium position, its acceleration is:

MCQ

- A zero
- B the greatest positive value
- C the greatest negative value
- D equal to the amplitude

- 20 If the frequency of a body moving with simple harmonic motion is  $f = 5.0$  Hz, then the periodic time of its motion equals:

MCQ

- A 0.20 s
- B 0.50 s
- C 5.0 s
- D 25 s

- 21 If the displacement of a body moving with simple harmonic motion is given by the relation  $x(t) = 0.04 \sin(10\pi t)$  m, **find the amplitude of the motion A, the periodic time T, the frequency f and the maximum speed  $v_{\max}$ .**

PROBLEM

- 22 A mass  $m = 0.40$  kg attached to a spring of constant  $k = 40 \text{ N}\cdot\text{m}^{-1}$  moves with simple harmonic motion. **Find the periodic time T and the frequency f**, given that  $\pi = 3.14$ .

PROBLEM

- 23 A body of mass  $m = 0.25$  kg moves with simple harmonic motion under the action of a restoring force that obeys the relation  $F = -kx$ , where  $k = 20 \text{ N}\cdot\text{m}^{-1}$ . If its displacement is  $x = -0.15$  m, **find the restoring force F and the acceleration a**, and determine the direction of each of them with respect to the equilibrium position.

PROBLEM

## Group C

- 24 When the displacement of a body in simple harmonic motion is equal to the amplitude and in the positive direction, the restoring force is:

MCQ

- A the greatest value it has, and directed towards the equilibrium position
- B zero
- C the greatest value it has, and directed away from the equilibrium position
- D constant and in the direction of the motion

25 A body moves with simple harmonic motion, and the periodic time of its motion is  $T = 0.80$  s. What is its frequency  $f$ ?

MCQ

A 0.80 Hz

B 1.25 Hz

C 2.50 Hz

D 8.00 Hz

26 If the displacement of a body moving with simple harmonic motion is given by the relation  $x(t) = 0.05 \sin(5\pi t)$  m, **find the amplitude of the motion A, the periodic time T, the frequency f and the maximum speed  $v_{\max}$ .**

PROBLEM

27 A mass  $m = 0.90$  kg attached to a spring of constant  $k = 40 \text{ N}\cdot\text{m}^{-1}$  oscillates with simple harmonic motion. **Find the periodic time T and the frequency f**, given that  $\pi = 3.14$ .

PROBLEM

28 A body of mass  $m = 0.40$  kg moves with simple harmonic motion under the action of a restoring force that obeys the relation  $F = -kx$ , where  $k = 32 \text{ N}\cdot\text{m}^{-1}$ . If its displacement is  $x = +0.125$  m, **find the restoring force F and the acceleration a**, and determine the direction of each of them with respect to the equilibrium position.

PROBLEM

# The Vertical Spring and the Simple Pendulum

Why the weight does not enter the periodic time of a spring, and the periodic time of a simple pendulum  $T = 2\pi\sqrt{L/g}$ .

Grade 11 · Egyptian Baccalaureate

28 questions

Step-by-step solutions

 $g = 9.8 \text{ m/s}^2 \cdot \pi = 3.14$ 

Mr. Gemy  
PHYSICS

## The toolkit for these two lessons

$$T = 2\pi \sqrt{m/k}$$

a spring pendulum — horizontal or vertical

$$T = 2\pi \sqrt{L/g}$$

a simple pendulum, small angle only

$$ke = mg$$

the static extension of a loaded vertical spring

$$T = t/n$$

time  $n$  complete oscillations, then divide

$$T \propto \sqrt{m}, T \propto 1/\sqrt{k}$$

for the spring pendulum

$$T \propto \sqrt{L}, T \propto 1/\sqrt{g}$$

for the simple pendulum

**Two things never appear in a periodic time.** The **amplitude** never appears in either formula, and the **mass of the bob** never appears in the pendulum formula. Whenever a question changes one of those two, the answer is “no change” — and whenever it changes  $m$ ,  $k$ ,  $L$  or  $g$ , use a **ratio** rather than a full substitution.

## LESSON 2-2 · CLASSWORK 1

A

### The vertical mass–spring system

- 1 A mass of magnitude  $0.20 \text{ kg}$  is fixed to a spring whose force constant is  $k = 80 \text{ N}\cdot\text{m}^{-1}$ , and it vibrates with simple harmonic motion. **Calculate the periodic time of the vibration**, using  $\pi = 3.14$ .

PROBLEM

- 2 For a body vibrating with simple harmonic motion on a spring, which of the following statements is correct for the periodic time  $T$ ?

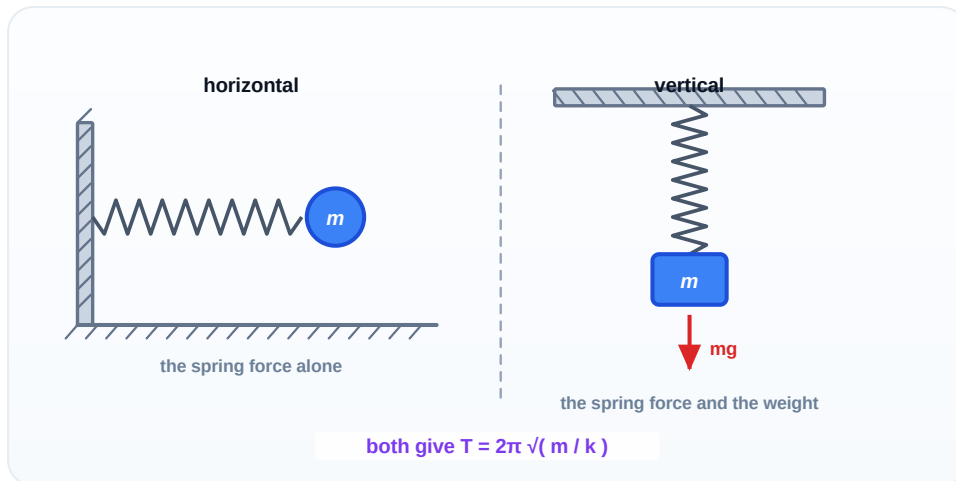


FIGURE 1

- A it depends on the acceleration due to gravity  $g$  only
- B  $T = 2\pi\sqrt{(k/m)}$
- C  $T = 2\pi\sqrt{(m/k)}$ , and it does not depend on the acceleration due to gravity  $g$
- D it increases with the increase of the amplitude  $A$  only

## LESSON 2-2 · HOME ASSIGNMENT 1

### B The role of the weight

- 3 In a system of a mass hung vertically from a spring, the equilibrium position changes because of the weight of the mass. **Explain why** the periodic time of the vibration is the same as that of a horizontal system having the same mass and the same spring constant, showing the role of the acceleration due to gravity in this.

EXPLAIN

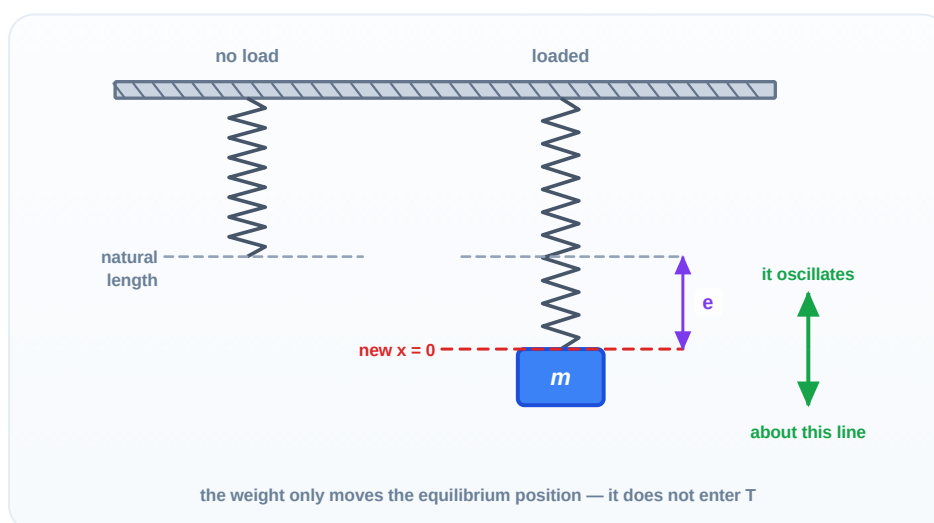


FIGURE 2

- 4 A mass was hung from a vertical spring and began to vibrate with simple harmonic motion about the equilibrium position. If the mass of the body is increased while the spring constant stays as it is, **how does the periodic time of the vibration change?** Explain your answer using the mathematical relation for the periodic time.

EXPLAIN

- 5 A mass of magnitude 0.50 kg was hung from a vertical spring whose force constant is  $k = 20 \text{ N/m}$ , and was left to move with simple harmonic motion. **Calculate the periodic time of the vibration**, taking  $\pi = 3.14$ .

PROBLEM

- 6 In an experiment on a spring pendulum, a mass of magnitude 0.40 kg was hung from a spring of force constant 40 N/m. If the mass is replaced by another one of magnitude 0.80 kg while the spring stays the same, what is the ratio between the new periodic time and the original periodic time?

MCQ

A 2 : 1

B  $\sqrt{2} : 1$

C 1 :  $\sqrt{2}$

D 1 : 2

## LESSON 2-3 · CLASSWORK 2

### C The simple pendulum

- 7 In the simple pendulum, **why** is its motion considered to be approximately simple harmonic when the angle of displacement is small?

EXPLAIN

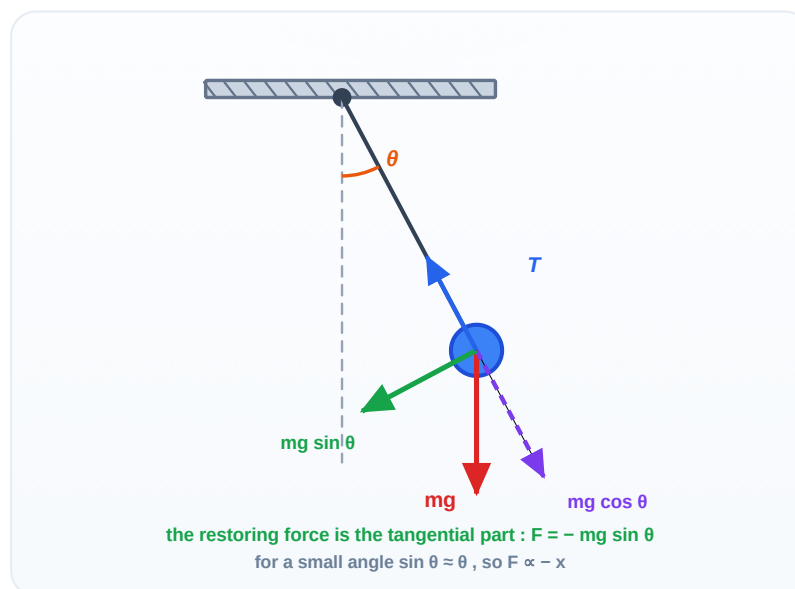


FIGURE 3

- 8 For a simple pendulum vibrating through a small angle, the periodic time  $T$  depends basically on:

MCQ

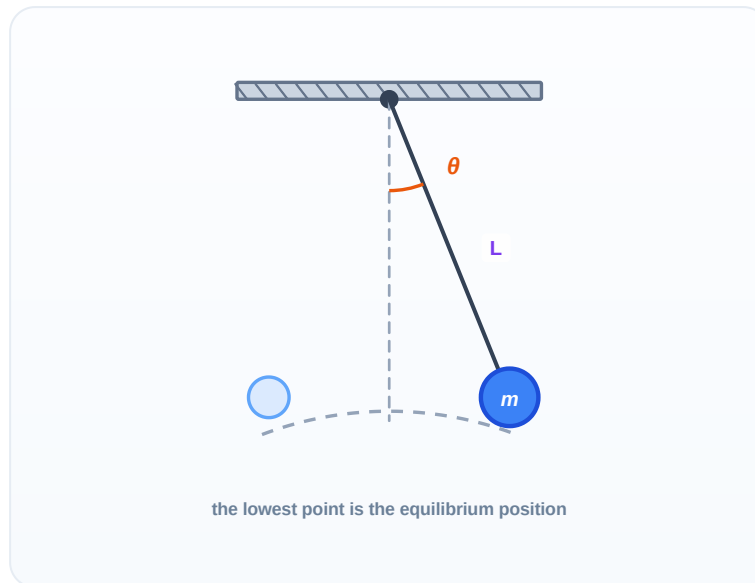


FIGURE 4

- A the mass of the bob and the amplitude of the vibration
- B the length of the string and the acceleration due to gravity
- C the mass of the bob and the length of the string
- D the amplitude of the vibration and the mass of the bob

- 9 A simple pendulum has a string of length  $L = 1.8$  m, the acceleration due to gravity is  $g = 9.8$  m/s<sup>2</sup>, and  $\pi = 3.14$ . **Calculate the periodic time  $T$  to the nearest two decimal places**, showing the law and the substitution.

PROBLEM

## LESSON 2-3 · HOME ASSIGNMENT 2

### D Measuring and changing $T$

- 10 If the length of the string of a simple pendulum is increased while the acceleration due to gravity of the Earth stays constant, **how does the periodic time of the pendulum change?** Give the reason.

EXPLAIN

- 11** Describe a suitable practical method for measuring the periodic time of a simple pendulum with greater accuracy, showing the role of timing a large number of oscillations.

EXPLAIN

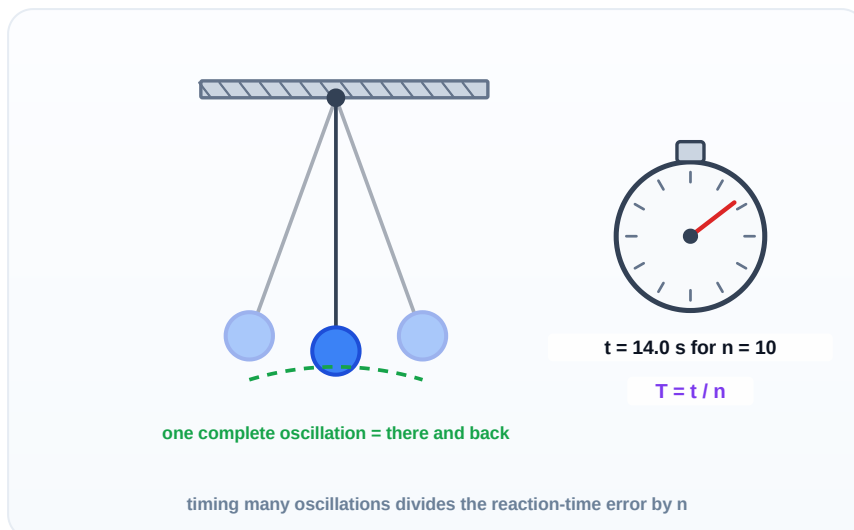


FIGURE 5

- 12** A simple pendulum took a time of magnitude 14.0 s to complete 10 full oscillations. Calculate the periodic time of one oscillation, then state the relation that connects the measured time with the number of oscillations and the periodic time.

PROBLEM

- 13** If the length of the string of a simple pendulum is increased to **four times** its value, while the acceleration due to gravity stays constant and the angle of vibration stays small, then the periodic time becomes:

MCQ

A half its value

B twice its value

C four times its value

D as it is, without change

#### WEEKLY ASSESSMENT

### E Groups A, B and C

#### Group A

- 14** A simple pendulum of length 1.0 m vibrates through small angles. If the length of the string is increased to 4.0 m while g stays constant, then the periodic time becomes:

MCQ

A  $T / 4$

B  $T / 2$

C  $2T$

D  $4T$

**15** A mass of magnitude 0.50 kg is attached to a spring whose constant is 20 N/m. The periodic time of the spring pendulum depends on:

MCQ

**A** the mass and the spring constant

**B** the amplitude only

**C** the acceleration due to gravity only

**D** the weight of the body only

**16** **Explain briefly why** the periodic time of a simple pendulum, at small angles, does not depend on the mass of the bob.

EXPLAIN

**17** **Calculate:** find the periodic time of a simple pendulum whose length is 0.40 m, if  $g = 9.8 \text{ m/s}^2$  and  $\pi = 3.14$ , writing the law and the unit.

PROBLEM

**18** **Compare**, in two or three lines, between the factors on which the periodic time of the simple pendulum depends and the factors on which the periodic time of the spring pendulum depends.

EXPLAIN

## Group B

**19** Two simple pendulums are in the same place; the length of the first is 0.25 m and the length of the second is 1.00 m. The ratio between their periodic times respectively equals:

MCQ

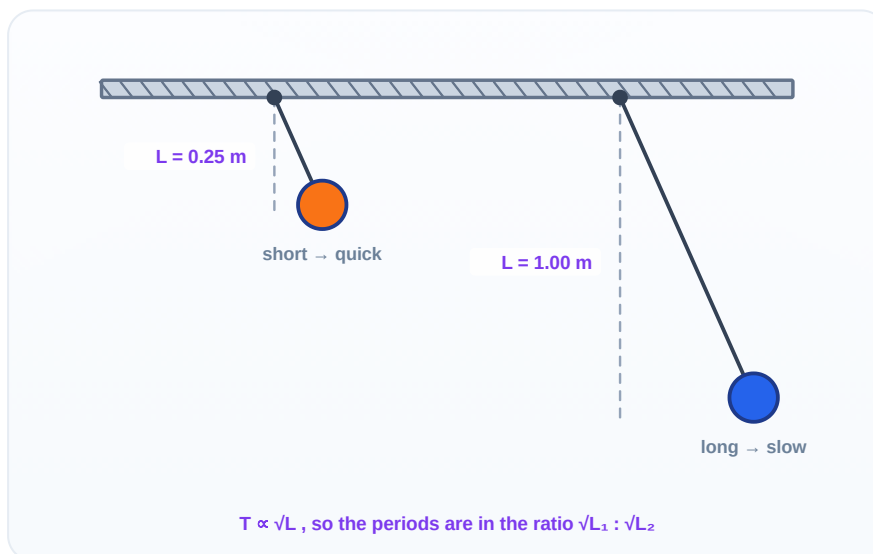


FIGURE 6

**A** 1 / 4

**B** 1 / 2

**C** 2

**D** 4

**20** If the mass of the body hung from a vertical spring is doubled while the spring constant stays as it is, then the periodic time:

MCQ

**A** decreases to half

**B** does not change

**C** increases by a factor of  $\sqrt{2}$

**D** becomes four times as large

**21** **State an important condition** that makes the motion of a simple pendulum approximately simple harmonic, and **state the effect** of increasing the small amplitude on the periodic time, approximately.

EXPLAIN

**22** **Calculate:** a body of mass 0.20 kg is attached to a horizontal spring whose constant is 80 N/m. Find the periodic time of the vibration using  $\pi = 3.14$ , showing the steps of the solution.

PROBLEM

**23** **Analyse the following situation:** a simple pendulum is moved to a place in which the acceleration due to gravity is smaller. Does the periodic time increase or decrease? Explain using the mathematical relation.

EXPLAIN

### Group C

**24** If the spring constant becomes four times its value while the mass stays constant, then the periodic time of the spring pendulum becomes:

MCQ

**A** 4T

**B** 2T

**C** T / 2

**D** T / 4

**25** Which of the following changes does **not** affect, approximately, the time of one cycle of a simple pendulum at small angles?

MCQ

**A** increasing the length of the string

**B** changing the acceleration due to gravity

**C** changing the mass of the bob

**D** using another place on the Earth

**26** **Explain:** on hanging a mass from a vertical spring the equilibrium position changes, but the periodic time does not depend on the magnitude of the static extension.

EXPLAIN

**27** **Calculate:** a simple pendulum whose length is 0.90 m vibrates through small angles, with  $g = 9.8 \text{ m/s}^2$  and  $\pi = 3.14$ . Find the periodic time to the nearest two decimal places.

PROBLEM

**28**

**Compare analytically:** if we want to double the periodic time, what do we do with the length of the simple pendulum? And what do we do with the mass of the spring pendulum while the spring constant stays fixed? State the relations used.

EXPLAIN

Mr. Gemy

Mr. Gemy

Mr. Gemy

Mr. Gemy

Mr. Gemy

Mr. Gemy

Mr. Gemy

Mr. Gemy

# Sinusoidal Waves

Wavelength, periodic time, frequency and the wave equation  $v = f\lambda$ , for transverse and longitudinal waves.

Grade 11 · Egyptian Baccalaureate

14 questions

Step-by-step solutions

 $v = f\lambda$ 

Mr. Gemy

PHYSICS

## The toolkit for this lesson

$$v = f\lambda$$

the wave equation

$$v = \lambda / T$$

the same thing, written with T

$$f = 1 / T$$

frequency and periodic time

**y – x graph**gives the amplitude A and the wavelength  $\lambda$ **y – t graph**

gives the amplitude A and the periodic time T

**transverse / longitudinal**particles  $\perp$  /  $\parallel$  to the travel direction

**A wave carries energy, not matter.** Every particle of the medium simply vibrates about its own fixed position; nothing travels along with the wave except the disturbance and the energy it carries. And one equation,  $v = f\lambda$ , answers almost every numerical question in this lesson — the only skill is deciding which two of the three quantities you have been given.

## CLASSWORK 1

A

### Frequency, wavelength and speed

- 1 A vibrating body completes one full oscillation in a periodic time of 0.20 s. **Find its frequency.**

PROBLEM

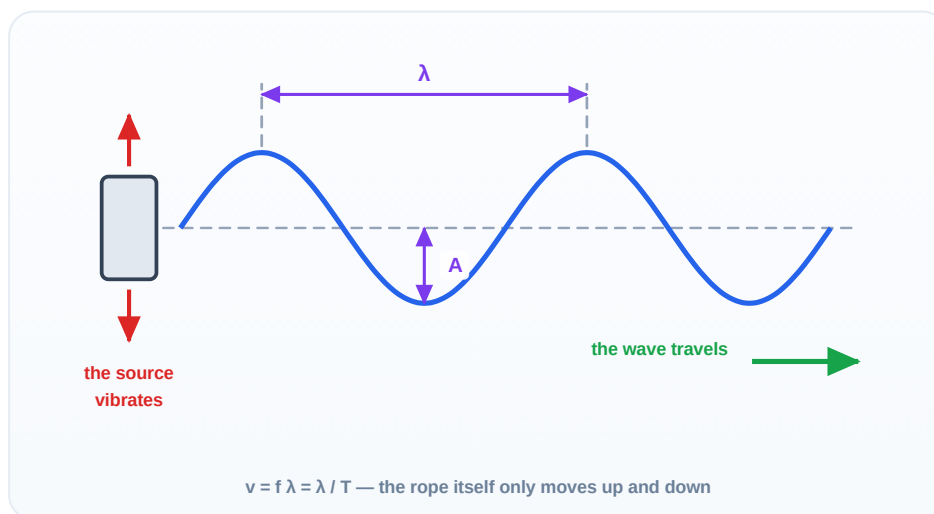


FIGURE 1

- 2 A sinusoidal wave has a wavelength of 0.2 m and a frequency of 0.3 Hz. **Calculate the speed with which the wave travels.**

PROBLEM

## What a wave carries

3 Which of the following is transported by a mechanical wave as it travels through the medium?

MCQ

A matter

B energy

C the particles of the medium

D the total displacement

4 A wave has a periodic time equal to 0.25 second. What is its frequency?

MCQ

A 0.25 hertz

B 2.5 hertz

C 4.0 hertz

D 0.4 hertz

5 A wave travels along a string; its periodic time is 0.10 s and its wavelength is 1.5 m.  
**Calculate the speed of the wave.**

PROBLEM

6 A wave pulse travels along a rope with a speed of 4.0 m/s and a wavelength of 0.80 m.  
**Find its frequency and its periodic time.**

PROBLEM

7 The opposite figure shows the vibration of the particles of a material medium for a longitudinal wave motion. **Using the scale shown in the figure, calculate the wavelength of the longitudinal wave.**

PROBLEM

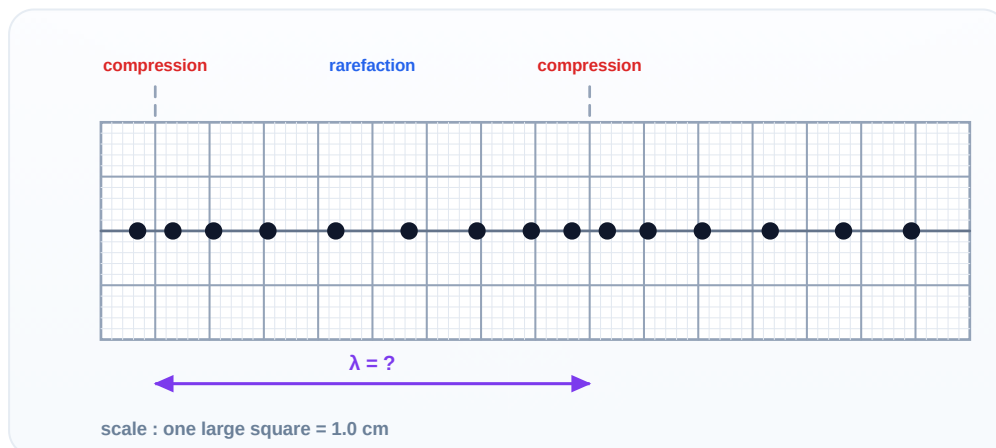


FIGURE 2

C

## Using the wave equation

- 8 A sinusoidal wave has a wavelength of 0.60 m, and its periodic time equals 0.20 s.  
**Calculate:**

PROBLEM

- a) the frequency  
b) the speed of the wave

- 9 A wave source vibrates with a periodic time of 0.25 s, and produces a wave whose wavelength is 1.2 m. **Calculate the speed of the wave.**

PROBLEM

D

## Reading the two graphs

- 10 Which of the graphs is used to determine **both** the amplitude and the wavelength of a wave motion made by a vibrating body, directly?

MCQ

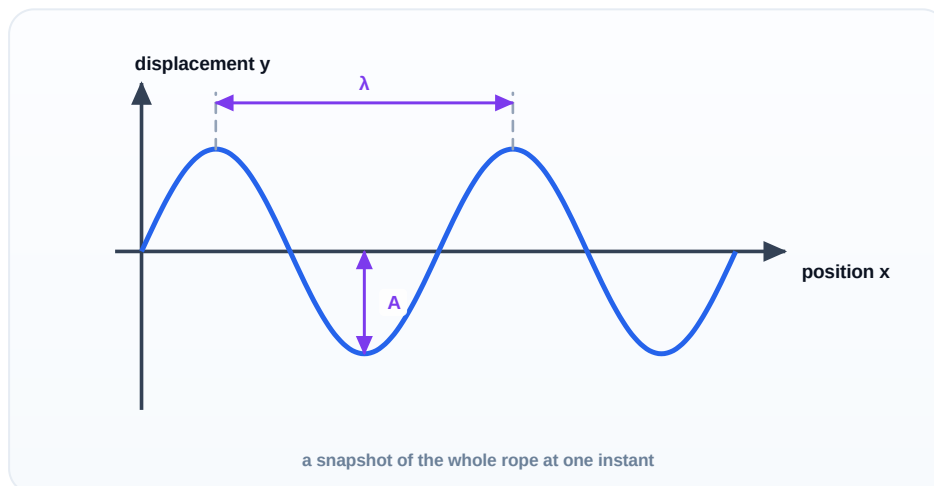


FIGURE 3

- A the displacement – time graph (  $y - t$  )  
B the displacement – position graph (  $y - x$  )  
C the velocity – time graph  
D the frequency – periodic time graph

11 In a longitudinal wave, the particles of the medium vibrate:

MCQ

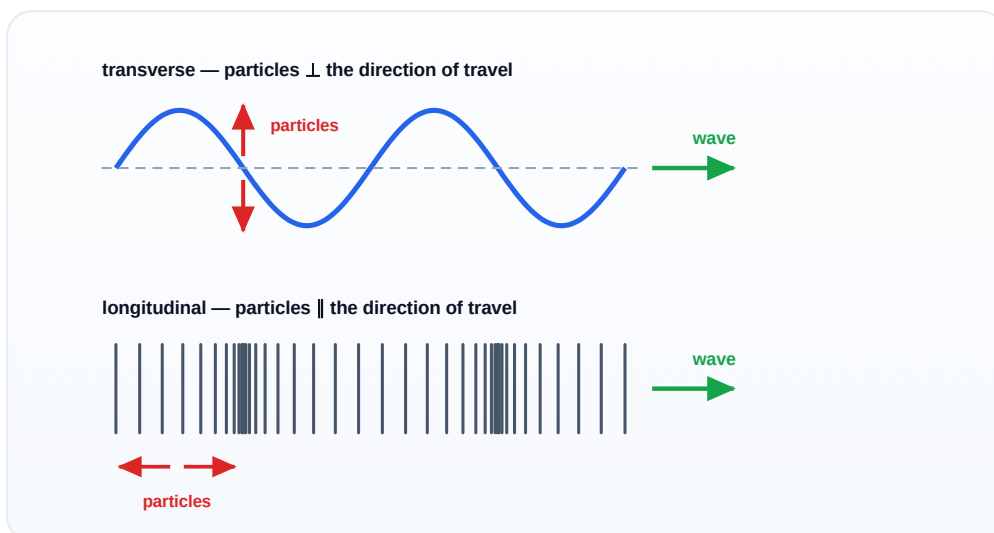


FIGURE 4

A perpendicular to the direction of propagation of the wave

B in a direction parallel to the direction of propagation of the wave

C in circular paths opposite to the direction of transfer of the energy

D with a constant speed equal to the speed of light

12 A sound wave travels through air with a speed of 340 m/s and a frequency of 500 Hz. Calculate its wavelength.

PROBLEM

13 The displacement – position graph of a sinusoidal wave shows a wavelength of 3.0 m and an amplitude of 0.50 m, while the displacement – time graph shows a periodic time of 0.60 s. Find the speed of the wave.

PROBLEM

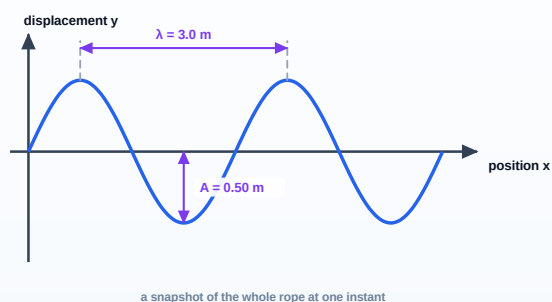


FIGURE 5

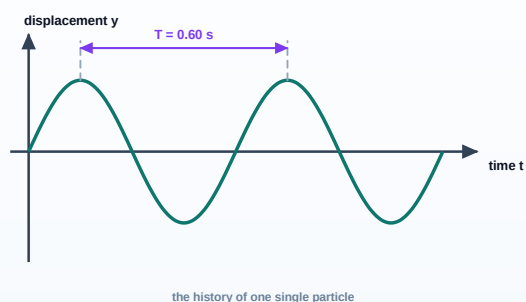


FIGURE 6

- 14 The opposite figure represents a sinusoidal wave motion. If the distance between the two points P and Q is 2 m, covered in a time interval of 0.1 s, **calculate:**

- a) the wavelength  
b) the speed of propagation of the wave

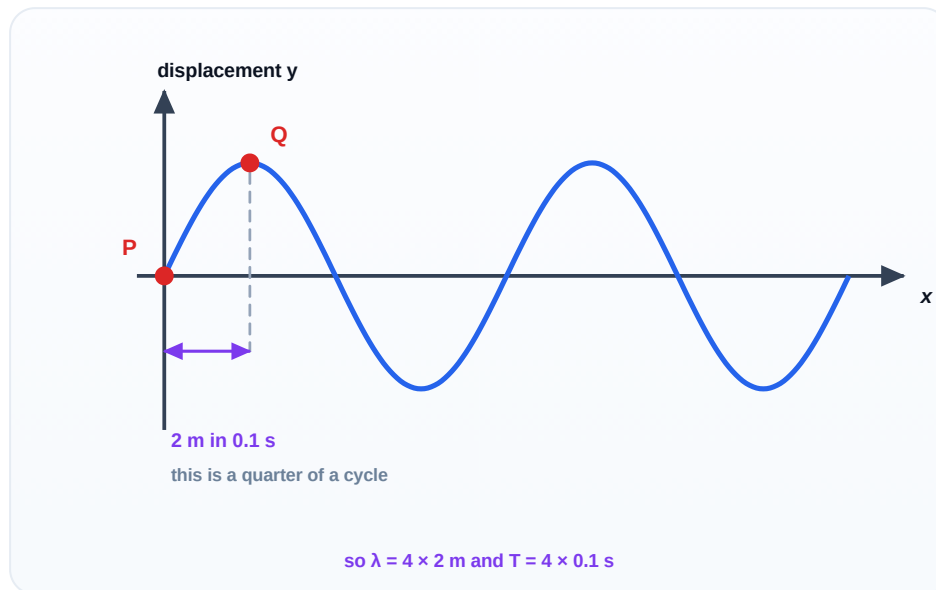


FIGURE 7

# Interference of Waves and Standing Waves

Coherent sources, path difference, constructive and destructive interference, nodes and antinodes — with the end-of-part review.

Grade 11 · Egyptian Baccalaureate

40 questions

Step-by-step solutions

 $\Delta = m\lambda$  or  $(m + \frac{1}{2})\lambda$ 

Mr. Gemy  
PHYSICS

## The toolkit for this lesson

$$\Delta = |S_2P - S_1P|$$

the path difference

$$\Delta = m\lambda$$

constructive — an **even** number of  $\lambda/2$ 

$$\Delta = (m + \frac{1}{2})\lambda$$

destructive — an **odd** number of  $\lambda/2$ 

$$A(\text{res}) = A_1 + A_2 \text{ or } |A_1 - A_2|$$

the resultant amplitude

node  $\leftrightarrow$  antinodespaced  $\lambda/4$  apart

$$N-N = A-A = \lambda/2$$

in a standing wave

**Count half-wavelengths, not wavelengths.** Divide the path difference by  $\lambda/2$ . An **even** answer means the two waves arrive in step  $\rightarrow$  constructive. An **odd** answer means they arrive exactly out of step  $\rightarrow$  destructive. And if the two **sources** themselves are in antiphase, the whole rule turns over.

## CLASSWORK 1

A

### Path difference and the type of interference

- 1 Two coherent sources, vibrating in phase, emit waves of wavelength 4.0 cm. At a point P, the path length from the first source is 10.0 cm and from the second source is 16.0 cm.

PROBLEM

- a) Calculate the path difference.  
b) Determine the number of half-wavelengths it contains, and state whether the interference at P is constructive or destructive.

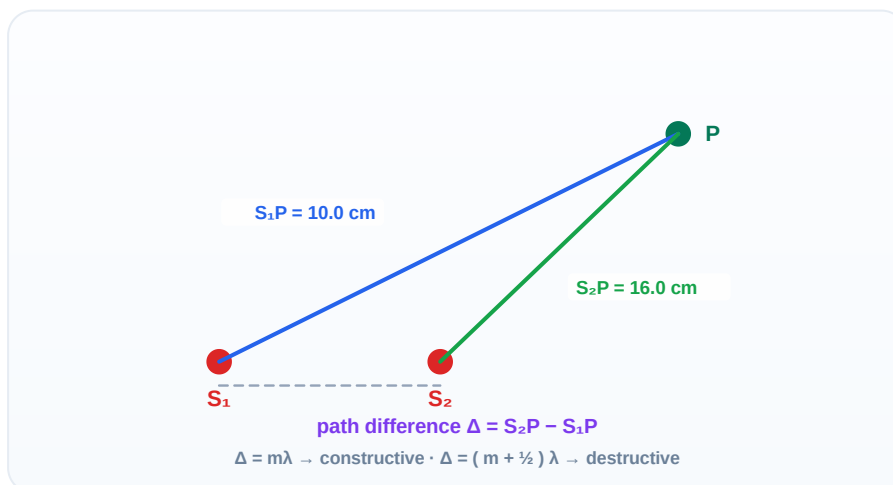


FIGURE 1

- 2 Two coherent sources  $S_1$  and  $S_2$ , a distance of 8.0 cm apart, vibrate in phase and emit waves of wavelength 2.0 cm. **Determine the number of destructive interference lines lying between  $S_1$  and  $S_2$ .**

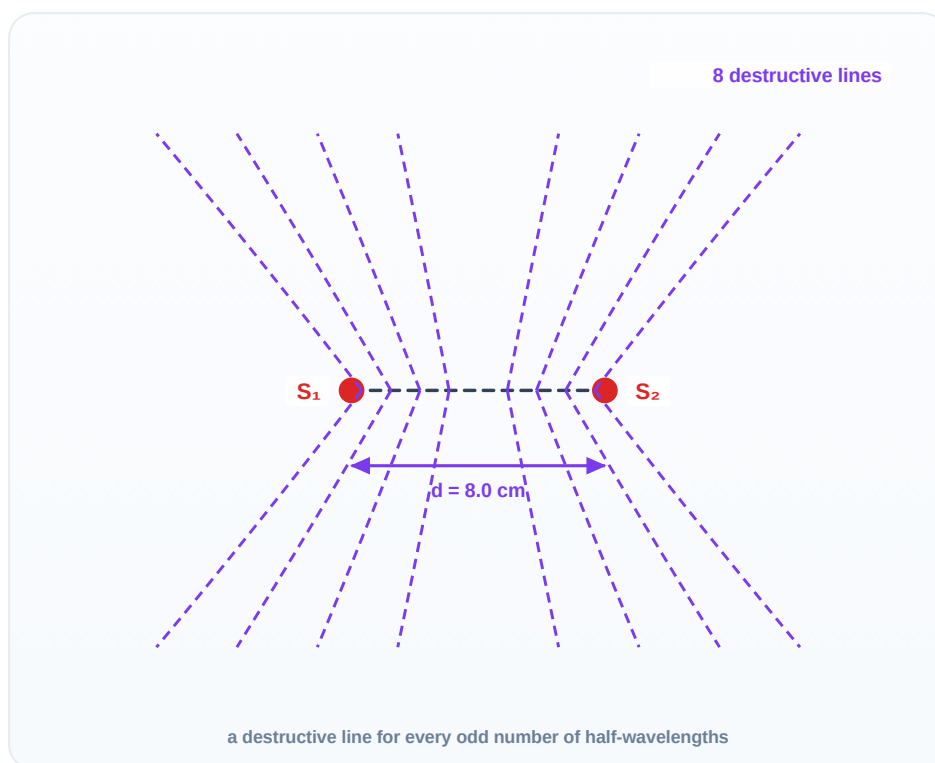


FIGURE 2

## HOME ASSIGNMENT 1

## B The principle of superposition

- 3 Two coherent sources vibrating in phase emit waves of wavelength  $\lambda$ . Constructive interference occurs at the points at which the path difference is equal to:

MCQ

A  $(m + \frac{1}{2}) \lambda$

B  $m \times \lambda$

C  $(2m + 1) \lambda / 2$

D  $\lambda / 4$

- 4 The principle of superposition states that the resultant displacement at a point where two waves interfere is equal to:

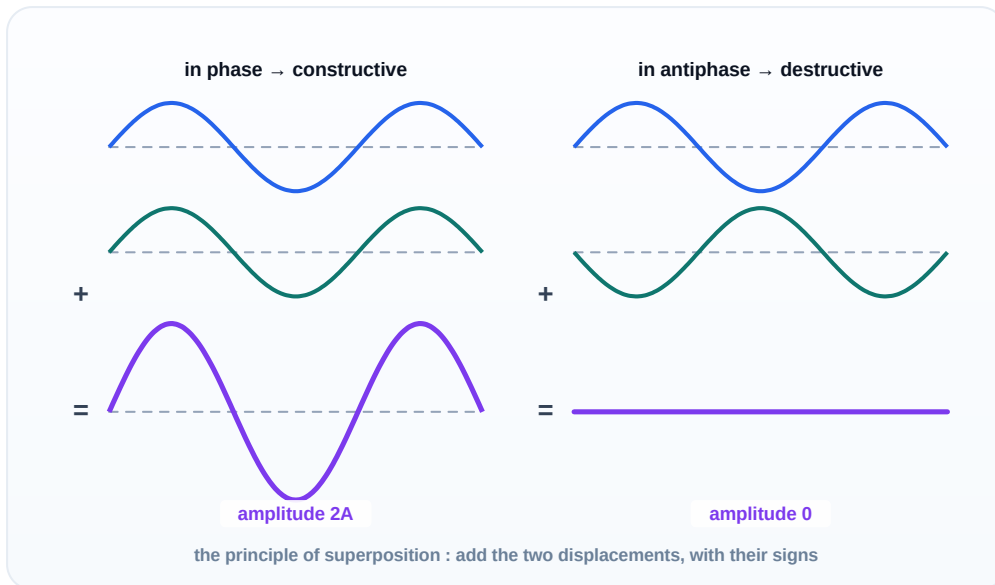


FIGURE 3

- A the product of their two displacements
- B the vector sum of their two individual displacements
- C the difference between their two frequencies
- D the average of their two amplitudes

- 5 Two coherent sources vibrating in phase produce waves of wavelength  $5.0\text{ cm}$  and amplitude  $1.0\text{ cm}$ . At a point  $P$ ,  $S_1P = 30.0\text{ cm}$  and  $S_2P = 37.5\text{ cm}$ . **Calculate the resultant amplitude at the point  $P$**  when:

- a) the two sources vibrate in phase.
- b) the two sources vibrate in antiphase (opposite phases).

- 6 Two waves of wavelength  $6.0\text{ cm}$  and amplitude  $0.40\text{ cm}$  are emitted in phase from the two sources  $S_1$  and  $S_2$ . For a point  $P$  at which  $S_1P = 20.0\text{ cm}$  and  $S_2P = 35.0\text{ cm}$ , **calculate:**

- a) the path difference.
- b) the resultant amplitude at  $P$ .

- 7 The opposite figure shows two waves spreading out in phase from the two sources  $S_1$  and  $S_2$ . What is the result of the superposition of the two waves at the two points P and Q? Justify your answer.

EXPLAIN

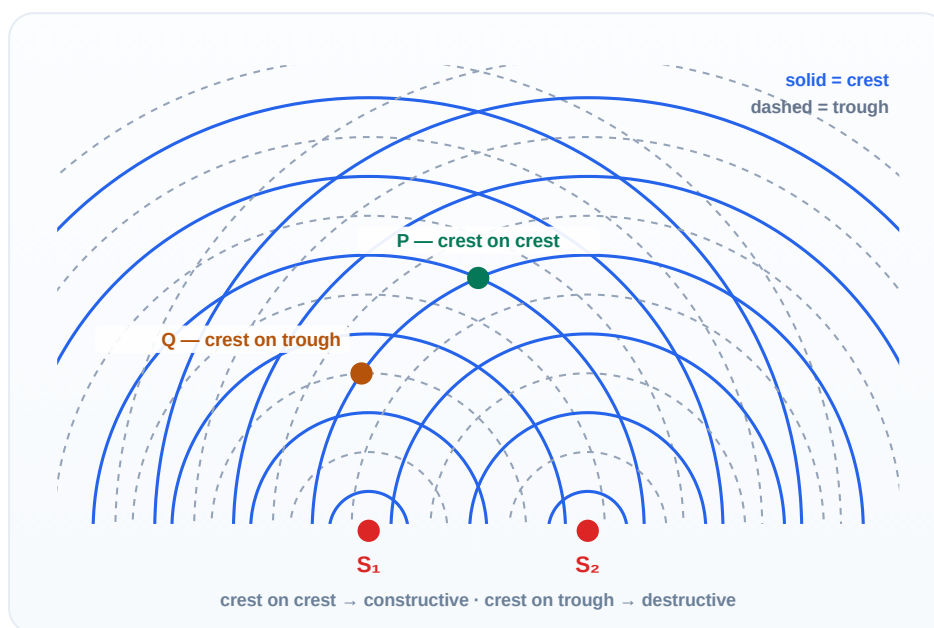


FIGURE 4

## CLASSWORK 2

### C Counting the half-wavelengths

- 8 Two coherent sources emit waves of wavelength 6.0 cm and amplitude 0.50 cm. At a point P,  $S_1P = 28.0$  cm and  $S_2P = 37.0$  cm. Determine the path difference and the number of half-wavelengths that it contains.

PROBLEM

- 9 Two coherent sources vibrating in phase are separated by a distance of 10 cm and emit waves of wavelength 2.5 cm. At a point P the path difference is 5.0 cm. Determine whether P is a point of constructive or destructive interference, and state the number of half-wavelengths.

PROBLEM

## HOME ASSIGNMENT 2

### D Nodes and the reversed rule

- 10 Two coherent sources vibrate in phase with a wavelength of 4.0 cm. If the path difference to a point P equals 6.0 cm, then the path difference contains:

MCQ

A 1.5 wavelengths

B 2.0 wavelengths

C 1.0 wavelength

D 0.5 wavelength

11 The node in a standing wave is defined as a point of:

MCQ

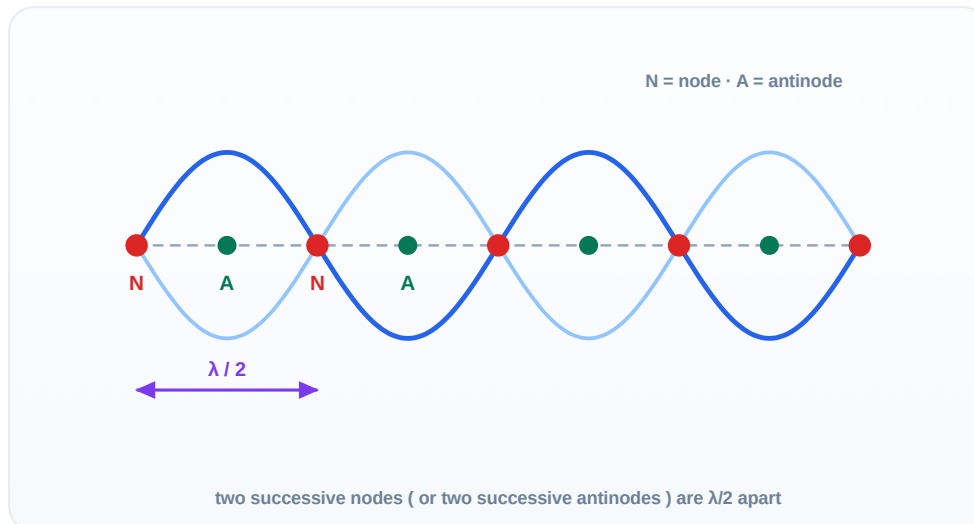


FIGURE 5

- ☐ A maximum displacement
- ☐ B a displacement equal to zero
- ☐ C maximum amplitude and energy
- ☐ D a changing frequency

12 Two coherent sources vibrating in phase emit waves of amplitude 0.40 cm and wavelength 3.0 cm. At a point Q the path difference is 9.0 cm. **Calculate the resultant amplitude at Q.**

PROBLEM

13 Two coherent sources vibrate in **antiphase** (opposite phases) with waves of wavelength 4.0 cm. At a point R the path difference is 8.0 cm. **Determine whether the point R undergoes constructive or destructive interference.**

PROBLEM

## Inside a pipe

- 14 The opposite figure shows a standing wave formed inside a pipe that is open at one end. Explain the nature of the vibration of the air particles near the point X.

EXPLAIN

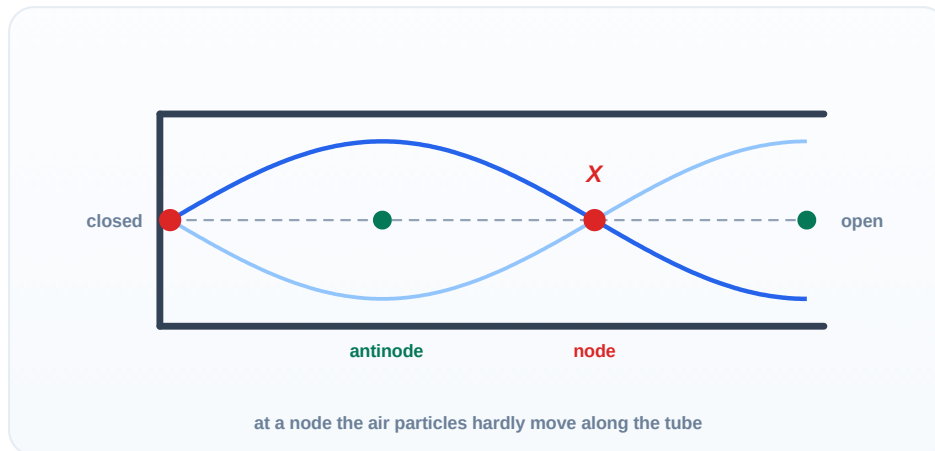


FIGURE 6

## Choose the correct answer

15 What happens to the speed of a transverse wave pulse travelling along a stretched string if the tension force in the string is increased?

MCQ

A it decreases

B it increases

C it stays without change

D it becomes zero

16 Increasing the amplitude of a wave while its frequency stays constant leads to:

MCQ

A an increase in the speed of the wave

B an increase in the energy transported by the wave

C a decrease in the wavelength

D an increase in the periodic time of the wave

17 Electromagnetic waves can travel through vacuum, while mechanical waves cannot, because:

MCQ

A electromagnetic waves have an infinite wavelength

B electromagnetic waves do not require material media

C electromagnetic waves are longitudinal waves

D electromagnetic waves travel more slowly than mechanical waves

18 During an earthquake, the primary seismic waves (P-waves) reach the observation station before the secondary waves (S-waves) because:

MCQ

A P-waves are transverse and travel with a greater speed

B P-waves are longitudinal and travel with a greater speed than S-waves

C S-waves travel through liquids only

D S-waves carry more energy than P-waves

19 Which of the following pairs contains transverse waves only?

MCQ

A sound waves and light waves

B light waves and waves along a stretched string

C waves along a stretched string and sound waves

D water surface waves and sound waves

20 The correct relation connecting the speed of a wave ( $v$ ), the frequency ( $f$ ) and the wavelength ( $\lambda$ ) is:

MCQ

A  $v = f / \lambda$

B  $v = f \times \lambda$

C  $v = \lambda / f$

D  $v = 1 / (f \times \lambda)$

21 Destructive interference occurs when two interfering waves arrive at a point with a phase difference of:

MCQ

A 0 (in phase)

B  $180^\circ$  (out of phase / in antiphase)

C  $360^\circ$

D  $90^\circ$

22 In an experiment on the interference of two waves, the formation of regions in which the resultant displacement is as large as possible is due to:

MCQ

A destructive interference

B constructive interference

C refraction of the waves

D reflection of the waves

23 A standing wave is formed as a result of the interference of two travelling waves that are characterised by:

MCQ

A different frequencies, travelling in the same direction

B the same frequency and amplitude, travelling in two opposite directions

C different amplitudes, travelling in two opposite directions

D the same wavelength, travelling in two perpendicular directions

24 The distance between two successive nodes (or two successive antinodes) in a standing wave equals:

MCQ

A a complete wavelength

B half a wavelength

C a quarter of a wavelength

D two wavelengths

25 If two coherent sources vibrating in phase produce waves of amplitude  $A$ , then the maximum resultant amplitude at a point of constructive interference is:

MCQ

A  $A$

B 0

C  $2A$

D  $4A$

26 What happens to the separation between the adjacent nodal lines in a ripple tank experiment when the frequency of vibration of the two sources is increased?

MCQ

A it increases

B it decreases

C it stays without change

D it becomes zero everywhere

## REVIEW

G

### Answer the following questions

27 What happens to the speed of a wave pulse travelling along a stretched rope when the tension force in the rope is increased?

EXPLAIN

28 What happens to the energy transported from a vibrating source producing a wave motion when the amplitude of the wave is doubled, its frequency staying constant?

EXPLAIN

29 What happens to the periodic time of a simple harmonic oscillator if its frequency is doubled?

EXPLAIN

30 What happens to the speed of a sound wave when it passes from a material medium into a perfect vacuum?

EXPLAIN

31 What happens to the speed of the observed wave if the frequency of the wave source is doubled while the properties of the medium stay as they are, without change?

EXPLAIN

32 What happens to the amplitude of a particle in a medium as a wave passes through it, as regards the **total displacement** of the particle after the wave has passed and the medium has settled?

EXPLAIN

33 What happens to the pitch (sharpness / frequency) of the musical note given out by a plucked string when the tension force in the string is increased?

EXPLAIN

34 What happens to the amplitude of vibration at a node on a stretched string on which a standing wave is formed?

EXPLAIN

35 What happens to the positions of constructive and destructive interference if the state of two coherent sources, which were vibrating in phase, is changed so that they vibrate in antiphase (a phase difference of  $180^\circ$ )?

EXPLAIN

**36** A tuning fork produces a wave of frequency 250 Hz which travels with a speed of 340 m/s in air. **Find the wavelength and the periodic time of the sound wave.**

PROBLEM

**37** A wave source produces 120 complete waves in 30 seconds along a stretched spring. If the distance between two successive crests is 0.5 m, **find:**

PROBLEM

a) the frequency.

b) the periodic time.

**38** A wave source produces 100 complete waves in 20 seconds along a stretched spring. If the distance between two successive crests is 0.4 m, **calculate the speed of the wave.**

PROBLEM

**39** Two loudspeakers separated by a distance of 2.0 m emit coherent sound waves, in phase, of wavelength 0.50 m. A listener stands at a place at which the path difference is 1.25 m. **Determine whether the listener hears a loud sound (constructive interference) or a faint sound (destructive interference).**

PROBLEM

**40** Two coherent sources vibrating in phase emit waves of wavelength 5.0 cm. At a point P the path difference is 12.5 cm. **Determine whether P is a point of constructive or destructive interference, and justify your answer.**

PROBLEM

# Refraction of Waves

Why a wave changes speed, wavelength and direction on crossing into a new medium — while its frequency never changes.

Mr. Gemy

PHYSICS

Grade 11 · Egyptian Baccalaureate

10 questions

Step-by-step solutions

f is the same in both media

## The toolkit for this lesson

$$v = f \lambda$$

and f is set by the **source**

$$f_1 = f_2$$

the frequency never changes on refraction

$$v_1/v_2 = \lambda_1/\lambda_2$$

so speed and wavelength change together

$$\sin i / \sin r = v_1 / v_2$$

the law of refraction

$$n = c / v$$

a slower medium has a larger n

**shallow water**

slower waves, shorter  $\lambda$ , same f

**One thing is fixed and everything else follows.** The **frequency** of a refracted wave is fixed by the source, so it is the same in both media. Since  $v = f\lambda$ , the speed and the wavelength must therefore change **by the same factor** — and it is that change of speed that bends the direction of travel.

## Sea waves approaching the shore

EXPLAIN

- 1 The figure shows the change in the shape, the direction, the behaviour and the properties of sea waves as they travel from deep water into shallow water near the shore. Using the figure and what you have studied about the properties of waves:

- Explain the changes that take place in each of: the speed of the waves, their wavelength, and their direction of propagation, as they pass from deep water into shallow water.
- Explain the reason for the bending of the waves and the adjustment of their direction so that they become parallel to the shore line near the beach.
- Deduce the physical concept that this phenomenon describes, and state its definition in your own words.

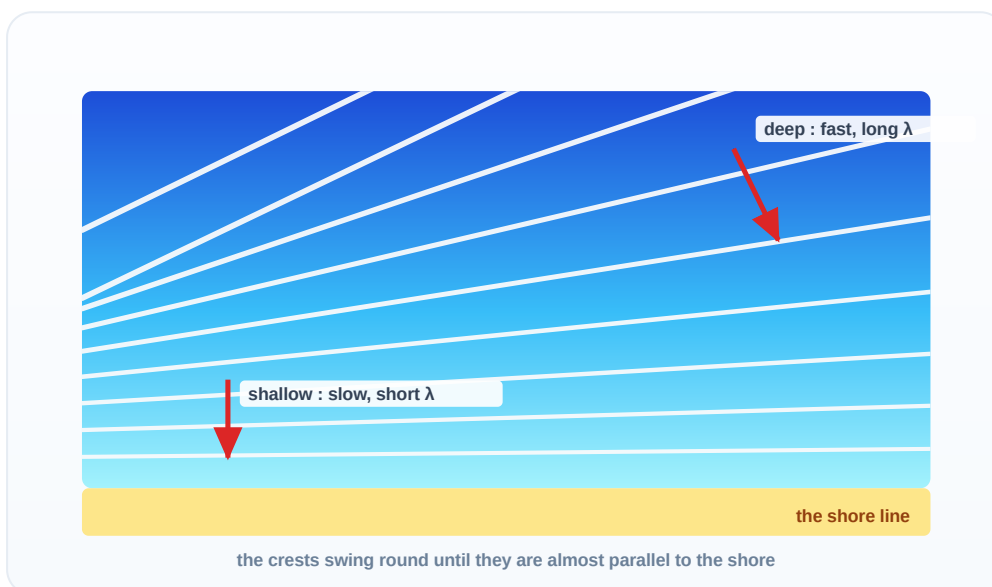


FIGURE 1

- 2 Which of the following changes take place in sea waves as they pass from deep water into shallow water near the shore?

MCQ

- A** their speed increases, their wavelength increases, and they continue in a straight line without any bending
- C** their speed stays constant, their wavelength decreases, and they bend away completely from the shore line

- B** their speed decreases, their wavelength decreases, and their direction bends so that they become parallel to the shore line
- D** their speed decreases, their wavelength increases, and they are completely reflected back towards the deep water

## Refraction in general

EXPLAIN

- 3 The figure shows the refraction of sea waves at different coastal shapes. Given that the depth of the water at the headland is less than its depth at the bay, **draw a table comparing the properties of the water waves at the “headland” and at the “bay”** as regards:

- a) the wave speed ( $v$ ).
- b) the wavelength ( $\lambda$ ).
- c) the wave frequency ( $f$ ).

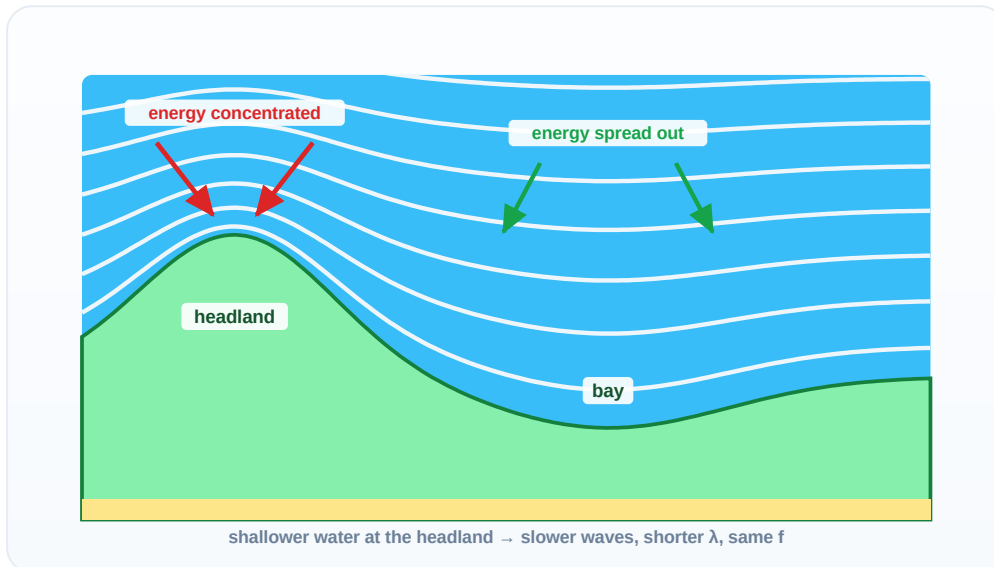


FIGURE 2

- 4 Which of the following statements correctly describes what happens to a light wave or a water wave when it is refracted while passing at an oblique angle from one medium into another medium of different properties?

MCQ

A the frequency of the wave changes, while its speed and its wavelength stay constant

B the speed of the wave and its wavelength change, while its frequency stays constant

C the speed of the wave, its wavelength and its frequency all increase

D the speed of the wave, its wavelength and its frequency all decrease

- 5 The graph shows the relation between the speed of a wave ( $v$ ) and its wavelength ( $\lambda$ ) for three waves from different sources, as the wave passes between media of different properties. **Which of the frequencies has the largest value?** Give the explanation.

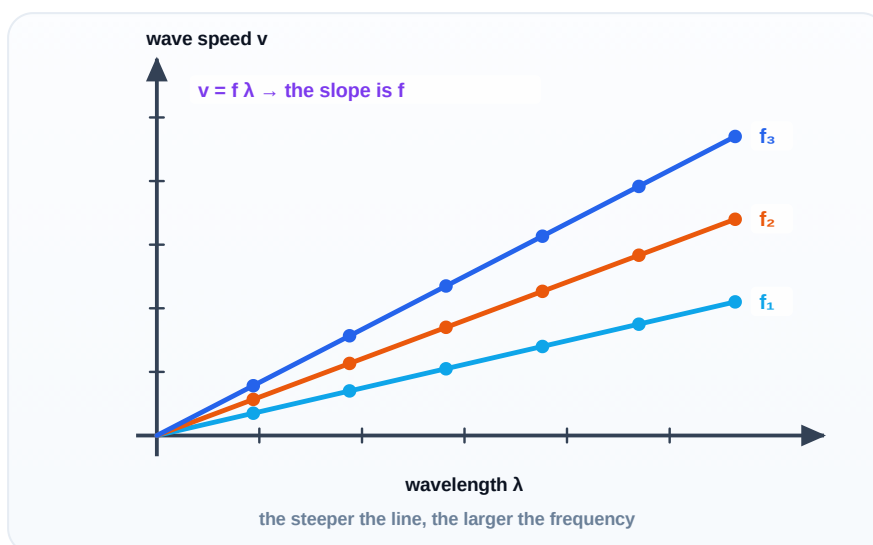


FIGURE 3

- 6 When a wave spreads through vacuum or through a medium, the locus of all the points that vibrate in the same phase is called the .....

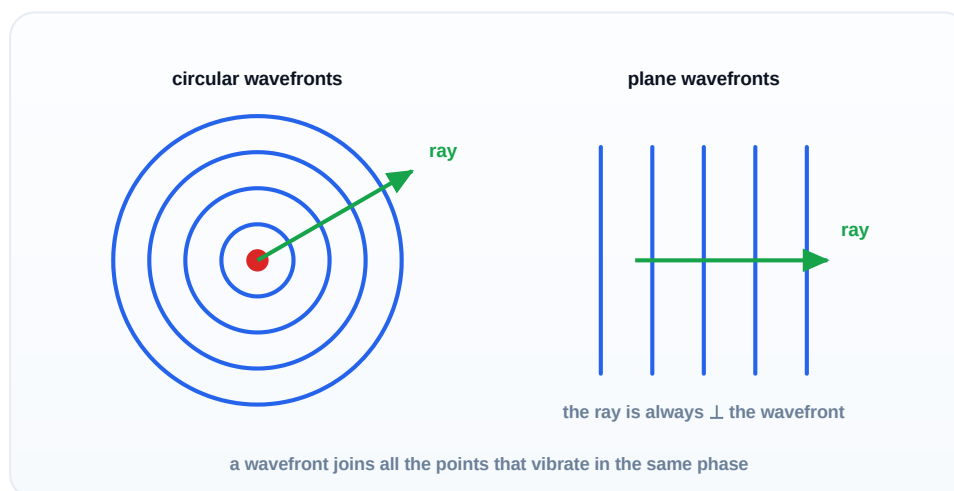


FIGURE 4

A frequency of the wave

B wavelength

C speed of the wave

D wavefront

- 7 A wave crosses a boundary separating two media as shown in the figure. **Calculate the following:**

- a) the ratio ( the frequency of the wave in medium 1 ) / ( the frequency of the wave in medium 2 ) = .....
- b) the ratio ( the wavelength of the wave in medium 1 ) / ( the wavelength of the wave in medium 2 ) = .....

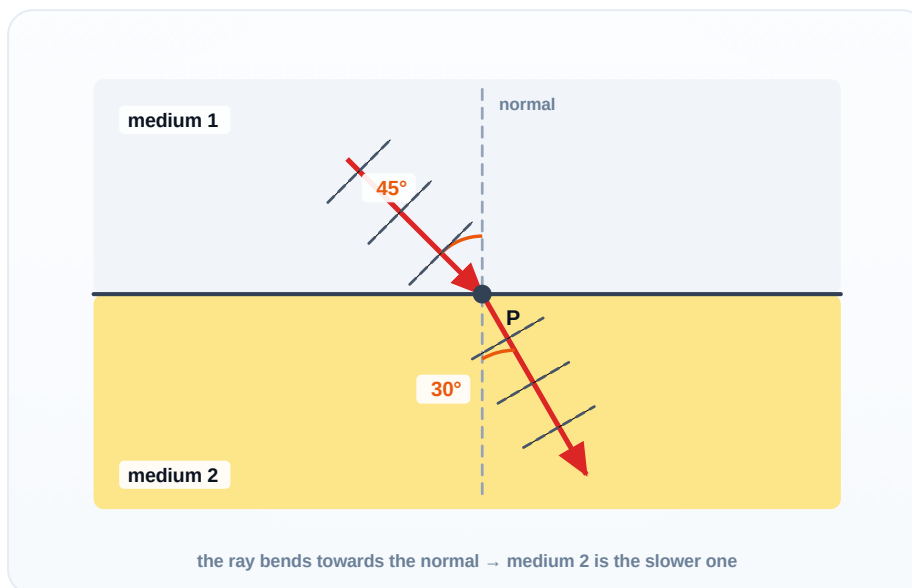


FIGURE 5

- 8 A wave of frequency 50 Hz travels in medium (1) with a speed of 300 m/s. On passing into medium (2) its speed changes to 600 m/s. Which of the following statements is **not** correct?

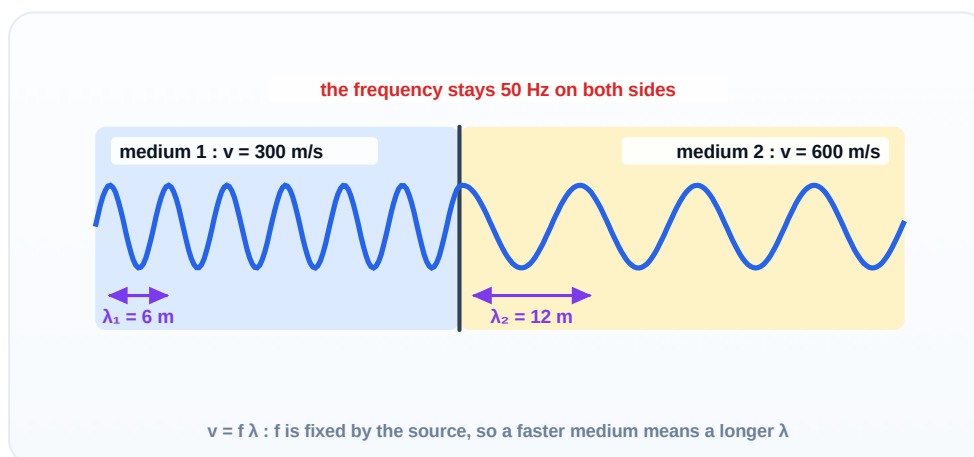


FIGURE 6

- A the frequency of the wave in medium (2) equals 50 Hz
- B the wavelength of the wave in medium (1) equals half its value in medium (2)
- C the angle of incidence of the wave in medium (1) equals half the angle of refraction in medium (2)
- D the refractive index of medium (1) equals twice the refractive index of medium (2)

**9** A light ray passes at an oblique angle from medium (1) into medium (2). Given that the absolute refractive index of medium (1) is greater than the absolute refractive index of medium (2), that is  $n_1 > n_2$ , which of the following physical changes correctly describes what happens to the light ray on entering medium (2)?

MCQ

**A** the ray is refracted towards the normal, and its speed and wavelength decrease

**B** the ray is refracted away from the normal, and its speed and wavelength increase

**C** the ray is refracted away from the normal, its speed increases and its frequency decreases

**D** the ray passes straight on without any deviation, and its speed and wavelength stay constant

**10** A light ray travels from material (A) into material (B), so that the angle of incidence in material (A) equals  $30^\circ$  and the angle of refraction in material (B) equals  $45^\circ$ . Which of the following statements correctly describes the behaviour of the light and the physical properties of the two materials?

MCQ

**A** the refractive index of (A) is greater than that of (B), and the speed of light in (A) is less than it is in (B)

**B** the refractive index of (A) is less than that of (B), and the speed of light in (A) is greater than it is in (B)

**C** the refractive index of (A) is greater than that of (B), and the speed of light in (A) is greater than it is in (B)

**D** the refractive index of (A) equals that of (B), and the frequency in (A) is doubled in (B)

# Interference of Sound and the Doppler Effect

Loud and quiet places in a room, Quincke's tube and noise cancelling — and why a siren changes its pitch as it passes you.

Grade 11 · Egyptian Baccalaureate

28 questions

Step-by-step solutions

 $\lambda = 4X$  ·  $v = f\lambda$ 

**Mr. Gemy**  
PHYSICS

## The toolkit for this lesson

$$\Delta d = m\lambda$$

loud — constructive

$$\Delta d = (m + \frac{1}{2})\lambda$$

quiet — destructive

$$\text{Quincke : } \lambda = 4X$$

pulling the slide by X adds 2X to the path

### approaching

 $\lambda$  shorter → f apparent is **higher**

### receding

 $\lambda$  longer → f apparent is **lower**

### v of sound

set by the air, never by the motion of the source

**Doppler changes what arrives, not what is sent.** The horn keeps sounding at the same frequency and the sound keeps travelling through the air at the same speed. What the motion does is squeeze the crests together in front of the source and stretch them out behind it — so only the **wavelength that reaches the listener**, and therefore the pitch they hear, is changed.

## Loud and quiet places

- 1 When you walk slowly across a wide courtyard while a continuous note is being played from two sources, what do you notice about the intensity of the sound you receive?

MCQ

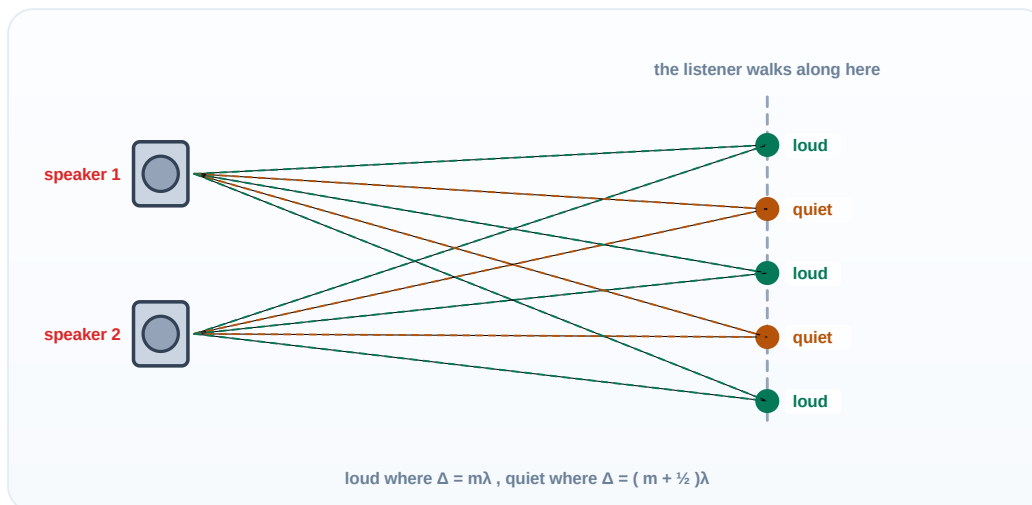


FIGURE 1

A hearing the same sound with the same intensity everywhere in the courtyard

B a gradual and continuous decrease in the intensity the further you move from the source

C a repeated pattern of rising and falling sound intensity

D a complete cutting off of the sound as soon as you move a few steps away from the source

- 2 **Compare** between constructive interference and destructive interference for sound waves coming from two coherent sources vibrating in the same phase, as regards:

EXPLAIN

- a) the intensity of the sound.  
b) the path difference.

## Where the sound disappears

- 3 **Explain the following:** when designing national theatres and large concert halls, sound engineers noticed that there are certain points among the seats at which the intensity of the music is very faint and muffled, in spite of the presence of large, highly efficient loudspeakers on both sides of the stage.

EXPLAIN

4 The figure shows an instrument used in physical measurements.

- What is the name of the instrument?
- What is its function?
- What is the scientific principle on which the idea of its operation is based?
- Explain briefly the way in which the instrument works.

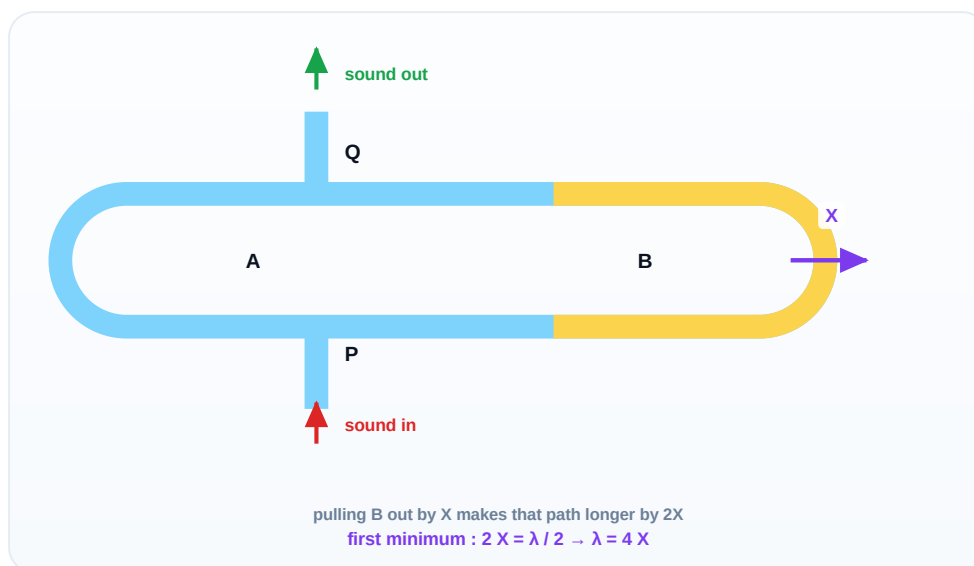


FIGURE 2

5 A student carries out an experiment in the physics laboratory to verify the phenomenon of the interference of sound waves, using loudspeakers and sensitive microphones. The students switched on one loudspeaker and moved the microphone slowly across the courtyard, then repeated the experiment with the two loudspeakers switched on together and sounding the same note. Which of the following correctly represents the change in the sound intensity measured by the microphone as it is moved across the courtyard in the two cases?

MCQ

**A** with one loudspeaker the intensity is exactly uniform, and with the two loudspeakers the intensity increases to several times its value, steadily, everywhere

**B** with one loudspeaker the intensity decreases gradually with distance, and with the two loudspeakers a repeated pattern of rise and fall in the intensity appears

**C** with one loudspeaker the sound is cut off completely, and with the two loudspeakers the sound becomes loud at one single point and faint everywhere else in the courtyard

**D** with one loudspeaker or with two, the level of the sound stays constant and does not change at any point

6 Destructive interference occurs between the waves coming from two coherent sources vibrating in the same phase when the path difference ( $\Delta d$ ) between them equals .....

MCQ

**A**  $\lambda$

**B**  $1.5 \lambda$

**C**  $2 \lambda$

**D**  $3 \lambda$

7 Two coherent sound sources vibrate in the same phase; the wavelength of the waves emitted by each of them is 2 m. Constructive interference occurs between the waves coming from them when the path difference ( $\Delta d$ ) between them equals .....

MCQ

A 1 m

B 2 m

C 3 m

D 5 m

## Group A

- 8 A light ray travels from material (X), whose refractive index is  $n_1 = 1.33$ , into material (Y), whose refractive index is  $n_2 = 1.5$ . Given that the speed of light in (X) is  $v_1$  and in (Y) is  $v_2$ , and that the wavelength in (X) is  $\lambda_1$  and in (Y) is  $\lambda_2$ , which of the following relations correctly expresses the behaviour of the light at this passage?

MCQ

A  $v_1 > v_2, \lambda_1 > \lambda_2$

B  $v_1 < v_2, \lambda_1 < \lambda_2$

C  $v_1 > v_2, \lambda_1 < \lambda_2$

D  $v_2 = v_1, \lambda_1 > \lambda_2$

- 9 Constructive interference occurs between the waves coming from two coherent sources vibrating in the same phase when the path difference ( $\Delta d$ ) between them equals .....

MCQ

A  $0.5 \lambda$

B  $\lambda$

C  $1.5 \lambda$

D  $2.5 \lambda$

- 10 **What happens** to the speed of sea waves, to their wavelength and to their direction of propagation, when they pass at an oblique angle from deep water into shallow water near the shore? Give the scientific explanation of each of them.

EXPLAIN

- 11** The figure shows Quincke's tube. The sound entering at P is split into two paths and is recombined at Q. When the tube B is pushed fully in, the two paths are equal in length. As the tube B is slowly pulled out, the first minimum in the sound intensity occurs when the tube is pulled out by an amount  $X = 10\text{ cm}$ . Taking the speed of sound as  $330\text{ m/s}$ , **calculate:**

- a) the wavelength of the sound, in metres.  
b) the frequency of the sound, in hertz.

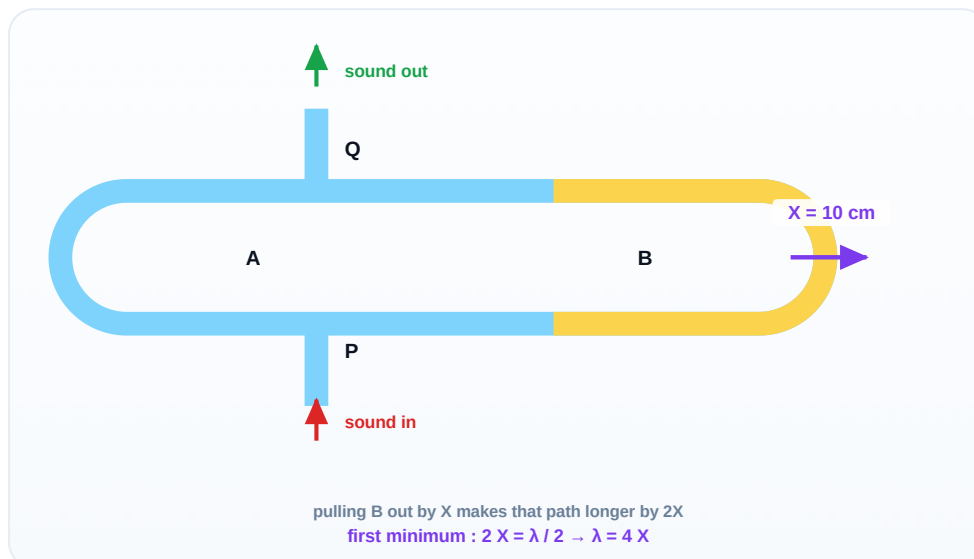


FIGURE 3

- 12 Explain:** why are the positions of **minimum** intensity of a wave, rather than the positions of maximum intensity, the strongest evidence that sound is a wave?

EXPLAIN

## WEEKLY ASSESSMENT

## D Group B

- 13** Ultrasound waves are used in medical examinations to follow the foetus and the internal organs. When an ultrasound wave passes from a soft tissue (such as muscle) into another tissue of higher density (such as bone), part of the wave is reflected and the other part is refracted inside the bone. Which of the following physical changes takes place in the refracted part of the wave as it enters the bone?

MCQ

**A** its frequency increases and its speed increases, while its wavelength decreases

**B** its frequency stays constant, and its speed and its wavelength increase

**C** its frequency and its speed decrease, while its wavelength increases

**D** its speed and its wavelength stay constant, while its frequency increases

- 14 Two coherent sound sources vibrate in the same phase; the wavelength of the waves emitted by each of them is 4 m. Destructive interference occurs between the waves coming from them when the path difference ( $\Delta d$ ) between them equals .....

MCQ

A 2 m

B 4 m

C 8 m

D 12 m

- 15 The figure shows Quincke's tube. The sound entering at P is split into two paths and is recombined at Q. When the tube B is pushed fully in, the two paths are equal in length. As the tube B is slowly pulled out, the first minimum in the sound intensity occurs when the tube is pulled out by an amount  $X = 12$  cm, the frequency of the sound being 1000 Hz. **Calculate:**

PROBLEM

- a) the wavelength of the sound, in metres.  
b) the speed of the sound, in metres per second.

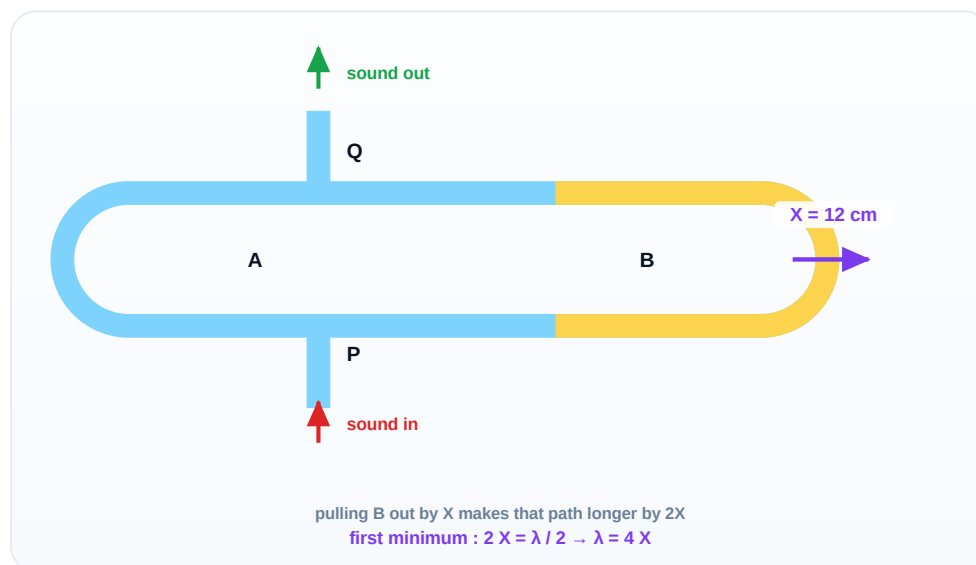


FIGURE 4

16 Explain briefly the way in which noise-cancelling headphones work.

EXPLAIN

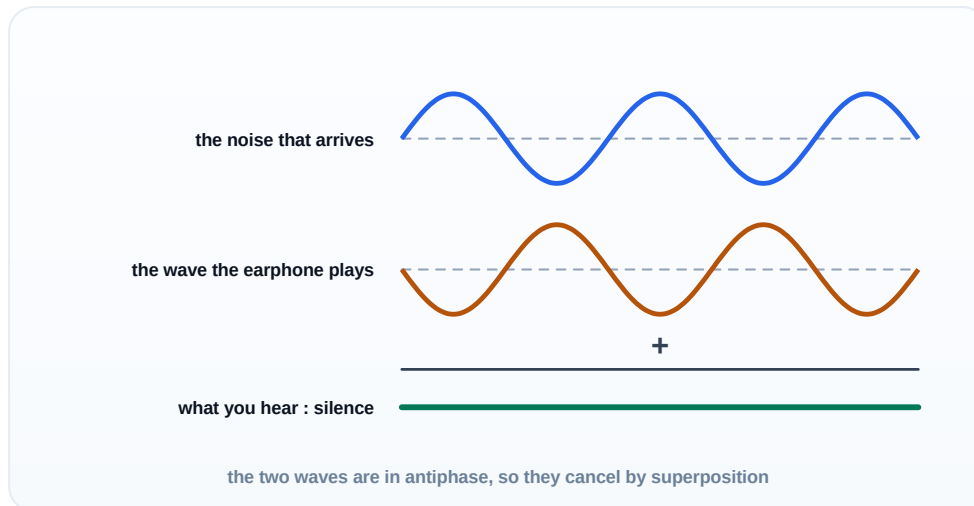


FIGURE 5

17 A wave crosses a boundary separating two media as shown in the figure. Calculate the **ratio** between the wavelength of the wave in medium 1 and the wavelength of the wave in medium 2.

PROBLEM

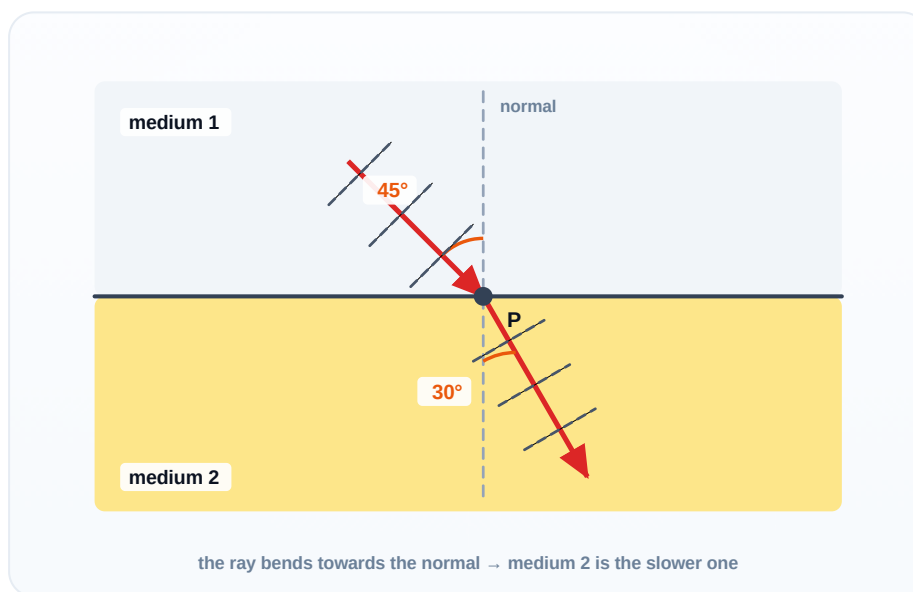


FIGURE 6

## Group C

- 18 A light wave of frequency  $6 \times 10^{14}$  Hz travels in the first medium with a speed of  $3 \times 10^8$  m/s. On being refracted and passing into the second medium its speed becomes  $2 \times 10^8$  m/s. Which of the following statements is **not** correct?

MCQ

- A the frequency of the wave in the second medium =  $6 \times 10^{14}$  Hz
- B the wavelength of the wave in the first medium =  $1.5 \times$  its wavelength in the second medium
- C the refractive index of the first medium =  $1.5 \times$  the refractive index of the second medium
- D the sine of the angle of incidence in the first medium =  $1.5 \times$  the sine of the angle of refraction in the second medium

- 19 Constructive interference occurs between the waves coming from two coherent sources vibrating in the same phase when the path difference ( $\Delta d$ ) between them equals .....

MCQ

- A an odd number of quarter wavelengths
- B an even number of quarter wavelengths
- C an odd number of half wavelengths
- D an even number of half wavelengths

- 20 The figure shows Quincke's tube. The sound entering at P is split into two paths and is recombined at Q. When the tube B is pushed fully in, the two paths are equal in length. As the tube B is slowly pulled out, the first minimum in the sound intensity occurs when the tube is pulled out by an amount X. Given that the frequency of the sound is 1200 Hz and the speed of sound is 360 m/s, **calculate**:

PROBLEM

- a) the wavelength of the sound, in metres.
- b) the magnitude of the distance X, in metres.

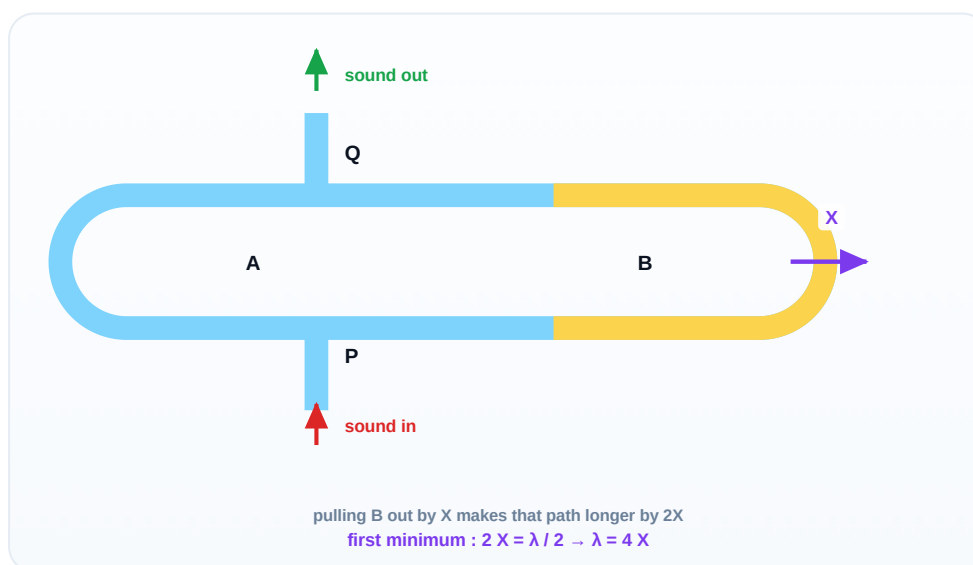


FIGURE 7

- 21 Explain:** why do noise-cancelling headphones reduce the steady hum of an engine much more effectively than a sudden sound such as the slamming of a door?

EXPLAIN

- 22** A wave crosses a boundary separating two media as shown in the figure. **Calculate:**

PROBLEM

- a) the ratio between the frequency of the wave in medium 1 and its frequency in medium 2.  
b) the ratio between the wavelength of the wave in medium 1 and its wavelength in medium 2.

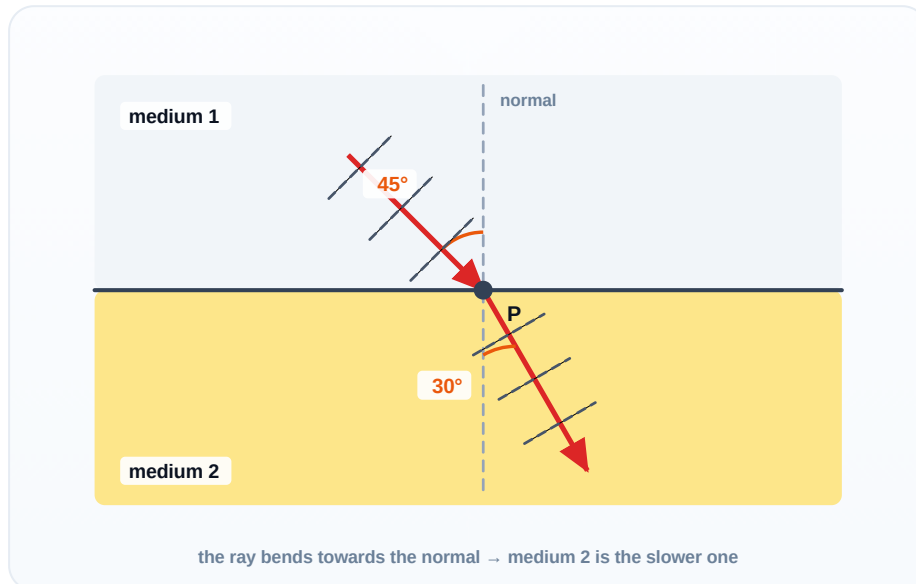


FIGURE 8

## A moving source of sound

- 23 The figure shows a student carrying a phone that gives out a note of constant frequency  $f$ , moving towards / away from another, stationary student (the listener). Using the figure and what you have studied about the properties of waves, **explain the changes that take place in the pitch of the note** during the approach and during the recession, as recorded by the listener.

EXPLAIN

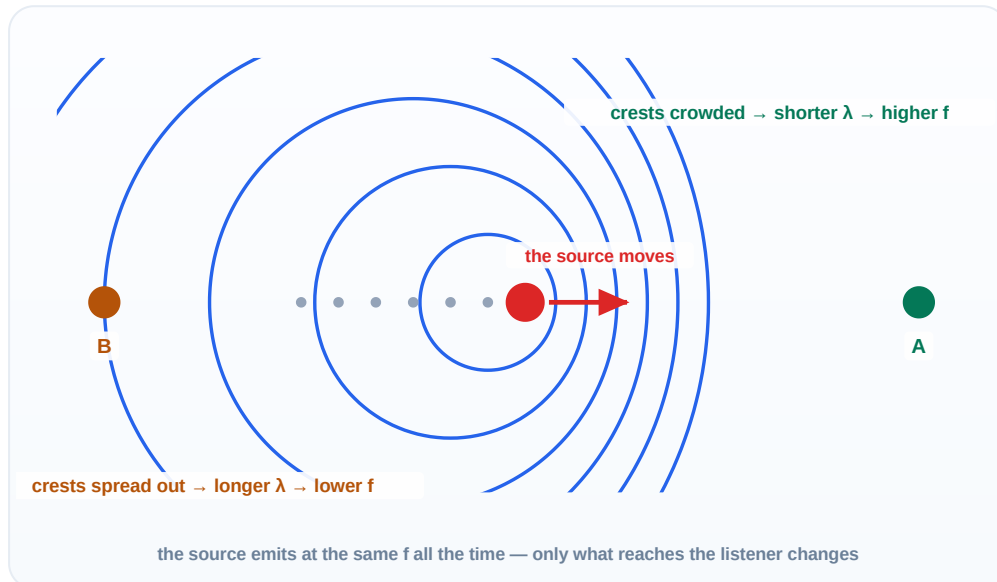


FIGURE 9

- 24 The student carrying the phone moved with a greater speed (a faster walk), about 4 m/s instead of 2 m/s, towards the stationary student. Which of the following changes takes place in the properties of the sound wave as recorded by the stationary student (the listener), compared with the first case?

MCQ

- A** the apparent frequency decreases and the sound becomes sharper
- B** the apparent frequency increases and the sound becomes even sharper
- C** the apparent frequency stays constant and only the speed of sound increases
- D** the apparent frequency decreases and the sound becomes deeper

## Approaching, passing, receding

- 25 While a student carrying a phone that gives out a note of constant frequency moves with a speed towards a stationary student, which of the following changes takes place in the apparent wavelength ( $\lambda$ ) of the sound waves in front of the phone, as received by the stationary student?

MCQ

A the wavelength increases because the speed of the source increases

B the wavelength decreases because the waves are given out closer together in front of the moving source

C the wavelength stays constant because the original frequency of the phone is constant

D the wavelength is doubled because the speed of sound in air increases

- 26 When a car sounding a horn of constant frequency  $f$  approaches, with a uniform speed, a person standing at the side of the road, then passes him and moves away from him: how does the apparent frequency of the sound ( $f'$ ) that the person notices at the instant the car passes exactly in front of him change, compared with the instant of its approach and the instant of its recession?

MCQ

A the frequency is as high as possible during the approach, equals the original frequency  $f$  at the instant of passing, and falls below  $f$  during the recession

B the frequency is as low as possible during the approach, equals the original frequency  $f$  at the instant of passing, and increases during the recession

C the frequency stays higher than the original frequency  $f$  continuously, at every instant

D the frequency stays equal to the original frequency  $f$  throughout the motion of the car

- 27 A police car sounding a siren of constant frequency  $f$  moves steadily, with a uniform speed  $v$ , towards a vertical mountain wall. What does the driver of the car notice as regards (the frequency of the direct sound coming from the car) and (the frequency of the sound reflected from the mountain, which he hears after its reflection)?

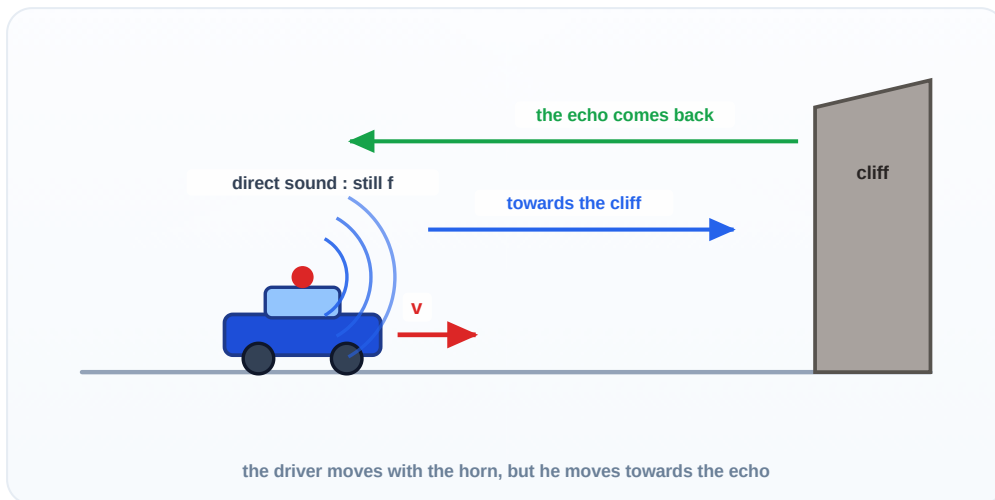


FIGURE 10

- A the direct frequency equals  $f$ , and the reflected frequency falls below  $f$
- B the direct frequency rises above  $f$ , and the reflected frequency equals  $f$
- C the direct frequency equals  $f$ , and the reflected frequency rises above  $f$
- D the direct frequency falls below  $f$ , and the reflected frequency falls below  $f$

- 28 A girl stands at the side of the road listening to the siren of a speeding ambulance. She noticed that the note of the sound was very sharp while the car was approaching her, but as soon as the car passed in front of her and moved away, she noticed that the note of the sound suddenly became deep.

- a) How does the physics of relative motion explain this sudden change in the pitch of the sound at the instant the car passes the listener?
- b) What change takes place in the speed of propagation of the sound in air during this whole process? Explain your answer.

# The Doppler Effect for Light

Red shift and blue shift,  $\Delta\lambda/\lambda = v/c$ , and how the pitch of a siren tells you which way a source is moving.

Mr. Gemy

Grade 11 · Egyptian Baccalaureate

22 questions

Step-by-step solutions

$c = 3 \times 10^8 \text{ m/s}$  ·  $v(\text{sound}) = 340 \text{ m/s}$

PHYSICS

## The toolkit for this lesson

$$\Delta\lambda / \lambda = v / c$$

the radial speed of a star or galaxy

**red shift**

$\lambda' > \lambda \rightarrow$  the source is moving **away**

**blue shift**

$\lambda' < \lambda \rightarrow$  the source is moving **towards** us

$$f' = f \cdot v / (v - v_s)$$

a sound source approaching

$$f' = f \cdot v / (v + v_s)$$

a sound source receding

**c is always c**

the speed of light never changes with the motion

**Ask one question first: is the gap closing or opening?** Closing gives a shorter wavelength — a higher pitch for sound, a blue shift for light. Opening gives a longer wavelength — a lower pitch, a red shift. Everything else in this lesson is arithmetic on top of that one idea.

## Red shift and blue shift

EXPLAIN

- 1 In an experiment simulating the Doppler effect for light in the astronomy laboratory, two detectors recorded the spectral state of two stars (X) and (Y):

1. The first detector found that the spectral line of star (X) had **increased** in wavelength by an amount  $\Delta\lambda$ .
2. The second detector found that the spectral line of star (Y) had **decreased** in wavelength by an amount  $\Delta\lambda$ .

- a) Determine which of them represents a “red shift” and which represents a “blue shift”.
- b) Explain how astronomers deduce, from these spectral changes, the speed and the direction of motion of the stars in the universe.

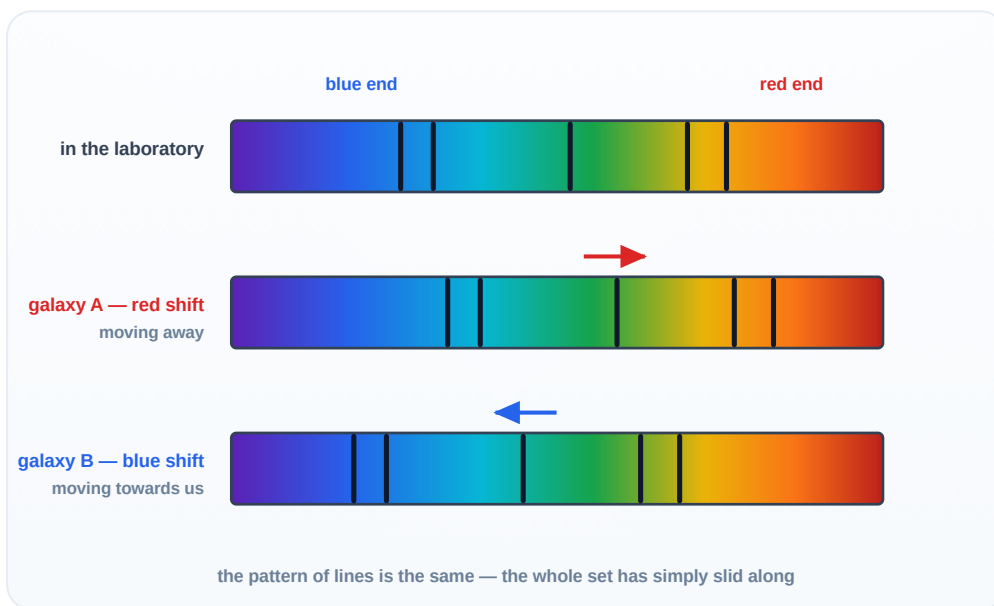


FIGURE 1

- 2 On observing light coming from a distant galaxy, a shift of its spectral lines towards the red colour is noticed. What is the physical reason for this shift, based on the Doppler effect?

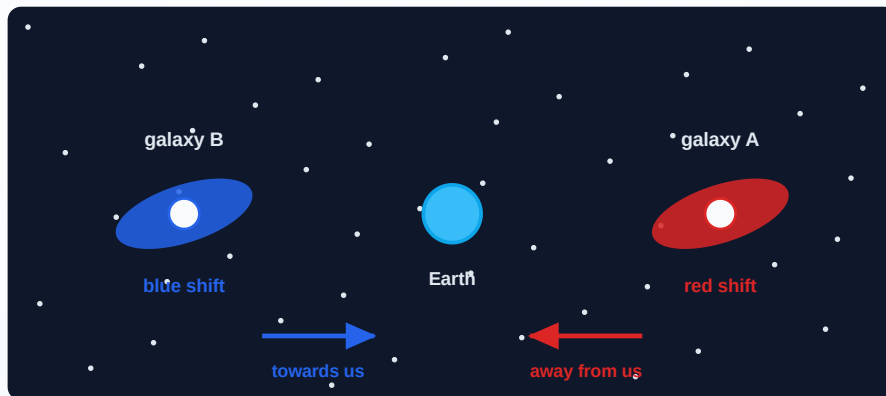
MCQ

- A the galaxy receding from the Earth leads to an increase in the wavelength
- B the galaxy approaching the Earth leads to an increase in the wavelength
- C the galaxy receding from the Earth leads to a decrease in the wavelength
- D the galaxy staying fixed in its position leads to an increase in the wavelength

## Measuring the speed of a galaxy

PROBLEM

- 3 The spectrum of the element hydrogen taken in the laboratory shows a spectral line at a wavelength  $\lambda = 656 \text{ nm}$ . On observing the same spectral line in light coming from a moving galaxy, the magnitude of the shift in the wavelength was found to equal  $\Delta\lambda = 1.312 \text{ nm}$ . **Calculate the radial speed ( $v$ ) of the galaxy relative to the observer**, given that the speed of light  $c = 3 \times 10^8 \text{ m/s}$ .



$\Delta\lambda / \lambda = v / c$  — the bigger the shift, the faster the galaxy

FIGURE 2

- 4 When a galaxy moves towards the planet Earth with a high speed, the apparent frequency of the light recorded by astronomers is:

MCQ

- A higher than the original frequency, and the lines shift towards the blue colour
- B lower than the original frequency, and the lines shift towards the red colour
- C higher than the original frequency, and the lines shift towards the red colour
- D lower than the original frequency, and the lines shift towards the blue colour

- 5 What happens to the speed of light ( $c$ ) in vacuum when it is emitted from a star moving with a great speed towards the observer?

MCQ

- A it stays constant and does not change
- B it increases by an amount equal to the speed of motion of the approaching star
- C it decreases by an amount equal to the speed of motion of the approaching star
- D it changes depending on the apparent frequency of the observed light

- 6 A car moving eastwards with a speed of 15 m/s gives out a sound of frequency 600 Hz, and a motorcycle moving westwards with a speed of 20 m/s passes beside it. **Find the frequency of the sound heard by the rider of the motorcycle after they have passed each other**, taking the speed of sound in air as 340 m/s.

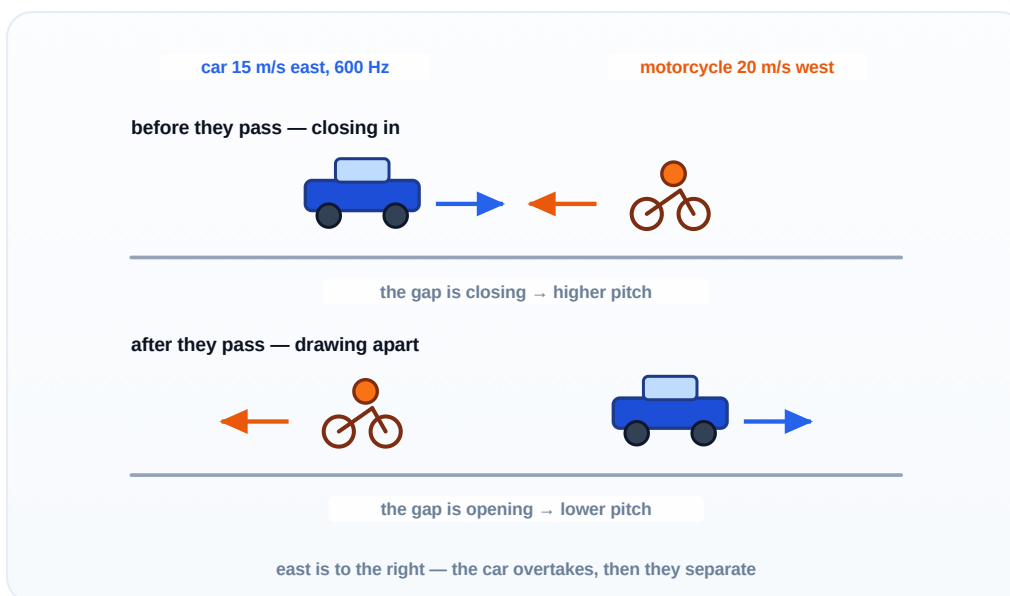


FIGURE 3

- 7 The phenomenon in which the observed frequency differs from the frequency of the source because of the relative motion between the source and the observer is called .....

MCQ

- A interference of waves
- B the Doppler effect
- C diffraction of waves
- D refraction of waves

- 8 An ambulance sounding its siren at a constant frequency moves with a uniform speed towards a person standing at the side of the road. What happens to the **intensity** of the sound (its loudness) and to the **pitch** of the sound (its frequency) as noticed by the stationary person while the car approaches him?

MCQ

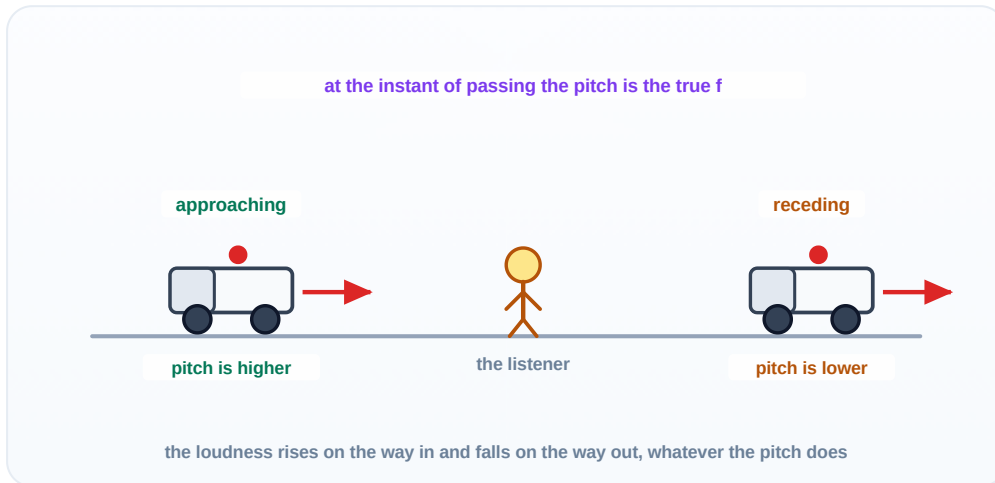


FIGURE 4

- A the intensity of the sound increases, and the apparent frequency increases
- B the intensity of the sound increases, and the apparent frequency decreases
- C the intensity of the sound decreases, and the apparent frequency increases
- D both the intensity and the frequency stay constant without change

- 9 The figure shows the observation of light coming from a galaxy; a shift of its spectral lines towards the red colour is noticed compared with the same spectrum in the laboratory. What is the physical reason for this shift, based on the Doppler effect?

MCQ

- A the galaxy receding from the Earth leads to an increase in the wavelength
- B the galaxy approaching the Earth leads to an increase in the wavelength
- C the galaxy receding from the Earth leads to a decrease in the wavelength
- D the galaxy staying fixed in its position leads to an increase in the wavelength

- 10 A sound source moving with a speed of 25 m/s approaches a stationary observer while giving out a sound of frequency 600 Hz. **Find the frequency of the sound heard by the observer**, given that the speed of sound in air is 340 m/s.

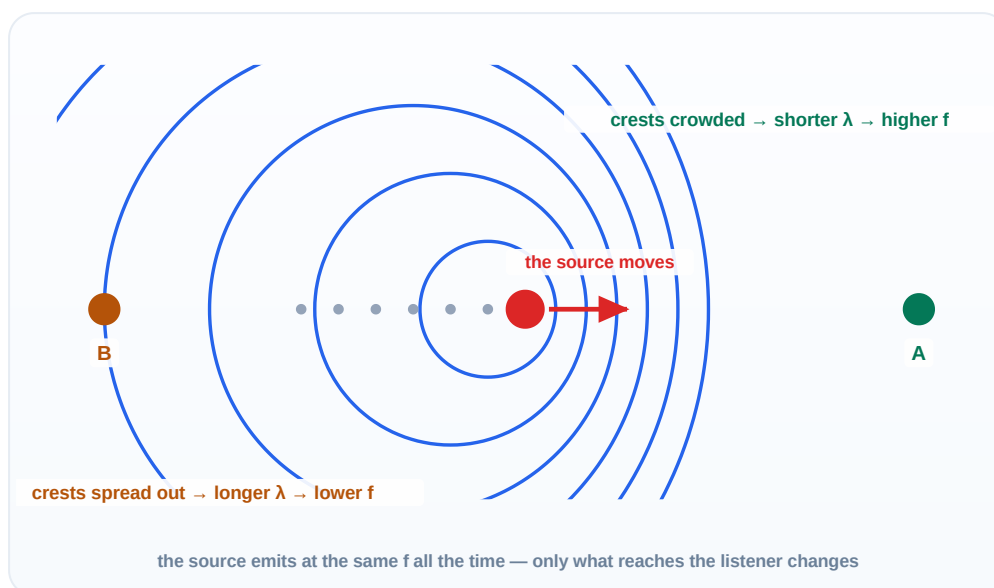


FIGURE 5

- 11 On observing the light coming from one of the distant galaxies, scientists noticed that all the colours of its spectrum were shifted slightly towards the red colour. **What do we conclude from this observation about the motion of the galaxy?** Explain the reason for this shift, based on the Doppler effect for light.

EXPLAIN

- 12 An increase is noticed in the sharpness of the whistle of a speeding train while it approaches the stopping station, while the sharpness of the sound decreases as soon as it passes the station and moves away from it. **Give the reason** for this observation, based on the concept of the Doppler effect.

EXPLAIN

## WEEKLY ASSESSMENT

## D Group B

- 13 When an ambulance sounding a horn of constant frequency passes a shore on which a listener is standing, and begins to move away from him with a uniform speed: how do the pitch of the sound (its apparent frequency) and the intensity of the sound measured by the listener change while the car is moving away?

MCQ

A the apparent frequency decreases, and the intensity of the sound decreases

B the apparent frequency increases, and the intensity of the sound increases

C the apparent frequency decreases, and the intensity of the sound increases

D the apparent frequency increases, and the intensity of the sound decreases

**14** If the wavelength of a spectral line observed for a galaxy equals ( $\lambda'$ ) and the reference wavelength in the laboratory equals ( $\lambda$ ), then the case that expresses the galaxy approaching with a radial speed ( $v$ ) is:

MCQ

**A**  $\lambda' < \lambda$ , and the spectral lines shift towards the blue colour

**B**  $\lambda' > \lambda$ , and the spectral lines shift towards the red colour

**C**  $\lambda' < \lambda$ , and the spectral lines shift towards the red colour

**D**  $\lambda' > \lambda$ , and the spectral lines shift towards the blue colour

**15** A sound source moving with a speed of 20 m/s recedes from a stationary observer while giving out a sound of frequency 600 Hz. **Find the frequency of the sound heard by the observer**, given that the speed of sound in air is 340 m/s.

PROBLEM

**16** On observing the electromagnetic spectrum of light coming from a distant galaxy, one of the astronomers noticed that the absorption spectrum lines of the element sodium had shifted towards the side of the lower frequencies (the red colour). **What conclusion does the astronomer reach about the direction of motion of this galaxy relative to the Earth?** Explain your answer.

EXPLAIN

**17** An ambulance approaches a person standing at the side of the road while sounding its siren, then passes him and moves away from him. **Compare** between the pitch (sharpness) of the sound that the person hears while the ambulance is approaching him, and the pitch of the sound after it has passed him and moved away.

EXPLAIN

#### WEEKLY ASSESSMENT

**E**

#### Group C

**18** When an ambulance sounding a horn of constant frequency  $f$  approaches a person standing at the side of the road, the apparent wavelength ( $\lambda$ ) of the sound waves that the person receives, compared with the original wavelength ( $\lambda$ ) of the wave in air, is:

MCQ

**A** larger; because of the piling up of the sound waves behind the source

**B** smaller; because of the compression of the sound waves in front of the source

**C** equal; because the frequency of the sound of the source is constant and does not change

**D** smaller; because of the increase in the speed of propagation of sound in air

19 The spectral shift of the light of galaxies towards the longer wavelengths is known as:

MCQ

A the red shift, and it indicates that the galaxy is receding from the Earth

B the blue shift, and it indicates that the galaxy is receding from the Earth

C the red shift, and it indicates that the galaxy is approaching the Earth

D the blue shift, and it indicates that the galaxy is approaching the Earth

20 A car moving eastwards with a speed of 15 m/s gives out a sound of frequency 600 Hz, and a motorcycle moving westwards with a speed of 20 m/s passes beside it. **Find the frequency of the sound heard by the rider of the motorcycle before they have passed each other**, taking the speed of sound in air as 340 m/s.

PROBLEM

21 The siren of a fire engine is heard at a high (sharp) frequency while it approaches a person standing at the side of the road, while the same siren is heard at a low (deep) frequency after it has passed him and moved away from him. **State the name of the physical phenomenon** responsible for this observation.

EXPLAIN

22 On observing the light spectrum coming from two different galaxies (A) and (B), it was noticed that the spectral lines of galaxy (A) were shifted towards the red colour, while the spectral lines of galaxy (B) were shifted towards the blue colour. **Explain the direction of motion of each of the two galaxies** relative to the observer.

EXPLAIN

# Refraction and Total Internal Reflection

The absolute refractive index, Snell's law, the critical angle, optical fibres and the blocking disc.

Grade 11 · Egyptian Baccalaureate

30 questions

Step-by-step solutions

 $\sin \theta_c = n_2 / n_1$ 

Mr. Gemy  
PHYSICS

## The toolkit for this lesson

$$n = c / v$$

the absolute refractive index

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

Snell's law of refraction

$$\sin \theta_c = n_2 / n_1$$

the critical angle,  $n_1 > n_2$ 

$$\sin \theta_c = 1 / n$$

when the outside medium is air

$$\lambda_1 / \lambda_2 = v_1 / v_2 = n_2 / n_1$$

f never changes

$$r = h \tan \theta_c$$

the radius of the blocking disc

**A denser medium is a slower medium.**  $n = c/v$ , so a bigger  $n$  always means a smaller  $v$ , a shorter  $\lambda$  and a ray that bends **towards** the normal. Going the other way — out of the denser medium — the ray bends away from the normal, and once the angle inside passes  $\theta_c$  it cannot get out at all.

## CLASSWORK 1

A

### Refraction between two media

- 1 The opposite diagram shows a light ray passing between two media A and B. When the ray passes from medium A into medium B, **what happens to the speed of light and to the wavelength?**

EXPLAIN

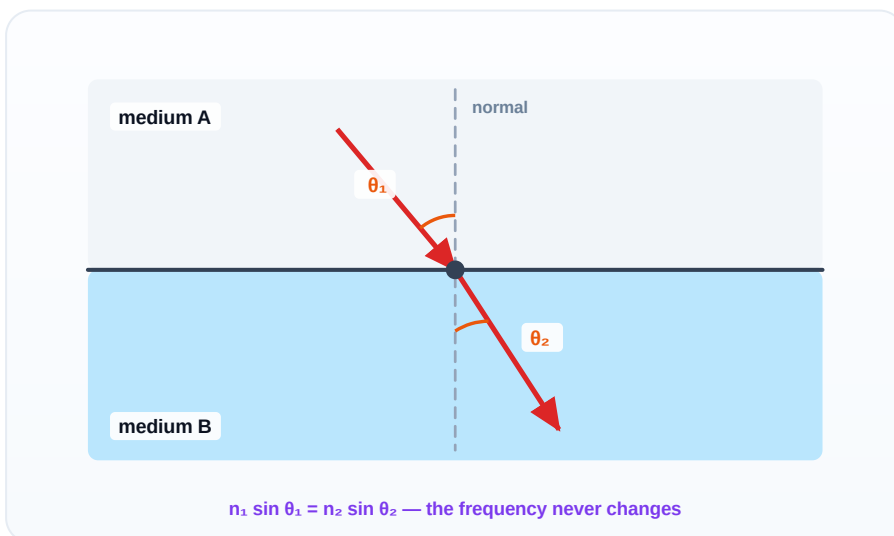


FIGURE 1

2 The absolute refractive index ( $n$ ) of a transparent medium is given by the relation .....

EXPLAIN

#### HOME ASSIGNMENT 1

### B The refractive index

3 Light falls from water into glass with an angle of incidence of  $55^\circ$ . Given that the relative refractive index between water and glass equals 1.15, **calculate the angle of refraction of the light in the glass.**

PROBLEM

4 **Compare between** the optically denser medium and the optically less dense medium as regards: the speed of light in it, and the absolute refractive index.

EXPLAIN

5 If the speed of light in air is  $3 \times 10^8$  m/s and its speed in glass is  $2 \times 10^8$  m/s, **calculate the absolute refractive index of the glass.**

PROBLEM

6 When light passes from a less optically dense medium into a more optically dense medium, then .....

MCQ

A both the frequency and the wavelength decrease

B the frequency increases and the wavelength decreases

C the frequency does not change and the wavelength increases

D the frequency does not change and the wavelength decreases

- 7 A light ray falls from medium (1), whose absolute refractive index is 1.3, into medium (2), whose absolute refractive index is 1.5, as in the figure. Which of the choices shows what will happen to both the wavelength and the speed of light in medium (2)?

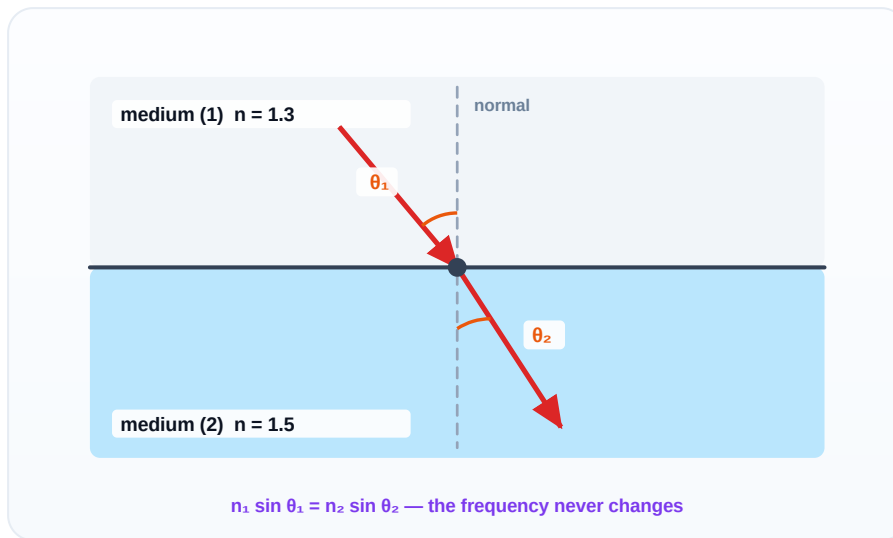


FIGURE 2

- A both the wavelength and the speed increase
- B both the wavelength and the speed decrease
- C the wavelength increases and the speed decreases
- D the wavelength decreases and the speed increases

## CLASSWORK 2

## C The critical angle

- 8 When light passes from a more optically dense medium into a less dense medium, **what is the greatest value of the angle of refraction** in the less dense medium?

EXPLAIN

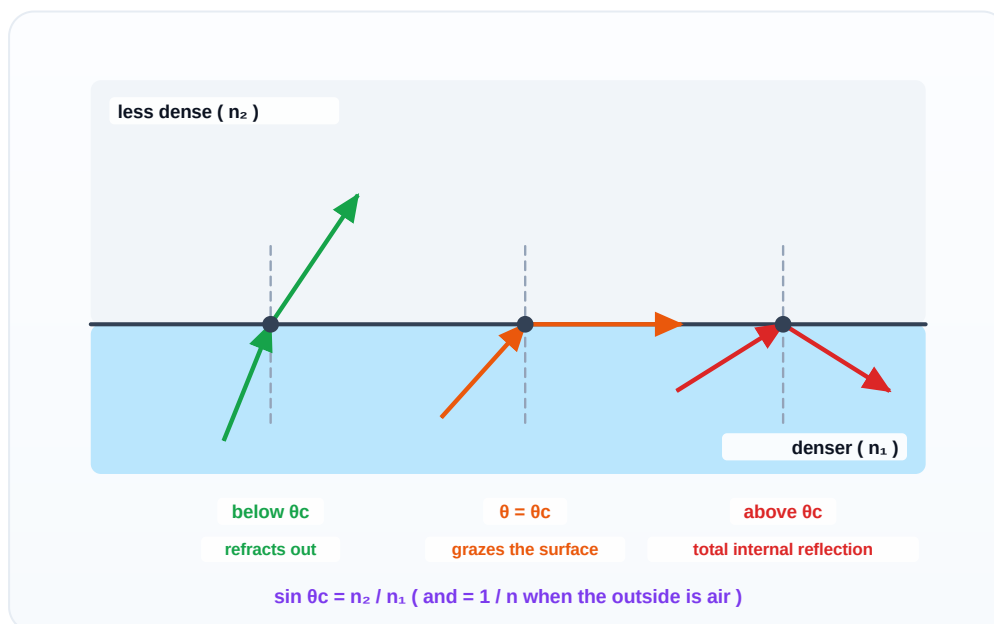


FIGURE 3

9 To calculate the critical angle between two media (  $n_1 > n_2$  ) we use the relation .....

EXPLAIN

## Total internal reflection

- 10 **Give the scientific reason:** the outer cladding of an optical fibre is made of a material whose refractive index is less than the refractive index of the inner core.

EXPLAIN

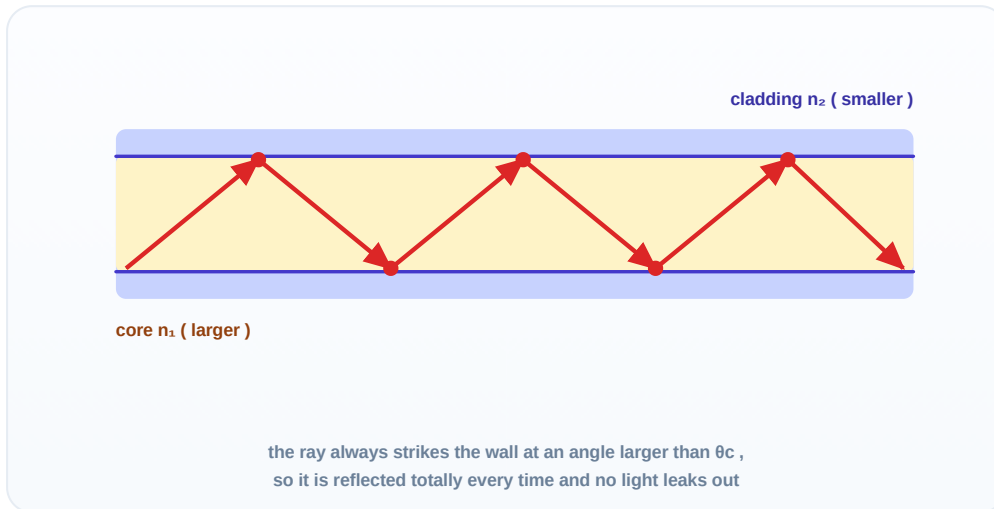


FIGURE 4

- 11 **What happens when** a narrow beam of light is directed upwards through the side of a glass of water towards the (water – air) surface, and the angle of incidence is gradually increased?

EXPLAIN

- 12 If the refractive index of the core in an optical fibre is 1.6 and that of the cladding is 1.4, then  $\sin \theta_c$  between them equals:

MCQ

A 1.4 / 1.6

B 1.6 / 1.4

C 1 / 1.6

D 1 / 1.4

- 13 Total internal reflection occurs when light passes from a medium:

MCQ

A of lower optical density into a medium of higher optical density

B of higher optical density into a medium of lower optical density

C into another of equal density

D that is opaque into one that is transparent

- 14 Given that the refractive index of glass is 1.5, which of the following figures shows the correct path of a light ray falling on the boundary separating the glass and the air at an angle of  $50^\circ$ ?

MCQ

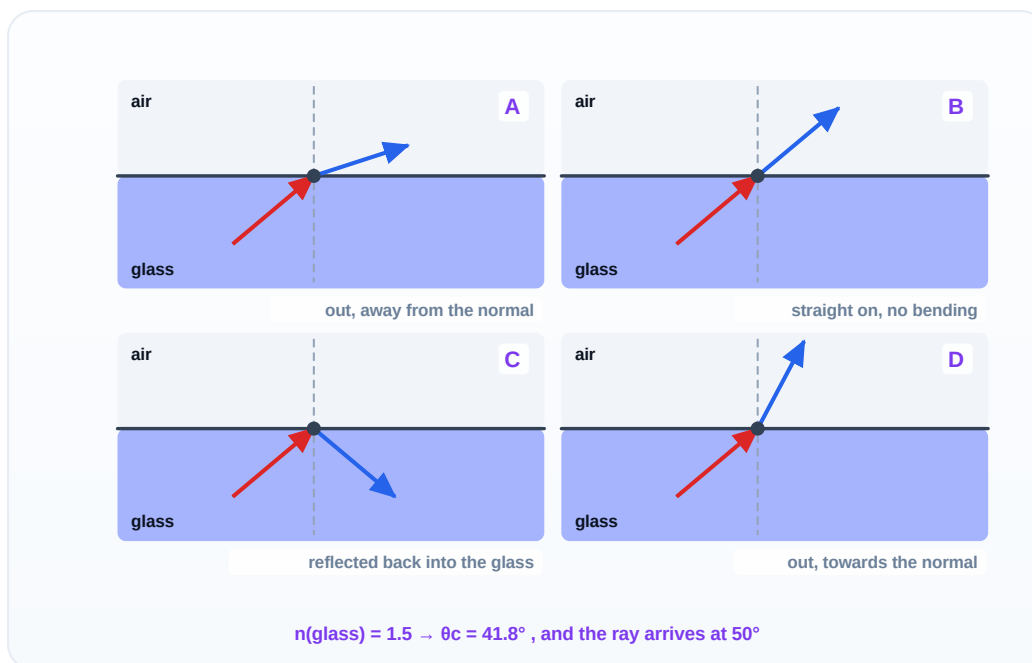


FIGURE 5

A figure A

B figure B

C figure C

D figure D

## WEEKLY ASSESSMENT

### E Group A

- 15 An optical fibre whose material has a refractive index of 2.1 is covered with an outer layer of cryolite. The refractive index of the outer layer that makes the critical angle between the two layers equal  $32^\circ$  is .....

MCQ

A 1.11

B 2.25

C 3.96

D 4.32

- 16 In the experiment of blocking light with a disc, if the water is replaced by another liquid of greater refractive index, then the area of the disc needed for the blocking:

MCQ

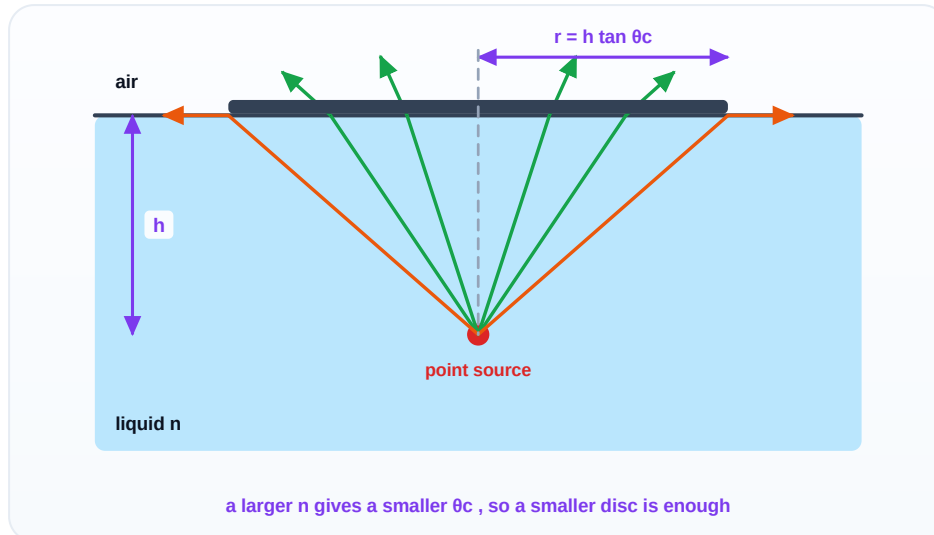


FIGURE 6

- A increases
- B decreases
- C does not change
- D becomes zero

- 17 Give the scientific reason: a light ray bends away from the normal when it passes from water into air.

EXPLAIN

- 18 What happens when a light ray falls from a more optically dense medium into a less optically dense medium at an angle of incidence equal to the critical angle?

EXPLAIN

- 19 Light whose speed in vacuum is  $c$  passes into two media A and B; its speed in A is  $0.8c$  and in B is  $0.6c$ . Calculate the critical angle when light passes between the two media.

PROBLEM

## WEEKLY ASSESSMENT

### F Group B

- 20 A light ray falls on the boundary separating two media whose absolute refractive indices are respectively  $n_1$  and  $n_2$ , where  $n_1 > n_2$ . What is the correct choice that expresses the condition required for total internal reflection to occur?

MCQ

- A the ray must fall in the medium  $n_1$  at an angle greater than the critical angle
- B the ray must fall in the medium  $n_2$  at an angle greater than the critical angle
- C the ray must fall in the medium  $n_1$  at an angle less than the critical angle
- D the ray must fall in the medium  $n_2$  at an angle less than the critical angle

21 If a light ray falls on a rectangular glass block with an angle of incidence equal to  $60^\circ$ , and the absolute refractive index of the glass =  $\sqrt{3}$ , then the angle of refraction of the light equals .....

MCQ

A  $90^\circ$

B  $60^\circ$

C  $30^\circ$

D  $45^\circ$

22 **What happens when** a light ray falls from a more optically dense medium into a less optically dense medium at an angle of incidence greater than the critical angle?

EXPLAIN

23 **Compare between** the more optically dense medium and the less optically dense medium as regards: the speed of light in it, and the absolute refractive index.

EXPLAIN

24 **Give the scientific reason:** the reflected ray inside the water does not disappear when light passes from water into air; rather it increases in brightness as the angle of incidence increases.

EXPLAIN

25 A light ray passes from a material into air. If the angle of incidence is  $30^\circ$  and the angle of refraction is  $50^\circ$ , **calculate the refractive index of the material.**

PROBLEM

#### WEEKLY ASSESSMENT

G

### Group C

26 If a light ray passes from a less optically dense medium into a medium of higher optical density, then the ratio between the wavelength of the incident light and the wavelength of the refracted light is .....

MCQ

A less than one

B greater than one

C equal to one

D zero

27 When light passes from a more optically dense medium into a less dense medium, then the greatest value of the angle of refraction in the less dense medium is .....

MCQ

A  $180^\circ$

B  $90^\circ$

C  $42^\circ$

D  $45^\circ$

28 **Compare between** the passage of light from air into water and its passage from water into air as regards: the frequency, and the wavelength.

EXPLAIN

**29** **Give the scientific reason:** the radius of the disc needed to block the light of a point source under the surface of a liquid decreases as the refractive index of that liquid increases.

EXPLAIN

**30** If the critical angle of glass with air is  $42^\circ$ , and the critical angle of water with air is  $49^\circ$ , **which of them is optically denser? And why, mathematically?**

PROBLEM

# Lenses and Images

## Young's Double-Slit Experiment

Converging and diverging lenses, the lens equation and magnification, and the interference of light through two narrow slits.

Grade 11 · Egyptian Baccalaureate

31 questions

Step-by-step solutions

 $\Delta y = \lambda L / d$ 

Mr. Gemy  
PHYSICS

### The toolkit for these two lessons

$$1/f = 1/a + 1/b$$

the lens equation,  $a$  = object distance

$$M = b/a = h'/h$$

the linear magnification

$$b > 0 \rightarrow \text{real, inverted}$$

$b < 0 \rightarrow$  virtual, erect

$$f > 0 \text{ convex} \cdot f < 0 \text{ concave}$$

the sign of the focal length

$$\Delta y = \lambda L / d$$

the fringe separation in Young's experiment

$$\Delta = m\lambda \rightarrow \text{bright}$$

$\Delta = (m + \frac{1}{2})\lambda \rightarrow \text{dark}$

**Two lessons, two pictures.** A lens bends **one** beam so that it meets again at an image — everything there comes out of  $1/f = 1/a + 1/b$ . Young's experiment splits **one** beam into two and lets them meet again as fringes — everything there comes out of the path difference. Keep the two pictures apart and the whole unit becomes easy.

### CLASSWORK 1

#### A Lesson 2–10 · the parts of a lens

1 What is the name of the lens that is thicker at its centre than at its edges?

EXPLAIN

2 The point at which rays parallel to the principal axis collect after being refracted by a convex lens is called:

MCQ

A the optical centre

B the principal focus

C the centre of curvature

D the pole of the lens

3 The straight line that passes through the optical centre of the lens and is perpendicular to its plane is known as:

MCQ

A the focal length

B the principal axis

C the radius

D the incident ray

## Lesson 2–10 · images formed by lenses

- 4 Using the ray diagram opposite, explain why the image formed by a concave lens is always virtual, erect and diminished.

EXPLAIN

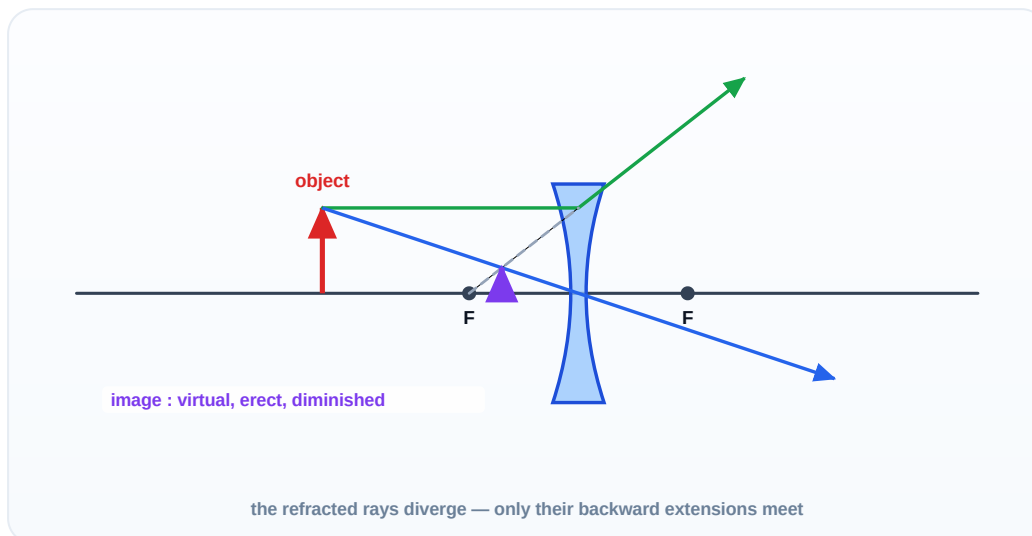


FIGURE 1 · A DIVERGING (CONCAVE) LENS

- 5 Draw the path of the rays for a magnifying (convex) lens used to look at an object placed at a small distance from it (inside the focal length), showing the ray that passes through the optical centre, the ray parallel to the principal axis, and how their extensions meet.

EXPLAIN

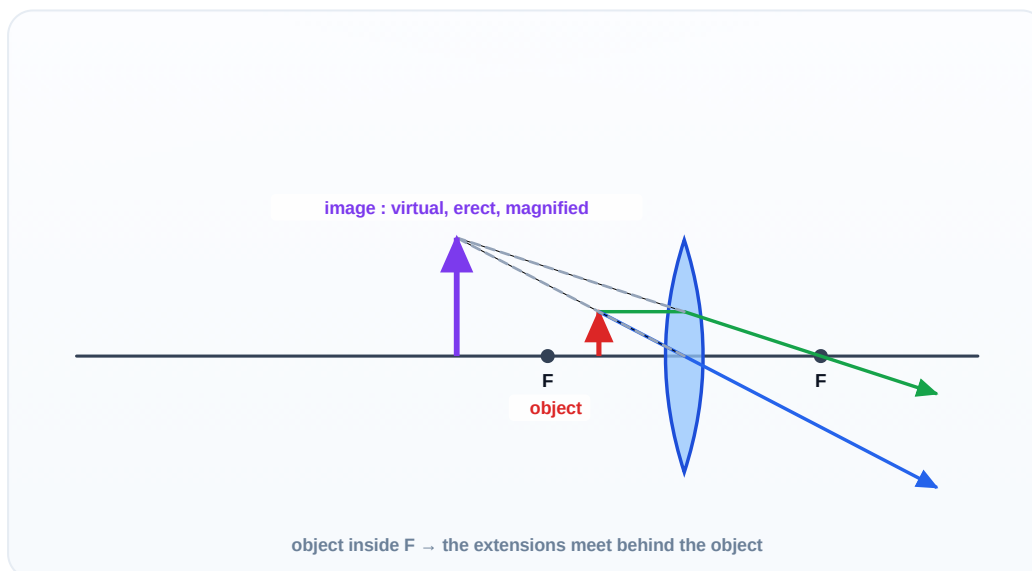


FIGURE 2 · A CONVEX LENS USED AS A MAGNIFIER

- 6 An object is placed at a distance  $a$  from a convex lens and a virtual image is formed. The lens is then replaced by a concave lens having the **same focal length ( $f$ )**, without moving the object, and another virtual image is formed. If the distance between the position of the first image and the position of the second image equals the focal length  $f$ , **find the object distance  $a$  in terms of  $f$ .**

PROBLEM

- 7 If an object is placed at a distance much greater than twice the focal length ( $2F$ ) of a convex lens, the image formed for it is:

MCQ

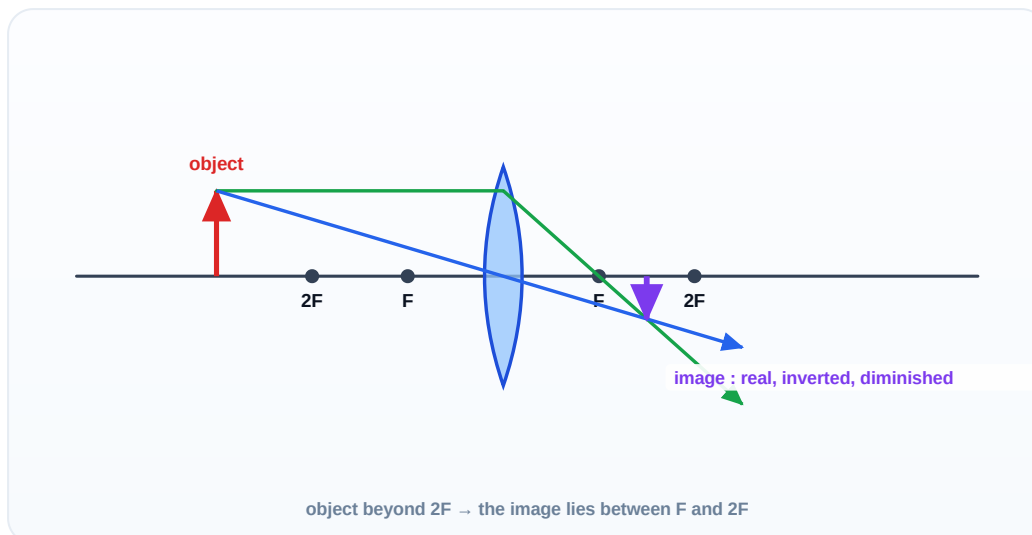


FIGURE 3 · AN OBJECT BEYOND  $2F$

- ☐ A real, inverted, diminished
- ☐ B real, inverted, magnified
- ☐ C virtual, erect, diminished
- ☐ D real, inverted, equal to the object

- 8 If the magnification ( $M = 0.5$ ), this means that the image is:

MCQ

- ☐ A magnified to twice the object
- ☐ B equal to the object
- ☐ C diminished to half the object
- ☐ D always inverted

## Lesson 2–11 · Young's experiment

- 9 The figure opposite shows Young's double-slit experiment. If the distance between two successive bright fringes is **5 mm**, **calculate the wavelength of the light used.**

PROBLEM

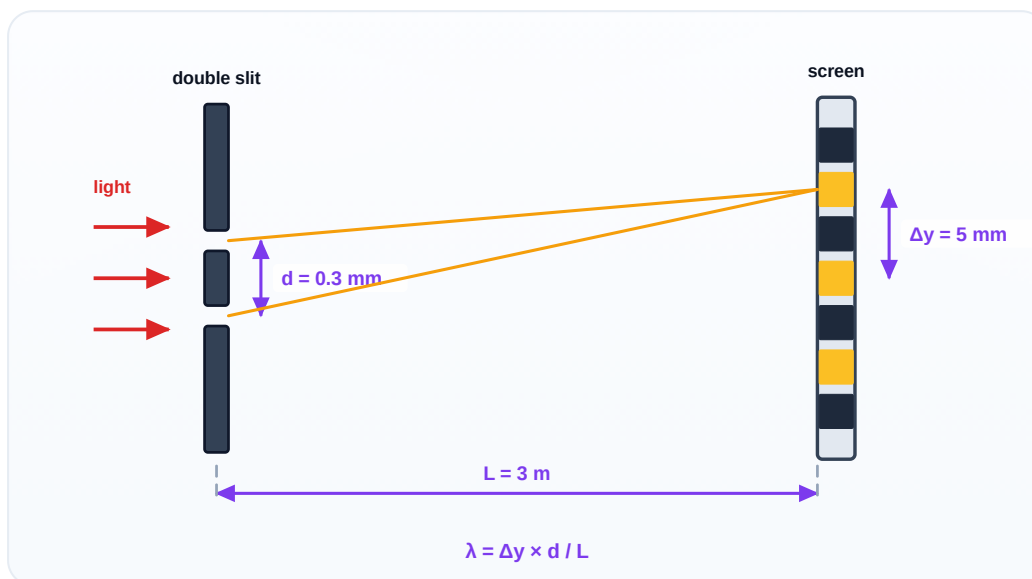


FIGURE 4 · THE DOUBLE-SLIT ARRANGEMENT

- 10 The wavelength  $\lambda$  of any monochromatic light in Thomas Young's double-slit experiment is determined from the relation ..... (R is the distance from the double slit to the screen)

MCQ

A  $\lambda = \Delta y R / d$

B  $R = \Delta y d / \lambda$

C  $\Delta y = \lambda d / R$

D  $d = \lambda \Delta y / R$

- 11 A monochromatic ray falls in Thomas Young's experiment and the distance between the two openings of the double slit is  $d_1$ . The double slit is then replaced by another one whose slit separation is **half** the first. The distance between two successive fringes of the same kind in the second case is:

MCQ

A  $\Delta y_1 / 2 = \Delta y_2$

B  $\Delta y_1 = \Delta y_2$

C  $4 \Delta y_1 = \Delta y_2$

D  $2 \Delta y_1 = \Delta y_2$

## Lesson 2–11 · fringes and their separation

- 12 Monochromatic light of wavelength 450 nm passes through two parallel slits separated by 0.3 mm, and the bright fringes can be observed on a screen 2 m away. **Calculate, in millimetres, the distance between two successive fringes of the same kind.**

PROBLEM

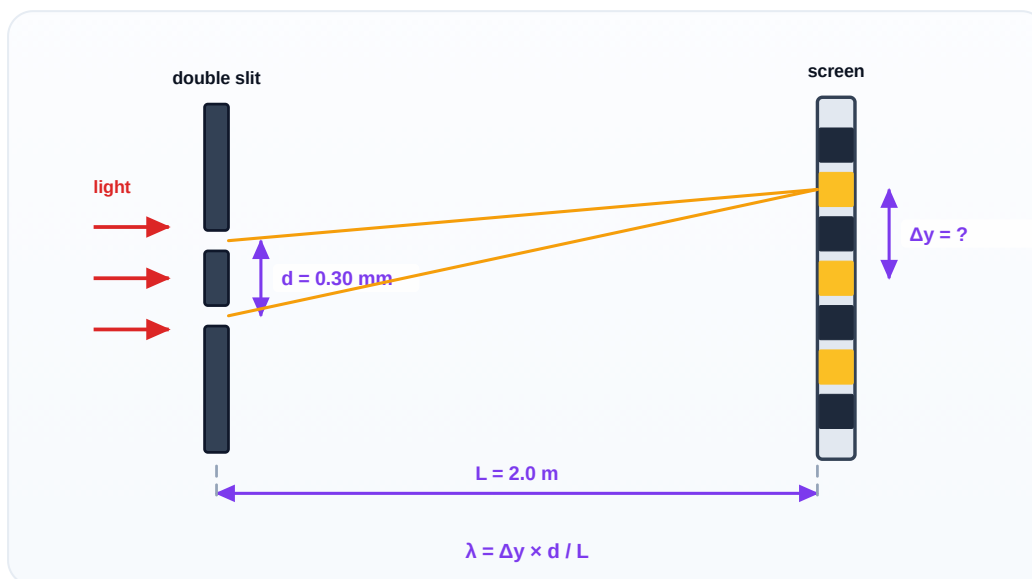


FIGURE 5 · NOT TO SCALE

- 13 From the figure opposite, which represents Young's double-slit experiment: with both slits open, **will two bright patches appear on the wall — one for each slit — or a row of many bright fringes? And where is the brightest one?**

EXPLAIN

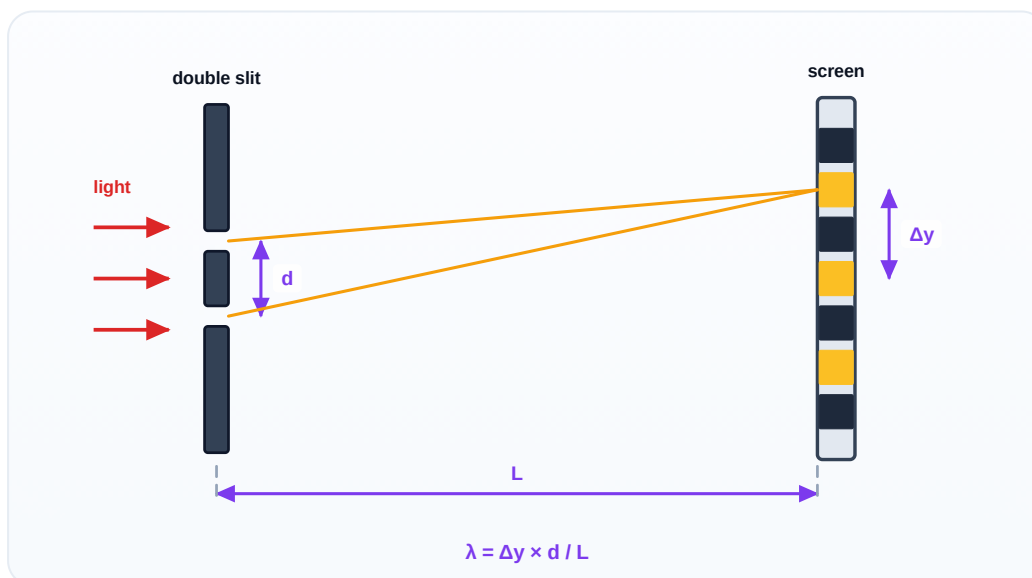


FIGURE 6 · BOTH SLITS OPEN

- 14 Monochromatic light of wavelength 600 nm was used in Young's experiment, and interference fringes were formed on a screen 1.2 m away. If the separation between successive bright fringes is 2.4 mm, **calculate the distance (d) between the two slits.**

PROBLEM

**15** Which of the following methods can increase the distance between the interference fringes in Young's double-slit experiment, using a green monochromatic light source?

MCQ

**A** decreasing the distance between the screen that receives the fringes and the two narrow slits

**B** increasing the distance between the source and the two narrow slits

**C** increasing the separating distance between the two narrow slits

**D** replacing the light source with a red monochromatic light source

**16** In Thomas Young's experiment, when the distance between the double-slit screen and the screen prepared to receive the fringes is doubled, then the distance between every two successive fringes of the same kind .....

MCQ

**A** doubles and the clarity of the fringes decreases

**B** doubles and the clarity of the fringes increases

**C** halves and the clarity of the fringes increases

**D** halves and the clarity of the fringes decreases

- 17 A convex lens whose focal length is 10 cm; an object is placed at a distance of 15 cm from it. The image formed is:

MCQ

- ☐ A virtual, magnified
- ☐ B real, magnified
- ☐ C real, diminished
- ☐ D real, equal to the object

- 18 Four students were allowed to choose among several sources of different wavelength, several double slits, and a suitable screen distance, in order to carry out the double-slit experiment. For which student will the fringes be the clearest (the most widely separated)?

MCQ

|   | the wavelength | the distance between the two coherent sources | the distance of the screen from the double slit |
|---|----------------|-----------------------------------------------|-------------------------------------------------|
| A | 500 nm         | 2 d                                           | 0.5 m                                           |
| B | 400 nm         | d                                             | 1 m                                             |
| C | 600 nm         | 3 d                                           | 1.5 m                                           |
| D | 450 nm         | 0.5 d                                         | 1.1 m                                           |

- ☐ A choice A
- ☐ B choice B
- ☐ C choice C
- ☐ D choice D

- 19 A student carries out an experiment to demonstrate the interference of light using the double-slit experiment as in the figure, but he found that the interference fringes observed on the screen are so close together that they cannot be told apart. **What changes could help the student to distinguish between the fringes? (two are enough)**

EXPLAIN

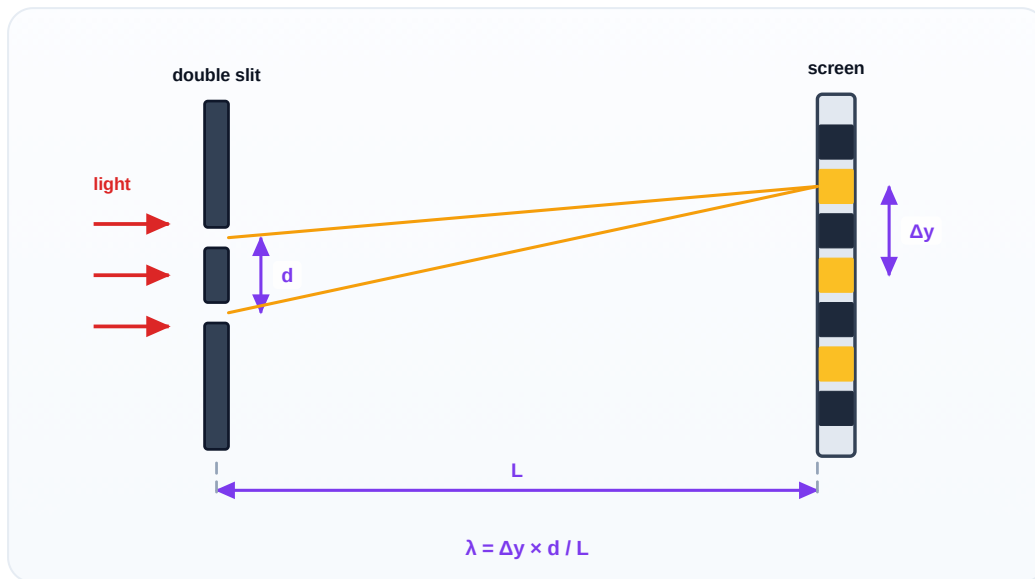


FIGURE 7 · THE FRINGES ARE TOO CLOSE TOGETHER

- 20 An object of length  $h$  is placed between a screen and a convex lens. When the position of the lens is changed, it is found that there are two positions that give a clear image on the screen. If the magnification at the first position is  $M_1 = 4$  and the magnification at the second position is  $M_2 = 0.25$ , **calculate the length of the first image and of the second image.**

PROBLEM

- 21 **How can you increase the distance between the interference fringes** in Young's double-slit experiment, using a green monochromatic light source?

EXPLAIN

## REVIEW

F

### Group B

- 22 An image is formed on a wall by a lens. If the distance between the lens and the wall is 30 cm and the magnification is  $M = 2$ , then **the type of the lens and the object distance** are:

MCQ

A concave,  $a = 15$  cm

B convex,  $a = 60$  cm

C convex,  $a = 15$  cm

D concave,  $a = 60$  cm

- 23 In an experiment on the interference of light by Young, the distance between the two narrow slits is 0.2 mm, the wavelength of the monochromatic light used is 600 nm, and the distance between the two narrow slits and the screen prepared to receive the fringes is 1 m. Then the distance between two successive fringes of the same kind equals .....

MCQ

A 3 mm

B 6 mm

C 30 mm

D 1 m

- 24 A convex lens gives, for an object placed in front of it, an erect image magnified 4 times. The object is then moved away from the lens through a distance of 6 cm, and the image becomes real, inverted and magnified 4 times as well. **Calculate the focal length of the lens.**

PROBLEM

- 25 A screen is placed at a distance of 90 cm from an object, and a convex lens of focal length 20 cm moves between the object and the screen. **Find all the values of the distance a between the lens and the object at which a clear image is formed on the screen.**

PROBLEM

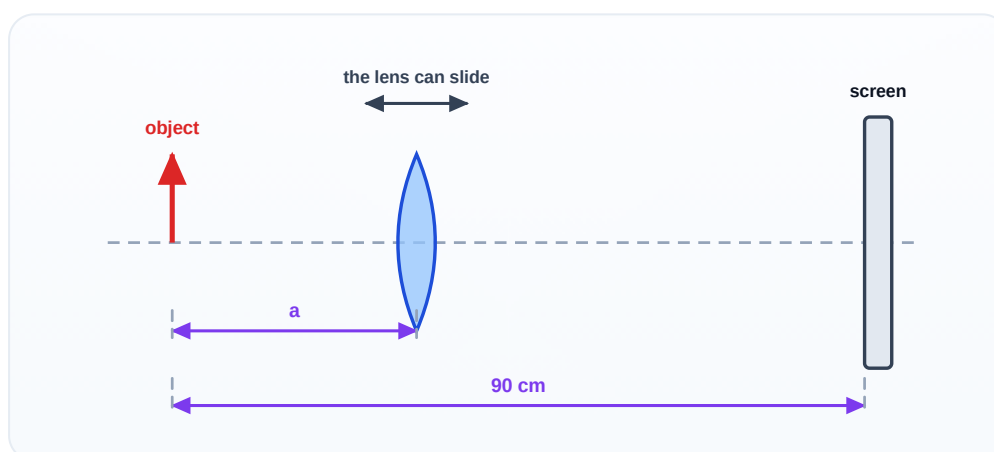


FIGURE 8 · THE LENS SLIDES BETWEEN OBJECT AND SCREEN

- 26 The drawing opposite shows the paths of two waves from the two slits  $S_1$  and  $S_2$  meeting at a point P. If the wavelength is  $\lambda$  and the distance  $S_2P$  exceeds  $S_1P$  by  $1.5\lambda$ , **what type of fringe is formed at the point P? And why?**

EXPLAIN

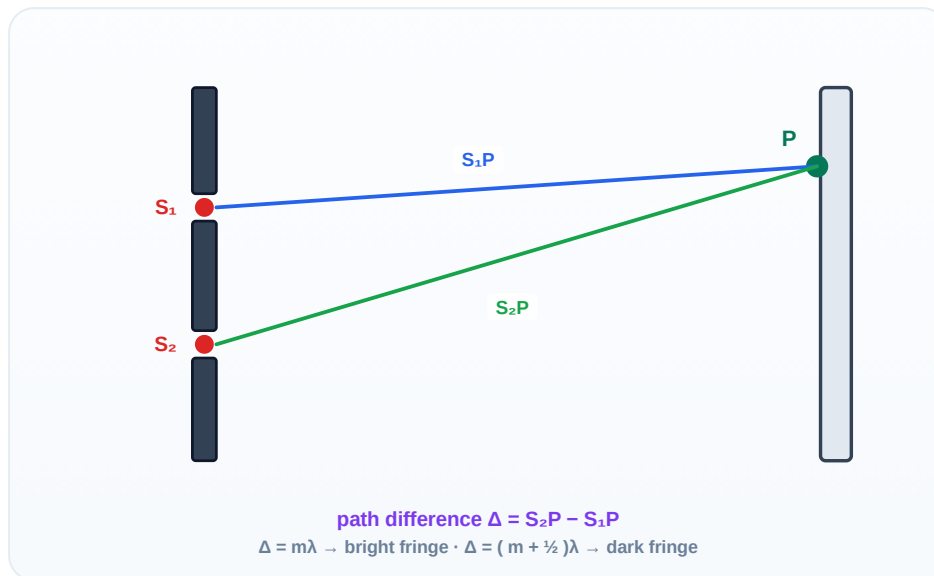


FIGURE 9 · THE TWO PATHS TO THE POINT P

## REVIEW

G

### Group C

- 27 When red light is used instead of blue light in Young's experiment, the distance between the fringes:

MCQ

- A** increases, because the wavelength of red is larger
- B** decreases, because the wavelength of red is smaller
- C** does not change
- D** the fringes disappear

- 28 A concave lens whose focal length is 20 cm; an object is placed at a distance of 20 cm from it. The image is:

MCQ

- A** at infinity
- B** virtual, erect, diminished
- C** real, equal to the object
- D** virtual, magnified

- 29 In the double-slit experiment, monochromatic light of wavelength  $6000 \text{ \AA}$  was used and fringes were formed on a screen at a distance ( $R$ ) from the double slit, the distance between every two successive bright fringes being  $\Delta y_1$ . If monochromatic light of wavelength  $4000 \text{ \AA}$  is then used and the distance between the double slit and the screen is doubled, the distance between every two successive bright fringes becoming  $\Delta y_2$ , **find the ratio  $\Delta y_1 / \Delta y_2$ .**

PROBLEM

**30** An object is placed beyond  $2F$  of a convex lens, then it moves with a uniform velocity  $v$  towards the focus  $F$ . **Prove that the speed of the image formed is variable, and that it increases as the object approaches the focus.**

PROBLEM

**31** **Give the scientific reason:** the central fringe in Young's double-slit experiment is always a bright fringe.

EXPLAIN

THE BACK OF THE BOOK

# Answer Key

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Every question, solved step by step

مفتاح الإجابة · الحل بالخطوات لكل سؤال



## Solutions — step by step

Every question is solved in full: the relation used, the substitution, then the result. Angles are given to one decimal place and  $g = 10 \text{ m/s}^2$  throughout.

### Part A — Classwork 1

- 1 resolving the eight vectors ·  $v_x = v \cos \theta$ ,  $v_y = v \sin \theta$  ( $\theta$  measured from the +x axis)

| Vector | Magnitude | Direction           | $v_x \text{ (m/s)}$ | $v_y \text{ (m/s)}$ |
|--------|-----------|---------------------|---------------------|---------------------|
| $V_1$  | 3 m/s     | along +x            | + 3.00              | 0                   |
| $V_2$  | 5 m/s     | along -x            | - 5.00              | 0                   |
| $V_3$  | 6 m/s     | along +y            | 0                   | + 6.00              |
| $V_4$  | 8 m/s     | along -y            | 0                   | - 8.00              |
| $V_5$  | 10 m/s    | 45° above +x        | + 7.07              | + 7.07              |
| $V_6$  | 6 m/s     | 60° above -x (120°) | - 3.00              | + 5.20              |
| $V_7$  | 6 m/s     | 60° below -x (240°) | - 3.00              | - 5.20              |
| $V_8$  | 10 m/s    | 20° from -y (290°)  | + 3.42              | - 9.40              |

A component is **positive** along +x (east) or +y (north) and **negative** along -x or -y. A vector lying on an axis has one component only.

- 2 resultant of each pair

- (a) the two vectors are along the same line, in opposite directions :

$$R = 8 - 6 = 2 \text{ m}, \text{ in the direction of the 8 m vector}$$

- (b) the two vectors are perpendicular :

$$R = \sqrt{6^2 + 8^2} = \sqrt{100} = 10 \text{ m}$$

$$\tan \alpha = 6 / 8 = 0.75 \rightarrow \alpha = 36.9^\circ \text{ measured from the 8 m vector}$$

- (c) the angle between them is  $30^\circ$  :

$$R = \sqrt{6^2 + 8^2 + 2 \times 6 \times 8 \times \cos 30^\circ}$$

$$= \sqrt{100 + 83.14} = \sqrt{183.14} = 13.53 \text{ m}$$

$$\tan \alpha = (8 \sin 30^\circ) / (6 + 8 \cos 30^\circ) = 4 / 12.93 = 0.309$$

$$\alpha = 17.2^\circ \text{ from the 6 m vector}$$

(a) 2 m · (b) 10 m at  $36.9^\circ$  · (c) 13.53 m at  $17.2^\circ$

### Part B — Home assignment 1

**3** aircraft climbing at  $60^\circ$

- a)  $v_x = v \cos \theta = 100 \times \cos 60^\circ = 100 \times 0.5 = 50 \text{ m/s}$   
b)  $v_y = v \sin \theta = 100 \times \sin 60^\circ = 100 \times 0.866 = 86.6 \text{ m/s}$   
c) the velocity is constant, so the vertical motion is uniform :  
 $h = v_y \times t = 86.6 \times 10 = 866 \text{ m}$

**$v_x = 50 \text{ m/s}$  ,  $v_y = 86.6 \text{ m/s}$  ,  $h = 866 \text{ m}$**

**4** magnitude and direction from the components

- 1)  $v = \sqrt{(v_x^2 + v_y^2)} = \sqrt{(3^2 + 4^2)} = \sqrt{25} = 5 \text{ m/s}$   
2)  $\tan \theta = v_y / v_x = 4 / 3 = 1.333 \rightarrow \theta = 53.1^\circ$

**$v = 5 \text{ m/s}$  at  $53.1^\circ$  above the horizontal**

**5** finding the angle of climb

the climb is at constant velocity, so the vertical motion is uniform :

$v_y = h / t = 500 / 10 = 50 \text{ m/s}$   
 $v_y = v \sin \theta \rightarrow \sin \theta = 50 / 100 = 0.5$   
 $\theta = 30^\circ$

**$\theta = 30^\circ$**

**6** resultant of each pair

- (a) perpendicular vectors :

$R = \sqrt{(6^2 + 8^2)} = 10 \text{ m}$   
 $\tan \alpha = 8 / 6 = 1.333 \rightarrow \alpha = 53.1^\circ$  from the 6 m vector

- (b) the angle between them is  $120^\circ$  :

$R = \sqrt{(6^2 + 8^2 + 2 \times 6 \times 8 \times \cos 120^\circ)}$   
 $= \sqrt{(100 - 48)} = \sqrt{52} = 7.21 \text{ m}$   
 $\tan \alpha = (8 \sin 120^\circ) / (6 + 8 \cos 120^\circ) = 6.93 / 2 = 3.464$   
 $\alpha = 73.9^\circ$  from the 6 m vector

**(a) 10 m at  $53.1^\circ$  · (b) 7.21 m at  $73.9^\circ$**

**7** initial speed from the maximum height

at the highest point the vertical component of the velocity = 0 :

$h = (v_0 \sin \theta)^2 / (2g)$   
 $45 = (v_0 \times \sin 30^\circ)^2 / (2 \times 10) = (0.5 v_0)^2 / 20$   
 $0.25 v_0^2 = 900 \rightarrow v_0^2 = 3600 \rightarrow v_0 = 60 \text{ m/s}$

**C) 60 m/s**

**Part C — Classwork 2**

### 8 the three cases of relative velocity

the general rule is a vector subtraction :  $v(B \text{ rel } A) = v(B) - v(A)$

First (same direction) :  $v(BA) = | v(B) - v(A) |$

Second (opposite directions) :  $v(BA) = v(B) + v(A)$

Third (perpendicular) :  $v(BA) = \sqrt{v(A)^2 + v(B)^2}$

In the second case one of the two velocities is negative, so subtracting it is the same as adding the two magnitudes. In the third case the subtraction is carried out with Pythagoras' theorem.

### 9 how A appears to an observer on B

$v(A \text{ rel } B) = v(A) - v(B)$

since  $v(A) > v(B)$  the result is positive, and it is smaller than  $v(A)$

The observer on B subtracts his own velocity from that of A, so only the **difference** between the two speeds is seen. This is why a car overtaking you on the motorway seems to crawl past.

**A appears to move in the same direction, but more slowly, at  $(v_A - v_B)$**

### 10 car A west, car B north

$v(A \text{ rel } B) = v(A) - v(B) = 60 \text{ (west)} + 80 \text{ (south)}$

the two components are perpendicular :

$|v(AB)| = \sqrt{60^2 + 80^2} = \sqrt{10000} = 100 \text{ m/s}$

$\tan \theta = 60 / 80 = 0.75 \rightarrow \theta = 36.9^\circ \text{ from the south direction}$

Reversing the velocity of B (north  $\rightarrow$  south) is what turns the subtraction into an ordinary vector addition.

**100 m/s ,  $53.1^\circ$  south of west**

## Part D — Home assignment 2

### 11 A twice as fast as B, opposite directions

take the direction of A as positive :  $v(A) = 2 v(B)$  ,  $v(B) = - v(B)$

$v(B \text{ rel } A) = v(B) - v(A) = - v(B) - 2 v(B) = - 3 v(B)$

The minus sign only tells us the direction: to an observer on A, body B rushes backwards at three times its own speed.

**speed of B relative to A =  $3 v_B (= 1.5 v_A)$  , opposite to the motion of A**

**12** the bus and the rain

for the passenger :  $v(\text{rain rel bus}) = v(\text{rain}) - v(\text{bus})$

so the apparent velocity has a horizontal component of 20 m/s pointing backwards, and a vertical component equal to the true speed of the rain.

a)  $\tan 40^\circ = 20 / v(\text{rain})$

$v(\text{rain}) = 20 / \tan 40^\circ = 20 / 0.839 = 23.8 \text{ m/s}$  (vertically downwards)

[ the apparent speed itself =  $20 / \sin 40^\circ = 31.1 \text{ m/s}$  ]

b) The passenger is a **moving observer**. His own velocity is subtracted from the velocity of every raindrop, which adds a backward horizontal component of 20 m/s to the vertical fall. The sum of the two is a slanted velocity, so the rain is seen falling at  $40^\circ$  to the vertical. A person standing still on the pavement sees the same rain falling vertically.

**$v(\text{rain}) = 23.8 \text{ m/s}$  vertically downwards**

**13** same direction

$v(AB) = v(A) - v(B)$  (plain subtraction, both velocities have the same sign)

**A)  $v_{AB} = v_A - v_B$**

**14** zero relative velocity

$v(A) - v(B) = 0 \rightarrow v(A) = v(B)$  in magnitude and direction

**B) equal speeds in the same direction**

**15** opposite directions

$v(1 \text{ rel } 2) = 30 - (-20) = 50 \text{ m/s}$

**C) 50 m/s**

**16** overtaking car

$v(A \text{ rel } B) = 40 - 25 = +15 \text{ m/s}$  (positive  $\rightarrow$  same direction)

**A) same direction at 15 m/s**

**17** bodies approaching

take right as positive :  $v(B) = -20$  ,  $v(A) = +30$

$v(B \text{ rel } A) = -20 - 30 = -50 \text{ m/s}$  (negative  $\rightarrow$  to the left)

**D) to the left at 50 m/s**

**18** perpendicular velocities

$v(B \text{ rel } A) = v(B) - v(A) = 40 \text{ (north)} + 30 \text{ (west)}$

$\tan \theta = 30 / 40 = 0.75 \rightarrow \theta = 36.9^\circ$  west of north , i.e.  $53.1^\circ$  north of west

**B)  $53.1^\circ$  north of west**

**19** the faster body seen from the slower one

$$v(A \text{ rel } B) = v(A) - v(B) > 0 \quad \text{and} \quad (v(A) - v(B)) < v(A)$$

**D) same direction, at a speed less than its actual speed**

**20** perpendicular velocities

$$v(B \text{ rel } A) = 40 \text{ (up)} + 30 \text{ (left)}$$

$$|v| = \sqrt{40^2 + 30^2} = \sqrt{2500} = 50 \text{ m/s}$$

**B) 50 m/s**

## Part E — Weekly assessment 1 · Group A

**21** very close speeds

$$v(A \text{ rel } B) = 20 - 21 = -1 \text{ m/s} \rightarrow \text{magnitude} = 1 \text{ m/s}$$

The minus sign means that A falls slowly behind B.

**D) 1 m/s**

**22** which pair gives the resultant

the given R points upwards and to the left, so it must have :

a component towards the west  $\rightarrow$  vector B directed west

a component towards the north  $\rightarrow$  vector A directed north

only diagram (a) has this pair; completing the parallelogram gives R.

**A) the pair in diagram (a)**

**23** which projectile rises higher

$$h(\text{max}) = (v \sin \theta)^2 / (2g)$$

the speed  $v$  is the same for both, so  $h$  depends on  $\sin \theta$  only,

and  $\sin \theta$  increases as  $\theta$  increases (up to  $90^\circ$ )

A larger angle puts a larger share of the same speed into the **vertical** component, so the body rises higher (although its horizontal range may be smaller).

**Projectile A , because it is launched at the larger angle**

**24** steering the ferry

the ferry must be aimed **upstream** at an angle  $\alpha$  to the line joining the two points, where

$$v(\text{boat}) \sin \alpha = v(\text{river}) \rightarrow \sin \alpha = v(\text{river}) / v(\text{boat})$$

then the upstream component exactly cancels the current, and what is left is

$$v(\text{resultant}) = \sqrt{v(\text{boat})^2 - v(\text{river})^2} \quad \text{straight across the river}$$

**aim upstream at  $\alpha = \sin^{-1} (v_{\text{river}} / v_{\text{boat}})$**

**25** car A east, car B south

$$v(A \text{ rel } B) = v(A) - v(B) = 30 \text{ (east)} + 40 \text{ (north)}$$

$$|v(AB)| = \sqrt{30^2 + 40^2} = 50 \text{ m/s}$$

$$\tan \theta = 40 / 30 = 1.333 \rightarrow \theta = 53.1^\circ$$

**50 m/s , 53.1° north of east**

## Group B

**26** angle from the maximum height

$$h = (v_0 \sin \theta)^2 / (2g)$$

$$20 = (40 \sin \theta)^2 / 20 \rightarrow (40 \sin \theta)^2 = 400$$

$$40 \sin \theta = 20 \rightarrow \sin \theta = 0.5 \rightarrow \theta = 30^\circ$$

**B) 30°**

**27** equal components

$$1) v = \sqrt{5^2 + 5^2} = \sqrt{50} = 7.07 \text{ m/s}$$

$$2) \tan \theta = 5 / 5 = 1 \rightarrow \theta = 45^\circ$$

Whenever the two components are equal the angle is  $45^\circ$ , whatever their size.

**v = 7.07 m/s at 45°**

**28** A is half as fast as B

$$v(A \text{ rel } B) = v(A) - v(B) = 0.5 v(B) - v(B) = -0.5 v(B)$$

To an observer on the faster body B, the slower body A is falling behind, so it seems to travel in the opposite direction.

**A appears to move backwards at half the speed of B (= its own speed)**

**29** equal speeds, same direction

$$v(A \text{ rel } B) = v - v = 0$$

Each body appears at rest to an observer on the other one — exactly like two cars keeping the same speed side by side on the road.

**the relative velocity is zero**

**30** two equal and opposite vectors

$$R = \sqrt{A^2 + A^2 + 2 A^2 \cos 180^\circ} = \sqrt{2A^2 - 2A^2} = 0$$

Equal magnitudes with opposite directions cancel completely, so the resultant vanishes for any two such vectors acting at the same point — no special condition is needed.

**the resultant is always zero**

## Group C

**31** equal speeds, opposite directions

$$v(A \text{ rel } B) = v - (-v) = 2v$$

**C) twice the speed of one of them**

**32** slower body seen from the faster one

$$v(B \text{ rel } A) = 20 - 30 = -10 \text{ m/s} \quad (\text{negative} \rightarrow \text{opposite direction})$$

**B) opposite direction at 10 m/s**

**33** components of the aircraft velocity

$$\text{a) } v_x = 80 \cos 30^\circ = 80 \times 0.866 = 69.28 \text{ m/s}$$

$$\text{b) } v_y = 80 \sin 30^\circ = 80 \times 0.5 = 40 \text{ m/s}$$

**$v_x = 69.28 \text{ m/s}$  ,  $v_y = 40 \text{ m/s}$**

**34** a faster ferry, same current

$$\text{time of crossing : } t = \text{width} / v(\text{boat})$$

$$\text{drift along the current : } d = v(\text{river}) \times t = v(\text{river}) \times \text{width} / v(\text{boat})$$

$v(\text{river})$  and the width do not change, while  $v(\text{boat})$  increases

The faster the ferry crosses, the less time the current has to carry it downstream, so the landing point moves closer to the point directly opposite the start.

**the drift decreases**

**35** raising the height reached in the same time

$$h = (v \sin \theta) t - \frac{1}{2} g t^2$$

for a fixed time  $t$ ,  $h$  grows only if the vertical component  $(v \sin \theta)$  grows :

Way 1 : increase the speed of projection  $v$  , keeping the angle  $\theta$  the same

Way 2 : increase the angle of projection  $\theta$  towards  $90^\circ$  , keeping  $v$  the same

**increase  $v$  , or increase  $\theta$  (both increase  $v \sin \theta$ )**



## Solutions — step by step

Every question is solved in full. Remember the golden rule: **the vertical motion decides the time, the horizontal motion decides the range.**

### Part A — Classwork 1

#### 1 reading the path of a horizontal projectile

- a) no force acts along the horizontal direction  $\rightarrow a(x) = 0$   
so  $v(x)$  is the same at 1, 2 and 3 (and at every point of the path)
- b) the vertical motion is a free fall :  $v(y) = g t$   
 $t$  is largest at point 3, so  $v(y)$  is greatest at point 3
- c)  $AB = v \times t$  (uniform horizontal motion)  
d)  $CB = \frac{1}{2} g t^2$  (free fall from rest, vertically)
- e) 1)  $t = \sqrt{(2h / g)}$  does not contain  $v \rightarrow$  the time does not change  
2)  $x = v t$ , with  $t$  unchanged  $\rightarrow$  the range is doubled

This is the heart of the lesson: the two motions are completely independent, and only the height decides how long the flight lasts.

**a) equal at 1, 2, 3 · b) point 3 · e) time unchanged, range doubled**

#### 2 projectile from 19.6 m at 14.7 m/s ( $g = 9.8 \text{ m/s}^2$ )

- a)  $h = \frac{1}{2} g t^2 \rightarrow t = \sqrt{(2h / g)} = \sqrt{(2 \times 19.6 / 9.8)} = \sqrt{4} = 2 \text{ s}$
- b)  $x = v t = 14.7 \times 2 = 29.4 \text{ m}$
- c)  $v(y) = g t = 9.8 \times 2 = 19.6 \text{ m/s}$   
 $v = \sqrt{(v^2 + v(y)^2)} = \sqrt{(14.7^2 + 19.6^2)} = \sqrt{600.25} = 24.5 \text{ m/s}$

**a) 2 s · b) 29.4 m · c) 24.5 m/s**

### Part B — Home assignment 1

#### 3 the horizontal motion

$a(x) = 0 \rightarrow$  the horizontal velocity never changes

**B) a constant velocity**

#### 4 doubling the launch speed

$t = \sqrt{(2h / g)}$  is independent of  $v \rightarrow t$  stays the same  
 $x = v t \rightarrow$  doubling  $v$  doubles  $x$

**C) doubling the horizontal range**

**5** height and range from the time of flight

$$h = \frac{1}{2} g t^2 = \frac{1}{2} \times 10 \times (2)^2 = 20 \text{ m}$$

$$x = v t = 4 \times 2 = 8 \text{ m}$$

$$h = 20 \text{ m}, x = 8 \text{ m}$$

**6** raising the point of projection

$$t = \sqrt{(2h / g)} \rightarrow \text{a larger } h \text{ gives a larger } t$$

$$x = v t, \text{ with } v \text{ unchanged} \rightarrow x \text{ increases}$$

The body keeps the same horizontal speed, but it now stays in the air for a longer time, so it travels further before landing.

**the horizontal range increases**

**7** projectile from a 45 m building

$$1) \quad t = \sqrt{(2h / g)} = \sqrt{(2 \times 45 / 10)} = \sqrt{9} = 3 \text{ s}$$

$$2) \quad x = v t = 20 \times 3 = 60 \text{ m}$$

$$t = 3 \text{ s}, x = 60 \text{ m}$$

## Part C — Classwork 2

**8** two projectiles from 20 m and 80 m

$$\text{first : } t_1 = \sqrt{(2 \times 20 / 10)} = \sqrt{4} = 2 \text{ s} \quad x_1 = 10 \times 2 = 20 \text{ m}$$

$$\text{second : } t_2 = \sqrt{(2 \times 80 / 10)} = \sqrt{16} = 4 \text{ s} \quad x_2 = 20 \times 4 = 80 \text{ m}$$

$$1) \quad t_1 : t_2 = 2 : 4 = 1 : 2$$

$$2) \quad x_1 : x_2 = 20 : 80 = 1 : 4$$

$$1) 1 : 2 \cdot 2) 1 : 4$$

**9** same height, greater speed

$$t = \sqrt{(2h / g)} \text{ does not contain } v \rightarrow \text{the time of flight does not change}$$

$$x = v t \rightarrow \text{the range increases in the same ratio as } v$$

Two bodies leaving the same table with different speeds hit the floor at the same instant — only the landing points are different.

**the time stays the same, the range increases**

## Part D — Home assignment 2

**10** the acceleration of a horizontal projectile

$$\text{horizontally : no force} \rightarrow a(x) = 0$$

$$\text{vertically : the weight only} \rightarrow a(y) = g$$

**A) zero acceleration during the horizontal motion**

**11** increasing the launch speed

$t = \sqrt{(2h / g)}$  is independent of  $v$

**B) the time of flight stays constant**

**12** the two projectiles  $h, v$  and  $4h, 2v$

$$1) \quad t_1 = \sqrt{(2h / g)} \quad t_2 = \sqrt{(2 \times 4h / g)} = 2 \sqrt{(2h / g)}$$

$$t_1 : t_2 = 1 : 2$$

$$2) \quad x_1 = v t_1 \quad x_2 = 2v \times t_2 = 2v \times 2 t_1 = 4 v t_1$$

$$x_1 : x_2 = 1 : 4$$

The time follows the **square root** of the height, so four times the height only doubles the time; the range then gains a second factor of 2 from the doubled speed.

**1) 1 : 2 · 2) 1 : 4**

**13** the billiard ball

$$t = \sqrt{(2h / g)} = \sqrt{(2 \times 0.60 / 10)} = \sqrt{0.12} = 0.35 \text{ s}$$

$$x = v t = 2.4 \times 0.35 = 0.83 \text{ m}$$

**t = 0.35 s , x = 0.83 m**

## Part E — Weekly assessment 2 · Group A

**14** from the tenth floor instead of the fourth

the tenth floor is higher  $\rightarrow h$  increases

$t = \sqrt{(2h / g)} \rightarrow t$  increases

$x = v t$ , with  $v$  unchanged  $\rightarrow x$  increases as well

**C) both the time and the range increase**

**15** the helicopter and the man

the package needs  $t = \sqrt{(2h / g)}$  to fall

in that time it travels horizontally  $x = v t = v \sqrt{(2h / g)}$

at the instant of release the helicopter is exactly above the release point, so the distance to the man is the hypotenuse of a right angle :

$$d = \sqrt{(x^2 + h^2)} = \sqrt{(v^2 \times 2h / g + h^2)} = \sqrt{(2v^2h / g + h^2)}$$

**B)  $\sqrt{(2v^2h / g + h^2)}$**

**16** height and range from  $t = 3 \text{ s}$

$$h = \frac{1}{2} g t^2 = \frac{1}{2} \times 10 \times 9 = 45 \text{ m}$$

$$x = v t = 6 \times 3 = 18 \text{ m}$$

**h = 45 m , x = 18 m**

**17** same height, smaller speed

$$t = \sqrt{(2h / g)} \rightarrow \text{unchanged}$$

$$x = v t \rightarrow \text{smaller } v \text{ gives a smaller } x$$

**the time stays the same, the range decreases**

**18** the billiard ball from 0.75 m

$$t = \sqrt{(2 \times 0.75 / 10)} = \sqrt{0.15} = 0.387 \text{ s}$$

$$x = v t = 1.5 \times 0.387 = 0.58 \text{ m}$$

**x = 0.58 m (t = 0.39 s)**

## Group B

**19** doubling the speed

$$t = \sqrt{(2h / g)} \text{ contains } h \text{ and } g \text{ only}$$

**C) it does not change**

**20** the vertical acceleration

$$\text{the only force acting is the weight, directed downwards} \rightarrow a(y) = g = 10 \text{ m/s}^2$$

**C) an acceleration equal to g during the vertical motion**

**21** raising the height from h to 4h

$$t_1 = \sqrt{(2h / g)} \quad t_2 = \sqrt{(2 \times 4h / g)} = 2 \sqrt{(2h / g)} = 2 t_1$$

$$x_1 = v t_1 \quad x_2 = v t_2 = 2 v t_1 = 2 x_1$$

Four times the height does not give four times the time, because the time depends on the **square root** of the height.

**both the time and the range are doubled**

**22** the projected coin and the dropped coin

vertically the two coins start with  $v(y) = 0$  and have the same  $a = g$  and the same h

$$h = \frac{1}{2} g t^2 \text{ gives the same } t \text{ for both}$$

the horizontal velocity of A does not affect its vertical motion

This is the classic experiment that proves the independence of the two motions: A lands far away from the table, B lands just below it, but both land at the same moment.

**the two coins reach the ground at the same instant**

**23** projectile from 30 m at 8 m/s

$$\text{a) } t = \sqrt{(2h / g)} = \sqrt{(2 \times 30 / 10)} = \sqrt{6} = 2.45 \text{ s}$$

$$\text{b) } x = v t = 8 \times 2.45 = 19.6 \text{ m}$$

**a) 2.45 s · b) 19.6 m**

## Group C

**24** halving the launch speed

$t$  is independent of  $v \rightarrow$  unchanged

$x = v t \rightarrow$  halving  $v$  halves  $x$

**C) halving the horizontal range**

**25** the velocity during the fall

$v(x) = v = \text{constant}$  (no horizontal force)

$v(y) = g t \rightarrow$  grows steadily with time

**D) its vertical velocity increases with time**

**26** finding the launch speed from the range

$t = \sqrt{(2h / g)} = \sqrt{(2 \times 22 / 10)} = \sqrt{4.4} = 2.10 \text{ s}$

$x = v t \rightarrow v = x / t = 35 / 2.10 = 16.7 \text{ m/s}$

**$v = 16.7 \text{ m/s}$**

**27** projecting from a greater height

$t = \sqrt{(2h / g)} \rightarrow$  a larger  $h$  gives a larger  $t$

$x = v t$ , with  $v$  unchanged  $\rightarrow x$  increases too

**both the time of flight and the horizontal range increase**

**28** range and height from  $t = 3 \text{ s}$

$x = v t = 3 \times 3 = 9 \text{ m}$

$h = \frac{1}{2} g t^2 = \frac{1}{2} \times 10 \times 9 = 45 \text{ m}$

**$x = 9 \text{ m}$ ,  $h = 45 \text{ m}$**



## Solutions — step by step

The whole lesson comes from two facts: the horizontal velocity never changes, and the vertical motion is a body thrown straight up with an initial speed  $v_0 \sin \theta$ .

### Part A — Classwork 1

**1** the relations that describe the whole path

- a)  $v(\theta x) = v_0 \cos \theta$  (stays the same all the way)  
 b)  $v(\theta y) = v_0 \sin \theta$

at position (1) , after a time  $t_1$  :

- c)  $v(x1) = v_0 \cos \theta$  (no horizontal force)  
 d)  $v(y1) = v_0 \sin \theta - g t_1$   
 e)  $v_1 = \sqrt{v(x1)^2 + v(y1)^2}$   
 f)  $X = (v_0 \cos \theta) t_1$   
 g)  $Y = (v_0 \sin \theta) t_1 - \frac{1}{2} g t_1^2$

- h)  $H = (v_0 \sin \theta)^2 / (2g) = v_0^2 \sin^2 \theta / (2g)$   
 i)  $R = v_0^2 \sin 2\theta / g$

at position (2) , the highest point :

- j)  $v(x2) = v_0 \cos \theta$  and  $v(y2) = 0$

At the highest point only the vertical component vanishes; the body keeps moving horizontally, which is why its speed there is  $v_0 \cos \theta$  and not zero.

### Part B — Home assignment 1

**2** the horizontal motion

$a(x) = 0 \rightarrow v(x)$  never changes

**B) a constant velocity**

**3** the horizontal acceleration

gravity acts vertically only, so it has no horizontal component

**A) zero**

**4** ball projected at 15 m/s ,  $60^\circ$  ( $g = 10$ )

$$v(\theta x) = 15 \cos 60^\circ = 15 \times 0.5 = 7.5 \text{ m/s}$$

$$v(\theta y) = 15 \sin 60^\circ = 15 \times 0.866 = 12.99 \text{ m/s}$$

a)  $t(\text{up}) = v(\theta y) / g = 12.99 / 10 = 1.3 \text{ s}$

b)  $T = 2 t(\text{up}) = 2.6 \text{ s}$

c)  $R = v(\theta x) \times T = 7.5 \times 2.6 = 19.5 \text{ m}$

d)  $H = v(\theta y)^2 / (2g) = 168.75 / 20 = 8.44 \text{ m}$

**a) 1.3 s · b) 2.6 s · c) 19.5 m · d) 8.44 m**

## Part C — Classwork 2

**5** doubling down on the time of flight

$$T = 2 v_0 \sin \theta / g \rightarrow T \text{ grows when } (v_0 \sin \theta) \text{ grows}$$

Way 1 : increase the speed of projection  $v_0$  , keeping  $\theta = 60^\circ$

Way 2 : increase the angle  $\theta$  towards  $90^\circ$  , keeping  $v_0$  the same

**increase  $v_0$  , or increase  $\theta$**

**6** increasing the maximum height

$$H = v_0^2 \sin^2 \theta / (2g)$$

Way 1 : increase  $v_0$  ( $H$  grows with the square of the speed)

Way 2 : increase  $\theta$  towards  $90^\circ$

**increase  $v_0$  , or increase  $\theta$**

**7** increasing the horizontal range

$$R = v_0^2 \sin 2\theta / g$$

Way 1 : increase  $v_0$

Way 2 : bring the angle closer to  $45^\circ$  (here : decrease  $\theta$  from  $60^\circ$  towards  $45^\circ$ ),  
because  $\sin 2\theta$  is a maximum at  $\theta = 45^\circ$

**increase  $v_0$  , or move  $\theta$  towards  $45^\circ$**

**8** the angle of maximum range

$$R = v_0^2 \sin 2\theta / g \text{ is a maximum when } \sin 2\theta = 1$$

$$2\theta = 90^\circ \rightarrow \theta = 45^\circ$$

At  $45^\circ$  the speed is shared equally between the horizontal and the vertical directions, which is the best compromise between staying long in the air and moving fast forward.

**$\theta = 45^\circ$**

**9** the angle that gives the same range

two angles give the same range when they are **complementary** :

$$\sin 2(90^\circ - \theta) = \sin (180^\circ - 2\theta) = \sin 2\theta$$

$$90^\circ - 60^\circ = 30^\circ$$

The  $30^\circ$  throw is flatter and faster, the  $60^\circ$  throw is higher and slower, and the two effects cancel exactly in the range.

$$\theta = 30^\circ$$

## Part D — Home assignment 2

**10** the upward vertical motion

$v(y) = v_0 \sin \theta - g t \rightarrow$  the vertical velocity decreases steadily  
while the acceleration stays constant ( $= g$ )

**B) a decreasing velocity**

**11** at the maximum height

the body stops rising, so  $v(y) = 0$

**A) zero**

**12** ball projected at 20 m/s ,  $60^\circ$  ( $g = 9.8$ )

$$v(0x) = 20 \cos 60^\circ = 10 \text{ m/s}$$

$$v(0y) = 20 \sin 60^\circ = 17.32 \text{ m/s}$$

$$\text{a) } t(\text{up}) = 17.32 / 9.8 = 1.77 \text{ s}$$

$$\text{b) } T = 2 \times 1.77 = 3.54 \text{ s}$$

$$\text{c) } R = 10 \times 3.54 = 35.4 \text{ m}$$

$$\text{d) } H = 17.32^2 / (2 \times 9.8) = 300 / 19.6 = 15.3 \text{ m}$$

**a) 1.77 s · b) 3.54 s · c) 35.4 m · d) 15.3 m**

## Part E — Weekly assessment 3 · Group A

**13** the speed at the top

at the highest point  $v(y) = 0$  , so

$$v = v(x) = v_0 \cos \theta$$

**D) the horizontal component of the initial velocity**

**14** the complementary angle

$$90^\circ - 30^\circ = 60^\circ \text{ gives the same range}$$

**C)  $60^\circ$**

**15** doubling the time of flight

$T = 2 v_0 \sin \theta / g$ , with  $\theta$  fixed at  $60^\circ$  :  $T$  is proportional to  $v_0$   
so the speed of projection must be doubled ( $v_0 \rightarrow 2 v_0$ )

Raising the angle cannot do it here:  $T$  would need  $\sin \theta$  to double, and  $2 \sin 60^\circ$  is greater than 1, which is impossible for any angle.

**double the initial speed of projection**

**16** an angle measured from the vertical

the angle with the horizontal =  $90^\circ - 60^\circ = 30^\circ$

$$R = v_0^2 \sin 2\theta / g = (30)^2 \times \sin 60^\circ / 10$$

$$R = 900 \times 0.866 / 10 = 77.9 \text{ m}$$

Always convert an angle given with the vertical into an angle with the horizontal before using the projectile relations.

**R = 77.9 m**

**17** projected from the top of a building

$$v(0x) = 15 \cos 30^\circ = 12.99 \text{ m/s}$$

$$v(0y) = 15 \sin 30^\circ = 7.5 \text{ m/s}$$

taking upwards as positive, the final vertical displacement is  $-50 \text{ m}$  :

$$-50 = 7.5 t - 5 t^2$$

$$5 t^2 - 7.5 t - 50 = 0 \quad \div 5 \quad \rightarrow \quad t^2 - 1.5 t - 10 = 0$$

$$t = [ 1.5 + \sqrt{(2.25 + 40)} ] / 2 = (1.5 + 6.5) / 2 = 4 \text{ s}$$

$T = 2 v_0 \sin \theta / g$  cannot be used here, because the body does not return to the level it started from.

**T = 4 s**

## Group B

**18** the vertical motion

$v(y)$  changes every second, while  $a = g$  is constant in size and direction

**B) a varying velocity and a constant acceleration**

**19** when the height equals the range

$$H = R$$

$$v_0^2 \sin^2 \theta / (2g) = v_0^2 \sin 2\theta / g = 2 v_0^2 \sin \theta \cos \theta / g$$

$$\sin^2 \theta = 4 \sin \theta \cos \theta \quad \rightarrow \quad \tan \theta = 4$$

$$\theta = 76^\circ$$

**C) 76°**

**20** doubling the maximum height

$H = v_0^2 \sin^2 \theta / (2g)$ , with  $\theta$  fixed :  $H$  is proportional to  $v_0^2$   
to double  $H$  :  $v_0^2$  must double  $\rightarrow v_0 \times \sqrt{2}$  ( about  $1.41 v_0$  )

Doubling the speed itself would make the height **four** times as large, not twice.

**multiply the speed of projection by  $\sqrt{2}$**

**21** time of flight of the football

$v(0y) = 25 \sin 60^\circ = 21.65 \text{ m/s}$

$T = 2 v(0y) / g = 2 \times 21.65 / 10 = 4.33 \text{ s}$

**T = 4.33 s**

**22** the maximum height above the ground

$v(0y) = 15 \sin 30^\circ = 7.5 \text{ m/s}$

height risen above the roof :  $H = v(0y)^2 / (2g) = 56.25 / 20 = 2.81 \text{ m}$

height above the ground =  $50 + 2.81 = 52.81 \text{ m}$

The relation for  $H$  always measures the rise **above the point of projection**, so the height of the building must be added afterwards.

**h(max) = 52.81 m above the ground**

## Group C

**23** the vertical velocity while rising

$v(y) = v_0 \sin \theta - g t$  keeps falling until it reaches zero at the top

**B) it decreases gradually until it becomes zero at the highest point**

**24** the angle of greatest range

$R = v_0^2 \sin 2\theta / g$  is a maximum when  $2\theta = 90^\circ$

**B)  $45^\circ$**

**25** doubling the horizontal range

$R = v_0^2 \sin 2\theta / g$ , with  $\theta$  fixed at  $60^\circ$  :  $R$  is proportional to  $v_0^2$   
to double  $R$  :  $v_0^2$  must double  $\rightarrow v_0 \times \sqrt{2}$

Changing the angle alone cannot double it: the largest possible value of  $\sin 2\theta$  is 1, while  $\sin 120^\circ$  is already 0.87, so the angle can add at most about 15 %.

**multiply the speed of projection by  $\sqrt{2}$**

**26** kicked at  $30^\circ$  to the vertical

the angle with the horizontal =  $90^\circ - 30^\circ = 60^\circ$

$$v(0y) = 20 \sin 60^\circ = 17.32 \text{ m/s}$$

$$H = v(0y)^2 / (2g) = 300 / 20 = 15 \text{ m}$$

$$H = 15 \text{ m}$$

**27** the range from the top of the building

$$v(0x) = 15 \cos 30^\circ = 12.99 \text{ m/s}$$

the total time of flight was found in question 17 :  $t = 4 \text{ s}$

$$x = v(0x) \times t = 12.99 \times 4 = 51.96 \text{ m}$$

$$x \approx 52 \text{ m}$$



## Solutions — step by step

Remember the two habits that save marks here: always use the **perpendicular** distance in a moment, and always resolve an inclined force before writing an equilibrium equation.

### Part A — Lesson 1-4 · Classwork 1

**1** the basic relation

moment = force  $\times$  perpendicular distance       $M = F d$

**B)  $M = F d$**

**2** what a moment measures

a force can move a body (translation) or turn it about a point (rotation);  
the moment measures the second effect

**B) the turning effect of the force about a point**

**3** the unit

$M = F \times d = \text{newton} \times \text{metre} = \text{N} \cdot \text{m}$

It is never written as a joule, even though a joule is also a newton  $\times$  a metre; work and moment are different physical quantities.

**C) newton  $\cdot$  metre**

**4** the law of the lever

formulated by Archimedes in antiquity

**B) Archimedes**

**5** a perpendicular force

the force is perpendicular to the arm, so the arm itself is the moment arm :

$M = F \times L = 5.0 \times 2.0 = 10 \text{ N} \cdot \text{m}$

**$M = 10 \text{ N} \cdot \text{m}$**

### Part B — Lesson 1-4 · Home assignment 1

**6** doubling the arm

$M = F d$  , with  $F$  constant  $\rightarrow M$  is proportional to  $d$

**B) it is doubled**

**7** where to push a door

pushing at the handle gives the largest perpendicular distance from the hinge, so it gives the largest moment

**B) the handle, far from the hinge**

**8** finding the arm

$$M = F d \rightarrow d = M / F = 12 / 4.0 = 3.0 \text{ m}$$

**A) 3.0 m**

**9** moment about O

$$M = F \times d = 3.0 \times 20 = 60 \text{ N}\cdot\text{m}$$

**M = 60 N·m**

## Part C — Lesson 1-5 · Classwork 2

**10** the meaning of equilibrium

equilibrium means zero resultant force, so the velocity does not change : the body is either at rest or moving uniformly in a straight line

**B) it stays at rest or moves with a constant velocity**

**11** the mathematical condition

$$\Sigma F = 0 \quad (\text{the vector sum of all the forces is zero})$$

**B)  $\Sigma F = 0$**

**12** how the condition is applied

a vector equation is equivalent to one equation for each axis :

$$\Sigma F(x) = 0 \quad \text{and} \quad \Sigma F(y) = 0$$

**B) on both axes x and y**

**13** the sagging wire

at the lowest point, the two tensions carry the weight between them :

$$2 T \sin \alpha = W \quad (\alpha = \text{the angle between the wire and the horizontal})$$

$$T = W / (2 \sin \alpha)$$

more sag  $\rightarrow \alpha$  becomes larger  $\rightarrow \sin \alpha$  becomes larger  $\rightarrow T$  becomes smaller  
measured from the pole (the vertical), the angle  $\theta$  becomes smaller.

This is why a clothes line that is pulled almost perfectly horizontal snaps so easily: as the angle with the horizontal goes to zero, the tension needed to carry the same weight grows without limit.

**a bigger sag makes  $\theta$  smaller and reduces the tension T**

## Part D — Lesson 1-5 · Home assignment 2

**14** the lamp on a cable and a rope

the lamp is in equilibrium, so for the vertical axis :

$$\Sigma F(y) = 0 \rightarrow T \cos 25^\circ - W = 0$$

the horizontal rope is horizontal, so it adds nothing to the vertical equation

**B) the weight of the lamp**

**15** the weight of the lamp

$$W = m g = 0.6 \times 9.8 = 5.88 \text{ N}$$

The mass in kilograms is not a weight; it must be multiplied by g to give a force in newtons.

**D) 5.88 N**

**16** raising the angle of the cables

$$2 T \sin \alpha = W \rightarrow T = W / ( 2 \sin \alpha )$$

$\alpha$  increases  $\rightarrow \sin \alpha$  increases  $\rightarrow T$  decreases

**B) decreases**

**17** which component carries the weight

$$\text{vertical axis : } \Sigma F(y) = 0 \rightarrow T \sin \theta = W$$

$$\text{horizontal axis : } \Sigma F(x) = 0 \rightarrow T \cos \theta \text{ is balanced by another horizontal force}$$

The horizontal component has no projection on the vertical axis at all, so it simply does not appear in the vertical equation — it is not "neglected", it is balanced separately by the other horizontal force.

**the vertical component  $T \sin \theta$  balances the weight**

## Part E — Weekly assessment 3 · Group A

**18** doubling the force

$$M = F d, \text{ with } d \text{ constant} \rightarrow M \text{ is proportional to } F$$

**A) it is doubled**

**19** the length of the spanner

$M = F d$  : a longer spanner gives a larger d, so the same force produces a larger moment

**B) longer**

**20** perpendicular force against an inclined force

$\theta = 90^\circ$  :  $M = F L$  (  $\sin 90^\circ = 1$  )  $\rightarrow$  the largest possible moment

$\theta < 90^\circ$  :  $M = F L \sin \theta$  (  $\sin \theta < 1$  )  $\rightarrow$  a smaller moment

Only the component of the force perpendicular to the arm turns the body; the component along the arm pulls or pushes it and has no moment.

**the perpendicular force always gives the larger moment**

**21** three ways to increase the moment

$M = F L \sin \theta$  , with  $F$  fixed :

- 1) use a longer arm  $L$
- 2) change the direction of the force so that  $\theta = 90^\circ$  (  $\sin \theta = 1$  )
- 3) apply the force at the point farthest from the axis of rotation,  
which is the same as increasing the perpendicular distance  $d = L \sin \theta$

**longer arm ,  $\theta = 90^\circ$  , or a point farther from the axis**

## Group B

**22** an inclined force on a short arm

$$M = F L \sin \theta = 4.0 \times 0.30 \times \sin 45^\circ$$

$$M = 4.0 \times 0.30 \times 0.71 = 0.85 \text{ N}\cdot\text{m}$$

The common trap here is to answer  $1.2 \text{ N}\cdot\text{m}$ , which is  $F \times L$  without  $\sin \theta$  — that value only applies when the force is perpendicular to the arm.

**A)  $0.85 \text{ N}\cdot\text{m}$**

**23** the balanced ruler

the two moments about the point of support are equal :

$$(1 \text{ coin}) \times 15 = (n \text{ coins}) \times 5$$

$$15 = 5n \rightarrow n = 3$$

**C) 3 coins**

**24** pushing a door near and far from the hinge

far from the hinge : large moment arm  $\rightarrow$  large moment  $\rightarrow$  the door turns easily

close to the hinge : small moment arm  $\rightarrow$  small moment  $\rightarrow$  the door hardly turns

**the same force gives a much larger moment when it is applied far from the hinge**

**25** why the moment changes with the point

$M = F \times d$ , and  $d$  is the perpendicular distance from the **chosen point** to the line of action of the force.

A and B are at different distances from that line, so each gives a different  $d$ , and a point lying on the line of action itself gives  $d = 0$  and  $M = 0$ .

The force never changes; what changes is the geometry between the force and the point you choose to turn about.

**because each point has its own perpendicular distance  $d$**

## Group C

**26** a longer spanner for the same moment

$M = F d$ , with  $M$  fixed  $\rightarrow F = M / d$

$d$  larger  $\rightarrow F$  smaller

**B) smaller**

**27** inclined force on a lever

$M = F L \sin \theta = 6.0 \times 0.50 \times \sin 30^\circ$

$M = 6.0 \times 0.50 \times 0.5 = 1.5 \text{ N}\cdot\text{m}$

**A) 1.5 N·m**

**28** the same force about two different points

1) about A :  $d(A) = 4.0 \sin 60^\circ = 4.0 \times 0.87 = 3.48 \text{ m}$

$M(A) = F \times d(A) = 4.0 \times 3.48 = 13.9 \text{ N}\cdot\text{m}$

the force turns the line anticlockwise about A

2) about B :  $d(B) = 3.0 \sin 60^\circ = 2.61 \text{ m}$

$M(B) = 4.0 \times 2.61 = 10.4 \text{ N}\cdot\text{m}$ , anticlockwise as well

**1) a) 13.9 N·m anticlockwise · 2) c) 10.4 N·m anticlockwise**

**29** the sign of a moment

positive moment  $\rightarrow$  anticlockwise rotation

negative moment  $\rightarrow$  clockwise rotation

The sign lets moments be added like ordinary numbers: a positive and a negative moment of the same size cancel each other exactly.

**the sign is only a label for the sense of rotation**

**30** sliding the force towards O

$M = F \times d$ , with  $F$  fixed in size and direction

as the point of application slides towards O, the perpendicular distance  $d$  gets smaller, so the moment decreases in the same ratio, until  $d = 0$  at O and the moment vanishes

At O the line of action of the force passes through the point itself, and a force whose line of action passes through a point can never turn the body about that point.

**the moment decreases steadily and becomes zero at O**



## Solutions — step by step

For every hanging body, write the two equilibrium equations separately: the horizontal one and the vertical one. Almost every question in this sheet falls out of those two lines.

### Part A — Power and efficiency

#### 1 what power measures

work tells how much energy was transferred;  
power tells how fast it was transferred

**B) the rate of doing work or of transferring energy**

#### 2 the relation

$P = \Delta W / \Delta t$  ( work done ÷ time taken )

**B)  $P = \Delta W / \Delta t$**

#### 3 the unit

1 watt = 1 joule / second

**B) the watt (W)**

#### 4 the two lifts

the two lifts raise the same load through the same height, so the useful work is the same :

$$W = m g h = 600 \times 9.8 \times 9.0 = 52\,920 \text{ J} \approx 5.3 \times 10^4 \text{ J}$$

lift A :  $P = W / t = 52\,920 / 24 = 2205 \text{ W}$

lift B :  $P = W / t = 52\,920 / 12 = 4410 \text{ W}$

Power is not "more work", it is the **same** work delivered in less time. Lift B finishes in half the time, so its power is twice as large.

**the same useful work; lift B has double the power**

### Part B — Home assignment 1

#### 5 the faster lift

same  $W$  , smaller  $t \rightarrow P = W / t$  is larger

**B) a greater power**

**6** useful output from the efficiency

$\eta = \text{useful output} / \text{total input}$

useful output =  $\eta \times \text{input} = 0.40 \times 250 = 100 \text{ J}$

**A) 100 J**

**7** the efficiency of the heater

$\eta = \text{useful power} / \text{input power} = 2040 / 2400 = 0.85 = 85 \%$

An efficiency greater than 100 % is impossible, which rules out the 117 % option immediately.

**A) 85 %**

**8** the power of an inclined force

$P = F v \cos \theta$

$\theta = 0^\circ : \cos \theta = 1 \rightarrow P = F v$  (the largest possible power)

$\theta \rightarrow 90^\circ : \cos \theta \rightarrow 0 \rightarrow P \rightarrow \text{zero}$

Only the component of the force along the motion,  $F \cos \theta$ , transfers energy to the body; a force perpendicular to the motion does no work at all.

**the power falls from its maximum  $F v$  down to zero**

## Part C — Equilibrium: review and weekly assessment

**9** the third force

$\Sigma F(y) = 0$

$+ 10 - 6 + F_3 = 0 \rightarrow F_3 = -4 \text{ N}$

the minus sign means: directed downwards

**B) 4 N downwards**

**10** the tension in the cable

the rope is horizontal, so the vertical equation contains the cable only :

$T \cos 25^\circ = W = 29.4 \text{ N}$

$T = 29.4 / 0.906 = 32.4 \text{ N}$

**B) 32.3 N**

**11** the horizontal components

$\Sigma F(x) = 0$  , and the two cables are symmetric, so the two horizontal components have the same size and opposite directions

**A) equal and opposite, so they cancel each other**

**12** cables closer to the horizontal

$$T = W / ( 2 \sin \alpha )$$

$\alpha$  decreases  $\rightarrow \sin \alpha$  decreases  $\rightarrow T$  increases

As the cables approach the horizontal,  $\sin \alpha$  approaches zero and the tension grows without limit — this is why a washing line pulled dead straight snaps so easily.

**A) increases**

**13** cables closer to the vertical

$$T = W / ( 2 \sin \alpha )$$

$\alpha$  increases  $\rightarrow \sin \alpha$  increases  $\rightarrow T$  decreases

**B) decreases**

**14** the weight of the metal piece

$$W = m g = 1.5 \times 9.8 = 14.7 \text{ N}$$

The spring constant and the angle are not needed for the weight; they are needed for the next question.

**B) 14.7 N**

**15** the force in each spring

the two springs are identical, so each carries the same force  $F$ .

each spring makes  $40^\circ$  with the **vertical**, so its vertical component is  $F \cos 40^\circ$  :

$$\Sigma F(y) = 0 \rightarrow 2 F \cos 40^\circ = W$$

$$2 F \times 0.77 = 14.7$$

$$F = 14.7 / 1.54 = 9.5 \text{ N}$$

**F  $\approx$  9.5 N in each spring**

**16** the tension from the horizontal rope

the cable has to balance both the weight and the pull of the rope :

$$\text{vertical} : T \cos \theta = W = m g = 4.0 \times 9.8 = 39.2 \text{ N}$$

$$\text{horizontal} : T \sin \theta = P = 15 \text{ N}$$

$$T = \sqrt{( W^2 + P^2 )} = \sqrt{( 39.2^2 + 15^2 )} = \sqrt{1761.6} = 42.0 \text{ N}$$

**T = 42.0 N**

**17** the angle of the cable

$$\tan \theta = P / W = 15 / 39.2 = 0.3827$$

$$\theta = 20.9^\circ$$

**$\theta \approx 20.9^\circ$  with the vertical**

### 18 reading the component diagram

vertical axis :  $\Sigma F(y) = 0 \rightarrow T_1 y + T_2 y - W = 0 \rightarrow T_1 y + T_2 y = W$

horizontal axis :  $\Sigma F(x) = 0 \rightarrow T_1 x = T_2 x$  (opposite directions)

The weight has no horizontal component at all, so nothing is left for the horizontal components to balance except each other.

**the vertical components carry the weight, the horizontal ones cancel**

### 19 equal angles against different angles

equal angles : by symmetry the two tensions are equal ,  $T = W / ( 2 \sin \alpha )$

different angles : the tensions are **not** equal; the cable that is closer to the vertical (the larger angle with the horizontal) carries the larger tension

**equal angles  $\rightarrow$  equal tensions ; different angles  $\rightarrow$  different tensions**

### 20 the shop sign on two cables

$$W = m g = 10 \times 9.8 = 98 \text{ N}$$

horizontal :  $T_1 \cos 35^\circ = T_2 \cos 65^\circ$

$$0.819 T_1 = 0.423 T_2 \rightarrow T_1 = 0.516 T_2$$

vertical :  $T_1 \sin 35^\circ + T_2 \sin 65^\circ = 98$

$$0.574 ( 0.516 T_2 ) + 0.906 T_2 = 98$$

$$1.202 T_2 = 98 \rightarrow T_2 = 81.5 \text{ N}$$

$$T_1 = 0.516 \times 81.5 = 42.1 \text{ N}$$

The steeper cable ( $65^\circ$ ) carries almost twice the tension, because it has to supply the larger share of the vertical support.

**$T_1 = 42.1 \text{ N}$  ,  $T_2 = 81.5 \text{ N}$**

### 21 the sum of the vertical components

$$\Sigma F(y) = 0 \rightarrow T_1 y + T_2 y = W = m g$$

**the sum equals the weight of the lamp**

### 22 a second book on the table

the table now has to support two books, so the normal force becomes :

$$F(N) = 2 m g \quad (\text{it doubles})$$

the weight of the first book is still  $m g$  , because it depends only on its own mass and on  $g$ .

The two forces have different **sources**: the weight is the gravitational pull of the Earth on that one book, while the normal force is a contact reaction from the table that adjusts itself to carry whatever rests on it.

**$F(N)$  doubles, the weight of the book itself does not change**



## Solutions — step by step

Two relations carry this whole sheet:  $\mathbf{J = \Delta p = F \Delta t}$  for a single body, and  $\mathbf{\Sigma p \text{ before} = \Sigma p \text{ after}}$  for a system of bodies.

### Part A — Lesson 1-7 · Classwork 1

#### 1 comparing two momenta

$$p = m v$$

$$p(A) = 40 \times 0.3 = 12 \text{ kg}\cdot\text{m/s}$$

$$p(B) = 60 \times 0.2 = 12 \text{ kg}\cdot\text{m/s}$$

They are equal in **magnitude** but opposite in direction, so the total momentum of the two persons together is zero.

**C) they are equal in magnitude**

#### 2 the phone on a hard floor and on a cushion

a) the phone arrives with the same speed, so the change in momentum  $\Delta p$  is the same.

$$F = \Delta p / \Delta t$$

the cushion makes the phone stop over a longer time  $\Delta t$ , so the same  $\Delta p$  is delivered by a smaller average force.

b)  $\Delta t$  doubled,  $\Delta p$  fixed  $\rightarrow$   $F$  becomes half its value

This is the physics behind airbags, crash helmets, knee pads and the way a fielder pulls his hands back while catching a ball.

**the cushion increases  $\Delta t$ , so  $F$  decreases ; doubling  $\Delta t$  halves  $F$**

#### 3 impulse when the direction changes

$$J = \Delta p = m v(\text{final}) - m v(\text{initial})$$

$$m v(\text{final}) = 0.10 \times 6 = 0.6 \text{ kg}\cdot\text{m/s} \quad \text{northwards}$$

$$m v(\text{initial}) = 0.10 \times 6 = 0.6 \text{ kg}\cdot\text{m/s} \quad \text{eastwards}$$

$$J = 0.6 \text{ (north)} - 0.6 \text{ (east)} = 0.6 \text{ (north)} + 0.6 \text{ (west)}$$

$$|J| = \sqrt{0.6^2 + 0.6^2} = 0.6\sqrt{2} = 0.85 \text{ N}\cdot\text{s}, \text{ directed north-west}$$

Even though the **speed** did not change, the velocity did, so there is a large impulse; momentum is a vector.

**B)  $0.6\sqrt{2}$  N·s north-west**

### Part B — Lesson 1-7 · Home assignment 1

#### 4 the airbag

$$F = \Delta p / \Delta t, \text{ with } \Delta p \text{ fixed}$$

$$\Delta t \times 3 \rightarrow F \div 3$$

**C) one third as large**

**5** reading the force–time curve

- a)  $J = F \times \Delta t$  for a constant force, and for a changing force the product becomes the **area enclosed between the curve and the time axis**.  
so : impulse = area under the force–time curve =  $\Delta p$  of the trolley
- b) a softer bumper lengthens the contact time and lowers the peak force, so the pulse becomes **wider and flatter** – but the area stays the same, because  $\Delta p$  has not changed.

**impulse = the area under the curve ; a softer bumper gives a wider, lower pulse of the same area**

**Part C — Lesson 1-8 · Classwork 1**

**6** the passenger walking in the boat

- a) both the boat and the passenger are at rest :  
 $p(\text{total}) = 0$
- b) the water exerts no horizontal force on the system, so momentum is conserved :  
 $m(p) v(p) + m(b) v(b) = 0 \rightarrow v(b) = - m(p) v(p) / m(b)$   
the minus sign means the boat moves backwards.

The boat is much heavier than the passenger, so its backward speed is much smaller than his forward speed.

**the total momentum stays zero, so the boat must move the opposite way**

**7** a perfectly inelastic collision

- a)  $\Sigma p (\text{before}) = \Sigma p (\text{after})$   
 $m_1 v_1 + m_2 v_2 = (m_1 + m_2) v$   
 $(0.50 \times 6.0) + (1.00 \times 0) = (1.50) v$   
 $3.0 = 1.5 v \rightarrow v = 2.0 \text{ m/s eastwards}$
- b) before :  $3.0 \text{ kg} \cdot \text{m/s}$       after :  $1.5 \times 2.0 = 3.0 \text{ kg} \cdot \text{m/s}$

The kinetic energy is **not** conserved here (9.0 J before, 3.0 J after): part of it becomes heat and sound. Momentum, however, is always conserved when no external force acts.

**$v = 2.0 \text{ m/s}$  ; the total momentum is the same before and after**

**Part D — Lesson 1-8 · Home assignment 1**

**8** a head-on collision

- take the direction of A as positive :
- $\Sigma p (\text{before}) = (2.0 \times +6.0) + (4.0 \times -2.0) = 12 - 8 = +4.0 \text{ kg} \cdot \text{m/s}$
- $\Sigma p (\text{after}) = (2.0 + 4.0) v = 6.0 v$
- $6.0 v = 4.0 \rightarrow v = +0.67 \text{ m/s}$  (the + sign : the direction of A)

**A) 0.67 m/s in the direction of A**

**9** a collision in two dimensions

take x along the original motion and y perpendicular to it.

$$\begin{aligned}y : \quad 0 &= 1.5 v(A) \sin 30^\circ - 3.0 v(B) \sin 60^\circ \\ 0.75 v(A) &= 2.598 v(B) \quad \rightarrow \quad v(A) = 3.46 v(B)\end{aligned}$$

$$\begin{aligned}x : \quad 1.5 \times 6.0 &= 1.5 v(A) \cos 30^\circ + 3.0 v(B) \cos 60^\circ \\ 9 &= 1.299 v(A) + 1.5 v(B) \\ 9 &= 1.299 ( 3.46 v(B) ) + 1.5 v(B) = 6.0 v(B) \\ v(B) &= 1.5 \text{ m/s} \quad \quad v(A) = 3.46 \times 1.5 = 5.2 \text{ m/s}\end{aligned}$$

Always start with the axis that has **zero** total momentum (here the y axis): it gives the simplest relation between the two unknowns.

$$v(A) = 5.2 \text{ m/s}, v(B) = 1.5 \text{ m/s}$$

## Part E — Weekly assessment · Group A

**10** the impulse on the tennis ball

$$J = \Delta p = m ( v(f) - v(i) ) = 0.060 ( 50 - 0 ) = 3.0 \text{ N}\cdot\text{s}$$

The contact time is not needed for the impulse itself; it is only needed if the **average force** is asked for (  $F = 3.0 / 4 \times 10^{-3} = 750 \text{ N}$  ).

$$\text{A) } 3.0 \text{ N}\cdot\text{s}$$

**11** doubling the collision time

$$F = \Delta p / \Delta t, \Delta p \text{ fixed} \quad \rightarrow \quad \Delta t \times 2 \text{ gives } F \div 2$$

$$\text{B) is halved}$$

**12** the explosion

the body was at rest, so  $\Sigma p \text{ (before)} = 0$

taking the right as positive :

$$0 = ( 5.0 \times -8.0 ) + ( 4.0 \times v(B) )$$

$$0 = -40 + 4 v(B) \quad \rightarrow \quad v(B) = +10 \text{ m/s}$$

In an explosion the momenta of the fragments always add up to the momentum before the explosion — here zero — so the two pieces must fly apart in opposite directions.

$$v(B) = 10 \text{ m/s towards the right}$$

**13** A leaves perpendicular to its original path

$$\begin{aligned}x : \quad 0.30 \times 4.0 &= 0 + 0.60 v(B) \cos 60^\circ \\1.2 &= 0.60 v(B) \times 0.5 = 0.30 v(B) \quad \rightarrow \quad v(B) = 4.0 \text{ m/s}\end{aligned}$$

$$\begin{aligned}y : \quad 0 &= 0.30 v(A) - 0.60 v(B) \sin 60^\circ \\0.30 v(A) &= 0.60 \times 4.0 \times 0.866 = 2.08 \quad \rightarrow \quad v(A) = 6.93 \text{ m/s}\end{aligned}$$

**Note for the teacher:** these particular data are not energy-consistent (the kinetic energy after the collision would be larger than before it). The question is meant as a drill on conservation of momentum in two dimensions, and the working above is the answer it asks for.

$$v(A) = 6.93 \text{ m/s}, v(B) = 4.0 \text{ m/s}$$

## Group B

**14** impulse when the ball is sent back

take the east as positive :

$$J = m ( v(f) - v(i) ) = 0.20 ( -3 - 5 ) = -1.6 \text{ N}\cdot\text{s}$$

the minus sign : the impulse is directed westwards

The two velocities must carry opposite signs; forgetting that gives 0.4 N·s, which is the commonest wrong answer here.

$$\text{C) } 1.6 \text{ N}\cdot\text{s westwards}$$

**15** halving the collision time

$$F = \Delta p / \Delta t \quad \rightarrow \quad \Delta t \div 2 \text{ gives } F \times 2$$

$$\text{A) is doubled}$$

**16** both bodies moving before the collision

$$\Sigma p \text{ (before)} = ( 0.40 \times 6.0 ) + ( 0.80 \times 1.5 ) = 2.4 + 1.2 = 3.6 \text{ kg}\cdot\text{m/s} \quad (\text{along } x)$$

$$\begin{aligned}y : \quad 0.40 v(A) \sin 53^\circ &= 0.80 v(B) \sin 30^\circ \\0.319 v(A) &= 0.40 v(B) \quad \rightarrow \quad v(A) = 1.25 v(B)\end{aligned}$$

$$\begin{aligned}x : \quad 0.40 v(A) \cos 53^\circ + 0.80 v(B) \cos 30^\circ &= 3.6 \\0.241 ( 1.25 v(B) ) + 0.693 v(B) &= 3.6 \\0.994 v(B) &= 3.6 \quad \rightarrow \quad v(B) = 3.62 \text{ m/s} \\v(A) &= 1.25 \times 3.62 = 4.53 \text{ m/s}\end{aligned}$$

**Note for the teacher:** as in question 13, the angles given here are not energy-consistent; the solution applies conservation of momentum exactly as the question intends.

$$v(A) = 4.53 \text{ m/s}, v(B) = 3.62 \text{ m/s}$$

**17** the recoil of the cannon

a) the cannon and the shell are both at rest :  $p(\text{total}) = 0$

b) no external horizontal force acts, so momentum is conserved :

$$0 = m(\text{shell}) v(\text{shell}) + M(\text{cannon}) v(\text{cannon})$$

$$v(\text{cannon}) = - m(\text{shell}) v(\text{shell}) / M(\text{cannon})$$

This is why a field gun is made heavy and anchored to the ground: the same shell momentum then produces a much smaller recoil speed.

**the cannon recoils backwards, and its recoil speed is inversely proportional to its mass**

## Group C

**18** a perpendicular change of velocity

$$p(\text{initial}) = 0.10 \times 8 = 0.8 \text{ kg}\cdot\text{m/s}$$

$$p(\text{final}) = 0.10 \times 6 = 0.6 \text{ kg}\cdot\text{m/s} , \text{ perpendicular to it}$$

$$|J| = \sqrt{(0.8^2 + 0.6^2)} = \sqrt{1.00} = 1.00 \text{ N}\cdot\text{s}$$

**C) 1.00 N·s**

**19** four times the collision time

$$F = \Delta p / \Delta t \rightarrow \Delta t \times 4 \text{ gives } F \div 4$$

**D) is reduced to one quarter**

**20** finding both the speed and the angle

$$x : 0.30 \times 5 = 0.60 v(B) \cos \theta$$

$$v(B) \cos \theta = 1.5 / 0.60 = 2.5$$

$$y : 0.30 \times 3 = 0.60 v(B) \sin \theta$$

$$v(B) \sin \theta = 0.9 / 0.60 = 1.5$$

$$v(B) = \sqrt{(2.5^2 + 1.5^2)} = \sqrt{8.5} = 2.92 \text{ m/s}$$

$$\tan \theta = 1.5 / 2.5 = 0.6 \rightarrow \theta = 31.0^\circ \text{ below the original line}$$

**B) 2.92 m/s ,  $\theta \approx 31^\circ$**

**21** comparing two force pulses

$J$  = area under the force–time graph

$$J_1 = F \times \Delta t$$

$$J_2 = (F / 2) \times (2 \Delta t) = F \times \Delta t$$

$$J_1 : J_2 = 1 : 1$$

Halving the force while doubling the time leaves the impulse untouched — exactly the trick that crumple zones and airbags use.

**$J_1 : J_2 = 1 : 1$**

**22** impulse from a triangular graph

$J = \text{area of the triangle} = \frac{1}{2} \times \text{base} \times \text{height}$

$$J = \frac{1}{2} \times 6 \text{ s} \times 6 \text{ N} = 18 \text{ N}\cdot\text{s}$$

$$J = \Delta p = m ( v - 0 )$$

$$18 = 1.50 v \quad \rightarrow \quad v = 12 \text{ m/s}$$

**C) 12 m/s**



## Solutions — step by step

In every collision start with the same line:  $\Sigma p \text{ before} = \Sigma p \text{ after}$ . Only after the velocities are known should you go on to compare the kinetic energies.

### Part A — Classwork 1

#### 1 what survives an inelastic collision

no external force acts on an isolated system  $\rightarrow$  momentum is always conserved  
the collision is inelastic  $\rightarrow$  part of the kinetic energy turns into internal energy, sound and deformation, so the kinetic energy decreases

The **total** energy is still conserved; it is only the **kinetic** part that goes down.

**C) the momentum is conserved while the kinetic energy decreases**

#### 2 the car and the truck

a)  $\Sigma p \text{ (before)} = \Sigma p \text{ (after)}$

$$(1000 \times 20) + (3000 \times 0) = (1000 + 3000) v$$

$$20\,000 = 4000 v \quad \rightarrow \quad v = 5 \text{ m/s eastwards}$$

b) the two bodies move together, so they never separate :

$$e = \text{separation speed} / \text{approach speed} = 0 / 20 = 0$$

$e = 0 \rightarrow$  a perfectly inelastic collision

c)  $KE \text{ (before)} = \frac{1}{2} \times 1000 \times 20^2 = 200\,000 \text{ J} = 200 \text{ kJ}$

$$KE \text{ (after)} = \frac{1}{2} \times 4000 \times 5^2 = 50\,000 \text{ J} = 50 \text{ kJ}$$

$$\text{loss} = 200 - 50 = 150 \text{ kJ}$$

Three quarters of the kinetic energy disappears here — it is exactly this energy that crushes the metal and heats it, which is why crumple zones are designed to absorb it.

**$v = 5 \text{ m/s}$  ,  $e = 0$  (perfectly inelastic) , loss = 150 kJ**

### Part B — Home assignment 1

#### 3 reading the type of collision

the kinetic energy decreased  $\rightarrow$  the collision is not perfectly elastic.

the two balls separated  $\rightarrow$  it is not perfectly inelastic either.

Sticking together is only the **extreme** case of an inelastic collision ( $e = 0$ ); any collision with  $0 < e < 1$  is inelastic while the bodies still separate.

**B) inelastic, and the two bodies need not stick together**

**4** two balls that stick together

$$\begin{aligned} \text{a) } (2.0 \times 6.0) + (3.0 \times 0) &= (2.0 + 3.0) v \\ 12 &= 5 v \quad \rightarrow \quad v = 2.4 \text{ m/s eastwards} \end{aligned}$$

$$\begin{aligned} \text{b) } \text{KE (before)} &= \frac{1}{2} \times 2.0 \times 6.0^2 = 36 \text{ J} \\ \text{KE (after)} &= \frac{1}{2} \times 5.0 \times 2.4^2 = 14.4 \text{ J} \\ \text{loss} &= 36 - 14.4 = 21.6 \text{ J} \end{aligned}$$

**$v' = 2.4 \text{ m/s}$  ,  $\text{loss} = 21.6 \text{ J}$**

**5** the ball and the wall

the wall does not move, so :

$$e = \text{speed of separation} / \text{speed of approach} = 6 / 10 = 0.6$$

e has no unit, and it always lies between 0 and 1 for a real collision; 0.6 tells us the collision is inelastic but the ball still bounces well.

**$e = 0.6$**

## Part C — Weekly assessment · Group A

**6** the clay and the trolley

$$\begin{aligned} (0.50 \times 8.0) + (1.50 \times 0) &= (0.50 + 1.50) v \\ 4.0 &= 2.0 v \quad \rightarrow \quad v = 2.0 \text{ m/s} \end{aligned}$$

**$v' = 2.0 \text{ m/s}$**

**7** describing the collision

the clay sticks to the trolley, so after the collision there is only one body moving with one velocity (a perfectly inelastic collision,  $e = 0$ )

Momentum is conserved here, which rules out option B; the kinetic energy is not, which rules out option A.

**C) the two masses move with one common speed after the collision**

**8** where the lost energy goes

the deformation of the metal and the rise of temperature are the record of the kinetic energy that was transferred into internal energy, sound and the work of permanent deformation

**B) internal energy, sound and deformation**

**9** using the coefficient of restitution

$$e = ( v'(B) - v'(A) ) / ( v(A) - v(B) )$$

$$0.50 = ( 3.0 - v'(A) ) / ( 4.0 - 0 )$$

$$3.0 - v'(A) = 0.50 \times 4.0 = 2.0$$

$$v'(A) = 3.0 - 2.0 = 1.0 \text{ m/s}$$

Ball A keeps moving forwards, but more slowly than B — which is exactly what must happen if the two are not to overlap after the collision.

**C) 1.0 m/s**

## Group B

**10** the percentage of energy lost

from question 4 : KE (before) = 36 J , KE (after) = 14.4 J

loss = 21.6 J

$$\text{percentage} = ( 21.6 / 36 ) \times 100 \% = 60 \%$$

Notice that the momentum did not change at all during the same collision — the two quantities behave completely differently.

**60 % of the kinetic energy is lost**

**11** the meaning of the sound

the sound carries energy away from the system, and that energy can only come from the kinetic energy of the two colliding bodies

**B) part of the kinetic energy is converted into sound energy**

## Group C

**12** the two trolleys

$$( 2.0 \times 4.0 ) + ( 3.0 \times 0 ) = ( 2.0 + 3.0 ) v$$

$$8.0 = 5.0 v \quad \rightarrow \quad v' = 1.6 \text{ m/s eastwards}$$

**v' = 1.6 m/s eastwards**

**13** the 30 J that vanished from the kinetic energy

energy is never destroyed; the 30 J left the kinetic store and appeared as internal energy (heat), sound and the work done in deforming the two bodies

Option C mixes two different quantities: joules cannot be subtracted from a momentum measured in kg·m/s, and in an isolated system the momentum does not change at all.

**B) it was converted into other forms inside the system and its surroundings**



## Solutions — step by step

Most of these questions are one substitution away from the answer once you have written down  $T$ ,  $\omega$  and  $v$ . Start every problem by finding the periodic time.

### Part A — Classwork 2

#### 1 what "uniform" really means

uniform circular motion : the **speed** ( a scalar ) is constant,  
but the **velocity** ( a vector ) changes, because its direction turns  
continuously  $\rightarrow$  there is an acceleration, directed towards the centre

**C) the speed is constant, but the direction of the velocity changes towards the centre**

#### 2 periodic time and angular velocity

- a)  $T = \text{total time} / \text{number of revolutions} = 2.0 / 10 = 0.20 \text{ s}$   
b)  $\omega = 2\pi / T = ( 2 \times 3.14 ) / 0.20 = 31.4 \text{ rad/s}$

**$T = 0.20 \text{ s}$  ,  $\omega = 31.4 \text{ rad/s}$**

#### 3 centripetal acceleration from $\omega$

$$a(c) = \omega^2 r = ( 3.0 )^2 \times 2.0 = 9 \times 2.0 = 18 \text{ m/s}^2$$

The mass is not needed for the **acceleration**; it is only needed for the force.

**$a(c) = 18 \text{ m/s}^2$**

### Part B — Home assignment 2

#### 4 periodic time and linear speed

- a)  $T = 5.0 / 20 = 0.25 \text{ s}$   
b)  $v = 2\pi r / T = ( 2 \times 3.14 \times 0.20 ) / 0.25 = 1.256 / 0.25 = 5.03 \text{ m/s}$

**$T = 0.25 \text{ s}$  ,  $v = 5.03 \text{ m/s}$**

#### 5 speed and force of the whirling ball

- a)  $v = \omega r = 3.0 \times 2.0 = 6.0 \text{ m/s}$   
b)  $F = m \omega^2 r = 0.50 \times 9 \times 2.0 = 9.0 \text{ N}$  (or  $F = m v^2 / r = 0.50 \times 36 / 2 = 9.0 \text{ N}$ )

**$v = 6.0 \text{ m/s}$  ,  $F = 9.0 \text{ N}$**

**6** the rotating spring

the spring supplies the centripetal force, and the radius is the **stretched** length :

$$r = \text{natural length} + \text{extension} = 0.10 + x$$

$$k x = m \omega^2 r$$

$$30 x = 0.50 \times (6.0)^2 \times (0.10 + x)$$

$$30 x = 18 (0.10 + x) = 1.8 + 18 x$$

$$12 x = 1.8 \quad \rightarrow \quad x = 0.15 \text{ m}$$

The commonest mistake here is to use  $r = 0.10 \text{ m}$ ; the spring is stretched while it turns, so the radius is  $0.25 \text{ m}$ .

**the extension = 0.15 m**

## Part C — Weekly assessment · Group A

**7** the definition of  $\omega$

$$\omega = \Delta\theta / \Delta t \quad (\text{radians per second})$$

**B) the angular displacement per unit time**

**8** the centripetal force

$$F = m a(c) \quad \text{and} \quad a(c) = v^2 / r \quad \rightarrow \quad F = m v^2 / r$$

**B)  $F = m v^2 / r$**

**9** the rotating disc

$$\text{a) } T = 2\pi / \omega = 2\pi / 4\pi = 0.5 \text{ s}$$

$$\text{b) } v = \omega r = 4\pi \times 0.30 = 4 \times 3.14 \times 0.30 = 3.77 \text{ m/s}$$

**$T = 0.5 \text{ s}$ ,  $v = 3.77 \text{ m/s}$**

**10** the washing machine

the wall of the drum pushes the clothes inwards and supplies the centripetal force that keeps them on the circular path.

the drops of water are not held by the wall: when a drop reaches a hole there is nothing left to push it towards the centre, so by its inertia it continues in a straight line along the tangent and escapes through the hole.

Nothing throws the water outwards; it simply keeps going straight while the drum keeps curving away from it.

**the wall cannot supply a centripetal force through the holes, so the water leaves along the tangent**

**11** acceleration and force from the periodic time

$$v = 2\pi r / T = (2 \times 3.14 \times 2.0) / 4.0 = 3.14 \text{ m/s}$$

$$\text{a) } a(c) = v^2 / r = (3.14)^2 / 2.0 = 9.86 / 2.0 = 4.93 \text{ m/s}^2$$

$$\text{b) } F = m a(c) = 3.0 \times 4.93 = 14.8 \text{ N}$$

**$a(c) = 4.93 \text{ m/s}^2$ ,  $F = 14.8 \text{ N}$**

## Group B

**12** the definition of T

T is the time of **one complete revolution**

**A) one revolution**

**13** the resultant force

the body is accelerating towards the centre, so by Newton's second law the resultant force must point towards the centre as well. no real force pushes the body outwards.

**B) there is no outward force; the resultant is always directed towards the centre**

**14** when the string breaks

while the string is intact it pulls the ball towards the centre and keeps bending its path. the instant it breaks, no horizontal force is left, so by Newton's first law the ball keeps the velocity it had at that instant : a straight line **tangent** to the circle, with the same speed.

It does not fly outwards along the radius because it never had a velocity in that direction; the velocity was always tangential.

**it moves in a straight line along the tangent, not radially outwards**

**15** two changes in the motion

a) the string breaks → the centripetal force disappears → the ball moves in a straight line along the tangent with a constant speed.

b) r increases with  $\omega$  constant :

$v = \omega r$  → the linear speed increases

$a(c) = \omega^2 r$  → the centripetal acceleration increases

$F = m \omega^2 r$  → the required force increases

**a) straight line along the tangent · b) v , a(c) and F all increase**

**16** periodic time, frequency and speed

a)  $T = 3.0 / 15 = 0.20 \text{ s}$

$f = 1 / T = 1 / 0.20 = 5.0 \text{ Hz}$

b)  $v = 2\pi r / T = ( 2 \times 3.14 \times 0.30 ) / 0.20 = 1.884 / 0.20 = 9.42 \text{ m/s}$

**T = 0.20 s , f = 5.0 Hz , v = 9.42 m/s**

## Group C

17 the two forms of  $a(c)$

$$a(c) = v^2 / r \quad \text{and} \quad v = \omega r \quad \rightarrow \quad a(c) = (\omega r)^2 / r = \omega^2 r$$

A)  $a(c) = \omega^2 r$  or  $v^2 / r$

18 the force on the ball inside the ring

the ball follows a circular arc, so its acceleration – and therefore the resultant force from the wall – must point towards the **centre** of that arc, which lies to the left of the ball in the figure.

The tangential arrows (c) and (d) would change the **speed** of the ball, and the outward arrow (a) would push it off the wall altogether.

B) the arrow directed towards the centre

19 constant speed but non-zero acceleration

acceleration = the rate of change of the **velocity**, and the velocity is a vector. in circular motion the direction of the velocity changes at every instant, so the velocity changes even though its magnitude does not  $\rightarrow a \neq 0$ , directed to the centre.

because the direction of the velocity keeps changing

20 two reasons

- a) the velocity is a vector: a change of direction alone is a change of velocity, so the body is accelerating even with a constant speed.
- b) the only real force acting is the inward one ( the string, the wall, gravity ... ). the outward "push" that a rotating passenger feels is the effect of his own **inertia** in a rotating frame, not a force acting on the body.

a) the direction of  $v$  changes · b) the outward feeling is inertia, not a real force

21 acceleration and force of the ball

$$a) \quad a(c) = v^2 / r = (2.0)^2 / 0.50 = 4.0 / 0.50 = 8.0 \text{ m/s}^2$$

$$b) \quad F = m a(c) = 0.40 \times 8.0 = 3.2 \text{ N}$$

$a(c) = 8.0 \text{ m/s}^2$ ,  $F = 3.2 \text{ N}$



## Solutions — step by step

For a vertical circle, remember that the **speed changes** around the loop (energy conservation), while for a conical pendulum the speed is constant and the two equilibrium equations do all the work.

### Part A — Classwork 1

#### 1 the minimum speed at the top

at the highest point both the tension and the weight point downwards, towards the centre :

$$T + mg = m v^2 / r$$

the slowest possible motion is the one in which the string is just about to go slack,  $T = 0$  :

$$mg = m v^2 / r \quad \rightarrow \quad v^2 = g r \quad \rightarrow \quad v(\min) = \sqrt{(g r)}$$

$$\text{B) } v(\min) = \sqrt{(g r)}$$

#### 2 why the water does not fall out

at the top the water needs a downward force of  $m v^2 / r$  to stay on its circular path. gravity supplies  $mg$ , and the bottom of the bucket supplies whatever is still missing. as long as  $v^2 / r \geq g$  ( that is  $v \geq \sqrt{(g r)}$  ), the weight is not more than what the circular path needs, so the water never gets the chance to fall away from the bucket.

The water **is** falling all the time — it simply falls along the circle, and the bucket falls with it.

**because gravity is being used up as the centripetal force needed for the circular path**

#### 3 tension at the lowest point

at the lowest point the tension points up (to the centre) and the weight points down :

$$T - mg = m v^2 / r$$

$$T = m ( g + v^2 / r ) = 0.50 ( 9.8 + 16 / 0.80 )$$

$$T = 0.50 ( 9.8 + 20 ) = 0.50 \times 29.8 = 14.9 \text{ N}$$

The tension is about three times the weight of the ball ( $mg = 4.9 \text{ N}$ ) — this is why strings snap at the bottom of the swing, never at the top.

$$\text{B) } T = 14.9 \text{ N}$$

### Part B — Home assignment 1

**4** the ball inside the conical vessel

the surface is smooth, so only two forces act : the weight  $mg$  and the normal force  $N$ , which is perpendicular to the slant side. with  $\theta$  measured between the axis and the slant :

a) vertical :  $N \sin \theta = mg \quad \rightarrow \quad N = mg / \sin \theta$

b) horizontal :  $N \cos \theta = m v^2 / r$   
 $( mg / \sin \theta ) \cos \theta = m v^2 / r \quad \rightarrow \quad v^2 = g r \cot \theta$   
and from the geometry of the cone :  $r = h \tan \theta$   
 $v^2 = g ( h \tan \theta ) \cot \theta = g h \quad \rightarrow \quad v = \sqrt{g h}$

The neat result is that the speed depends on the **height** only, not on the angle of the cone nor on the mass.

**$N = mg / \sin \theta$  ,  $v = \sqrt{g h}$**

**5** the minimum speed for a taut string

$v(\min) = \sqrt{g r} = \sqrt{9.8 \times 0.80} = \sqrt{7.84} = 2.8 \text{ m/s}$

**$v(\min) = 2.8 \text{ m/s}$**

**6** where the water spills first

at the top the only force available is the weight, and the path needs  $m v^2 / r$ .  
if  $v^2 / r < g$  the weight is more than the circle needs, so the water leaves the circular path and falls out.

at the bottom the bucket can push the water inwards as hard as necessary  
(  $T - mg = m v^2 / r$  ), so there is no tendency to spill there.

**because the top is the only place where gravity alone has to supply the centripetal force**

## Part C — Classwork 2

**7** tension at the lowest point

$T = m ( g + v^2 / r ) = 0.40 ( 9.8 + 16 / 0.60 )$   
 $T = 0.40 ( 9.8 + 26.7 ) = 0.40 \times 36.5 = 14.6 \text{ N}$

**C) 14.6 N**

**8** the forces on a conical pendulum

only two forces act on the bob :

the weight  $mg$  , vertically downwards

the tension  $T$  , along the string

the bob has no vertical acceleration, so the vertical parts must cancel :

$$T \cos \theta = mg$$

the bob does accelerate horizontally, towards the centre of its circle :

$$T \sin \theta = m v^2 / r$$

The weight has no horizontal part at all, so the only candidate left for the centripetal force is the horizontal component of the tension.

**the horizontal component  $T \sin \theta$  supplies the centripetal force**

**9** the equations for the conical vessel

here  $\theta$  is measured between the slant side and the **horizontal**, so the normal force makes the angle  $\theta$  with the vertical :

a) vertical equilibrium :  $N \cos \theta = mg$

b) centripetal force :  $N \sin \theta = m v^2 / r$

Dividing the second equation by the first gives  $\tan \theta = v^2 / (r g)$ , which is the same relation as for the conical pendulum.

**$N \cos \theta = mg$  ,  $N \sin \theta = m v^2 / r$**

## Part D — Home assignment 2

**10** the tension in the string

the ball has no vertical acceleration :

$$T \cos 30^\circ = mg = 0.5 \times 9.8 = 4.9 \text{ N}$$

$$T = 4.9 / 0.866 = 5.66 \text{ N}$$

**B) 5.66 N**

**11** tension and normal force together

the radius of the circle is  $r = l \sin \theta$

a) horizontal :  $T \sin \theta = m \omega^2 r = m \omega^2 l \sin \theta$

$$T = m \omega^2 l \quad (\text{the angle cancels out})$$

b) vertical :  $T \cos \theta + N = mg$

$$N = mg - T \cos \theta = m (g - \omega^2 l \cos \theta)$$

As  $\omega$  grows,  $N$  falls; when  $\omega^2 l \cos \theta$  reaches  $g$  the ball leaves the surface altogether and the problem becomes an ordinary conical pendulum.

**$T = m \omega^2 l$  ,  $N = m (g - \omega^2 l \cos \theta)$**

**12** the large force at the bottom of the loop

at the top ( just keeping contact ) :  $N \approx 0$  and  $v(\text{top})^2 = g r$

from the top to the bottom the car falls a height  $2r$ , so by conservation of energy :

$$v(\text{bottom})^2 = v(\text{top})^2 + 4 g r = g r + 4 g r = 5 g r$$

at the bottom :  $N - mg = m v(\text{bottom})^2 / r = 5 mg$

$$N = 6 mg$$

Two effects add up: the car is moving much faster at the bottom, and the track must also carry the whole weight there instead of being helped by it.

**the normal force at the bottom is about six times the weight**

## Part E — Weekly assessment · Group A

**13** the direction of the resultant at the top

the body is moving on a circle, so its acceleration – and the resultant force – must point towards the centre, which at the highest point means straight down.

**C) directly towards the centre of the circle**

**14** where the tension is largest

bottom :  $T = m ( g + v^2/r )$       top :  $T = m ( v^2/r - g )$

the speed is also largest at the bottom, so both effects push  $T$  to its maximum there.

**B) at the lowest point of the path**

**15** judging the statement

the statement is **wrong**.

the minimum speed at the top is obtained by letting the tension ( or the normal force ) fall to **zero**, not grow to infinity :

$$T = 0 \rightarrow mg = m v^2 / r \rightarrow v(\text{min}) = \sqrt{g r}$$

an infinite force is physically impossible, and a larger force would mean a larger speed, not a smaller one.

**wrong — the correct condition is  $T = 0$ , giving  $v(\text{min}) = \sqrt{g r}$**

**16** the change of velocity between two positions

the speed is the same at X and at Y, but the **directions** differ.

$\Delta v = v(Y) - v(X)$  , which is drawn by reversing  $v(X)$  and adding it to  $v(Y)$ .

the resulting arrow points **into** the circle, along the bisector of the two positions, and for a quarter of a circle its length is  $v\sqrt{2}$ .

This is the geometric reason why the acceleration of uniform circular motion points to the centre.

**$\Delta v$  points towards the inside of the path, with magnitude  $v\sqrt{2}$  for a quarter circle**

**17** halving the radius of the path

from A to B the body rises a height equal to the radius, so by energy conservation :

$$v(B)^2 = v_0^2 - 2 g r$$

if  $r$  becomes  $r/2$  with  $v_0$  unchanged :

$$v(B)^2 = v_0^2 - g r \quad ( \text{ a smaller amount is subtracted } )$$

A smaller loop means less height to climb, so less kinetic energy is converted into potential energy on the way up.

**the speed at B increases**

## Group B

**18** the condition used in the derivation

the string can only pull; the slowest circular motion is the one in which it is just about to go slack,  $T = 0$ , leaving gravity alone to bend the path.

**B) zero ( $T = 0$ )**

**19** why the loop is teardrop-shaped

a circular loop would need a very large radius at the bottom entry, giving the riders a violent centripetal acceleration there. the teardrop ( clothoid ) shape changes the radius of curvature gradually, so  $a(c) = v^2 / r$  stays within safe limits everywhere.

**B) to change the radius of curvature gradually and control the centripetal acceleration**

**20** the analysis of the conical pendulum

the weight  $mg$  is vertical, so it has **no horizontal component** at all.

the bob moves in a horizontal circle, so the resultant force must be horizontal, pointing to the centre. the only horizontal force available is  $T \sin \theta$ .

the vertical component  $T \cos \theta$  is fully used up in balancing the weight.

**$T \sin \theta$  supplies the centripetal force ;  $T \cos \theta$  balances the weight**

**21** two changes

a) too slow at the top :  $v^2 / r < g$  , so the weight is greater than the force the circular path needs. the water can no longer follow the circle – it leaves the bucket and falls as a projectile.

b) increasing  $\theta$  :  $r = l \sin \theta$  increases ,  $T = mg / \cos \theta$  increases, and since  $\tan \theta = v^2 / ( r g )$  the speed must increase as well.

**a) the water spills out · b)  $r$  ,  $T$  and the required speed all increase**

**22** the minimum speed at the top

$$v(\min) = \sqrt{ g r } = \sqrt{ 9.8 \times 1.0 } = 3.13 \text{ m/s}$$

The mass does not appear: a heavy ball and a light ball need exactly the same minimum speed.

**$v(\min) = 3.13 \text{ m/s}$**

## Group C

**23** the component that turns the bob

the circle is horizontal, so the centripetal force must be horizontal :  $T \sin \theta$ .

**B)  $T \sin \theta$  (the horizontal component)**

**24** both forces pointing the same way at the top

at the highest point the tension and the weight are both directed downwards, i.e. both towards the centre, so they add :

$$T + mg = m v^2 / r \quad \rightarrow \quad T = m ( v^2 / r - g )$$

gravity is therefore **helping** to bend the path, which is why the tension is at its smallest there, and why a minimum speed exists at all.

**they add together, so T is smallest at the top and vanishes at  $v = \sqrt{g r}$**

**25** the centrifugal misconception

there is no outward force acting on the water. the correct picture is inertia : the water tends to keep moving in a **straight line** ( Newton's first law ), and the bucket must keep pulling it inwards to bend that line into a circle. when the bucket is too slow at the top, the inward force available ( the weight ) is larger than the circle needs, so the water follows a projectile path and separates from the bucket – it is not thrown outwards; it is simply left behind.

**no outward force exists — the water only keeps its straight-line inertia**

**26** two reasons

- a) the teardrop shape lets the radius of curvature change gradually, keeping the centripetal acceleration  $v^2 / r$  on the riders within safe limits.
- b) at the lowest point  $T = m ( g + v^2 / r )$  : the tension must both carry the weight and bend the path, and the speed is also largest there.

**a) to control the g-forces · b) T carries the weight and bends the path together**

**27** the conical pendulum with numbers

a)  $T \cos 30^\circ = mg = 0.30 \times 9.8 = 2.94 \text{ N}$   
 $T = 2.94 / 0.866 = 3.39 \text{ N}$

b)  $r = l \sin 30^\circ = 0.80 \times 0.5 = 0.40 \text{ m}$   
 $T \sin 30^\circ = m v^2 / r$   
 $3.39 \times 0.5 = 0.30 v^2 / 0.40$   
 $1.70 = 0.75 v^2 \quad \rightarrow \quad v^2 = 2.26 \quad \rightarrow \quad v = 1.50 \text{ m/s}$

A quicker route for (b) is  $v^2 = g r \tan \theta = 9.8 \times 0.40 \times 0.577 = 2.26$ .

**T = 3.39 N , v = 1.50 m/s**



## Solutions — step by step

Two habits make this whole lesson easy. For **gravitation**, always substitute in SI units and collect the powers of ten at the very end. For **Kepler**, never substitute G and M when a **ratio** will do:  $T_1^2/T_2^2 = r_1^3/r_2^3$ .

### Part A — Classwork 1

- 1 the force between two identical masses 1 m apart

Newton's law of universal gravitation :

$$F = G m_1 m_2 / r^2$$

the two masses are identical,  $m_1 = m_2 = m$ , and the separation is  $r = 1 \text{ m}$  :

$$F = G \times m \times m / (1)^2 = G m^2$$

Notice that the force depends on the **product** of the masses, never on their sum — which is why choice (C) is the classic trap.

**A)  $F = G m^2$**

- 2 two 60 kg bodies, 3.0 m apart

$$F = G m_1 m_2 / r^2$$

$$F = ( 6.67 \times 10^{-11} \times 60 \times 60 ) / (3.0)^2$$

$$F = ( 6.67 \times 10^{-11} \times 3600 ) / 9$$

$$F = 6.67 \times 10^{-11} \times 400$$

$$F = 2.67 \times 10^{-8} \text{ N}$$

About 27 nanonewtons — roughly the weight of a grain of dust. Gravitation is by far the **weakest** of the fundamental forces; it only becomes noticeable when one of the masses is astronomically large.

**$F = 2.67 \times 10^{-8} \text{ N}$**

- 3 the periodic time of a planet at 9.0 AU

Kepler's third law, written for this planet and for the Earth :

$$T^2 / r^3 = T_e^2 / r_e^3$$

for the Earth  $T_e = 1.0 \text{ year}$  and  $r_e = 1.0 \text{ AU}$ , so the right-hand side equals 1 :

$$T^2 = r^3 = (9.0)^3 = 729$$

$$T = \sqrt{729} = 27 \text{ years}$$

With years and astronomical units the law collapses to  $T = (a)^{3/2}$ . Here  $9^{3/2} = (\sqrt{9})^3 = 3^3 = 27$ .

**$T = 27 \text{ years}$**

### Part B — Home assignment 1

#### 4 the gravitational field of the Earth and its field lines

**Definition.** The gravitational field of the Earth is the region of space surrounding it in which any body placed at a point feels a force of attraction directed towards the centre of the Earth. Its strength at a point is defined as the force acting on **unit mass** placed there,  $g = F / m$ , measured in N/kg.

**Field lines.** The field is drawn as a set of lines that are **radial** and point **inwards**, towards the centre of the Earth, because that is the direction of the force on a test mass. The **tangent** to a line at any point gives the direction of the field there, and the **crowding** of the lines measures the strength: the lines are closest together near the surface (strong field) and spread further apart as the distance increases (weak field), which is the geometrical picture of the inverse-square law. Field lines never cross, because the field at a point has one single direction. Close to the surface, over a small region, the lines look practically parallel and equally spaced — this is why  $g$  is taken as a constant 9.8 N/kg in everyday problems.

a region of space in which any mass experiences a gravitational force of attraction towards the Earth

#### 5 two 40 kg bodies, 3.0 m apart

$$F = G m_1 m_2 / r^2$$

$$F = ( 6.67 \times 10^{-11} \times 40 \times 40 ) / (3.0)^2$$

$$F = ( 6.67 \times 10^{-11} \times 1600 ) / 9$$

$$F = 6.67 \times 10^{-11} \times 177.8$$

$$F = 1.19 \times 10^{-8} \text{ N}$$

$$F = 1.19 \times 10^{-8} \text{ N}$$

#### 6 the comet: nearest speed compared with farthest speed

Kepler's second law (equal areas in equal times) gives, at the two ends of the major axis, where the velocity is perpendicular to the radius :

$$v(\text{near}) \times r(\text{near}) = v(\text{far}) \times r(\text{far})$$

so the speed ratio is the **inverse** of the distance ratio :

$$v(\text{near}) / v(\text{far}) = r(\text{far}) / r(\text{near}) = 2.5 / 1.5$$

$$v(\text{near}) / v(\text{far}) = 5 / 3 \approx 1.67$$

This is conservation of **angular momentum** in disguise:  $m v r$  stays constant because the gravitational pull is always directed along the radius, so it exerts no moment about the Sun.

$$v(\text{near}) : v(\text{far}) = 5 : 3 \rightarrow \text{the comet is about 1.67 times faster at the nearest point}$$

### Part C — Classwork 2

#### 7 the statement of Kepler's third law

the square of the periodic time is directly proportional to the cube of the semi-major axis :

$$T^2 \propto r^3 \quad \text{or} \quad T^2 / r^3 = \text{constant}$$

C) the cube of the semi-major axis (  $r^3$  )

**8** the periodic time of a planet at 7.0 AU

$$T^2 / r^3 = 1 \quad (\text{using years and astronomical units})$$

$$T^2 = (7.0)^3 = 343$$

$$T = \sqrt{343} = 18.5 \text{ years}$$

A useful check: the planet is 7 times further out than the Earth but takes 18.5 times longer, not 7 times — the period grows faster than the radius.

$$T \approx 18.5 \text{ years}$$

**9** why a geostationary satellite has a 24-hour period

“Geostationary” means the satellite appears **motionless** to an observer standing on the ground. For that to happen the satellite and the observer must have the same **angular velocity**,  $\omega = 2\pi / T$ . If the two angular velocities are equal, the two periodic times must be equal as well, so  $T$  must equal the period of rotation of the Earth about its own axis (one sidereal day  $\approx 24$  h). Any other period would make the satellite drift steadily eastwards or westwards relative to the ground. Two extra conditions follow: the orbit must lie in the **plane of the equator** (otherwise the satellite would swing north and south), and the satellite must move in the **same sense** as the rotation of the Earth. Kepler’s third law then fixes the radius of that orbit at about  $42 \times 10^3$  km from the centre of the Earth, roughly  $36 \times 10^3$  km above the surface.

**so that it turns through exactly the same angle as the Earth in the same time, and therefore stays above one fixed point**

## Part D — Home assignment 2

**10** tripling the separating distance

the law itself does not change :

$$F = G m_1 m_2 / r^2 \quad \rightarrow \quad F \propto 1 / r^2$$

with the masses unchanged, replacing  $r$  by  $3r$  :

$$F' = G m_1 m_2 / (3r)^2 = (G m_1 m_2 / r^2) / 9 = F / 9$$

Tripling the distance does not make the force three times smaller — it makes it **nine** times smaller. This is what “inverse square” means in practice.

**D) inversely proportional to the square of the distance ( the force drops to  $F / 9$  )**

**11** why  $T^2/r^3$  is the same for all the moons of Jupiter

for a moon of mass  $m$  in a circular orbit of radius  $r$  around Jupiter  
( of mass  $M$  ), the gravitational pull is the centripetal force :

$$G M m / r^2 = m v^2 / r \quad \text{and} \quad v = 2\pi r / T$$

$$G M / r^2 = ( 4\pi^2 r^2 / T^2 ) / r = 4\pi^2 r / T^2$$

$$T^2 / r^3 = 4\pi^2 / ( G M )$$

The mass of the moon **cancels out** of the equation, and the radius of its own orbit is not in the final expression either. All four Galilean moons share the same central body, so they all share the same constant — even though Io is close in with a period of 1.8 days and Callisto is far out with a period of 16.7 days.

**$T^2/r^3 = 4\pi^2/(G M)$  — it contains only  $G$  and the mass of Jupiter, so it is the same for every moon**

**12** planets and asteroids: why distant bodies are slower

for any body orbiting the Sun :

$$T^2 = (4\pi^2 / G M) r^3 \quad \rightarrow \quad T \propto r^{3/2}$$

compare an asteroid in the belt (  $r \approx 2.8 \text{ AU}$  ) with the Earth (  $r = 1.0 \text{ AU}$  ) :

$$T = (2.8)^{3/2} \approx 4.7 \text{ years}$$

There are **two** effects working in the same direction. First, the circumference of the orbit is larger, so there is simply more distance to cover. Second, the Sun's pull is weaker far away, so the orbital speed itself is smaller ( $v = \sqrt{GM/r}$ ).

Combining a longer path with a slower speed makes the period rise faster than the radius — exactly the  $r^{3/2}$  dependence of Kepler's third law. Mercury takes 88 days, the Earth 365 days, and Neptune 165 years.

**because  $T$  grows as  $r^{3/2}$ , a body that is further out has both a longer path and a slower speed, so its periodic time is much longer**

## Part E — Weekly assessment, Group A

**13** the statement of the law of gravitation

$$F = G m_1 m_2 / r^2$$

so  $F \propto m_1 m_2$  ( the product, not the sum ) and  $F \propto 1 / r^2$

**B) directly proportional to the product of the masses and inversely proportional to  $r^2$**

**14** the field strength at a height above the surface

the distance in the law is always measured from the **centre** of the body, not from its surface. a point at a height  $r$  above the surface of a sphere of radius  $R$  lies

at a distance (  $R + r$  ) from the centre, so :

$$g = G M / (R + r)^2$$

Choice (A) is what the formula becomes only when  $r$  is measured from the centre. Read the question carefully: "above the surface" always means add  $R$ .

**B)  $g = G M / (R + r)^2$**

**15** Kepler's second law and angular momentum

angular momentum about the Sun :  $L = m v r$  ( with  $v \perp r$  )

the gravitational pull acts along the line joining the planet to the Sun, so its moment about the Sun is zero  $\rightarrow$   $L$  is conserved :

$$m v_1 r_1 = m v_2 r_2 \quad \rightarrow \quad v r = \text{constant}$$

**The link.** In a short time  $\Delta t$  the planet moves a distance  $v\Delta t$ , and the thin triangle it sweeps has area  $\approx \frac{1}{2} r (v \Delta t)$ . So the rate of sweeping area is  $\frac{1}{2} v r = L / 2m$ . Because no external moment acts about the Sun,  $L$  is constant, therefore the area swept per second is constant — which is precisely the second law. The physical consequence is that the planet must move **fastest at perihelion** (small  $r$ ) and **slowest at aphelion** (large  $r$ ).

**the line joining a planet to the Sun sweeps out equal areas in equal intervals of time**

**16** why a larger orbit means a longer period

from  $T^2 = (4\pi^2 / GM) r^3$  :

$$T \propto r^{3/2}$$

while for uniform motion in a straight line at a fixed speed,  $t = d / v$  ,  
so the time would only be **directly** proportional to the distance.

In uniform linear motion doubling the distance doubles the time, because the speed is the same. In orbital motion doubling the radius multiplies the time by  $2^{3/2} \approx 2.83$ , because the speed itself drops as  $1/\sqrt{r}$ .

**the period rises as  $r^{3/2}$ , faster than the simple proportionality  $t \propto d$  of uniform linear motion**

**17** the comet with  $r(\text{far}) = 5 r(\text{near})$

a) Kepler's second law at the two ends of the major axis :

$$v(\text{near}) \times r(\text{near}) = v(\text{far}) \times r(\text{far})$$

$$v(\text{far}) / v(\text{near}) = r(\text{near}) / r(\text{far}) = r(\text{near}) / (5 r(\text{near})) = 1/5 = 0.2$$

b)  $v(\text{near}) = 30 \text{ km/s}$  :

$$v(\text{far}) = 0.2 \times 30 = 6 \text{ km/s}$$

Halley's comet behaves exactly like this: it races past the Sun at about 54 km/s and crawls at under 1 km/s out beyond Neptune.

**a)  $v(\text{far}) : v(\text{near}) = 1 : 5$    b)  $v(\text{far}) = 6 \text{ km/s}$**

## Part E — Weekly assessment, Group B

**18** what  $T^2$  is proportional to

$$T^2 \propto r^3$$

**B) the cube of the length of the semi-major axis ( $r^3$ )**

**19** the name of the 24-hour equatorial satellite

a period equal to the rotation period of the Earth, an equatorial orbit,  
and the same sense of rotation → the satellite appears fixed in the sky

**B) a geostationary satellite**

**20** why N/kg is the same as  $\text{m/s}^2$

start from the definition of the field strength and from Newton's second law :

$$g = F / m \quad \text{and} \quad F = m a$$

substituting the unit of force,  $1 \text{ N} = 1 \text{ kg} \cdot \text{m/s}^2$  :

$$1 \text{ N} / \text{kg} = (1 \text{ kg} \cdot \text{m/s}^2) / \text{kg} = 1 \text{ m/s}^2$$

The two units describe the **same** quantity seen from two sides. N/kg is the **field** point of view — the pull on each kilogram;  $\text{m/s}^2$  is the **motion** point of view — the acceleration that pull produces on a freely falling body. Because the mass cancels in  $a = F/m = g$ , every body falls with the same acceleration, which is why a feather and a hammer land together on the Moon.

**$1 \text{ N/kg} \equiv 1 \text{ m/s}^2$**

**21** two changes and their effects

a)  $F \propto 1 / r^2$  , so replacing  $r$  by  $2r$  :

$$F' = G m_1 m_2 / (2r)^2 = F / 4$$

b) a larger circular orbit means a larger  $r$ , so from

$$v = \sqrt{(G M / r)} \quad \rightarrow \quad v \text{ decreases}$$

$$T^2 \propto r^3 \quad \rightarrow \quad T \text{ increases}$$

Part (b) is the reason a geostationary satellite at 42 000 km moves at only about 3.1 km/s, while the International Space Station at 6 800 km races round at 7.7 km/s.

**a) the force falls to a quarter of its value    b) the orbital speed decreases and the periodic time increases**

**22** two 80 kg bodies, 2.0 m apart

$$F = G m_1 m_2 / r^2$$

$$F = ( 6.67 \times 10^{-11} \times 80 \times 80 ) / (2.0)^2$$

$$F = ( 6.67 \times 10^{-11} \times 6400 ) / 4$$

$$F = 6.67 \times 10^{-11} \times 1600$$

$$F = 1.07 \times 10^{-7} \text{ N}$$

$$F = 1.07 \times 10^{-7} \text{ N}$$

## Part E — Weekly assessment, Group C

**23** the definition of the gravitational field strength

$$g = F / m \quad \rightarrow \quad \text{the force acting on each unit of mass, in N/kg}$$

**B) the gravitational force acting per unit mass placed at that point**

**24** the position of the Sun in Kepler's first law

every planet moves in an **ellipse**, and the Sun occupies **one of the two foci** of that ellipse – never the centre.

The other focus is empty. For most planets the ellipse is so nearly circular that the two foci sit very close together — which is why a circle is a good first approximation for the Earth.

**B) at one of the two foci of the elliptical orbit**

**25** two 100 kg bodies, 5.0 m apart

$$F = G m_1 m_2 / r^2$$

$$F = ( 6.67 \times 10^{-11} \times 100 \times 100 ) / (5.0)^2$$

$$F = ( 6.67 \times 10^{-11} \times 10^4 ) / 25$$

$$F = 6.67 \times 10^{-11} \times 400$$

$$F = 2.67 \times 10^{-8} \text{ N}$$

$$F = 2.67 \times 10^{-8} \text{ N}$$

**26** two reasons

**a)** A geostationary satellite has a periodic time equal to the period of rotation of the Earth and orbits in the plane of the equator in the same sense. Its angular velocity  $\omega = 2\pi/T$  therefore matches that of the point below it, so the line joining the two never turns: the satellite stays fixed above that single point and a dish on the ground can be aimed once and left alone.

**b)** The gravitational pull of the Sun acts along the radius, so it has no moment about the Sun and the angular momentum  $m v r$  of the comet is conserved. Since  $v r$  must stay constant, the smallest  $r$  goes with the largest  $v$  — the comet is at its fastest at the point of closest approach. This is Kepler's second law seen from the side of the forces.

**a) equal angular velocities    b) conservation of angular momentum (  $v r = \text{constant}$  )**

**27** the period of the planet whose axis is 4.0 times larger

Kepler's third law as a ratio, so  $G$  and  $M$  never appear :

$$T_1^2 / T_2^2 = r_1^3 / r_2^3 = ( r_1 / r_2 )^3 = (4.0)^3 = 64$$

$$T_1 / T_2 = \sqrt{64} = 8$$

$$T_1 = 8 \times T_2 = 8 \times 2.0 = 16 \text{ years}$$

Both planets orbit the **same** star, which is what allows the constant  $4\pi^2/GM$  to cancel. Always look for this shortcut before reaching for a calculator.

**$T_1 = 16 \text{ years}$**



## Solutions — step by step

**One idea runs through the whole lesson.** Because  $F = -kx$ , the acceleration is  $a = -(k/m)x$ . So  $a$  and  $x$  are always **opposite in sign**, both vanish together at  $x = 0$ , and both reach their maximum together at  $x = \pm A$ . Every conceptual question in this lesson is an application of that one sentence.

### Part A — Classwork 1

#### 1 the vibrating ruler

a) The pitch of a note is decided by its **frequency**, and the frequency of the ruler is fixed by the properties of the system itself — the free length, the stiffness of the steel and the mass that is vibrating. It is **not** fixed by how far the end swings. This is the defining property of simple harmonic motion: the periodic time is **independent of the amplitude** (isochronism). Air resistance removes energy, so the amplitude dies away and the note becomes **quieter**, but every swing still takes the same time, so the note keeps the same pitch until it fades out.

b) Shortening the projecting part makes the ruler much **harder to bend** — the effective force constant  $k$  becomes larger — while the length of steel that is actually moving becomes smaller, so the effective mass  $m$  becomes smaller as well. Since  $f = (1/2\pi) \sqrt{k/m}$ , both changes push the frequency **up**, and a higher frequency is heard as a sharper note. It is exactly the same reason why a short guitar string, or a string pressed close to the bridge, gives a higher note.

**a) because the frequency of a vibrating body does not depend on the amplitude b) because a shorter free length is stiffer, so the frequency rises**

#### 2 why the force and the acceleration are greatest at the extreme

the restoring force obeys :

$$F = -kx \quad \text{and therefore} \quad a = F/m = -(k/m)x$$

so both  $F$  and  $a$  are proportional to the **displacement**  $x$ , not to the speed.

at the point of maximum displacement  $x = \pm A$ , so  $|F|$  and  $|a|$  are greatest there.

Choice (A) reverses cause and effect: a zero speed does not **make** the force large, the two simply happen at the same point. Choice (C) is wrong because at the extreme the energy is entirely **potential**, not kinetic, and choice (D) is wrong because mass never changes.

**B) because the restoring force depends on the displacement, which is greatest at this point**

### Part B — Home assignment 1

#### 3 quadrupling the mass

$$\begin{aligned} \text{a) } T &= 2\pi \sqrt{m/k} \quad \rightarrow \quad T \propto \sqrt{m} \quad (\text{same spring, } k \text{ fixed}) \\ T_2 / T_1 &= \sqrt{4m/m} = \sqrt{4} = 2 \quad \rightarrow \quad \text{the periodic time } \mathbf{increases} \end{aligned}$$

$$\text{b) } T_2 = 2 \times T_1 = 2 \times 1.2 = 2.4 \text{ s}$$

Notice that  $k$  cancels in the ratio, which is why part (a) can be answered without a single number. A heavier mass has more inertia, so it is slower to be turned round by the same spring and each cycle takes longer.

**a)  $T_2 : T_1 = 2 : 1$  b)  $T_2 = 2.4 \text{ s}$**

4 does  $n$  times the mass give  $n$  times the periodic time?

$T = 2\pi \sqrt{m/k}$  contains  $\sqrt{m}$ , not  $m$ , so :

$m \rightarrow nm$  gives  $T \rightarrow \sqrt{n} \times T$

for example  $n = 4$  gives  $\sqrt{4} = 2$  ( not 4 ), and  $n = 9$  gives 3 ( not 9 ).

To **double** the periodic time you must make the mass **four times** as large. The square root is what makes the mass-spring system so forgiving: even a big change of mass produces only a modest change of timing.

**no — multiplying the mass by  $n$  multiplies the periodic time by  $\sqrt{n}$  only**

5 one complete cycle, instant by instant

the key relation is  $a = -(k/m)x$ , together with  $E = KE + PE = \text{constant}$  :

at  $x = 0$  all the energy is kinetic  $\rightarrow v$  is greatest,  $a = 0$

at  $x = \pm A$  all the energy is potential  $\rightarrow v = 0$ ,  $|a|$  is greatest

The body is never “pushed along” by the spring at  $x = 0$  — it passes through with the spring exerting no force at all, carried by its own inertia. Everything the spring does is to slow it down on the way out and speed it up on the way back.

| instant    | displacement $x$ | speed $v$              | acceleration $a$       |
|------------|------------------|------------------------|------------------------|
| $t = 0$    | 0                | greatest, towards $+x$ | 0                      |
| $t = T/4$  | $+A$             | 0                      | greatest, towards $-x$ |
| $t = T/2$  | 0                | greatest, towards $-x$ | 0                      |
| $t = 3T/4$ | $-A$             | 0                      | greatest, towards $+x$ |
| $t = T$    | 0                | greatest, towards $+x$ | 0                      |

**$v$  is greatest at  $x = 0$  and zero at  $x = \pm A$ ;  $a$  is zero at  $x = 0$  and greatest at  $x = \pm A$**

6 the instant of release from full compression

at the instant of release the displacement has its greatest magnitude,  $x = -A$  :

$$F = -kx = -k(-A) = +kA$$

a positive force means a force towards the **right**, that is towards  $x = 0$ , and its magnitude  $kA$  is the largest value the restoring force ever takes.

Choice (C) is the classic mistake: the body is momentarily at rest, but **rest does not mean no force**. It is precisely because the force is largest here that the body will not stay at rest for more than an instant.

**B) towards the right, towards the equilibrium position, and greatest at this instant**

7 increasing the amplitude of a 0.9 kg mass on a 40 N/m spring

the periodic time of a mass-spring system does not contain the amplitude at all :

$$T = 2\pi \sqrt{m/k} = 2 \times 3.14 \times \sqrt{0.9/40}$$

$$T = 6.28 \times \sqrt{0.0225} = 6.28 \times 0.15$$

$$T = 0.942 \approx 0.94 \text{ s} \quad (\text{unchanged by the larger amplitude})$$

A larger amplitude means the body travels further, but it also travels **faster** (a larger restoring force). The two effects cancel exactly, which is what makes this motion so useful for clocks.

**C) the periodic time does not change, and its value is about 0.94 s**

## Part C — Classwork 2

- 8 the angular frequency of the 0.50 kg block

$$\omega = \sqrt{k / m}$$

$$\omega = \sqrt{20 / 0.50} = \sqrt{40}$$

$$\omega = 6.32 \text{ rad/s}$$

**Note on the printed question.** The book writes the relation as  $\omega = \sqrt{m/k}$ ; this is a misprint. The angular frequency is  $\omega = \sqrt{k/m}$ , which is the one used here. A quick check settles it:  $\omega$  must get **larger** for a stiffer spring and **smaller** for a heavier mass, and only  $k/m$  does that. Note also that  $\sqrt{m/k}$  would come out as 0.158, which has the units of a time, not of rad/s.

$$\omega = 6.32 \text{ rad/s}$$

- 9 the periodic time of the 0.20 kg block on an 80 N/m spring

$$T = 2\pi \sqrt{m / k}$$

$$T = 2 \times 3.14 \times \sqrt{0.20 / 80}$$

$$T = 6.28 \times \sqrt{0.0025} = 6.28 \times 0.05$$

$$T = 0.314 \text{ s}$$

**Note on the printed question.** The four options printed in the book (61.0 s, 13.0 s, 36.0 s and 62.1 s) do not match these data at all — they are hundreds of times too large, and the given values cannot produce any of them. The question has therefore been set here as a straight calculation. A stiff spring with a light block vibrates several times a second, so a periodic time of a fraction of a second is exactly what we should expect.

$$T = 0.314 \text{ s} \approx 0.31 \text{ s}$$

## Part D — Home assignment 2

- 10 the periodic time of the 0.30 kg system

$$T = 2\pi \sqrt{m / k}$$

$$T = 2 \times 3.14 \times \sqrt{0.30 / 30}$$

$$T = 6.28 \times \sqrt{0.01} = 6.28 \times 0.1$$

$$T = 0.628 \text{ s}$$

$$T = 0.628 \text{ s}$$

- 11 finding the amplitude from  $x = A \sin(\omega t)$

$$x = A \sin(\omega t)$$

$$0.15 = A \sin(2 \times \pi/4) = A \sin(\pi/2)$$

$$\sin(\pi/2) = 1 \quad (\text{the angle is in radians, } \pi/2 \text{ rad} = 90^\circ)$$

$$0.15 = A \times 1 \quad \rightarrow \quad A = 0.15 \text{ m}$$

At this particular instant the body happens to be exactly at the end of its swing, so its displacement is the amplitude. Keep the calculator in radian mode for this kind of question.

$$A = 0.15 \text{ m}$$

**12** the kinetic energy at the equilibrium position

the total energy is constant :  $E = KE + PE$

at  $x = 0$  the spring is neither stretched nor compressed, so  $PE = 0$  :

$$KE = E - 0 = E \quad (\text{and the speed there is } v(\text{max}) = \omega A)$$

The equilibrium position is the one place where the body owns all of the energy of the system as energy of motion — and it is therefore the place where it is moving fastest.

**D)  $KE = E$**

**13** finding the mass from the periodic time

square both sides of the relation and make  $m$  the subject :

$$T = 2\pi \sqrt{(m/k)} \rightarrow T^2 = 4\pi^2 m/k$$

$$m = k T^2 / (4\pi^2)$$

$$m = 20 \times (1.256)^2 / (4 \times (3.14)^2)$$

$$m = 20 \times 1.5775 / 39.44 = 31.55 / 39.44$$

$$m = 0.80 \text{ kg}$$

A neat shortcut: 1.256 s is exactly  $4 \times 0.314$  s, and 0.314 s belongs to  $\sqrt{(m/k)} = 0.05$ . So  $\sqrt{(m/k)}$  here is 0.2, giving  $m/k = 0.04$  and  $m = 0.8$  kg.

**A)  $m = 0.8$  kg**

## Part E — Weekly assessment, Group A

**14** the speed at the maximum displacement

at  $x = \pm A$  the body reverses its direction, so it must pass through rest :

$$\text{all the energy is potential there} \rightarrow KE = 0 \rightarrow v = 0$$

**B) zero**

**15** the frequency for  $T = 0.25$  s

$$f = 1/T = 1/0.25 = 4 \text{ Hz}$$

**C)  $f = 4$  Hz**

**16** reading  $x(t) = 0.06 \sin(8\pi t)$

compare with the standard form  $x = A \sin(\omega t)$  :

$$A = 0.06 \text{ m} \quad \omega = 8\pi \text{ rad/s}$$

$$T = 2\pi / \omega = 2\pi / 8\pi = 0.25 \text{ s}$$

$$f = 1/T = 4 \text{ Hz}$$

$$v(\text{max}) = \omega A = 8\pi \times 0.06 = 0.48\pi = 1.51 \text{ m/s}$$

No calculator is needed for  $T$ : the  $\pi$  in  $\omega$  always cancels the  $\pi$  in  $2\pi$ , so  $T = 2/8 = 0.25$  s straight away.

**$A = 0.06 \text{ m}$  ,  $T = 0.25 \text{ s}$  ,  $f = 4 \text{ Hz}$  ,  $v(\text{max}) = 1.51 \text{ m/s}$**

17 T and f for  $m = 0.20 \text{ kg}$  on  $k = 80 \text{ N/m}$

$$T = 2\pi \sqrt{(m / k)} = 6.28 \times \sqrt{(0.20 / 80)}$$

$$T = 6.28 \times \sqrt{0.0025} = 6.28 \times 0.05 = 0.314 \text{ s}$$

$$f = 1 / T = 1 / 0.314 = 3.18 \text{ Hz}$$

**T = 0.314 s , f = 3.18 Hz**

18 the force and the acceleration at  $x = +0.10 \text{ m}$

$$F = - k x = - 40 \times (+0.10) = - 4.0 \text{ N}$$

$$a = F / m = - 4.0 / 0.50 = - 8.0 \text{ m/s}^2$$

the minus sign means the direction is opposite to the displacement,  
that is **towards** the equilibrium position.

Always quote the sign as well as the size. The minus sign in  $F = -kx$  is the whole physics of the lesson: it is what turns a push into an **oscillation** instead of an escape.

**F = 4.0 N and a = 8.0 m/s<sup>2</sup>, both directed towards the equilibrium position**

## Part E — Weekly assessment, Group B

19 the acceleration at the equilibrium position

$$a = - (k / m) x \quad \text{and at the equilibrium position } x = 0 :$$

$$a = 0 \quad (\text{the spring exerts no force there})$$

**A) zero**

20 the periodic time for  $f = 5.0 \text{ Hz}$

$$T = 1 / f = 1 / 5.0 = 0.20 \text{ s}$$

**A) T = 0.20 s**

21 reading  $x(t) = 0.04 \sin(10\pi t)$

$$A = 0.04 \text{ m} \quad \omega = 10\pi \text{ rad/s}$$

$$T = 2\pi / 10\pi = 0.2 \text{ s}$$

$$f = 1 / 0.2 = 5 \text{ Hz}$$

$$v(\text{max}) = \omega A = 10\pi \times 0.04 = 0.4\pi = 1.26 \text{ m/s}$$

**A = 0.04 m , T = 0.20 s , f = 5 Hz , v(max) = 1.26 m/s**

22 T and f for  $m = 0.40 \text{ kg}$  on  $k = 40 \text{ N/m}$

$$T = 6.28 \times \sqrt{(0.40 / 40)} = 6.28 \times \sqrt{0.01}$$

$$T = 6.28 \times 0.1 = 0.628 \text{ s}$$

$$f = 1 / 0.628 = 1.59 \text{ Hz}$$

**T = 0.628 s , f = 1.59 Hz**

**23** the force and the acceleration at  $x = -0.15$  m

$$F = -kx = -20 \times (-0.15) = +3.0 \text{ N}$$

$$a = F/m = +3.0 / 0.25 = +12 \text{ m/s}^2$$

the plus sign here means the direction is the positive one – and since the body is on the negative side, that is again **towards** the equilibrium position.

The sign of  $F$  is always **opposite** to the sign of  $x$ . That is the only thing you have to watch when the displacement is given as a negative number.

**$F = 3.0 \text{ N}$  and  $a = 12 \text{ m/s}^2$ , both directed towards the equilibrium position**

## Part E — Weekly assessment, Group C

**24** the restoring force at  $x = +A$

$$F = -kx = -kA \quad (\text{largest magnitude, negative direction})$$

so the force has its greatest value, and it points back towards  $x = 0$ .

**A) the greatest value it has, directed towards the equilibrium position**

**25** the frequency for  $T = 0.80$  s

$$f = 1/T = 1/0.80 = 1.25 \text{ Hz}$$

**B)  $f = 1.25 \text{ Hz}$**

**26** reading  $x(t) = 0.05 \sin(5\pi t)$

$$A = 0.05 \text{ m} \quad \omega = 5\pi \text{ rad/s}$$

$$T = 2\pi / 5\pi = 0.4 \text{ s}$$

$$f = 1/0.4 = 2.5 \text{ Hz}$$

$$v(\text{max}) = \omega A = 5\pi \times 0.05 = 0.25\pi = 0.785 \text{ m/s}$$

**$A = 0.05 \text{ m}$ ,  $T = 0.40 \text{ s}$ ,  $f = 2.5 \text{ Hz}$ ,  $v(\text{max}) = 0.785 \text{ m/s}$**

**27**  $T$  and  $f$  for  $m = 0.90$  kg on  $k = 40$  N/m

$$T = 6.28 \times \sqrt{(0.90 / 40)} = 6.28 \times \sqrt{0.0225}$$

$$T = 6.28 \times 0.15 = 0.942 \text{ s}$$

$$f = 1/0.942 = 1.06 \text{ Hz}$$

This is the same system as question 7 — a good place to check that you get the same answer by both routes.

**$T = 0.942 \text{ s}$ ,  $f = 1.06 \text{ Hz}$**

**28** the force and the acceleration at  $x = +0.125$  m

$$F = -kx = -32 \times (+0.125) = -4.0 \text{ N}$$

$$a = F/m = -4.0 / 0.40 = -10 \text{ m/s}^2$$

**$F = 4.0 \text{ N}$  and  $a = 10 \text{ m/s}^2$ , both directed towards the equilibrium position**



## Solutions — step by step

**Work with ratios.** Almost every question in these two lessons changes one quantity and asks what happens to  $T$ . Write the law, see which power the quantity carries, and take the square root:  $T \propto \sqrt{m}$ ,  $T \propto 1/\sqrt{k}$ ,  $T \propto \sqrt{L}$ ,  $T \propto 1/\sqrt{g}$ . Nothing else is needed.

### Part A — Lesson 2-2, Classwork 1

- 1 the periodic time of the 0.20 kg mass on an 80 N/m spring

$$T = 2\pi \sqrt{(m / k)}$$

$$T = 2 \times 3.14 \times \sqrt{(0.20 / 80)}$$

$$T = 6.28 \times \sqrt{0.0025} = 6.28 \times 0.05$$

$$T = 0.314 \text{ s}$$

Do the division inside the root **first**. Taking each square root separately and then dividing is the commonest slip here.

$$T = 0.314 \text{ s}$$

- 2 which statement is true for  $T$

the periodic time of a spring pendulum is :

$$T = 2\pi \sqrt{(m / k)}$$

it contains only  $m$  and  $k$  – there is no  $g$  and no amplitude in it.

choice (B) is upside down :  $\sqrt{(k/m)}$  is  $\omega$ , not  $T$ .

Hanging the spring vertically brings gravity into the problem, but only to **move** the equilibrium position. Once the body oscillates about that new position,  $g$  has no further part to play — which is the subject of question 3.

$$\text{C) } T = 2\pi\sqrt{(m/k)}, \text{ and it does not depend on } g$$

### Part B — Lesson 2-2, Home assignment 1

- 3 why the vertical system has the same periodic time as the horizontal one

at the new equilibrium position the spring is already stretched by  $e$ , and :

$$k e = m g \quad \rightarrow \quad e = m g / k$$

now measure the displacement  $x$  from that new position. the stretch is  $(e + x)$ ,

so the resultant force on the body is :

$$F = m g - k (e + x) = (m g - k e) - k x = 0 - k x = -k x$$

which is exactly the horizontal equation, so  $T = 2\pi \sqrt{(m / k)}$  as well.

**The role of  $g$ .** Gravity does one job only: it decides **where** the equilibrium position is ( $e = mg/k$ ). It does not change the **stiffness** of the restoring force, and the stiffness is what sets the timing. That is why the same mass on the same spring keeps time identically lying on a bench or hanging from the ceiling — and why it would keep the same time on the Moon.

$$\text{because the weight is exactly cancelled by the initial stretch, leaving } F = -kx \text{ again}$$

**4** increasing the mass at constant  $k$

$$T = 2\pi \sqrt{m/k} \quad \rightarrow \quad T \propto \sqrt{m}$$

so a larger  $m$  gives a larger  $T$  : the vibration becomes slower.

quantitatively, multiplying  $m$  by  $n$  multiplies  $T$  by  $\sqrt{n}$  :

$$m \rightarrow 4m \quad \text{gives} \quad T \rightarrow 2T \qquad m \rightarrow 9m \quad \text{gives} \quad T \rightarrow 3T$$

The extra mass adds inertia without adding any restoring force, so the same spring needs longer to turn the body round at each end of the swing.

**the periodic time increases, in proportion to  $\sqrt{m}$**

**5** the 0.50 kg mass on a 20 N/m vertical spring

$$T = 2\pi \sqrt{m/k}$$

$$T = 6.28 \times \sqrt{0.50 / 20} = 6.28 \times \sqrt{0.025}$$

$$T = 6.28 \times 0.1581$$

$$T = 0.993 \approx 0.99 \text{ s}$$

The spring is vertical, but as question 3 showed, that changes nothing in the formula —  $g$  never appears.

**$T \approx 0.99 \text{ s}$**

**6** doubling the mass on the same spring

$$T \propto \sqrt{m} \quad (\text{the spring is unchanged})$$

$$T(\text{new}) / T(\text{old}) = \sqrt{0.80 / 0.40} = \sqrt{2}$$

so the ratio of the new periodic time to the original one is  $\sqrt{2} : 1$

$\sqrt{2} \approx 1.41$ , so the vibration becomes about 41 % slower — not twice as slow. Doubling the periodic time would need **four** times the mass.

**B)  $\sqrt{2} : 1$**

## Part C — Lesson 2–3, Classwork 2

**7** why a small swing of a pendulum is simple harmonic

resolve the weight along the string and along the tangent of the path :

along the string :  $mg \cos \theta$  ( balanced by the tension )

along the tangent :  $mg \sin \theta$  ( this is the restoring force )

so  $F = -mg \sin \theta$

for a small angle measured in radians,  $\sin \theta \approx \theta = x / L$  , so :

$$F \approx -mgx / L = - (mg / L) x \quad \rightarrow \quad F \propto -x$$

**This is the whole approximation.** The definition of simple harmonic motion is that the restoring force is proportional to the displacement and opposite to it. The exact pendulum force goes with  $\sin \theta$ , which is **not** proportional to  $\theta$  — but for small angles the two are almost equal (at  $10^\circ$ ,  $\sin \theta$  and  $\theta$  differ by only about 0.5 %). Comparing with  $F = -kx$  gives an effective  $k = mg/L$ , and substituting in  $T = 2\pi\sqrt{m/k}$  gives  $T = 2\pi\sqrt{L/g}$  — with the mass cancelling out.

**because for a small angle  $\sin \theta \approx \theta$ , which turns  $F = -mg \sin \theta$  into  $F = -(mg/L)x$**

8 what  $T$  of a simple pendulum depends on

$$T = 2\pi \sqrt{L / g}$$

the only two quantities in the formula are the length of the string  $L$  and the acceleration due to gravity  $g$ . the mass of the bob and the amplitude are absent.

**B) the length of the string and the acceleration due to gravity**

9 the periodic time for  $L = 1.8$  m

$$T = 2\pi \sqrt{L / g}$$

$$T = 2 \times 3.14 \times \sqrt{1.8 / 9.8}$$

$$T = 6.28 \times \sqrt{0.1837}$$

$$T = 6.28 \times 0.4286$$

$$T = 2.6914 \approx 2.69 \text{ s}$$

A useful benchmark to remember: a pendulum about 1 m long has a periodic time of about 2 s, so a 1.8 m pendulum taking a little under 2.7 s is exactly the right size of answer.

$$T = 2.69 \text{ s}$$

## Part D — Lesson 2–3, Home assignment 2

10 lengthening the string

$$T = 2\pi \sqrt{L / g} \quad \rightarrow \quad T \propto \sqrt{L} \quad (g \text{ fixed})$$

so a longer string gives a longer periodic time – but only as the square root, so  $L$  must be made four times larger to double  $T$ .

A longer string means a gentler arc: the restoring force per unit displacement,  $mg/L$ , becomes smaller, so the bob is pulled back more weakly and takes longer to return.

**the periodic time increases, in proportion to  $\sqrt{L}$**

11 measuring the periodic time accurately

method :

- 1 set the pendulum swinging through a **small** angle ( less than about  $10^\circ$  )
- 2 start the stopwatch as the bob passes the **lowest point**, where it is fastest
- 3 count  $n = 10$  ( or 20 ) **complete** oscillations, there and back
- 4 read the total time  $t$  , then  $T = t / n$
- 5 repeat and take the mean

**Why  $n$  oscillations?** The uncertainty of a stopwatch reading comes almost entirely from the experimenter's reaction time, roughly 0.2 s, and that error is the **same** whether you time one oscillation or twenty. Dividing the total time by  $n$  divides the error by  $n$  as well: timing 20 swings turns a 0.2 s error into a 0.01 s error. Starting at the lowest point helps for the same reason — the bob is moving fastest there, so the instant of passing is the easiest to judge.

**$T = t / n$  , timing  $n$  complete oscillations rather than a single one**

**12** 10 oscillations in 14.0 s

$$T = t / n$$

$$T = 14.0 / 10$$

$$T = 1.40 \text{ s}$$

Reading it backwards,  $t = n T$  is a quick way to predict how long an experiment will take.

**$T = 1.40 \text{ s}$  · the relation is  $T = t / n$**

**13** four times the length

$$T \propto \sqrt{L}$$

$$T(\text{new}) / T(\text{old}) = \sqrt{(4L / L)} = \sqrt{4} = 2$$

$$T(\text{new}) = 2 T(\text{old})$$

**B) twice its value**

## Part E — Weekly assessment, Group A

**14** from 1.0 m to 4.0 m

$$T \propto \sqrt{L}$$

$$\rightarrow T(\text{new}) = \sqrt{(4.0 / 1.0)} \times T = 2 T$$

**C) 2T**

**15** what  $T$  of a spring pendulum depends on

$$T = 2\pi \sqrt{(m / k)}$$

only  $m$  and  $k$  appear – not the amplitude, not  $g$ , not the weight.

**A) the mass and the spring constant**

**16** why the mass of the bob does not matter

the restoring force along the tangent is proportional to the mass :

$$F = - m g \sin \theta$$

but the acceleration it produces divides by that same mass :

$$a = F / m = - g \sin \theta \quad (\text{no } m \text{ left})$$

It is the same reason why all bodies fall freely with the same acceleration. A heavier bob is pulled back harder, but it is also that much harder to move, and the two effects cancel exactly — which is why  $T = 2\pi\sqrt{(L/g)}$  has no  $m$  in it.

**because the mass appears in both the restoring force and the inertia, and cancels out**

**17** the periodic time for  $L = 0.40 \text{ m}$

$$T = 2\pi \sqrt{(L / g)}$$

$$T = 6.28 \times \sqrt{(0.40 / 9.8)} = 6.28 \times \sqrt{0.0408}$$

$$T = 6.28 \times 0.2020$$

$$T = 1.2688 \approx 1.27 \text{ s}$$

The unit matters:  $L$  in metres and  $g$  in  $\text{m/s}^2$  make  $L/g$  a time squared, so the square root is a time in **seconds**.

**$T = 1.27 \text{ s}$**

**18** the two periodic times compared

simple pendulum :  $T = 2\pi \sqrt{L / g}$  → L and g only

spring pendulum :  $T = 2\pi \sqrt{m / k}$  → m and k only

neither of them contains the amplitude.

The clearest difference is the **mass**: it is absent from the pendulum and present in the spring. The second is **gravity**: a pendulum clock runs slow on a mountain and would run very slow on the Moon, while a spring-driven watch keeps the same time everywhere.

**the pendulum depends on L and g; the spring depends on m and k; neither depends on the amplitude**

## Part E — Weekly assessment, Group B

**19** two pendulums of 0.25 m and 1.00 m

$$T \propto \sqrt{L}$$

$$T_1 / T_2 = \sqrt{0.25 / 1.00} = \sqrt{0.25} = 0.5$$

$$T_1 : T_2 = 1 : 2 \quad \text{that is } 1/2$$

The lengths are in the ratio 1 : 4, so the periodic times are in the ratio 1 : 2 — the square root always shrinks the ratio.

**B) 1 / 2**

**20** doubling the hung mass

$$T \propto \sqrt{m} \quad \rightarrow \quad T(\text{new}) = \sqrt{2} \times T(\text{old})$$

**C) it increases by a factor of  $\sqrt{2}$**

**21** the condition, and the effect of a larger small amplitude

the condition : the angle of swing must be **small** ( about  $10^\circ$  or less ),  
so that  $\sin \theta \approx \theta$  and the restoring force becomes proportional to x.

the effect :  $T = 2\pi \sqrt{L/g}$  contains no amplitude, so within this range  
increasing the small amplitude leaves the periodic time practically unchanged.

Strictly the periodic time does creep up as the angle grows — by roughly 0.2 % at  $10^\circ$  and about 2 % at  $30^\circ$  — because  $\sin \theta$  falls behind  $\theta$ . That is precisely why the condition is written as “small angle”, and why pendulum clocks are built to swing through only a few degrees.

**a small angle of swing; and increasing the small amplitude does not change T appreciably**

**22** the 0.20 kg body on an 80 N/m horizontal spring

$$T = 2\pi \sqrt{m / k}$$

$$T = 6.28 \times \sqrt{0.20 / 80} = 6.28 \times \sqrt{0.0025}$$

$$T = 6.28 \times 0.05$$

$$T = 0.314 \text{ s}$$

**T = 0.314 s**

**23** moving the pendulum to a place of smaller  $g$

$$T = 2\pi \sqrt{L / g} \quad \rightarrow \quad T \propto 1 / \sqrt{g}$$

$g$  is in the **denominator**, so a smaller  $g$  gives a **larger**  $T$  :  
the pendulum swings more slowly and a pendulum clock would lose time.

On the Moon  $g$  is about  $1.6 \text{ m/s}^2$ , roughly one sixth of its value here, so  $\sqrt{g}$  is about 2.5 times smaller and the same pendulum would take about 2.5 times as long for each swing.

**the periodic time increases, because  $T$  is inversely proportional to  $\sqrt{g}$**

## Part E — Weekly assessment, Group C

**24** four times the spring constant

$$T = 2\pi \sqrt{m / k} \quad \rightarrow \quad T \propto 1 / \sqrt{k}$$

$$T(\text{new}) = T / \sqrt{4} = T / 2$$

A stiffer spring pulls back harder for the same displacement, so the body is turned round sooner and the vibration is quicker.

**C)  $T / 2$**

**25** the change that does not matter

$$T = 2\pi \sqrt{L / g}$$

$L$  appears,  $g$  appears – and a different place on the Earth changes  $g$  slightly.  
the mass of the bob appears nowhere in the formula.

**C) changing the mass of the bob**

**26** why the static extension does not enter  $T$

at the new equilibrium position :  $ke = mg$   
measuring  $x$  from that position, the resultant force is :

$$F = mg - k(e + x) = -kx$$

the terms  $mg$  and  $ke$  cancel, so  $e$  has disappeared from the equation of motion,  
and  $T = 2\pi \sqrt{m / k}$  contains no  $e$ .

The static extension tells you **where** the body hangs at rest; the spring constant tells you **how quickly** it comes back.  
Only the second of those sets the timing. Notice, though, that  $e$  is a handy way of **measuring**  $k$  in the laboratory:  $k = mg/e$ .

**because  $mg$  and  $ke$  cancel at the equilibrium position, leaving  $F = -kx$  with no  $e$  in it**

**27** the periodic time for  $L = 0.90 \text{ m}$

$$T = 2\pi \sqrt{L / g}$$

$$T = 6.28 \times \sqrt{0.90 / 9.8} = 6.28 \times \sqrt{0.0918}$$

$$T = 6.28 \times 0.3030$$

$$T = 1.9031 \approx 1.90 \text{ s}$$

**$T = 1.90 \text{ s}$**

**28** what to change in order to double T

simple pendulum :

$$T \propto \sqrt{L} \quad \rightarrow \quad 2 T \text{ needs } \sqrt{(L' / L)} = 2 \quad \rightarrow \quad L' = 4 L$$

spring pendulum ( k fixed ) :

$$T \propto \sqrt{m} \quad \rightarrow \quad 2 T \text{ needs } \sqrt{(m' / m)} = 2 \quad \rightarrow \quad m' = 4 m$$

Both answers are “four times” for the same reason — the quantity you are changing sits under a square root, so you must multiply it by the **square** of the factor you want in T.

**make the length of the pendulum four times as large; make the mass on the spring four times as large**



## Solutions — step by step

**One equation, three faces.**  $v = f\lambda$ ,  $f = 1/T$  and  $v = \lambda/T$  are the same statement written three ways. Read the question, write down which two quantities you were handed, and the third one follows in a single line.

### Part A — Classwork 1

- 1 the frequency for a periodic time of 0.20 s

$$f = 1 / T$$

$$f = 1 / 0.20$$

$$f = 5 \text{ Hz}$$

The frequency is simply how many complete oscillations happen each second. One oscillation every fifth of a second means five of them every second.

$$f = 5 \text{ Hz}$$

- 2 the speed of a wave with  $\lambda = 0.2 \text{ m}$  and  $f = 0.3 \text{ Hz}$

$$v = f \lambda$$

$$v = 0.3 \times 0.2$$

$$v = 0.06 \text{ m/s}$$

Each second the source sends out 0.3 of a wavelength, and each wavelength is 0.2 m long, so the front of the wave advances 0.06 m in that second. That is what  $v = f\lambda$  is saying.

$$v = 0.06 \text{ m/s}$$

### Part B — Home assignment 1

- 3 what a mechanical wave transports

each particle of the medium vibrates about its own fixed position and stays there.  
what moves forward is the disturbance, and with it the **energy** of the vibration.

A cork floating on a ripple bobs up and down, it does not travel across the pond. The matter stays put; only the energy is passed on from one particle to the next.

**B) energy**

- 4 the frequency for  $T = 0.25 \text{ s}$

$$f = 1 / T = 1 / 0.25 = 4.0 \text{ Hz}$$

**C) 4.0 hertz**

5 a wave on a string,  $T = 0.10$  s and  $\lambda = 1.5$  m

$$v = \lambda / T$$

$$v = 1.5 / 0.10$$

$$v = 15 \text{ m/s}$$

You could also find  $f = 1/0.10 = 10$  Hz first, then  $v = f\lambda = 10 \times 1.5 = 15$  m/s. The two routes are the same equation.

$$v = 15 \text{ m/s}$$

6 a pulse of  $v = 4.0$  m/s and  $\lambda = 0.80$  m

$$v = f \lambda \quad \rightarrow \quad f = v / \lambda$$

$$f = 4.0 / 0.80 = 5 \text{ Hz}$$

$$T = 1 / f = 1 / 5$$

$$T = 0.20 \text{ s}$$

$$f = 5 \text{ Hz}, T = 0.20 \text{ s}$$

7 the wavelength of the longitudinal wave from the figure

in a longitudinal wave the wavelength is the distance between the centres of two **successive compressions** ( or two successive rarefactions ).

from the figure, the two compression centres are 8 large squares apart :

$$\lambda = 8 \times 1.0 \text{ cm}$$

$$\lambda = 8.0 \text{ cm} = 0.080 \text{ m}$$

**Note on the printed figure.** The figure in the book is drawn on a squared grid but no numerical scale is printed beside it, so the question cannot be answered from the book as it stands. The figure has been redrawn here with the scale stated explicitly (one large square = 1.0 cm). The physics is unchanged: find two successive compressions, count the squares between their centres, and multiply by the scale. Measuring between a compression and the **next rarefaction** would give only half a wavelength — that is the usual mistake.

$$\lambda = 8.0 \text{ cm} = 0.080 \text{ m}$$

## Part C — Classwork 2

8  $\lambda = 0.60$  m and  $T = 0.20$  s

a)  $f = 1 / T$

$$f = 1 / 0.20 = 5 \text{ Hz}$$

b)  $v = f \lambda$

$$v = 5 \times 0.60$$

$$v = 3.0 \text{ m/s}$$

Check it the short way:  $v = \lambda/T = 0.60/0.20 = 3.0$  m/s, the same answer without finding  $f$  at all.

$$\text{a) } f = 5 \text{ Hz} \quad \text{b) } v = 3.0 \text{ m/s}$$

9 a source of  $T = 0.25$  s producing  $\lambda = 1.2$  m

$$v = \lambda / T$$

$$v = 1.2 / 0.25$$

$$v = 4.8 \text{ m/s}$$

Notice that the source decides the **frequency** and the medium decides the **speed**; the wavelength is then whatever  $v/f$  happens to be.

$$v = 4.8 \text{ m/s}$$

## Part D — Home assignment 2

10 which graph gives  $A$  and  $\lambda$

the  $y - x$  graph is a **snapshot** of the whole medium at one instant, so :

the height of a crest above the axis  $\rightarrow$  the amplitude  $A$

the distance between two crests  $\rightarrow$  the wavelength  $\lambda$

the  $y - t$  graph follows one single particle, so it gives  $A$  and  $T$ , not  $\lambda$ .

The two graphs look identical, and that is exactly the trap. Always read the **horizontal axis** first: if it is a distance, the repeat length is  $\lambda$ ; if it is a time, the repeat length is  $T$ .

**B) the displacement – position graph (  $y - x$  )**

11 the vibration of the particles in a longitudinal wave

longitudinal  $\rightarrow$  each particle vibrates **along** the same line on which the wave travels, producing compressions and rarefactions ( sound is the example ).

transverse  $\rightarrow$  each particle vibrates **across** that line ( a rope, light ).

**B) in a direction parallel to the direction of propagation of the wave**

12 the wavelength of a 500 Hz sound wave

$$v = f \lambda \quad \rightarrow \quad \lambda = v / f$$

$$\lambda = 340 / 500$$

$$\lambda = 0.68 \text{ m}$$

This is about the length of your forearm — and it is why a 500 Hz sound bends easily round a doorway, while light, with a wavelength a million times smaller, does not.

$$\lambda = 0.68 \text{ m}$$

13 reading the speed from the two graphs

the  $y - x$  graph gives  $\lambda = 3.0$  m ( and  $A = 0.50$  m )

the  $y - t$  graph gives  $T = 0.60$  s

$$v = \lambda / T$$

$$v = 3.0 / 0.60$$

$$v = 5.0 \text{ m/s}$$

The amplitude 0.50 m is not needed for the speed — it is there to see whether you know that the amplitude never enters  $v = f\lambda$ .

$$v = 5.0 \text{ m/s}$$

**14** P and Q a quarter of a cycle apart

from the figure, P is at the equilibrium level and Q is at the next crest, so PQ is a **quarter** of a complete wave :

a)  $\lambda / 4 = 2 \text{ m}$   
 $\lambda = 4 \times 2 = 8 \text{ m}$

b)  $T / 4 = 0.1 \text{ s} \quad \rightarrow \quad T = 0.4 \text{ s}$   
 $v = \lambda / T = 8 / 0.4$   
 $v = 20 \text{ m/s}$

The quarter-cycle is the whole question. From the equilibrium level to the next crest is a quarter of a wave, from crest to the next crest is a whole wave, and from crest to trough is half a wave. Get that right and both parts follow at once. As a check,  $f = 1/0.4 = 2.5 \text{ Hz}$  and  $v = f\lambda = 2.5 \times 8 = 20 \text{ m/s}$ .

**a)  $\lambda = 8 \text{ m}$    b)  $v = 20 \text{ m/s}$**



## Solutions — step by step

The same three lines solve almost every problem here. (1) Find the path difference  $\Delta$ . (2) Divide by  $\lambda/2$  to count the half-wavelengths. (3) Even  $\rightarrow$  constructive, odd  $\rightarrow$  destructive — then reverse the verdict if the two sources are in antiphase.

### Part A — Classwork 1

1 path difference for  $\lambda = 4.0$  cm

a)  $\Delta = S_2P - S_1P$

$$\Delta = 16.0 - 10.0$$

$$\Delta = 6.0 \text{ cm}$$

b)  $\lambda / 2 = 4.0 / 2 = 2.0$  cm

$$\text{number of half-wavelengths} = 6.0 / 2.0 = 3 \quad (\text{an odd number})$$

$$\text{equivalently } \Delta = 1.5 \lambda = (1 + \frac{1}{2}) \lambda \rightarrow \text{destructive}$$

Three half-wavelengths means the two waves arrive exactly out of step: a crest of one meets a trough of the other, and if the amplitudes are equal the point stays permanently still.

a)  $\Delta = 6.0$  cm    b) 3 half-wavelengths  $\rightarrow$  destructive interference

2 the destructive lines between two sources 8.0 cm apart

a destructive line needs an **odd** number of half-wavelengths :

$$\lambda / 2 = 2.0 / 2 = 1.0 \text{ cm}$$

$$\Delta = 1.0, 3.0, 5.0, 7.0 \text{ cm} \quad (\text{odd multiples of } 1.0 \text{ cm})$$

between the two sources the path difference can only run from 0 up to  $d = 8.0$  cm, so  $\Delta = 9.0$  cm and beyond is impossible. that gives 4 values on one side of the centre line, and 4 mirror values on the other side :

$$\text{total} = 4 + 4 = 8 \text{ destructive lines}$$

A quick formula for in-phase sources: the number of nodal lines is  $2 \times$  the whole-number part of  $(d/\lambda + \frac{1}{2}) = 2 \times \text{int}(4 + 0.5) = 8$ . Note that the central line, where  $\Delta = 0$ , is always a **constructive** line, never a nodal one.

8 destructive (nodal) lines

### Part B — Home assignment 1

3 the condition for constructive interference

the two waves must arrive in step, so the extra path must hold a **whole** number of wavelengths :

$$\Delta = m \lambda \quad (m = 0, 1, 2, 3 \dots)$$

Choices (A) and (C) are the same thing written two ways, and both are the condition for **destructive** interference, not constructive.

B)  $m \times \lambda$

**4** the principle of superposition

where two waves meet, the displacement of the medium at each instant is the **algebraic (vector) sum** of the two displacements the waves would produce separately – taking the signs into account.

Two equal crests give  $2A$ ; a crest meeting an equal trough gives  $A - A = 0$ . That single rule of signs is the whole of interference.

**B) the vector sum of their two individual displacements**

**5** the resultant amplitude at P for  $\lambda = 5.0$  cm

$$\Delta = 37.5 - 30.0 = 7.5 \text{ cm}$$

$$\Delta / \lambda = 7.5 / 5.0 = 1.5 \quad \rightarrow \quad \Delta = ( 1 + \frac{1}{2} ) \lambda$$

$$\text{number of half-wavelengths} = 7.5 / 2.5 = 3 \quad ( \text{ odd } )$$

a) sources in phase  $\rightarrow$  destructive

$$A(\text{res}) = | A_1 - A_2 | = | 1.0 - 1.0 | = 0$$

b) sources in antiphase  $\rightarrow$  the verdict is reversed  $\rightarrow$  constructive

$$A(\text{res}) = A_1 + A_2 = 1.0 + 1.0 = 2.0 \text{ cm}$$

Part (b) is the whole point of the question. Putting the sources in antiphase adds an extra half-wavelength of its own to the journey, so every point that was quiet becomes loud and every point that was loud becomes quiet — the whole pattern simply swaps over.

**a)  $A = 0$    b)  $A = 2.0$  cm**

**6** path difference and amplitude for  $\lambda = 6.0$  cm

a)  $\Delta = 35.0 - 20.0$

$$\Delta = 15.0 \text{ cm}$$

b)  $\lambda / 2 = 3.0 \text{ cm}$

$$\text{number of half-wavelengths} = 15.0 / 3.0 = 5 \quad ( \text{ odd } )$$

$\rightarrow$  destructive , and the amplitudes are equal :

$$A(\text{res}) = 0.40 - 0.40 = 0$$

Check it the other way as well:  $15.0 / 6.0 = 2.5$  wavelengths, and any “point five” answer is always destructive for in-phase sources.

**a)  $\Delta = 15.0$  cm   b)  $A(\text{res}) = 0$  (destructive)**

**7** the superposition at P and at Q

read the figure :

at P a crest of  $S_1$  falls on a crest of  $S_2$  ( P lies on the centre line,  
so the two paths are equal and  $\Delta = 0$  )

at Q a crest of one source falls on a trough of the other

**At P.** Two solid (crest) circles cross there, so the two waves arrive in step and the displacements add: the water rises to  $2A$  and then falls to  $-2A$ . P lies on the perpendicular bisector of  $S_1S_2$ , where  $\Delta = 0$  — the central bright line of every interference pattern.

**At Q.** A solid circle of one source crosses a dashed (trough) circle of the other, so the path difference is an odd number of half-wavelengths. The crest of one wave is cancelled by the trough of the other, and if the amplitudes are equal the point Q stays permanently at rest.

**P is a point of constructive interference; Q is a point of destructive interference**

## Part C — Classwork 2

**8** path difference and half-wavelengths for  $\lambda = 6.0$  cm

$$\Delta = 37.0 - 28.0 = 9.0 \text{ cm}$$

$$\lambda / 2 = 3.0 \text{ cm}$$

$$\text{number of half-wavelengths} = 9.0 / 3.0 = 3 \quad (\text{odd} \rightarrow \text{destructive})$$

The amplitude 0.50 cm was not needed here — but if it were asked for, equal amplitudes and destructive interference would give a resultant of zero.

**$\Delta = 9.0$  cm , containing 3 half-wavelengths  $\rightarrow$  destructive interference**

**9** constructive or destructive for  $\lambda = 2.5$  cm

$$\lambda / 2 = 1.25 \text{ cm}$$

$$\text{number of half-wavelengths} = 5.0 / 1.25 = 4 \quad (\text{even})$$

or, more directly :

$$\Delta / \lambda = 5.0 / 2.5 = 2 \quad (\text{a whole number of wavelengths})$$

The separation of 10 cm is not needed for the verdict; it only limits how far out the pattern can extend.

**P is a point of constructive interference, containing 4 half-wavelengths**

## Part D — Home assignment 2

**10** how many wavelengths are in 6.0 cm

$$\Delta / \lambda = 6.0 / 4.0 = 1.5$$

And 1.5 wavelengths is  $(1 + \frac{1}{2})\lambda$ , so this point is a destructive one — a useful thing to notice even though the question only asked for the number.

**A) 1.5 wavelengths**

**11** the definition of a node

a standing wave is the interference of two identical waves travelling in opposite directions. at a **node** the two always arrive out of step, so they cancel :

the displacement stays zero at all times

The opposite point is the **antinode**, where the two waves always arrive in step and the displacement swings between  $+2A$  and  $-2A$ . A node and the next antinode are  $\lambda/4$  apart.

**B) a point of a displacement equal to zero**

**12** the resultant amplitude at Q for  $\lambda = 3.0$  cm

$$\Delta / \lambda = 9.0 / 3.0 = 3 \quad (\text{a whole number} \rightarrow \text{constructive})$$

$$A(\text{res}) = A_1 + A_2 = 0.40 + 0.40$$

$$A(\text{res}) = 0.80 \text{ cm}$$

Counting half-wavelengths gives the same verdict:  $9.0 / 1.5 = 6$ , an even number.

**A(res) = 0.80 cm**

**13** antiphase sources with  $\Delta = 8.0$  cm

$$\Delta / \lambda = 8.0 / 4.0 = 2 \quad (\text{a whole number of wavelengths})$$

for sources **in phase** that would be constructive.

but the sources here are in **antiphase**, which adds an extra half a wave to the difference, so the verdict is turned over :

R is a point of **destructive** interference

A neat way to keep it straight: total difference = path difference + source phase difference. Here  $2\lambda$  from the path plus  $\lambda/2$  from the sources gives  $2.5\lambda$  — an odd number of half-wavelengths, so destructive.

**R undergoes destructive interference**

## Part E — Standing waves

**14** the air particles near the point X

X lies where the two envelope curves cross, so X is a **node** :

the two waves travelling in opposite directions always arrive there out of step, and their displacements cancel at every instant.

**Displacement and pressure are out of step.** At a displacement node the air layers on either side move towards X together and then away from X together, so the air at X itself never moves along the tube — but it is alternately squeezed and stretched, which makes the node a point of **maximum pressure variation**. The opposite holds at an antinode: the air there swings backwards and forwards with the greatest amplitude, while the pressure hardly changes. In a pipe closed at one end the closed end is always a displacement node and the open end is always an antinode.

**X is a node: the air particles there stay practically at rest, with zero displacement amplitude — while the pressure variation there is at its maximum**

## Part F — Review, choose the correct answer

15 tension and the speed of a pulse

for a stretched string :  $v = \sqrt{F / \mu}$

a larger tension  $F$  gives a larger  $v$  (  $v \propto \sqrt{F}$  ).

B) it increases

16 a larger amplitude at constant frequency

the energy carried by a wave is proportional to the **square** of the amplitude :

$$E \propto A^2$$

the speed and the wavelength are set by the medium and the frequency, not by  $A$ .

B) an increase in the energy transported by the wave

17 why electromagnetic waves cross a vacuum

a mechanical wave is a disturbance passed from particle to particle, so it needs a material medium. an electromagnetic wave is an oscillating electric and magnetic field, which needs no medium at all.

B) electromagnetic waves do not require material media

18 why P-waves arrive first

P-waves are **longitudinal** ( compressions and rarefactions ) and travel faster than the **transverse** S-waves through the rocks of the Earth.

It is the delay between the two arrivals that lets a seismologist work out how far away the earthquake was.

B) P-waves are longitudinal and travel with a greater speed than S-waves

19 which pair is transverse only

light → transverse ( electromagnetic )

a wave on a stretched string → transverse

sound → **longitudinal**, so any pair containing sound is ruled out.

B) light waves and waves along a stretched string

20 the wave equation

$$v = f \times \lambda$$

B)  $v = f \times \lambda$

21 the phase difference for destructive interference

the two waves must arrive exactly out of step : crest against trough.

that is half a cycle, which is a phase difference of  $180^\circ$  (  $\pi$  radians ).

B)  $180^\circ$  ( out of phase / in antiphase )

**22** regions of the largest resultant displacement

the displacements add instead of cancelling, which is **constructive** interference.

**B) constructive interference**

**23** how a standing wave is formed

two waves of the **same frequency and the same amplitude** travelling in **opposite** directions along the same line interfere and produce a pattern of fixed nodes and antinodes that does not travel.

**B) the same frequency and amplitude, travelling in two opposite directions**

**24** the distance between two successive nodes

in a standing wave :

$$\text{node to node} = \lambda / 2$$

$$\text{antinode to antinode} = \lambda / 2$$

$$\text{node to the next antinode} = \lambda / 4$$

**B) half a wavelength**

**25** the maximum resultant amplitude

at a constructive point the two displacements add :

$$A(\text{res}) = A + A = 2A$$

And at a destructive point with equal amplitudes the result is  $A - A = 0$ , which is why choice (B) is there.

**C) 2A**

**26** raising the frequency in a ripple tank

the speed is fixed by the water, so :

$$f \text{ increases} \rightarrow \lambda = v / f \text{ decreases}$$

the spacing of the nodal lines is proportional to  $\lambda$ , so it becomes smaller.

This is exactly why a higher-pitched sound gives a finer interference pattern, and why blue light gives narrower fringes than red light in Young's experiment.

**B) it decreases**

## Part G — Review, answer the following

**27** tension and the speed of a pulse on a rope

$$v = \sqrt{(F / \mu)} \rightarrow v \propto \sqrt{F}$$

Four times the tension gives only twice the speed.

**the speed increases — in proportion to the square root of the tension**

**28** doubling the amplitude

$$E \propto A^2$$

$$A \rightarrow 2A \quad \text{gives} \quad E \rightarrow (2)^2 E = 4 E$$

This is why a sound twice as “loud” in amplitude carries four times the energy.

**the energy transported becomes four times as large**

**29** doubling the frequency of an oscillator

$$T = 1 / f$$

$$f \rightarrow 2f \quad \text{gives} \quad T \rightarrow T / 2$$

**the periodic time is halved**

**30** sound entering a perfect vacuum

sound is a **mechanical** wave : it needs particles of matter to pass the disturbance along. a perfect vacuum has none, so the wave cannot propagate at all and simply stops at the boundary.

This is the classic bell-in-a-vacuum-jar demonstration: as the air is pumped out the bell is seen to be ringing but is no longer heard.

**the wave does not travel at all — its speed in vacuum is zero, because sound needs a material medium**

**31** doubling the source frequency in the same medium

the speed of a wave is fixed by the **medium**, not by the source :

$v$  stays the same

what changes is the wavelength :

$$\lambda = v / f \quad \rightarrow \quad f \rightarrow 2f \quad \text{gives} \quad \lambda \rightarrow \lambda / 2$$

A common mistake is to think a “faster vibration” makes a faster wave. It does not — it makes a **shorter** one.

**the speed stays without change; it is the wavelength that halves**

**32** the particle after the wave has passed

each particle of the medium only vibrates about its own fixed position; it is not carried along. when the wave has gone by and the medium settles :

the amplitude falls back to zero , and the total displacement = zero

The wave transports **energy** across the medium, never matter. A floating cork bobs and then rests exactly where it started.

**the particle returns to its original position — its total displacement is zero**

**33** tightening a plucked string

$$v = \sqrt{F / \mu} \quad \text{increases}$$

$$f = v / (2 L) \quad \text{increases as well ( } L \text{ unchanged )}$$

It is precisely what a guitarist does when tuning: tightening the peg raises the pitch, loosening it lowers the pitch.

**the pitch rises — the note becomes sharper**

**34** the amplitude at a node

at a node the two waves always arrive out of step and cancel exactly :  
amplitude = 0 at all times

**the amplitude of vibration at a node is zero — the point stays permanently at rest**

**35** turning the sources into antiphase

changing the sources to antiphase adds an extra half a wavelength to every journey, so every condition is turned over :

points that were constructive become destructive

points that were destructive become constructive

The **spacing** of the lines does not change, because  $\lambda$  and the separation of the sources are untouched; only the labels on them are exchanged. In particular the centre line, which used to be the brightest, becomes a nodal line.

**the two sets of positions swap over completely — the whole pattern is inverted**

**36** a 250 Hz tuning fork

$$v = f \lambda \quad \rightarrow \quad \lambda = v / f$$

$$\lambda = 340 / 250 = 1.36 \text{ m}$$

$$T = 1 / f = 1 / 250$$

$$T = 0.004 \text{ s} = 4 \times 10^{-3} \text{ s}$$

**$\lambda = 1.36 \text{ m}$  ,  $T = 4 \times 10^{-3} \text{ s}$**

**37** 120 waves in 30 s

a)  $f = \text{number of waves} / \text{time}$

$$f = 120 / 30$$

$$f = 4 \text{ Hz}$$

b)  $T = 1 / f = 1 / 4$

$$T = 0.25 \text{ s}$$

The 0.5 m between successive crests is the wavelength; it is not needed for either part here, but it would give  $v = f\lambda = 4 \times 0.5 = 2.0 \text{ m/s}$ .

**a)  $f = 4 \text{ Hz}$    b)  $T = 0.25 \text{ s}$**

**38** 100 waves in 20 s with  $\lambda = 0.4 \text{ m}$

$$f = 100 / 20 = 5 \text{ Hz}$$

$$v = f \lambda = 5 \times 0.4$$

$$v = 2.0 \text{ m/s}$$

**$v = 2.0 \text{ m/s}$**

**39** the listener between two loudspeakers

$$\Delta / \lambda = 1.25 / 0.50 = 2.5$$

$$\Delta = ( 2 + \frac{1}{2} ) \lambda \quad ( \text{an odd number of half-wavelengths : } 5 )$$

→ the two sounds arrive exactly out of step and cancel

Walking a little to one side would bring the listener to a place where  $\Delta$  is a whole number of wavelengths, and the sound would become loud again. This is why the loud and quiet patches in a hall move as you walk about.

**the listener hears a faint sound — destructive interference**

**40** P with  $\Delta = 12.5$  cm and  $\lambda = 5.0$  cm

$$\Delta / \lambda = 12.5 / 5.0 = 2.5$$

or, counting half-wavelengths :

$$\lambda / 2 = 2.5 \text{ cm} \quad \text{and} \quad 12.5 / 2.5 = 5 \quad ( \text{an odd number} )$$

For sources vibrating in phase, any path difference ending in “.5 of a wavelength” is destructive, and any whole number of wavelengths is constructive.

**P is a point of destructive interference, because the path difference contains 5 half-wavelengths (  $\Delta = 2.5\lambda$  )**



## Solutions — step by step

Start every refraction question the same way. Write  $f_1 = f_2$ . Then  $v = f\lambda$  forces  $\lambda$  to follow the speed: a slower medium means a shorter wavelength, and the ray bends **towards** the normal.

### Part A — Classwork 1

1 sea waves running into shallow water

a) **The three changes.** The speed of a water wave depends on the **depth**: the shallower the water the slower the wave, so on moving inshore **v decreases**. The frequency is set by whatever produced the waves far out at sea, so **f does not change**. Since  $v = f\lambda$  and  $f$  is fixed, the wavelength must fall in the same proportion as the speed:  **$\lambda$  decreases**, which is why the crests are drawn closer and closer together in the figure. The **direction** swings round until the crests are almost parallel to the shore line.

b) **Why they bend.** A crest arriving obliquely does not meet the shallow water all at once. The end of the crest that is already in shallow water slows down, while the other end is still in deeper water and keeps moving faster. The faster end therefore swings forward about the slower one, exactly as a marching line wheels about its inner end, and the crest is turned until it lies along the depth contours — that is, along the shore line.

c) **The concept.** This is **refraction**: the change in the direction of a wave when it passes obliquely from one medium into another in which its speed is different, its frequency staying the same.

a) **v decreases,  $\lambda$  decreases, f stays the same, and the direction bends towards the shore** b) **because the shallow end of each crest slows down first** c) **refraction**

2 what changes from deep to shallow water

depth decreases  $\rightarrow$  v decreases

f is fixed by the source, so from  $v = f\lambda$  :

$$\lambda = v / f \quad \rightarrow \quad \lambda \text{ decreases as well}$$

and the oblique crests swing round towards the shore line.

Choice (D) is the tempting wrong one: it gets the speed right but then makes  $\lambda$  increase, which would need the frequency to fall — and the frequency cannot change.

**B) the speed decreases, the wavelength decreases, and the direction bends to become parallel to the shore line**

### Part B — Home assignment 1

### 3 the headland compared with the bay

the water at the headland is **shallower**, so the waves there are slower :

$$v(\text{headland}) < v(\text{bay})$$

the frequency is the same everywhere ( the same source out at sea ) :

$$f(\text{headland}) = f(\text{bay})$$

so from  $\lambda = v / f$  :

$$\lambda(\text{headland}) < \lambda(\text{bay})$$

Look at the figure: because the crests slow down over the shallow water off the headland, they bend round it from both sides. The rays therefore **converge** on the headland and **diverge** in the bay. That is why headlands are attacked and worn away by the sea while bays stay calm and collect sand — the wave energy per metre of coast is far greater at the headland.

| property             | at the headland (shallow) | at the bay (deeper) |
|----------------------|---------------------------|---------------------|
| wave speed $v$       | smaller                   | larger              |
| wavelength $\lambda$ | shorter                   | longer              |
| frequency $f$        | the same                  | the same            |
| wave energy          | concentrated              | spread out          |

**the headland has the smaller  $v$  and the shorter  $\lambda$ , and the same  $f$  as the bay**

### 4 what refraction changes and what it does not

the source sets the frequency, and the boundary cannot create or destroy cycles :

$$f_1 = f_2 \quad (\text{the frequency never changes})$$

the new medium sets a new speed, so from  $v = f \lambda$  with  $f$  fixed :

$$v \text{ changes} \rightarrow \lambda \text{ changes by the same factor}$$

Think of the boundary as a turnstile: the same number of crests arrive every second as leave every second, otherwise crests would pile up at the boundary. That is the whole reason why  $f$  cannot change.

**B) the speed of the wave and its wavelength change, while its frequency stays constant**

### 5 which frequency is the largest

the wave equation  $v = f \lambda$  can be read as the equation of a straight line :

$$v = f \times \lambda \rightarrow (y = \text{slope} \times x)$$

so for each source the graph of  $v$  against  $\lambda$  is a straight line through the origin whose **slope is the frequency**.

the steepest of the three lines is the one belonging to  $f_3$ .

Read one point off any line to get a number: if a line passes through  $\lambda = 4.0 \text{ m}$  at  $v = 200 \text{ m/s}$ , then  $f = v/\lambda = 50 \text{ Hz}$ . The three lines all pass through the origin, which is the graphical way of saying that a wave of zero wavelength would have zero speed.

**$f_3$  has the largest value, because its line has the greatest slope and the slope of a  $v - \lambda$  graph is the frequency**

**6** the name of the locus of points in the same phase

all the points of a **wavefront** have been disturbed by the source for the same length of time, so they are all at the same stage of their vibration. near a point source the wavefronts are circles ( or spheres ) ; far away they are practically straight lines ( or planes ). the ray is always  $\perp$  the wavefront.

**D) the wavefront**

**7** the two ratios at the boundary

from the figure :  $i = 45^\circ$  in medium 1  $r = 30^\circ$  in medium 2

a) the frequency is set by the source and cannot change at a boundary :

$$f_1 / f_2 = 1$$

b) since  $f$  is the same,  $\lambda \propto v$ , and the law of refraction gives :

$$\lambda_1 / \lambda_2 = v_1 / v_2 = \sin i / \sin r$$

$$\lambda_1 / \lambda_2 = \sin 45^\circ / \sin 30^\circ = 0.7071 / 0.5$$

$$\lambda_1 / \lambda_2 = 1.41 \quad ( = \sqrt{2} )$$

The ray bends **towards** the normal ( $45^\circ \rightarrow 30^\circ$ ), so medium 2 is the slower, optically denser one — and a slower medium always carries the shorter wavelength, which is exactly what the ratio 1.41 says.

**a)  $f_1 : f_2 = 1 : 1$    b)  $\lambda_1 : \lambda_2 = \sqrt{2} : 1 \approx 1.41$**

**8** which statement is not correct

check each statement against  $f_1 = f_2 = 50 \text{ Hz}$  and  $\lambda = v / f$  :

$$\lambda_1 = 300 / 50 = 6 \text{ m}$$

$$\lambda_2 = 600 / 50 = 12 \text{ m}$$

(A)  $f$  in medium 2 = 50 Hz ✓ correct

(B)  $\lambda_1 = 6 \text{ m} = \text{half of } \lambda_2 = 12 \text{ m}$  ✓ correct

(D)  $n = c / v$ , so  $n_1 / n_2 = v_2 / v_1 = 600/300 = 2$  ✓ correct

(C) the law of refraction gives

$$\sin i / \sin r = v_1 / v_2 = 1 / 2, \text{ that is } \sin r = 2 \sin i$$

which is **not** the same as  $i = r / 2$  ✗ wrong

The relation between the two angles involves their **sines**, not the angles themselves. Test it with numbers: for  $i = 20^\circ$ ,  $\sin r = 2 \sin 20^\circ = 0.684$ , so  $r = 43.2^\circ$  — not  $40^\circ$ .

**C) — the statement that the angle of incidence equals half the angle of refraction**

**9** entering a medium of smaller refractive index

$$n_1 > n_2 \quad \text{and} \quad n = c / v \quad \rightarrow \quad v_1 < v_2$$

so the ray speeds up on entering medium 2, which means it bends **away** from the normal. with  $f$  unchanged,  $\lambda = v / f$  increases as well.

Choice (C) says the frequency decreases — the one thing refraction can never do. Remember the pair: **into a denser medium** means slower, shorter  $\lambda$ , bending towards the normal; **into a less dense medium** means faster, longer  $\lambda$ , bending away from it.

**B) the ray is refracted away from the normal, and its speed and wavelength increase**

**10** comparing the two materials A and B

apply the law of refraction between the two materials :

$$n(A) \sin i = n(B) \sin r$$

$$n(A) \sin 30^\circ = n(B) \sin 45^\circ$$

$$n(A) \times 0.5 = n(B) \times 0.7071$$

$$n(A) / n(B) = 0.7071 / 0.5 = 1.41 \quad \rightarrow \quad n(A) > n(B)$$

and since  $v = c / n$  :

$$\text{a larger } n \text{ means a smaller } v \quad \rightarrow \quad v(A) < v(B)$$

A quick sanity check without any algebra: the ray bends **away** from the normal on going from A into B ( $30^\circ \rightarrow 45^\circ$ ), so it must have **sped up** — B is the faster, less dense material. Everything else follows from that one observation.

**A) the refractive index of (A) is greater than that of (B), and the speed of light in (A) is less than in (B)**



## Solutions — step by step

**Two rules carry the whole lesson.** For interference, divide the path difference by  $\lambda/2$ : an even answer is loud, an odd answer is quiet. For Doppler, ask one question only — is the gap between source and listener **closing** or **opening**? Closing means a higher pitch, opening means a lower one.

### Part A — Classwork 2

#### 1 walking across the courtyard

the two sources are coherent and in phase, so at each point the path difference decides what is heard :

$$\Delta d = m\lambda \quad \rightarrow \quad \text{the two waves arrive in step} \quad \rightarrow \quad \text{loud}$$

$$\Delta d = (m + \frac{1}{2})\lambda \quad \rightarrow \quad \text{they arrive out of step} \quad \rightarrow \quad \text{quiet}$$

as you walk,  $\Delta d$  changes steadily, so loud and quiet places alternate.

This is the single most convincing classroom demonstration that sound is a **wave**: two loudspeakers, both switched on, can produce places that are **quieter** than one loudspeaker alone. Nothing made of particles could do that.

**C) a repeated pattern of rising and falling sound intensity**

#### 2 constructive compared with destructive

both depend on the same quantity — the path difference  $\Delta d$  — and on nothing else, as long as the two sources are coherent and vibrate in the same phase.

Note that energy is not destroyed at the quiet places. It is simply redistributed: whatever is missing at the quiet points turns up at the loud points, so the total energy in the room is exactly what the two loudspeakers put into it.

|                                | constructive interference                  | destructive interference                                    |
|--------------------------------|--------------------------------------------|-------------------------------------------------------------|
| the sound heard                | loud — maximum intensity                   | faint or silent — minimum intensity                         |
| the path difference $\Delta d$ | $m\lambda$ ( a whole number of $\lambda$ ) | $(m + \frac{1}{2})\lambda$ ( an odd number of $\lambda/2$ ) |
| the two waves arrive           | in step — crest on crest                   | out of step — crest on trough                               |
| the resultant amplitude        | $A_1 + A_2$                                | $ A_1 - A_2 $                                               |

**a) loud for constructive, faint for destructive    b)  $\Delta d = m\lambda$  for constructive,  $(m + \frac{1}{2})\lambda$  for destructive**

### Part B — Home assignment 2

### 3 the dead seats in a concert hall

the two loudspeakers are fed by the same amplifier, so they are **coherent** and in phase. a seat at which the distances from the two loudspeakers differ by

$$\Delta d = (m + \frac{1}{2}) \lambda$$

receives the two waves exactly out of step, and they cancel each other.

The effect is worst for low notes, whose wavelengths are metres long, so the quiet patches are large enough to swallow a whole seat. Concert halls are designed with this in mind — delaying one channel slightly, or scattering the sound from the walls, breaks the coherence and washes the dead spots out.

**because those seats lie on lines of destructive interference between the two loudspeakers**

### 4 Quincke's tube

the sound entering at P is split into two paths – the fixed arm A and the sliding arm B – and the two parts meet again at Q.

pulling B out by a distance X lengthens that path by **2X** ( out and back ), so the path difference becomes  $\Delta d = 2X$ , and :

$$\text{first minimum : } 2X = \lambda / 2 \quad \rightarrow \quad \lambda = 4 X$$

**d) How it works.** A single source of sound is sent in at P. The tube splits it into two parts that travel along the two arms and are brought together again at a detector (or the ear) at Q. Because both parts come from the same source they are perfectly coherent. Sliding B out changes the length of one path only, so the path difference  $\Delta d = 2X$  can be dialled to any value. The listener hears the sound rise and fall; at the **first** minimum  $2X = \lambda/2$ , so  $\lambda = 4X$ , and successive minima come every further half wavelength, that is every extra  $\lambda/4$  of pull. Knowing  $\lambda$  and either f or v gives the other from  $v = f\lambda$ .

**a) Quincke's tube b) it measures the wavelength (and hence the speed or the frequency) of sound c) the interference of sound waves d) see the note below**

### 5 one loudspeaker compared with two

one loudspeaker → no second wave to interfere with, so the intensity simply falls off steadily as the microphone moves further away.

two loudspeakers → the path difference changes as the microphone moves, giving alternate maxima and minima.

The first half of the statement matters as much as the second: it rules out any explanation based on the room alone, because the pattern only appears when the **second** source is switched on.

**B) with one loudspeaker the intensity decreases gradually with distance, and with two a repeated pattern of rise and fall appears**

### 6 the path difference for destructive interference

destructive needs an **odd** number of half-wavelengths :

$$\Delta d = 0.5\lambda, 1.5\lambda, 2.5\lambda, \dots$$

of the four choices, only  $1.5\lambda$  is of this form (  $\lambda$ ,  $2\lambda$  and  $3\lambda$  are whole numbers of wavelengths, which give constructive interference ).

**B)  $1.5\lambda$**

7 constructive interference with  $\lambda = 2 \text{ m}$

constructive needs a whole number of wavelengths :

$$\Delta d = m \lambda = m \times 2 = 2, 4, 6, \dots \text{ m}$$

of the four choices only 2 m is a whole multiple of 2 m.

Check the others: 1 m and 3 m are odd multiples of  $\lambda/2 = 1 \text{ m}$ , so they are destructive; 5 m is  $2.5\lambda$ , also destructive.

**B) 2 m**

## Part C — Weekly assessment, Group A

8 light from  $n = 1.33$  into  $n = 1.5$

$n = c / v \rightarrow$  a larger  $n$  means a smaller  $v$

$$n_1 = 1.33 < n_2 = 1.5 \rightarrow v_1 > v_2$$

the frequency does not change, so  $\lambda = v / f$  follows the speed :

$$\lambda_1 > \lambda_2$$

Speed and wavelength always move **together**; the frequency never moves at all. Any choice in which  $v$  and  $\lambda$  go opposite ways, such as (C), can be rejected at a glance.

**A)  $v_1 > v_2$  and  $\lambda_1 > \lambda_2$**

9 the path difference for constructive interference

constructive needs a whole number of wavelengths :  $\Delta d = m \lambda$

$0.5\lambda$  ,  $1.5\lambda$  and  $2.5\lambda$  are all odd numbers of half-wavelengths  $\rightarrow$  destructive.

**B)  $\lambda$**

10 sea waves entering shallow water

the speed of a water wave depends on the **depth** :

depth decreases  $\rightarrow v$  decreases

the frequency is fixed by the source far out at sea :

$f$  unchanged  $\rightarrow \lambda = v / f$  decreases

and the end of each crest that reaches the shallow water first slows down while the other end is still moving faster, so the crest is turned round.

This is refraction. The three facts hang together in one chain: the depth sets the speed, the source sets the frequency, and  $v = f\lambda$  then fixes the wavelength. The bending is just the sideways consequence of one end of the crest being slower than the other.

**$v$  decreases,  $\lambda$  decreases,  $f$  is unchanged, and the direction bends until the crests are almost parallel to the shore**

**11** Quincke:  $X = 10 \text{ cm}$  and  $v = 330 \text{ m/s}$

a) pulling the slide out by  $X$  lengthens that path by  $2X$ , so at the first minimum :

$$2X = \lambda / 2 \quad \rightarrow \quad \lambda = 4X$$

$$\lambda = 4 \times 0.10$$

$$\lambda = 0.40 \text{ m}$$

b)  $v = f \lambda \quad \rightarrow \quad f = v / \lambda$

$$f = 330 / 0.40$$

$$f = 825 \text{ Hz}$$

Take care with the factor of **4**. The commonest mistake is to write  $\lambda = 2X$ , forgetting either that the sound travels the extension twice, or that the **first** minimum needs only half a wavelength of path difference.

**a)  $\lambda = 0.40 \text{ m}$    b)  $f = 825 \text{ Hz}$**

**12** why the quiet points are the stronger evidence

a loud point could be explained without any wave idea at all – two sources simply deliver more energy than one. but a **quiet** point cannot :

switching on a **second** source makes the sound **disappear**.

If sound were a stream of particles, adding a second stream could only ever make the sound louder. Cancellation is a property that belongs to waves alone, which is why the minima, not the maxima, settle the argument.

**because two sources producing silence can only be explained by waves cancelling, whereas two sources producing a loud sound could be explained by simple addition of energy**

## Part D — Weekly assessment, Group B

**13** ultrasound entering bone

the frequency is set by the transducer and cannot change at a boundary :

$f$  unchanged

sound travels **faster** in a denser, stiffer material such as bone :

$v$  increases  $\rightarrow \lambda = v / f$  increases as well

Be careful: for **sound**, a stiffer and denser medium such as bone is the **faster** one (sound goes at about  $1500 \text{ m/s}$  in soft tissue and about  $3500 \text{ m/s}$  in bone). That is the opposite of the rule for **light**, where a denser medium is the slower one.

**B) its frequency stays constant, and its speed and its wavelength increase**

**14** destructive interference with  $\lambda = 4 \text{ m}$

$$\lambda / 2 = 2 \text{ m}$$

destructive needs an odd number of half-wavelengths :

$$\Delta d = 2, 6, 10, \dots \text{ m}$$

of the four choices only  $2 \text{ m}$  is of this form (  $4, 8$  and  $12 \text{ m}$  are whole numbers of wavelengths  $\rightarrow$  constructive ).

**A)  $2 \text{ m}$**

**15** Quincke:  $X = 12 \text{ cm}$  and  $f = 1000 \text{ Hz}$

$$\begin{aligned} \text{a) } \lambda &= 4 X = 4 \times 0.12 \\ \lambda &= 0.48 \text{ m} \end{aligned}$$

$$\begin{aligned} \text{b) } v &= f \lambda = 1000 \times 0.48 \\ v &= 480 \text{ m/s} \end{aligned}$$

**Note on the printed data.** The answer that follows from the book's numbers is 480 m/s, but the speed of sound in air at ordinary temperatures is about 340 m/s, so these figures do not describe sound in air. Treat it as an exercise in the method: the relation  $\lambda = 4X$  and then  $v = f\lambda$  are what is being tested, and both are applied correctly here.

**a)  $\lambda = 0.48 \text{ m}$    b)  $v = 480 \text{ m/s}$**

**16** how noise-cancelling headphones work

a small microphone on the outside of the earcup picks up the incoming noise.  
the electronics invert it – that is, produce a wave of the same amplitude and the same frequency but **180° out of phase** – and the loudspeaker in the earcup plays it into the ear.

by the principle of superposition the two add to ( almost ) zero.

This is destructive interference put to work. Cancellation can never be perfect, because the inverted wave has to be produced in the microsecond or two before the noise reaches the eardrum — which is exactly the point of the next question.

**the headphone plays a wave in antiphase with the noise, so the two cancel by superposition**

**17** the wavelength ratio at the boundary

from the figure :  $i = 45^\circ$  in medium 1 ,  $r = 30^\circ$  in medium 2  
the frequency is the same in both media, so  $\lambda \propto v$  , and :

$$\lambda_1 / \lambda_2 = v_1 / v_2 = \sin i / \sin r$$

$$\lambda_1 / \lambda_2 = \sin 45^\circ / \sin 30^\circ = 0.7071 / 0.5$$

$$\lambda_1 / \lambda_2 = 1.41 \quad ( = \sqrt{2} )$$

**$\lambda_1 : \lambda_2 = \sqrt{2} : 1 \approx 1.41$**

## Part E — Weekly assessment, Group C

**18** which statement is not correct

$$\lambda_1 = v_1 / f = 3 \times 10^8 / 6 \times 10^{14} = 5.0 \times 10^{-7} \text{ m}$$

$$\lambda_2 = v_2 / f = 2 \times 10^8 / 6 \times 10^{14} = 3.33 \times 10^{-7} \text{ m}$$

$$n = c / v : \quad n_1 = 3 \times 10^8 / 3 \times 10^8 = 1 , \quad n_2 = 3 \times 10^8 / 2 \times 10^8 = 1.5$$

(A)  $f$  is the same in both media

✓ correct

(B)  $\lambda_1 / \lambda_2 = 5.0 / 3.33 = 1.5$

✓ correct

(D)  $\sin i / \sin r = v_1 / v_2 = 1.5$

✓ correct

(C)  $n_1 = 1$  and  $n_2 = 1.5$  , so  $n_2 = 1.5 n_1$  – the other way round ✗ wrong

The trap is that the ratio 1.5 really does appear three times in this question — for the wavelengths, for the sines and for the refractive indices — but for the refractive indices it points the **other way**, because  $n$  is inversely proportional to  $v$ .

**C) — the refractive index of the first medium is not 1.5 times that of the second; it is the second that is 1.5 times the first**

**19** constructive interference in words

$$\Delta d = m \lambda = m \times (2 \times \lambda/2) = 2m \times \lambda/2$$

so a whole number of wavelengths is the same thing as an **even** number of half-wavelengths.

And destructive interference is an **odd** number of half-wavelengths. Counting in half-wavelengths is the quickest way to answer every question of this kind.

**D) an even number of half wavelengths**

**20** Quincke:  $f = 1200 \text{ Hz}$  and  $v = 360 \text{ m/s}$

a)  $v = f \lambda \quad \rightarrow \quad \lambda = v / f$

$$\lambda = 360 / 1200$$

$$\lambda = 0.30 \text{ m}$$

b) at the first minimum :  $2X = \lambda / 2 \quad \rightarrow \quad X = \lambda / 4$

$$X = 0.30 / 4$$

$$X = 0.075 \text{ m} = 7.5 \text{ cm}$$

This is question 11 run backwards: there  $\lambda = 4X$  gave the wavelength from the pull, here  $X = \lambda/4$  gives the pull from the wavelength.

**a)  $\lambda = 0.30 \text{ m}$    b)  $X = 0.075 \text{ m} = 7.5 \text{ cm}$**

**21** a steady hum compared with a sudden bang

the headphone has to **measure** the incoming sound and then produce its exact mirror image before the sound reaches the eardrum.

a steady hum repeats itself thousands of times a second, so the next cycle is completely predictable and can be cancelled accurately.

a door slam is a single short pulse with no repeat and a very wide range of frequencies – by the time the electronics have analysed it, it has already gone.

It is also a matter of wavelength. Low, steady hums have long wavelengths, so the small phase errors caused by the few centimetres between the microphone and the eardrum hardly matter. For the high frequencies inside a bang, those same few centimetres can be a whole wavelength, and the “cancelling” wave might even add to the noise.

**because a steady hum is repetitive and therefore predictable, while a sudden bang is over before the headphone can produce the cancelling wave**

**22** the two ratios at the boundary

from the figure :  $i = 45^\circ$  in medium 1 ,  $r = 30^\circ$  in medium 2

a) the frequency is fixed by the source, and a boundary cannot change it :

$$f_1 / f_2 = 1$$

b)  $\lambda_1 / \lambda_2 = v_1 / v_2 = \sin i / \sin r$

$$\lambda_1 / \lambda_2 = \sin 45^\circ / \sin 30^\circ = 0.7071 / 0.5 = 1.41$$

**a)  $f_1 : f_2 = 1 : 1$    b)  $\lambda_1 : \lambda_2 \approx 1.41$**

## Part F — The Doppler effect, Classwork 1

**23** the pitch heard as the phone approaches and recedes

the phone keeps giving out the same frequency  $f$ , and the sound keeps travelling through the air at the same speed  $v$ . what changes is the **spacing** of the crests that reach the listener :

- approaching → each new crest is sent from a point nearer the listener,  
so the crests are crowded :  $\lambda$  is shorter,  $f' = v/\lambda$  is higher
- receding → each new crest is sent from a point further away,  
so the crests are stretched :  $\lambda$  is longer,  $f'$  is lower

**What does not change.** Three things stay exactly as they were: the frequency the phone produces, the speed of sound in the air, and the number of crests sent out per second. Only the number of crests **arriving** per second is altered, because the listener and the crests are closing on one another. Note also that the change is a step from high to low as the phone goes past — not a gradual slide.

**the listener hears a pitch higher than  $f$  while the phone approaches and lower than  $f$  while it recedes**

**24** doubling the speed of the moving source

the faster the source closes the gap, the more tightly the crests in front of it are crowded together :

$$\lambda' = (v - v(s)) / f \quad \rightarrow \text{a larger } v(s) \text{ gives a smaller } \lambda'$$
$$f' = v / \lambda' \quad \rightarrow \text{so } f' \text{ is larger still}$$

The pitch was already raised at 2 m/s; walking at 4 m/s simply raises it further. A useful check on any Doppler answer: the direction of the shift is fixed by whether the gap is closing or opening, and the **size** of the shift grows with the speed.

**B) the apparent frequency increases and the sound becomes even sharper**

## Part G — The Doppler effect, Home assignment 1

**25** the apparent wavelength in front of the moving source

in the time of one cycle the source sends out one crest and then moves forward a distance  $v(s) T$  before sending the next one, so the gap between them is :

$$\lambda' = v T - v(s) T = (v - v(s)) / f < \lambda$$

Choice (D) contains the standard misconception: the motion of the source has **no effect whatever** on the speed of sound in air, which is fixed by the air itself.

**B) the wavelength decreases, because the waves are given out closer together in front of the moving source**

**26** approaching, passing, and receding

- while approaching : the gap is closing →  $f' > f$  ( highest )
- at the instant of passing : the car is moving straight **across** the line of sight, so the distance is momentarily neither growing nor shrinking →  $f' = f$
- while receding : the gap is opening →  $f' < f$

Only the part of the velocity **along the line joining the source to the listener** counts. At the moment of passing, the whole of the car's velocity is at right angles to that line, so there is no shift at all — which is why the drop in pitch sounds so sudden.

**A) highest during the approach, equal to  $f$  at the instant of passing, and below  $f$  while receding**

**27** the police car and the mountain wall

the **direct** sound : the driver sits with the siren, so there is no relative motion between them at all → he hears the true frequency  $f$ .

the **reflected** sound : it is shifted **twice** –

- 1 the car is approaching the wall, so the wall receives a raised frequency
- 2 the wall re-emits that raised frequency, and the car is now approaching those returning crests as well

→ the echo comes back at a frequency clearly higher than  $f$ .

The two shifts combine to give  $f' = f (v + v(\text{car})) / (v - v(\text{car}))$ . The driver therefore hears **two** different pitches at once, the true one from his own siren and a higher one from the cliff, and the small difference between them produces audible beats — a practical way of measuring the car's speed.

**C) the direct frequency equals  $f$ , and the reflected frequency rises above  $f$**

**28** the ambulance siren passing the girl

**a)** The pitch heard depends on how fast the gap between the siren and the ear is changing. On the way in, that gap is closing at nearly the full speed of the ambulance, so the crests pile up and the note is sharp. At the instant the ambulance is level with the girl its velocity is entirely **across** the line of sight, so the gap is momentarily not changing and she hears the true note. A moment later the gap is opening at nearly the full speed, the crests are stretched out, and the note is deep. Because the ambulance sweeps past in a fraction of a second, the whole reversal is squeezed into that instant and is heard as a sudden drop.

**b)** The speed of sound in air is decided by the air alone — essentially by its temperature — and is about 340 m/s throughout. It is completely unaffected by how fast the source is moving. The motion changes the **wavelength** of the sound in the air, and therefore the frequency that arrives, but never the speed at which it travels.

**a) the component of the velocity along the line of sight reverses at the instant of passing, so the shift flips from + to –   b) the speed of sound in the air does not change at all**



## Solutions — step by step

One sentence covers the whole lesson. A **closing** gap squeezes the waves: shorter  $\lambda$ , higher  $f$ , blue shift. An **opening** gap stretches them: longer  $\lambda$ , lower  $f$ , red shift. For light,  $\Delta\lambda/\lambda = v/c$  turns the size of the shift into a speed.

### Part A — Classwork 2

1 the two stars X and Y

star X :  $\lambda$  increased  $\rightarrow$  the light has been stretched  $\rightarrow$  **red shift**

star Y :  $\lambda$  decreased  $\rightarrow$  the light has been squeezed  $\rightarrow$  **blue shift**

**b) How astronomers use it.** Every element gives out (or absorbs) light at a fixed set of wavelengths, and that pattern is measured in the laboratory. When the same pattern is found in the light of a star, the whole set of lines is seen to have slid along the spectrum without changing its **spacing** — which is what proves it is a Doppler shift rather than a different element. The **direction** of the slide gives the direction of motion: towards the red for a receding star, towards the blue for an approaching one. The **size** of the slide gives the speed, from  $\Delta\lambda/\lambda = v/c$ . Note that this measures only the **radial** component of the velocity — motion straight across the line of sight produces no shift at all.

|                   | red shift                       | blue shift                       |
|-------------------|---------------------------------|----------------------------------|
| the wavelength    | $\lambda' > \lambda$ ( longer ) | $\lambda' < \lambda$ ( shorter ) |
| the frequency     | lower than $f$                  | higher than $f$                  |
| the motion        | the source is receding          | the source is approaching        |
| the speed from it | $v = c \Delta\lambda / \lambda$ | $v = c \Delta\lambda / \lambda$  |

a) X shows a red shift (moving away) and Y shows a blue shift (moving towards us)    b) from the size of  $\Delta\lambda : v = c \Delta\lambda / \lambda$

2 the reason for a red shift

red is the **long** wavelength end of the visible spectrum, so a shift towards the red means  $\lambda' > \lambda$  — the waves have been stretched out.

waves are stretched when the gap between source and observer is **opening**.

Choices (B) and (C) each get one half right and one half wrong, which is exactly what makes them tempting. Fix the pair in your memory: **red = receding = longer**.

A) the galaxy receding from the Earth leads to an increase in the wavelength

### Part B — Home assignment 2

**3** the radial speed of the galaxy

$$\begin{aligned}\Delta\lambda / \lambda &= v / c && \rightarrow && v = c \Delta\lambda / \lambda \\ v &= ( 3 \times 10^8 ) \times 1.312 / 656 \\ 1.312 / 656 &= 0.002 \\ v &= ( 3 \times 10^8 ) \times 0.002 \\ v &= 6 \times 10^5 \text{ m/s} && = && 600 \text{ km/s}\end{aligned}$$

The two wavelengths are both in nanometres, so the units cancel in the ratio and there is no need to convert either of them. Notice how small the shift is — 0.2 % — and yet it corresponds to 600 km/s, because it is being compared with the speed of light.

$$v = 6 \times 10^5 \text{ m/s} = 600 \text{ km/s}$$

**4** a galaxy approaching at high speed

the gap is **closing**, so the waves are squeezed :

$$\lambda' < \lambda \quad \text{and} \quad f' = c / \lambda' > f$$

a shorter wavelength is nearer the blue end of the spectrum.

**A) higher than the original frequency, and the lines shift towards the blue colour**

**5** the speed of light from a moving star

the speed of light in vacuum is the **same for every observer**, whatever the motion of the source or of the observer :

$$c = 3 \times 10^8 \text{ m/s} \quad \text{always}$$

the motion changes  $\lambda$  and  $f$ , and they change together so that  $c = f \lambda$  still holds.

This is the one place where light differs completely from sound. The speed of **sound** is fixed by the air, so a wind can change it; the speed of **light** in vacuum is a constant of nature and is the foundation of the special theory of relativity.

**A) it stays constant and does not change**

**6** the motorcycle rider after they pass

after passing, the car goes on eastwards and the motorcycle goes on westwards, so the two are **separating** : both motions lower the pitch.

$$\begin{aligned}f' &= f \times ( v - v_o ) / ( v + v_s ) \\ f' &= 600 \times ( 340 - 20 ) / ( 340 + 15 ) \\ f' &= 600 \times 320 / 355 \\ f' &= 540.8 \approx 541 \text{ Hz}\end{aligned}$$

**Which sign goes where.** The observer's speed is **subtracted** on the top when the observer is running away; the source's speed is **added** on the bottom when the source is running away. Both effects lower the pitch here, which is why the answer is well below 600 Hz. **Note on the printed question:** the speed of sound is not stated in this item; 340 m/s has been used, the value the book gives in the other questions of this same lesson.

$$f' \approx 541 \text{ Hz}$$

**7** the name of the phenomenon

a difference between the observed frequency and the frequency of the source, caused purely by relative motion, is the **Doppler effect**.

**B) the Doppler effect**

**8** loudness and pitch as the ambulance approaches

two separate things are happening :

the car is getting **closer** → the same energy is spread over a smaller area → the sound is louder

the gap is **closing** → the waves are squeezed → higher  $f'$

Keep loudness and pitch apart in your mind. Loudness depends on the **distance**, so it rises all the way in and falls all the way out. Pitch depends on whether the distance is **changing**, so it is high on the way in, true at the instant of passing, and low on the way out.

**A) the intensity of the sound increases, and the apparent frequency increases**

**9** the red shift of a galaxy

red → longer  $\lambda$  →  $\lambda' > \lambda$  → the gap is opening  
→ the galaxy is moving away from the Earth.

**A) the galaxy receding from the Earth leads to an increase in the wavelength**

**10** a source approaching at 25 m/s

the source is approaching, so the wavelength in front of it is shortened :

$$f' = f \times v / (v - v_s)$$

$$f' = 600 \times 340 / (340 - 25)$$

$$f' = 600 \times 340 / 315$$

$$f' = 647.6 \approx 648 \text{ Hz}$$

Check the direction before you check the arithmetic: the source is coming closer, so the answer **must** be above 600 Hz. If your calculator gives less than 600, you have put the minus sign in the wrong place.

**$f' \approx 648 \text{ Hz}$**

**11** all the colours shifted towards the red

the whole pattern of lines has moved to longer wavelengths :

$\lambda' > \lambda$  → the gap is opening

The key point is that **all** the lines moved, and by the same **fraction** of their own wavelength. A change of the source itself would move some lines and not others; only motion slides the whole pattern in step. This is how Hubble discovered that almost every distant galaxy is receding, and that the further away a galaxy is the faster it goes — the evidence that the universe is expanding.

**the galaxy is moving away from us, and the faster it recedes the bigger the shift**

**12** the train's whistle at the station

approaching : each crest is sent from a point nearer the station, so the crests arrive crowded →  $\lambda'$  is shorter →  $f'$  is higher – a sharp note

receding : each crest is sent from a point further away, so they arrive stretched out →  $\lambda'$  is longer →  $f'$  is lower – a deep note

The whistle itself never changes. Neither does the speed of sound in the air. The only thing that changes is the **spacing of the crests that reach the ear**.

**the Doppler effect: the gap closes on the way in (higher pitch) and opens on the way out (lower pitch)**

## Part D — Weekly assessment, Group B

**13** the ambulance moving away

the car is getting **further** → the sound is fainter

the gap is **opening** → the waves are stretched → lower  $f'$

**A) the apparent frequency decreases, and the intensity of the sound decreases**

**14** the case that means the galaxy is approaching

approaching → the gap is closing → the waves are squeezed :

$\lambda' < \lambda$  and a shorter  $\lambda$  is nearer the **blue** end

**A)  $\lambda' < \lambda$ , and the spectral lines shift towards the blue colour**

**15** a source receding at 20 m/s

the source is receding, so the wavelength behind it is stretched :

$$f' = f \times v / (v + v_s)$$

$$f' = 600 \times 340 / (340 + 20)$$

$$f' = 600 \times 340 / 360$$

$$f' = 566.7 \approx 567 \text{ Hz}$$

Compare with question 10: there the source approached at 25 m/s and the pitch rose by about 48 Hz; here it recedes at 20 m/s and the pitch falls by about 33 Hz. The shift is always towards the observer's side of 600 Hz.

**$f' \approx 567 \text{ Hz}$**

**16** the sodium lines shifted to lower frequencies

lower frequency → longer wavelength → a **red** shift

→ the gap between the galaxy and the Earth is opening.

The sodium lines are a good marker precisely because their laboratory wavelengths are known to great accuracy (the familiar orange-yellow pair at about 589 nm), so even a small displacement can be measured with confidence and converted into a speed with  $\Delta\lambda/\lambda = v/c$ .

**the galaxy is moving away from the Earth**

**17** the pitch before and after the ambulance passes

approaching : the gap is closing  $\rightarrow \lambda'$  shorter  $\rightarrow f' > f$  ( sharp )

at the instant of passing : the motion is across the line of sight  $\rightarrow f' = f$

receding : the gap is opening  $\rightarrow \lambda'$  longer  $\rightarrow f' < f$  ( deep )

The drop is heard as a sudden step rather than a slide, because the whole reversal from “closing fast” to “opening fast” happens in the fraction of a second in which the ambulance sweeps past.

**the pitch is higher than the true note while approaching and lower than it after passing, and the change is sudden**

## Part E — Weekly assessment, Group C

**18** the apparent wavelength in front of an approaching source

in the time of one cycle the source emits one crest and then moves forward before emitting the next, so the crests in front of it are closer together :

$$\lambda' = (v - v_s) / f < \lambda$$

Choice (D) contains the standard misconception. The speed of sound in air is decided by the air alone; the motion of the source changes the **spacing** of the crests, never their speed.

**B) smaller; because of the compression of the sound waves in front of the source**

**19** the name of a shift to longer wavelengths

longer  $\lambda \rightarrow$  towards the red end  $\rightarrow$  a **red shift**

and stretching happens when the gap is opening  $\rightarrow$  the galaxy is receding.

**A) the red shift, and it indicates that the galaxy is receding from the Earth**

**20** the motorcycle rider before they pass

before passing, the car is still to the west of the motorcycle : the car drives east towards it and the motorcycle drives west towards the car, so the two are **closing** – both motions raise the pitch.

$$f' = f \times (v + v_o) / (v - v_s)$$

$$f' = 600 \times (340 + 20) / (340 - 15)$$

$$f' = 600 \times 360 / 325$$

$$f' = 664.6 \approx 665 \text{ Hz}$$

Put this beside question 6, which is the same pair of vehicles after they have passed: the rider hears about 665 Hz on the way in and about 541 Hz on the way out — a drop of some 124 Hz in an instant, which is why such a pass sounds so dramatic. **Note on the printed question:** the speed of sound is again not stated here, so 340 m/s has been used.

**$f' \approx 665 \text{ Hz}$**

**21** the name of the phenomenon

the pitch changes only because the fire engine is **moving** relative to the listener – the siren itself is unaltered.

**the Doppler effect**

**22** the two galaxies A and B

galaxy A : lines shifted towards the **red** →  $\lambda' > \lambda$  → stretched  
→ the gap is opening → A is moving **away** from the observer

galaxy B : lines shifted towards the **blue** →  $\lambda' < \lambda$  → squeezed  
→ the gap is closing → B is moving **towards** the observer

Almost every galaxy we can see shows a red shift, so a blue-shifted galaxy such as B is unusual and means it is close enough for its own motion within the local group to beat the general expansion. The Andromeda galaxy is the famous example — it is blue-shifted, and is approaching us at about 110 km/s.

**A is receding from the observer and B is approaching the observer**



## Solutions — step by step

Everything follows from  $n = c/v$ . Decide first which medium is the denser one; then the speed, the wavelength and the direction of bending all follow automatically. And remember that  $\theta_c$  only exists for light trying to **leave** the denser medium.

### Part A — Classwork 1

#### 1 passing from medium A into medium B

the figure shows the ray bending **towards** the normal in B, so B is the optically denser medium :

$$n(B) > n(A) \quad \text{and} \quad n = c / v \quad \rightarrow \quad v(B) < v(A)$$

the frequency is set by the source and cannot change at a boundary, so :

$$\lambda = v / f \quad \rightarrow \quad \lambda(B) < \lambda(A)$$

The three always move together: slower, shorter, bent towards the normal. The one quantity that never moves is the frequency — the boundary cannot create or destroy wave crests.

|                              | optically denser medium         | optically less dense medium       |
|------------------------------|---------------------------------|-----------------------------------|
| the speed of light $v$       | smaller                         | larger                            |
| the absolute index $n = c/v$ | larger                          | smaller                           |
| the wavelength $\lambda$     | shorter                         | longer                            |
| the frequency $f$            | the same                        | the same                          |
| a ray entering it            | bends <b>towards</b> the normal | bends <b>away</b> from the normal |

**both the speed of light and the wavelength decrease in B, while the frequency stays the same**

#### 2 the definition of the absolute refractive index

the absolute refractive index of a medium is the ratio of the speed of light in vacuum to its speed in that medium :

$$n = c / v$$

it can also be written from the law of refraction, for light going from vacuum :

$$n = \sin(\text{angle in vacuum}) / \sin(\text{angle in the medium})$$

and, since  $f$  is unchanged,  $n = \lambda(\text{vacuum}) / \lambda(\text{medium})$

Because nothing travels faster than  $c$ ,  $v$  is always less than or equal to  $c$  and so  $n \geq 1$ . For air  $n \approx 1.0003$ , which is why air is treated as vacuum in school problems.

**$n = c / v$  — a pure number, never less than 1**

### Part B — Home assignment 1

**3** water into glass at  $55^\circ$

the relative refractive index from water to glass is :

$$n(\text{water} \rightarrow \text{glass}) = \sin i / \sin r = 1.15$$

$$\sin r = \sin 55^\circ / 1.15$$

$$\sin r = 0.8192 / 1.15 = 0.7123$$

$$r = 45.4^\circ \approx 45^\circ$$

The relative index is greater than 1, so the glass is the denser of the two and the ray must bend **towards** the normal — and indeed  $45^\circ$  is smaller than  $55^\circ$ . If your answer had come out bigger than  $55^\circ$  you would know the ratio had been used upside down.

$$r \approx 45^\circ$$

**4** denser compared with less dense

$n = c / v \rightarrow$   $n$  and  $v$  always move in **opposite** directions

|                              | optically denser medium         | optically less dense medium       |
|------------------------------|---------------------------------|-----------------------------------|
| the speed of light $v$       | smaller                         | larger                            |
| the absolute index $n = c/v$ | larger                          | smaller                           |
| the wavelength $\lambda$     | shorter                         | longer                            |
| the frequency $f$            | the same                        | the same                          |
| a ray entering it            | bends <b>towards</b> the normal | bends <b>away</b> from the normal |

**the denser medium has the smaller speed of light and the larger refractive index**

**5** the refractive index of the glass

$$n = c / v$$

$$n = (3 \times 10^8) / (2 \times 10^8)$$

$$n = 1.5$$

A refractive index is a ratio of two speeds, so it has no unit. 1.5 is the standard value for ordinary glass, and it makes the critical angle with air about  $41.8^\circ$  — a number worth remembering.

$$n = 1.5$$

**6** entering a denser medium

the frequency is fixed by the source :  $f$  unchanged

the denser medium is the slower one :  $v$  decreases

$$\lambda = v / f \rightarrow \lambda \text{ decreases as well}$$

**D) the frequency does not change and the wavelength decreases**

7 from  $n = 1.3$  into  $n = 1.5$

$$n_2 = 1.5 > n_1 = 1.3 \quad \rightarrow \text{medium (2) is the denser one}$$

$$n = c / v \quad \rightarrow \quad v_2 < v_1$$

$$\lambda = v / f \text{ ( } f \text{ fixed )} \rightarrow \lambda_2 < \lambda_1$$

**Note on the printed question.** In the book the four choices are given as small diagrams rather than as sentences; they have been written out here in words, with the same meaning, so that the question can be answered from the text alone.

**B) both the wavelength and the speed decrease**

## Part C — Classwork 2

8 the greatest angle of refraction

going into the **less** dense medium the ray bends away from the normal, so the angle of refraction is always larger than the angle of incidence.

the refracted ray can at most run along the surface itself :

$$\text{greatest angle of refraction} = 90^\circ$$

and the angle of incidence that produces it is the **critical angle**  $\theta_c$ .

Beyond that there is no refracted ray at all: every ray arriving at more than  $\theta_c$  is reflected back into the denser medium, which is total internal reflection.

**$90^\circ$  — the refracted ray grazes along the surface**

9 the relation for the critical angle

at the critical angle the refracted ray grazes the surface, so  $\theta_2 = 90^\circ$  :

$$n_1 \sin \theta_c = n_2 \sin 90^\circ = n_2$$

$$\sin \theta_c = n_2 / n_1 \quad (n_1 > n_2)$$

and when the second medium is air ( $n_2 \approx 1$ ) :

$$\sin \theta_c = 1 / n$$

**$\sin \theta_c = n_2 / n_1$  ( $= 1/n$  with air outside)**

## Part D — Home assignment 2

10 why the cladding has the smaller refractive index

total internal reflection can only happen when light tries to leave the **denser** medium, so the core must be the denser of the two :

$$n(\text{core}) > n(\text{cladding})$$

then a critical angle exists at the core-cladding boundary, and every ray that strikes it at more than  $\theta_c$  is reflected back into the core.

If the cladding were the denser one there would be **no** critical angle at all, and light would escape at every bounce. The cladding also keeps the core surface clean and prevents light from leaking into a neighbouring fibre.

**so that a critical angle exists at the core-cladding boundary and the light is totally reflected instead of leaking out**

**11** raising the angle inside a glass of water

the light is going from water ( denser ) to air ( less dense ) :

small angles → most of the light refracts out, bending away from the normal,  
and a weak reflected ray stays in the water

$\theta = \theta_c \approx 49^\circ$  → the refracted ray grazes along the surface (  $90^\circ$  ) and the  
reflected ray is at its brightest so far

$\theta > \theta_c$  → the refracted ray disappears completely : **total internal reflection**, all the light stays in the water

**the refracted ray bends further and further away until, at  $\theta_c$ , it grazes the surface and then vanishes — beyond  $\theta_c$  all the light is reflected back into the water**

**12**  $\sin \theta_c$  for a 1.6 / 1.4 fibre

$$\sin \theta_c = n_2 / n_1 \quad ( n_1 = \text{core} , n_2 = \text{cladding} )$$

$$\sin \theta_c = 1.4 / 1.6$$

$$\sin \theta_c = 0.875 \quad \rightarrow \quad \theta_c = 61^\circ$$

The smaller index always goes on top — otherwise the sine would come out greater than 1, which is impossible, and that is exactly how you can spot the wrong choices (B) and (D).

**A) 1.4 / 1.6**

**13** when total internal reflection happens

two conditions are needed together :

- 1 the light must be going from the **denser** medium to the less dense one
- 2 the angle of incidence must be **greater** than the critical angle

**B) from a medium of higher optical density into a medium of lower optical density**

**14** the correct path at  $50^\circ$  in glass

find the critical angle for glass and air :

$$\sin \theta_c = 1 / n = 1 / 1.5 = 0.6667$$

$$\theta_c = 41.8^\circ$$

the ray arrives at  $50^\circ$ , and  $50^\circ > 41.8^\circ$  :

→ no light gets out — the ray is reflected back into the glass, with the angle of reflection equal to the angle of incidence.

Figures A and D would need a refracted ray to exist, which it does not above  $\theta_c$ , and B would need the two media to have the same index. **Note on the printed question:** the four choices in the book are diagrams without letters; they are drawn here as four labelled panels so that the answer can be given as a letter.

**C) — the ray is totally reflected back into the glass at  $50^\circ$**

**Part E — Weekly assessment, Group A**

15 the cryolite cladding for  $\theta_c = 32^\circ$

$$\begin{aligned}\sin \theta_c &= n_2 / n_1 & \rightarrow & n_2 = n_1 \sin \theta_c \\ n_2 &= 2.1 \times \sin 32^\circ \\ n_2 &= 2.1 \times 0.5299 \\ n_2 &= 1.11\end{aligned}$$

The other three choices are all **larger** than 2.1, and a cladding denser than the core would make total internal reflection impossible — so they can be rejected before any arithmetic is done.

A)  $n = 1.11$

16 a liquid of greater refractive index

$$\begin{aligned}\sin \theta_c &= 1 / n & \rightarrow & \text{a larger } n \text{ gives a smaller } \theta_c \\ r &= h \tan \theta_c & \rightarrow & \text{a smaller } \theta_c \text{ gives a smaller } r \\ \text{area} &= \pi r^2 & \rightarrow & \text{and therefore a smaller area}\end{aligned}$$

The escaping light is confined to a cone of half-angle  $\theta_c$ . Making the liquid denser narrows that cone, so a smaller lid is enough to cover it.

B) it decreases

17 why a ray bends away going from water to air

$$\begin{aligned}n(\text{water}) &\approx 1.33 > n(\text{air}) \approx 1 \\ n &= c / v & \rightarrow & v(\text{air}) > v(\text{water})\end{aligned}$$

the light speeds up as it enters the air. from  $n_1 \sin \theta_1 = n_2 \sin \theta_2$  :

$$\sin \theta_2 = (n_1 / n_2) \sin \theta_1 > \sin \theta_1 \quad \rightarrow \quad \theta_2 > \theta_1$$

A picture that helps: think of the wavefront as a line of marchers. The end that reaches the faster medium first pulls ahead, and the whole line swings round — away from the normal when the new medium is the faster one.

**because air is the optically less dense medium, so the light travels faster in it and the ray turns away from the normal**

18 incidence exactly at the critical angle

$$n_1 \sin \theta_c = n_2 \sin \theta_2 \quad \text{with } \theta_2 = 90^\circ$$

the refracted ray leaves along the surface itself, at  $90^\circ$  to the normal, and is extremely faint; at the same time the reflected ray inside the denser medium is at its brightest so far.

**the refracted ray grazes along the surface at  $90^\circ$  — the borderline case just before total internal reflection**

19 the critical angle between A and B

$$n = c / v :$$

$$n(A) = c / 0.8c = 1.25$$

$$n(B) = c / 0.6c = 1.667$$

B is the denser one, so the light must be going from B into A :

$$\sin \theta_c = n(A) / n(B) = 1.25 / 1.667$$

$$\sin \theta_c = 0.75$$

$$\theta_c = 48.6^\circ$$

Notice the short cut:  $\sin \theta_c = n(A)/n(B) = v(B)/v(A) = 0.6c/0.8c = 0.75$ , so the speeds can be used directly without working out either index.

$$\theta_c \approx 48.6^\circ$$

## Part F — Weekly assessment, Group B

20 the condition for total internal reflection

the light must start in the **denser** medium, which is  $n_1$  here, and must arrive at the boundary at more than the critical angle :

$$\text{the ray is in } n_1 \quad \text{and} \quad \theta > \theta_c$$

Choices (B) and (D) put the ray in the **less** dense medium, where no critical angle even exists; choice (C) has the inequality the wrong way round, and would simply give ordinary refraction.

**A) the ray must fall in the medium  $n_1$  at an angle greater than the critical angle**

21 refraction into glass of  $n = \sqrt{3}$  at  $60^\circ$

$$n = \sin i / \sin r \quad \rightarrow \quad \sin r = \sin i / n$$

$$\sin r = \sin 60^\circ / \sqrt{3} = 0.8660 / 1.7321$$

$$\sin r = 0.5$$

$$r = 30^\circ$$

Choice (A),  $90^\circ$ , is impossible for light **entering** a denser medium, and (B) would mean no bending at all. The ray must bend towards the normal, so the answer has to be less than  $60^\circ$ .

**C)  $30^\circ$**

22 incidence greater than the critical angle

above  $\theta_c$  the law of refraction would need  $\sin \theta_2 > 1$ , which is impossible, so no refracted ray can exist :

→ all the light is reflected back into the denser medium, obeying the ordinary law of reflection ( angle of reflection = angle of incidence ).

This is the only kind of reflection that is complete. An ordinary mirror always absorbs a few per cent, which is why optical fibres and reflecting prisms in binoculars use total internal reflection instead of mirrors.

**total internal reflection takes place — 100 % of the light is reflected back inside the denser medium**

23 denser compared with less dense, again

$$n = c / v \quad \rightarrow \quad \text{larger } n \leftrightarrow \text{smaller } v$$

|                              | optically denser medium         | optically less dense medium       |
|------------------------------|---------------------------------|-----------------------------------|
| the speed of light $v$       | smaller                         | larger                            |
| the absolute index $n = c/v$ | larger                          | smaller                           |
| the wavelength $\lambda$     | shorter                         | longer                            |
| the frequency $f$            | the same                        | the same                          |
| a ray entering it            | bends <b>towards</b> the normal | bends <b>away</b> from the normal |

the denser medium: smaller  $v$ , larger  $n$ . the less dense medium: larger  $v$ , smaller  $n$

24 why the internally reflected ray gets brighter

at every boundary the light divides into a refracted part and a reflected part. as  $\theta$  grows towards  $\theta_c$ , less and less energy goes into the refracted ray, so more and more is left for the reflected one. at  $\theta = \theta_c$  the refracted ray vanishes and beyond it **all** the energy is reflected.

So the reflected ray was always there; it is simply too faint to notice at small angles. This is the same effect that makes a lake look transparent when you look straight down and like a mirror when you look across it at a glancing angle.

because the share of the energy taken by the refracted ray falls as the angle grows, so the share left for the reflected ray rises — reaching 100 % at and beyond  $\theta_c$

25 the index from  $30^\circ$  inside to  $50^\circ$  in air

the light goes from the material into air :

$$n \sin i = n(\text{air}) \sin r = 1 \times \sin r$$

$$n \times \sin 30^\circ = \sin 50^\circ$$

$$n \times 0.5 = 0.7660$$

$$n = 1.53$$

The ray bent away from the normal ( $30^\circ \rightarrow 50^\circ$ ), so the material must be the denser one and  $n$  must come out greater than 1 — a quick check that the fraction was not written upside down.

$$n \approx 1.53$$

## Part G — Weekly assessment, Group C

26 the ratio of the two wavelengths

going into a denser medium the light slows down, so the wavelength shortens :

$$\lambda(\text{incident}) / \lambda(\text{refracted}) = v_1 / v_2 = n_2 / n_1$$

and  $n_2 > n_1$ , so the ratio is **greater than one**.

B) greater than one

**27** the greatest angle of refraction

the refracted ray bends away from the normal and can at most lie along the surface itself :

greatest angle of refraction =  $90^\circ$

Choice (C),  $42^\circ$ , is the critical angle of glass with air — an angle of **incidence**, not of refraction. It is put there to catch anyone answering from memory.

**B)  $90^\circ$**

**28** air to water compared with water to air

the frequency is fixed by the source in both directions :

$f$  is the same going either way

the wavelength follows the speed :

air  $\rightarrow$  water :  $v$  decreases  $\rightarrow \lambda$  decreases , ray bends towards the normal

water  $\rightarrow$  air :  $v$  increases  $\rightarrow \lambda$  increases , ray bends away from the normal

**Note on the printed questions.** This exact comparison appears twice in Group C of the book (as questions 3 and 5, word for word); the two have been merged into this single item.

**the frequency is unchanged in both cases; the wavelength decreases going into the water and increases coming out of it**

**29** why a denser liquid needs a smaller disc

$\sin \theta_c = 1 / n \quad \rightarrow \quad n \text{ larger} \quad \rightarrow \quad \theta_c \text{ smaller}$

$r = h \tan \theta_c \quad \rightarrow \quad \theta_c \text{ smaller} \quad \rightarrow \quad r \text{ smaller}$

Worth seeing with numbers: for water ( $n = 1.33$ )  $\theta_c = 48.8^\circ$  and  $r = 1.14 h$ , while for a liquid with  $n = 1.6$   $\theta_c = 38.7^\circ$  and  $r = 0.80 h$  — the disc is about 30 % smaller in radius and only half the area.

**because a larger  $n$  gives a smaller critical angle, which narrows the cone of escaping light, so a smaller disc covers it**

**30** glass with  $\theta_c = 42^\circ$  against water with  $\theta_c = 49^\circ$

$\sin \theta_c = 1 / n \quad \rightarrow \quad n = 1 / \sin \theta_c$

glass :  $n = 1 / \sin 42^\circ = 1 / 0.6691 = 1.49$

water :  $n = 1 / \sin 49^\circ = 1 / 0.7547 = 1.33$

$1.49 > 1.33 \quad \rightarrow \quad \text{the glass is the denser one}$

The rule to carry away: **the smaller the critical angle, the denser the medium**. A diamond, with  $n = 2.42$ , has a critical angle of only  $24^\circ$ , which is why so much light is trapped inside it and why it sparkles as it does.

**the glass is optically denser, because  $n = 1/\sin\theta_c$  and the smaller critical angle gives the larger refractive index**



## Solutions — step by step

**Two equations run this whole handout.** For every lens question write  $1/f = 1/a + 1/b$  and  $M = b/a$ , put  $f$  negative for a concave lens and  $b$  negative for a virtual image, and the algebra does the rest. For every Young question write  $\Delta y = \lambda L/d$  and ask only which of the three letters on the right has changed.

### Part A — Classwork 1

**1** the lens that is thicker at the centre

a lens that is thicker at its centre than at its edges is the  
convex ( converging ) lens

parallel rays that strike it are bent inwards and really meet at a point,  
so its principal focus is a real focus and its focal length is positive.

The mirror image of this rule is just as useful: a lens that is **thinner** at its centre than at its edges is the concave (diverging) lens, its rays spread apart, its focus is virtual and its focal length is taken as negative.

the convex (converging) lens

**2** where parallel rays collect

rays parallel to the principal axis are refracted by a convex lens and meet  
at one point on the axis ; that point is the principal focus ( F ).  
the distance from the optical centre to F is the focal length f.

Do not confuse the three points on the axis. The **optical centre** is the middle of the lens itself and a ray through it goes straight on; the **centre of curvature** is the centre of the sphere the surface was cut from; the **principal focus** is where the parallel rays actually gather.

choice B — the principal focus

**3** the line through the optical centre

the straight line through the optical centre, perpendicular to the plane of  
the lens, is the principal axis.

it is the reference line from which every distance ( a , b , f , 2f ) is measured.

**Note on the printed lesson.** In the book the questions of this classwork are numbered 1, 3, 4 — number 2 is missing from the printed page. Nothing is lost; the numbering here has simply been made continuous.

choice B — the principal axis

### Part B — Home assignment 1

#### 4 why a concave lens always gives a virtual, erect, diminished image

take the two standard rays from the top of the object :

ray 1 : parallel to the axis, then refracted so that it seems to come from the principal focus F on the same side as the object.

ray 2 : straight through the optical centre, undeviated.

after the lens the two rays diverge , so they can never really cross :

→ only their backward extensions meet → the image is VIRTUAL

both extensions meet above the axis, on the same side as the object :

→ the image is ERECT

algebraically, with f negative and a positive :

$$1/b = -1/f - 1/a \quad \rightarrow \quad b = -af / (a + f)$$

$$M = |b| / a = f / (a + f) < 1 \quad \text{for every positive } a$$

→ the image is DIMINISHED , and it always lies between F and the lens

The last line is the real proof:  $f/(a+f)$  is a fraction whose denominator is bigger than its numerator for **any** positive a, so M can never reach 1. This is exactly why a diverging lens is used in spectacles for short sight, and in a door viewer — it shrinks a wide scene into a small erect picture.

**always virtual, erect and diminished — whatever the object distance**

#### 5 the ray diagram of the magnifying glass

the object is placed inside the focal length :  $a < f$

ray 1 : parallel to the axis, then refracted through the far focus F'.

ray 2 : straight through the optical centre O.

after the lens the two refracted rays are still diverging , so they are traced backwards ( dashed ) until their extensions meet.

$$1/b = 1/f - 1/a, \quad \text{and} \quad a < f \quad \rightarrow \quad b \text{ is negative}$$

$$b = -af / (f - a) \quad M = f / (f - a) > 1$$

The closer the object creeps towards F, the smaller  $(f - a)$  becomes and the larger M grows — which is exactly what you feel when you slide a magnifying glass until the print suddenly leaps in size just before it blurs.

| position of the object | position of the image   | nature of the image        |
|------------------------|-------------------------|----------------------------|
| beyond 2F              | between F and 2F        | real, inverted, diminished |
| at 2F                  | at 2F                   | real, inverted, equal      |
| between F and 2F       | beyond 2F               | real, inverted, magnified  |
| at F                   | at infinity             | no image is formed         |
| inside F               | same side as the object | virtual, erect, magnified  |

**the extensions meet on the object side → a virtual, erect, magnified image**

**6** the two virtual images a focal length apart

convex lens, virtual image ( so  $a < f$  ) :

$$1/b_1 = 1/f - 1/a = (a - f) / (a f)$$

$$|b_1| = a f / (f - a) \quad \text{on the object side}$$

concave lens of the same  $f$  (  $f$  is negative ) :

$$1/b_2 = -1/f - 1/a = - (a + f) / (a f)$$

$$|b_2| = a f / (a + f) \quad \text{also on the object side}$$

both images lie on the same side, so the distance between them is

$$\begin{aligned} |b_1| - |b_2| &= a f / (f - a) - a f / (a + f) \\ &= a f [ (a + f) - (f - a) ] / (f - a)(f + a) \\ &= 2 a^2 f / (f^2 - a^2) \end{aligned}$$

this distance is given equal to  $f$  :

$$2 a^2 f / (f^2 - a^2) = f$$

$$2 a^2 = f^2 - a^2$$

$$3 a^2 = f^2$$

$$a = f / \sqrt{3} = ( \sqrt{3} / 3 ) f \approx 0.58 f$$

The answer had to come out smaller than  $f$ , and it does — a convex lens can only make a virtual image when the object is inside the focus. Notice also that the convex image is the farther of the two:  $a f / (f - a)$  is always bigger than  $a f / (a + f)$ , which is why the subtraction was taken in that order.

$$a = f / \sqrt{3} \approx 0.58 f$$

**7** an object far beyond  $2F$

$$1/b = 1/f - 1/a, \quad \text{with } a \text{ very large}$$

$$1/a \rightarrow \text{very small} \rightarrow b \rightarrow f$$

so the image settles just outside the focus, between  $F$  and  $2F$  :

$$M = b / a \approx f / a \ll 1$$

$b$  is positive  $\rightarrow$  the image is real and inverted

This is the camera and the eye. A very distant object throws its image practically **at** the focus, tiny and upside down — which is why the sensor of a camera sits one focal length behind the lens when you photograph a far-away mountain.

| position of the object | position of the image   | nature of the image        |
|------------------------|-------------------------|----------------------------|
| beyond $2F$            | between $F$ and $2F$    | real, inverted, diminished |
| at $2F$                | at $2F$                 | real, inverted, equal      |
| between $F$ and $2F$   | beyond $2F$             | real, inverted, magnified  |
| at $F$                 | at infinity             | no image is formed         |
| inside $F$             | same side as the object | virtual, erect, magnified  |

**choice A — real, inverted and diminished**

8 the meaning of  $M = 0.5$

$$M = h' / h = 0.5 \quad \rightarrow \quad h' = 0.5 h$$

the image height is half the object height.

M answers only one question: **how big**. Whether the image is erect or inverted is carried by the **sign**, and the printed  $M = 0.5$  carries no sign, so nothing at all can be said here about the orientation — which is why choice D is wrong.

**choice C — the image is diminished to half the object**

## Part C — Classwork 2

9 the wavelength from the fringe separation

from the figure :  $d = 0.3 \text{ mm} = 0.3 \times 10^{-3} \text{ m}$  ,  $L = 3 \text{ m}$

given :  $\Delta y = 5 \text{ mm} = 5 \times 10^{-3} \text{ m}$

$$\Delta y = \lambda L / d \quad \rightarrow \quad \lambda = \Delta y \cdot d / L$$

$$\lambda = ( 5 \times 10^{-3} \times 0.3 \times 10^{-3} ) / 3$$

$$\lambda = ( 1.5 \times 10^{-6} ) / 3$$

$$\lambda = 5 \times 10^{-7} \text{ m} = 500 \text{ nm}$$

A sanity check you should always make: visible light runs from about 400 nm (violet) to about 700 nm (red). An answer of 500 nm sits comfortably in the middle, in the green. If a double-slit calculation ever hands you microns or nanometres by the thousand, a power of ten has slipped.

**$\lambda = 5 \times 10^{-7} \text{ m} = 500 \text{ nm}$  (green light)**

10 which relation gives  $\lambda$

the fringe separation in Young's experiment is

$$\Delta y = \lambda R / d$$

rearranged, the three correct forms are

$$\lambda = \Delta y \cdot d / R$$

$$R = \Delta y \cdot d / \lambda$$

$$d = \lambda R / \Delta y$$

testing the four printed choices :

A)  $\lambda = \Delta y R / d$       $\times$      ( R and d are interchanged )

B)  $R = \Delta y d / \lambda$       $\checkmark$

C)  $\Delta y = \lambda d / R$       $\times$      ( d and R are interchanged )

D)  $d = \lambda \Delta y / R$       $\times$      (  $\Delta y$  and R are interchanged )

**Note on the printed question.** The wording asks for the relation that gives  $\lambda$ , which points at choice A — but A has R and d the wrong way round and is wrong. The only sound statement among the four is B. Keep the master form  $\Delta y = \lambda R / d$  in your head and read the others off it:  $\lambda$  and R are always **upstairs** together, d is always **downstairs**.

**choice B — it is the only algebraically correct relation of the four**

11 halving the slit separation

$$\Delta y = \lambda R / d \quad \rightarrow \quad \Delta y \propto 1 / d \quad ( \lambda \text{ and } R \text{ unchanged} )$$

$$d_2 = d_1 / 2$$

$$\Delta y_2 / \Delta y_1 = d_1 / d_2 = d_1 / ( d_1 / 2 ) = 2$$

$$\Delta y_2 = 2 \Delta y_1$$

Inverse proportion is the trap here. Squeezing the two slits **closer** pushes the fringes **farther** apart — the same reason a very narrow gap spreads a wave over a wide angle. Everything that makes the geometry tighter makes the pattern wider.

**choice D —  $2 \Delta y_1 = \Delta y_2$ , the fringes spread to twice their separation**

## Part D — Home assignment 2

### 12 the fringe separation for 450 nm

$$\lambda = 450 \text{ nm} = 450 \times 10^{-9} \text{ m} = 4.5 \times 10^{-7} \text{ m}$$

$$d = 0.3 \text{ mm} = 3 \times 10^{-4} \text{ m}, \quad L = 2.0 \text{ m}$$

$$\Delta y = \lambda L / d$$

$$\Delta y = (4.5 \times 10^{-7} \times 2.0) / (3 \times 10^{-4})$$

$$\Delta y = (9 \times 10^{-7}) / (3 \times 10^{-4})$$

$$\Delta y = 3 \times 10^{-3} \text{ m} = 3 \text{ mm}$$

Three millimetres between fringes on a screen two metres away is typical for a school set-up, and it is why the slits have to be a fraction of a millimetre apart. Open them to a millimetre and  $\Delta y$  drops to under a millimetre — the pattern becomes a blur.

$$\Delta y = 3 \text{ mm}$$

### 13 two patches, or a row of fringes?

each slit is narrow enough to diffract the light, so the two beams spread out and overlap on the wall; being cut from one wavefront they are coherent (a constant phase difference).

wherever the path difference is  $\Delta = m \lambda \rightarrow$  constructive  $\rightarrow$  bright

wherever the path difference is  $\Delta = (m + \frac{1}{2}) \lambda \rightarrow$  destructive  $\rightarrow$  dark

these conditions repeat right across the wall, so the result is a row of many bright and dark fringes, not two separate patches.

at the centre (point 0)  $S_1O = S_2O \rightarrow \Delta = 0 \rightarrow$  always bright

Two patches is what you would get from two **independent** lamps, because their phase difference would change millions of times a second and the fringes would wash out. The whole point of Young's single source lighting both slits is to make the two beams coherent.

**a row of many bright fringes separated by dark ones, and the brightest is the central fringe opposite the midpoint between the two slits**

### 14 finding the slit separation

$$\lambda = 600 \text{ nm} = 6 \times 10^{-7} \text{ m}, \quad L = 1.2 \text{ m}, \quad \Delta y = 2.4 \text{ mm} = 2.4 \times 10^{-3} \text{ m}$$

$$\Delta y = \lambda L / d \rightarrow d = \lambda L / \Delta y$$

$$d = (6 \times 10^{-7} \times 1.2) / (2.4 \times 10^{-3})$$

$$d = (7.2 \times 10^{-7}) / (2.4 \times 10^{-3})$$

$$d = 3 \times 10^{-4} \text{ m} = 0.3 \text{ mm}$$

Notice how small  $d$  has to be. The slit separation comes out about 500 wavelengths — and that is exactly why Young's experiment had to wait for careful ruling: any wider and the fringes crowd into an unreadable band.

$$d = 3 \times 10^{-4} \text{ m} = 0.3 \text{ mm}$$

**15** how to widen the fringes

$$\Delta y = \lambda L / d$$

to make  $\Delta y$  bigger : increase  $\lambda$  , increase  $L$  , or decrease  $d$ .

checking the printed choices :

- A) decrease  $L$  →  $\Delta y$  decreases ✗
- B) source-to-slit distance does not appear in the law ✗
- C) increase  $d$  →  $\Delta y$  decreases ✗
- D) green → red ,  $\lambda$  larger →  $\Delta y$  increases ✓

Red is the long-wavelength end of the visible spectrum ( $\approx 700$  nm against  $\approx 530$  nm for green), so simply swapping the colour stretches the pattern by about a third. Choice B is the distractor worth naming: the distance from the lamp to the slits changes the **brightness**, never the fringe spacing.

**choice D — replace the green source by a red one**

**16** doubling the screen distance

$$\Delta y = \lambda L / d \quad \rightarrow \quad \Delta y \propto L$$

$$L_2 = 2 L_1 \quad \rightarrow \quad \Delta y_2 = 2 \Delta y_1$$

but the same amount of light is now spread over twice the area, so each fringe is fainter and its edges are less sharp :

→ the separation doubles , the clarity ( contrast ) decreases

This is the trade-off every experimenter meets: you can always spread a pattern out, but you pay for the space with brightness. Doubling  $L$  spreads the same energy over four times the area, so the fringes are about four times dimmer.

**choice A — the separation doubles and the clarity of the fringes decreases**

## Part E — Review, Group A

**17** convex lens,  $f = 10$  cm,  $a = 15$  cm

$$1/f = 1/a + 1/b$$

$$1/b = 1/10 - 1/15 = ( 3 - 2 ) / 30 = 1 / 30$$

$$b = + 30 \text{ cm} \quad ( \text{positive} \rightarrow \text{real, on the far side} )$$

$$M = b / a = 30 / 15 = 2 \quad ( \text{magnified} )$$

Read the position first and the answer follows without any arithmetic:  $f = 10$  so  $2F = 20$ , and the object at 15 cm sits **between F and 2F** — the one region that always gives a real, inverted, magnified image. This is the projector setting.

| position of the object | position of the image   | nature of the image        |
|------------------------|-------------------------|----------------------------|
| beyond 2F              | between F and 2F        | real, inverted, diminished |
| at 2F                  | at 2F                   | real, inverted, equal      |
| between F and 2F       | beyond 2F               | real, inverted, magnified  |
| at F                   | at infinity             | no image is formed         |
| inside F               | same side as the object | virtual, erect, magnified  |

**choice B — a real, magnified (and inverted) image, 30 cm from the lens**

**18** which student gets the clearest fringes

the fringes are easiest to tell apart when  $\Delta y = \lambda L / d$  is largest.

taking  $d$  as the common unit :

A)  $\Delta y = 500 \times 10^{-9} \times 0.5 / (2 d) = 1.25 \times 10^{-7} / d$

B)  $\Delta y = 400 \times 10^{-9} \times 1.0 / (1 d) = 4.00 \times 10^{-7} / d$

C)  $\Delta y = 600 \times 10^{-9} \times 1.5 / (3 d) = 3.00 \times 10^{-7} / d$

D)  $\Delta y = 450 \times 10^{-9} \times 1.1 / (0.5 d) = 9.90 \times 10^{-7} / d \quad \leftarrow \text{the largest}$

Do not be tempted by the longest wavelength alone. Choice C has the reddest light of the four and still loses, because its slits are six times farther apart than D's. In  $\lambda L/d$  all three letters vote, and here the small  $d$  carries the day.

**choice D — 450 nm, 0.5 d, 1.1 m gives about 2.5 times the fringe separation of choice B**

**19** the fringes are too close together

$$\Delta y = \lambda L / d$$

so  $\Delta y$  can be made larger by

1. moving the screen farther from the double slit ( increase  $L$  ) ,
2. using a double slit whose two openings are closer ( decrease  $d$  ) ,
3. using light of a longer wavelength – swap the red filter for nothing less than red, i.e. use the reddest source available ( increase  $\lambda$  ).

The figure shows a red filter already in place, so the wavelength is close to the visible maximum and little more can be won there. The two changes with real room to move are  $L$  and  $d$  — which is why they are the two answers a marker will be looking for.

**any two of: increase the slit–screen distance  $L$ , use a double slit with a smaller separation  $d$ , or use light of a longer wavelength  $\lambda$**

**20** the two lens positions

the magnification is defined by

$$M = h' / h \quad \rightarrow \quad h' = M h$$

first position :  $h'_1 = M_1 h = 4 h$

second position :  $h'_2 = M_2 h = 0.25 h = h / 4$

The pair is no accident: for a fixed object–screen distance the two lens positions are mirror images of each other (object and image distances simply swap), so the two magnifications are always reciprocals — 4 and  $\frac{1}{4}$ . Multiply them and you get exactly 1, which is a quick check on any answer of this type.

**the first image is 4 h long and the second is 0.25 h long**

**21** widening the fringes with a green source

$$\Delta y = \lambda L / d$$

to increase  $\Delta y$ , change any one of the three :

- increase  $L$  – move the screen farther from the double slit ,
- decrease  $d$  – use a double slit with the two openings closer together ,
- increase  $\lambda$  – replace the green source with a red one  
(  $\lambda_{\text{red}} \approx 700 \text{ nm} > \lambda_{\text{green}} \approx 530 \text{ nm}$  ).

The one thing that will **not** work is turning the lamp up. Intensity changes how bright the fringes are, not where they fall; the geometry of  $\Delta y = \lambda L/d$  contains no term for the power of the source.

**increase  $L$ , decrease  $d$ , or move to a longer wavelength (green  $\rightarrow$  red)**

**22** the image on the wall

an image caught on a wall is a real image, and only a convex lens can make one → the lens is convex.

$$b = 30 \text{ cm} \quad (\text{lens-to-wall distance})$$

$$M = b / a = 2 \quad \rightarrow \quad a = b / 2 = 30 / 2$$

$$a = 15 \text{ cm}$$

$$\text{check : } 1/f = 1/15 + 1/30 = 3/30 \quad \rightarrow \quad f = 10 \text{ cm} \quad (\text{a sensible convex lens})$$

The single word “wall” settles half the question before any algebra: a concave lens produces only virtual images, and a virtual image cannot be caught on a screen. That kills choices A and D immediately.

**choice C — a convex lens with the object 15 cm away**

**23** Young with  $d = 0.2 \text{ mm}$ ,  $\lambda = 600 \text{ nm}$ ,  $L = 1 \text{ m}$

$$\lambda = 6 \times 10^{-7} \text{ m}, \quad d = 0.2 \text{ mm} = 2 \times 10^{-4} \text{ m}, \quad L = 1 \text{ m}$$

$$\Delta y = \lambda L / d$$

$$\Delta y = (6 \times 10^{-7} \times 1) / (2 \times 10^{-4})$$

$$\Delta y = 3 \times 10^{-3} \text{ m} = 3 \text{ mm}$$

Choice D (1 m) is there to catch a slip in the powers of ten, and choice C (30 mm) a slip of one power. The habit that protects you is to turn every length into metres in the first line, before anything is divided.

**choice A —  $\Delta y = 3 \text{ mm}$**

**24** the lens that magnifies 4 times twice over

first position – erect, magnified  $4 \times$  → the image is virtual,  $b$  is negative :

$$M = |b| / a_1 = 4 \quad \rightarrow \quad |b| = 4 a_1$$

$$1/f = 1/a_1 - 1/(4 a_1) = 3 / (4 a_1) \quad \rightarrow \quad a_1 = 3 f / 4$$

second position – real, inverted, magnified  $4 \times$  →  $b$  is positive :

$$M = b / a_2 = 4 \quad \rightarrow \quad b = 4 a_2$$

$$1/f = 1/a_2 + 1/(4 a_2) = 5 / (4 a_2) \quad \rightarrow \quad a_2 = 5 f / 4$$

the object was moved away by 6 cm :

$$a_2 - a_1 = 6$$

$$5f/4 - 3f/4 = 6$$

$$2f/4 = 6 \quad \rightarrow \quad f / 2 = 6$$

$$f = 12 \text{ cm}$$

$$\text{check : } a_1 = 9 \text{ cm} \quad (\text{inside } F = 12, \text{ virtual } \checkmark) \quad a_2 = 15 \text{ cm} \quad (\text{between } F \text{ and } 2F \checkmark)$$

The two positions straddle the focus, one on each side, and that is what makes the problem work: the same lens gives the same size of image twice, once virtually (inside  $F$ ) and once really (between  $F$  and  $2F$ ). The check at the end is worth writing every time — it confirms both positions land in the regions the wording described.

**$f = 12 \text{ cm}$  — the object moves from 9 cm to 15 cm**

**25** the two lens positions for a fixed 90 cm

let the object-to-lens distance be  $a$  ; the screen is 90 cm from the object,  
so the lens-to-screen distance is  $b = 90 - a$

$$1/f = 1/a + 1/b$$

$$1/20 = 1/a + 1/(90 - a)$$

$$1/20 = (90 - a + a) / [a(90 - a)]$$

$$1/20 = 90 / [a(90 - a)]$$

$$a(90 - a) = 1800$$

$$a^2 - 90a + 1800 = 0$$

$$a = [90 \pm \sqrt{(8100 - 7200)}] / 2 = (90 \pm 30) / 2$$

$$a = 60 \text{ cm} \quad \text{or} \quad a = 30 \text{ cm}$$

The two roots add up to 90, and that is the whole story: the object and image distances simply swap between the two positions (30 & 60, then 60 & 30). One gives a magnified image ( $M = 2$ ), the other a diminished one ( $M = \frac{1}{2}$ ) — this is the standard “displacement method” for measuring a focal length.

**$a = 30 \text{ cm}$  or  $60 \text{ cm}$  — two lens positions give a sharp image**

**26** the fringe at P when  $\Delta = 1.5 \lambda$

$$\Delta = S_2P - S_1P = 1.5 \lambda$$

write it in the standard form :

$$1.5 \lambda = (1 + \frac{1}{2}) \lambda \quad \rightarrow \quad m = 1$$

this matches the destructive condition  $\Delta = (m + \frac{1}{2}) \lambda$

so the two waves arrive at P exactly out of phase (half a cycle apart),  
a crest of one meeting a trough of the other, and they cancel.

The rule to carry into the exam: divide the path difference by  $\lambda$ . A **whole** number (0, 1, 2 ...) means bright; a whole number **and a half** (0.5, 1.5, 2.5 ...) means dark. Here 1.5 is the second dark fringe out from the centre.

**a dark fringe — the path difference is an odd number of half wavelengths, so the two waves arrive out of phase and interfere destructively**

## Part G — Review, Group C

**27** red light instead of blue

$$\Delta y = \lambda L / d \quad \rightarrow \quad \Delta y \propto \lambda \quad (L \text{ and } d \text{ unchanged})$$

$$\lambda_{\text{red}} \approx 700 \text{ nm} \quad \lambda_{\text{blue}} \approx 450 \text{ nm}$$

$$\lambda_{\text{red}} > \lambda_{\text{blue}} \quad \rightarrow \quad \Delta y \text{ increases}$$

Roughly,  $\Delta y$  grows by the ratio  $700/450 \approx 1.6$ , so the pattern stretches by about half again. This is also why a white-light double slit shows a white centre with coloured edges: every colour makes its own pattern, all sharing the same central fringe.

**choice A — the separation increases, because the wavelength of red light is larger**

**28** concave lens,  $f = 20$  cm,  $a = 20$  cm

for a concave lens  $f$  is negative :  $f = -20$  cm

$$1/f = 1/a + 1/b$$

$$-1/20 = 1/20 + 1/b$$

$$1/b = -1/20 - 1/20 = -2/20 = -1/10$$

$$b = -10 \text{ cm} \quad (\text{negative} \rightarrow \text{virtual, on the object side})$$

$$M = |b| / a = 10 / 20 = 0.5 \quad (\text{diminished})$$

$b$  negative with  $a$  positive  $\rightarrow$  erect

Choice A is the trap: "the object is at the focal distance" sends the image to infinity only for a **convex** lens. A concave lens has no such special position — it gives the same three adjectives for every object distance, as question 4 proved.

**choice B — virtual, erect and diminished, 10 cm from the lens**

**29** the ratio of the two fringe separations

$$\lambda_1 = 6000 \text{ \AA} = 6 \times 10^{-7} \text{ m} \quad R_1 = R$$

$$\lambda_2 = 4000 \text{ \AA} = 4 \times 10^{-7} \text{ m} \quad R_2 = 2R$$

the slit separation  $d$  is the same in both cases :

$$\Delta y_1 = \lambda_1 R_1 / d = 6000 R / d$$

$$\Delta y_2 = \lambda_2 R_2 / d = 4000 \times 2 R / d = 8000 R / d$$

$$\Delta y_1 / \Delta y_2 = 6000 / 8000$$

$$\Delta y_1 / \Delta y_2 = 3 / 4 = 0.75$$

Because it is a ratio, the ångströms never have to be converted — the units cancel, and so does  $d$  and the unknown  $R$ . Always look for that before reaching for a calculator: two of the three letters in  $\lambda R/d$  dropped out here on their own.

$$\Delta y_1 / \Delta y_2 = 3 : 4$$

**30** why the image speeds up near the focus

for a convex lens  $1/f = 1/a + 1/b \rightarrow b = a f / (a - f)$

differentiate  $b$  with respect to  $a$  ( $f$  is a constant) :

$$db/da = [f(a - f) - af] / (a - f)^2$$

$$db/da = -f^2 / (a - f)^2$$

the object moves at a uniform speed  $v = |da/dt|$

so the speed of the image is

$$v' = |db/dt| = |db/da| \times |da/dt|$$

$$v' = f^2 / (a - f)^2 \times v$$

and since  $M = f / (a - f)$  :

$$v' = M^2 v$$

as the object approaches  $F$ ,  $(a - f) \rightarrow 0$ , so  $M$  grows without limit :

$\rightarrow v'$  depends on  $a$ , therefore it is VARIABLE

$\rightarrow v'$  grows as the object nears the focus, and  $\rightarrow \infty$  at  $a = f$

The minus sign in  $db/da$  carries its own meaning: as the object comes **towards** the lens the image runs **away** from it. And because the factor is  $M^2$ , not  $M$ , the effect is dramatic — at  $M = 10$  the image is already moving a hundred times faster than the object. This is exactly why focusing a camera on something very close is so delicate.

**$v' = M^2 v$  — the image speed depends on the object position, so it is variable, and it increases without limit as the object approaches  $F$**

**31** why the central fringe is always bright

the central point O lies on the perpendicular bisector of  $S_1S_2$  , so

$$S_1O = S_2O$$

$$\Delta = S_2O - S_1O = 0 = 0 \times \lambda \quad \rightarrow \quad m = 0$$

a zero path difference means the two waves arrive exactly in phase ,  
crest on crest, for every wavelength in the beam ,  
so they reinforce and the centre is always the brightest fringe.

“For every wavelength” is the part worth remembering. Every other fringe sits at a position that depends on  $\lambda$ , so in white light they spread into colours; only the central fringe is fixed at  $\Delta = 0$  for all of them, which is why it comes out white and why it is used as the reference point of the whole pattern.

**because the two distances from the slits to the centre are equal, the path difference there is zero for every wavelength — the waves always arrive in phase and interfere constructively**